

Theories of deliberated choice

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1. Introduction

Acts of choice are the basis of much work in economics. Such acts have been used, among others, to give an observable basis to the concept of preference [Samuelson, 1983].

Most of these approaches consider only acts of choice that are performed spontaneously in the normal course of action of an individual, or that are given a specific frame depending on the psychological aspect that is to be studied. As an example of the former kind, typical willingness to pay studies asks the subject whether she prefers an object at some price or another object at another price, describing the objects in a way that is considered immediately understandable by the subject (or simply showing the objects). Examples of the latter kind include the celebrated work of Kahneman and Tversky [Bell et al., 1988, Kahneman, 2012].

On the other hand, since at least the seminal work of Fishkin [2011], another sort of judgment involving subjective desirability is emerging as also worth studying. These are judgments given by subjects after arguments in favor and against different possible stance have been considered. Such judgments may differ from immediate judgments of desirability given in natural conditions. Indeed, individuals may change opinion on the desirability of some course of action, and therefore revise their choice, while learning about the properties of objects, the consequences of some acts (or empirical knowledge useful to estimate the likelihood of some consequences), or the logical or empirical impossibilities between consequences. This is particularly likely to happen when choosing among non everyday objects or acts whose consequences are not well known by the subject. This article more generally considers all sorts of influences, including arguments, provided they can be targeted to an individual and the choice after exposure be observed. TODO: rename argument to influence throughout.

While numerous articles have discussed the concept of deliberation in the last decades, how to observe a deliberated judgment in a systematic and reproducible fashion remains unsettled. There is a need to define this concept sufficiently precisely that, first, it is clear what counts as a deliberated judgment (i.e., when “sufficient” [Meinard and Cailloux, 2020] or “correct” deliberation has happened); and second, it is clear which part of the claim is empirical and which part stems from an unfalsifiable conception of what “deliberated” means. This permits to study deliberated judgments empirically, meaning here that disagreements about the deliberated judgment of a set of individuals in a given context can be solved, at least in principle, by an empirical experiment, provided the very conception of what constitutes a deliberated judgment is shared. This article is interested in defining the concept of deliberated judgement of a subject and show how one can study it in an empirical manner. The “classical” counterpart of deliberated judgments, that does not involve confrontation with arguments, will henceforth be called the natural judgment. As we deal with choice acts, we will also name the concept “deliberated choice”, as opposed to “natural choice”.

While the concept of deliberation in the deliberative democracy literature usually refers to multiple individuals debating, this article considers deliberation as related to a single individual: an individual deliberates (in the sense of: carefully weighs) the stance she considers most appropriate while being confronted to arguments. The deliberated judgment of an individual is thus defined with no requirement that the arguments be uttered by peers in a debate; they can be given by an automaton, for example. In this sense, deliberated judgments come close to what Rawls [1999] calls reflective equilibrium (referring to an idea of Goodman [1983]), a concept that has much inspired the present proposal.

To define deliberated judgments, one needs to start with the set of arguments under consideration, and the way the subject gets to know them, called hereafter the “exposure protocol”. As a result, we need always talk about the deliberated judgment *given some exposure protocol*. The notion of argument invoked here is an extremely large one, including texts, images, sounds, experiences (a hiking trip), and so on: anything that can be transmitted to a subject qualifies as a possible argument. In particular, such an “argument” need not have the logical structure of what is considered a proper argument in argumentation theory. This important feature of this proposal permits to avoid taking a position on what is a correct argument, and permits to study wide enough conceptions of deliberated judgment, including those involving no paternalism [Cailloux and Meinard, 2020]: anything that influences the judgment of a subject, as judged by the subject herself and not by any external standard, may a priori be considered worth including in at least some conceptions of deliberated judgment, thus including whatever some could consider as “fallacies” or “bad reasons”.

The exposure protocol defines a set of argument and, importantly, how arguments are sent to the subject and her subsequent choice observed. There is no explicit dependency on the internal process that occurs in the subject’s mind while she (possibly) processes the arguments, as it is not desirable to assume that the internal process could be observed. We consider subjects as black boxes and only require a capacity of observation of their choices after exposure to arguments. Depending on the exposure protocol and

the subject, it is of course possible for the arguments to never affect any choice of the subject, a conclusion that may be interesting in its own right.

The deliberated judgment of a subject is defined as a function of the exposure protocol, as accepting every claims that resists every counter-arguments. The notion of resistance is itself defined according to the exposure protocol.

Studying judgments resulting from exposure to arguments includes a difficulty that is not present with natural choice experiments: exposing an individual to arguments may change the individual's stance, an effect that cannot be erased in order to submit the individual to an unrelated sequence of arguments. The observer thus cannot in general assume that multiple unrelated sequences of arguments can be tested on a given individual. This difficulty will be worked around by considering general theories of deliberated judgments, that apply to sets of individuals. This is methodologically similar to defining theories of how objects break. One can break a given (unrepairable) object only once, so such theories must apply to sets of objects if one wants to be able to test multiple ways of falsifying the theories.

For conceptual clarity, I will focus on the case of a single binary decision to take, akin to a choice situation facing a bundle of two courses of action, exactly one of which must be adopted.

2. Exposure protocol and deliberated choice

I assume given two antagonistic acts, constituting the decision at stake, such as “buy this object for the price of \$15” and “do not buy this object for the price of \$15”, or “take this job offer” and “decline this job offer”. Let the set $P = \{\varphi, \varphi', 0\}$ denote the set of possible preferences, where φ denote a strict preference for the first act against the other one, φ' conversely, and 0 denotes indifference (both acts are considered equally worth).

2.1. Exposure protocol

An exposure protocol is a tuple $(I, \mathcal{A}, \rightsquigarrow)$, where I is a set of individuals, $\mathcal{A}$ is a set of arguments, and $\rightsquigarrow$, “leads to”, is the behavior of the individuals as a function of the arguments they have been exposed to.

While I and $\mathcal{A}$ are fundamental sets whose elements are considered elementary, a few notations and concepts are required to define $\rightsquigarrow$. Throughout the article, $\mathbb{N}$ designates the natural numbers (including 0) and $\mathbb{N}^* = \mathbb{N} \setminus \{0\}$ designates the positive numbers. Given $j, l \in \mathbb{Z}$, the notation $\llbracket j, l \rrbracket$ represents $\{k \in \mathbb{Z} \mid j \leq k \leq l\}$ (thus $\forall j \in \mathbb{N}^* : \llbracket j, 0 \rrbracket = \emptyset$). Define $\mathcal{A}^{<\mathbb{N}} = \bigcup_{k \in \mathbb{N}} \mathcal{A}^{\llbracket 1, k \rrbracket}$ as the finite sequences of arguments, including the empty sequence. A generic element of $\mathcal{A}^{<\mathbb{N}}$ (a possibly empty finite sequence of arguments) is denoted by α and called an argumentative path, or simply a path. Its length is denoted by $\#\alpha \in \mathbb{N}$ and its elements, $\alpha(k)$ or α_k . Thus, given an argumentative path $\alpha \in \mathcal{A}^{<\mathbb{N}}$, $\forall k \in \llbracket 1, \#\alpha \rrbracket : \alpha(k) = \alpha_k \in \mathcal{A}$, and $\alpha = \emptyset \Leftrightarrow \#\alpha = 0$. Throughout the text, functions are set-theoretically considered as sets of pairs (identified with their graph), so that for

example a path is seen as a subset of $\mathbb{N} \times \mathcal{A}$. A path $\alpha \in \mathcal{A}^{<\mathbb{N}}$ is a prefix of a path $\alpha' \in \mathcal{A}^{<\mathbb{N}}$ whenever $\alpha \subseteq \alpha'$, thus, $\forall k \in \llbracket 1, \#\alpha \rrbracket : \alpha_k = \alpha'_k$. Given any $\alpha \in \mathcal{A}^{<\mathbb{N}}$, $\llbracket \alpha \rrbracket = \{\alpha|_{\llbracket 1, k \rrbracket} \mid k \in \llbracket 0, \#\alpha \rrbracket\}$ denotes the set of prefixes of α .

The function $\rightsquigarrow : I \times \mathcal{A}^{<\mathbb{N}} \rightarrow P$ associates to each individual and path an element of P , $\rightsquigarrow(i, \alpha)$ representing the choice of i following exposure to the arguments of α , in order. That is, $\rightsquigarrow(i, \alpha) = \varphi$ denotes that i chooses the first act after exposure to α , $\rightsquigarrow(i, \alpha) = \varphi'$ denotes i choosing the second act and $\rightsquigarrow(i, \alpha) = 0$ is used to represent indifference of i . The notation $\alpha \rightsquigarrow_i p$, pronounced “ α leads i to p ”, will be used to mean $\rightsquigarrow(i, \alpha) = p$, considering $\rightsquigarrow_i \subseteq \mathcal{A}^{<\mathbb{N}} \times P$ as a binary relation. As usual with binary relations, $\alpha \not\rightsquigarrow_i p$ means $\rightsquigarrow(i, \alpha) \neq p$ or equivalently $\rightsquigarrow(i, \alpha) \in P \setminus \{p\}$.

2.2. Deliberated choice

The deliberated choice of i given the exposure protocol $(I, \mathcal{A}, \rightsquigarrow)$ is denoted $P_i \subseteq P$. Its formal definition requires the notion of decisive argument.

Given $\alpha \in \mathcal{A}^{<\mathbb{N}}$, $R(\alpha) = \alpha(\llbracket 1, \#\alpha \rrbracket)$ denotes the range of α , that is, all arguments contained in the sequence α . Let $\alpha_{end} = \alpha(\llbracket \max(1, \#\alpha - 1), \#\alpha \rrbracket)$ denote the set containing the last two arguments of the sequence α if the sequence has at least two elements; the unique argument of the sequence if it has exactly one element; and the empty set if it is empty.

Given $i \in I, p \in P, a \in \mathcal{A}$, define $a \hookrightarrow_i p$, pronounced “ a leads definitely i to p ”, iff $\forall \alpha \in \mathcal{A}^{<\mathbb{N}} \mid a \in \alpha_{end} : \alpha \rightsquigarrow_i p$. When $a \hookrightarrow_i p$, a is said to be a decisive argument for p from i ’s point of view given the exposure protocol. Another interpretation of this definition is that a then resists every counter-arguments, in the sense that no argument a' given at the end of any sequence of arguments, in presence of a , leads the individual to $p' \neq p$ (formally, $\nexists a' \in \mathcal{A}, \alpha \in \mathcal{A}^{<\mathbb{N}} \mid a, a' \in \alpha_{end} \wedge \alpha \rightsquigarrow_i p'$).

The deliberated choice of i is then considered as the set of propositions supported by decisive arguments.

Definition 1 (Deliberated choice). $\forall i \in I, p \in P : p \in P_i \Leftrightarrow [\exists a \in \mathcal{A} \mid a \hookrightarrow_i p]$.

It follows from the consideration of “end” as the last *two* arguments that the whole set of decisive arguments can argue for at most a single proposition, thus, that the deliberated choice cannot hold more than one proposition.

Theorem 1 (Deliberated choice singularity). $\forall i \in I : \#P_i \leq 1$.

Proof. Assume that $\exists a, a' \in \mathcal{A}, p, p' \in P \mid a \hookrightarrow_i p \wedge a' \hookrightarrow_i p'$. Since $a \hookrightarrow_i p$, $(a, a') \rightsquigarrow_i p$, and since $a' \hookrightarrow_i p'$, $(a, a') \rightsquigarrow_i p'$, hence $p' = p$. $\square$

The intended semantics is that P_i represents the preference of i over the proposed set of two acts, after deliberation, given the exposure protocol. In other words, if $P_i = \{\varphi\}$, the individual considers the first act strictly better than the second one, after deliberation (and conversely for $P_i = \{\varphi'\}$); if $P_i = \{0\}$, the deliberated judgment of i is that both acts are equally worth; and if $P_i = \emptyset$, no act is deliberately strictly better than the other one and they are also not deliberately exactly equivalent to each other.

This latter possibility may look unpleasant, but it is a fact that facing some difficult problems, individuals sometimes are undecided, and it is an a priori empirical possibility that the individual remains undecided (or even, *becomes* undecided) when considering multiple arguments. Also, being undecided is not the same state as being decidedly indifferent about two issues.

Given an behavior function $\rightsquigarrow$, deliberated choices are represented by an element $\pi(\rightsquigarrow) \subseteq I \times P$ defined as $\pi_i(\rightsquigarrow) = \pi(\rightsquigarrow)(i) = P_i$. Theorem 1 shows that $\pi(\rightsquigarrow)$ is right unique (meaning that $\forall i \in I : \#\pi(\rightsquigarrow)(i) \leq 1$). Equivalently, note that $(P \cup \{\emptyset\})^I$, where $\{\emptyset\}$ is used as a sentinel value for individuals with no deliberated judgment, can be interpreted as the right unique relations over $I \times P$ (discarding those individuals with no deliberated judgment); and then the deliberated choices are an element $\pi(\rightsquigarrow) \in (P \cup \{\emptyset\})^I$.

The deliberated judgment of a subject depends on the exposure protocol, that is, on the set of arguments and the way they are submitted to the subject, and not only on the set of arguments, because the very way that they are transmitted to the individual may affect her judgment differently. For example, reading an image description and showing an image to a subject may impress them differently [Railton, 2003].

2.3. An example and a discussion about observability

The following toy example is inspired by the following argument from Railton [2003, p. 59].

An experiment in cognitive psychology reveals that when subjects observe two individuals taking a test, one of whom answers a high percentage of the early questions correctly but then falters, while the other does poorly initially but then answers a high percentage of the later questions correctly, the first is usually viewed as more able, even though in the end each answers the same number of questions correctly in all. If the subjects had been asked which test-taker they would prefer as a math tutor, they likely would have answered “The first,” and perhaps they would also have been willing to pay a premium to secure his services. But if they were to become convinced of the influence of sheer order on their evaluations of relative ability, they presumably would no longer want their initial preference for the first test-taker to influence their choices – or willingness to pay – in this way.

Example 1 (Math tutor). Let I be a set of individuals desiring to evaluate two candidates $\{m_1, m_2\}$ for math tutoring. Consider one individual $i \in I$. Let $\rightsquigarrow(i, a_1)$ represent i being shown a test of candidate m_1 wherein m_1 answers 70% of the first half of questions correctly but then only 40% of the other half correctly. Let $\rightsquigarrow(i, a_2)$ represent i being shown a test of candidate m_2 wherein m_2 answers 45% of the first half of questions correctly, then 75% of the other half correctly. Let $\rightsquigarrow(i, a_e)$ represent i being given the following argument. “When tested, m_1 answers 70% of the first half of questions correctly but then only 40% of the other half correctly; whereas m_2 answers 45% of the first half of questions correctly, then 75% of the other half correctly. It follows that m_1

answers on average less well than m_2 . Observing the candidates take the tests may yield a contrary impression; this is a well-known psychological phenomenon depending on the order of the questions that the candidates answer correctly.” Let φ_1 represent choosing m_1 and φ_2 choosing m_2 (for this example we use $P = \{\varphi_1, \varphi_2, 0\}$).

If the prediction described in the quote applies to i , the relation $\rightsquigarrow_i$ satisfies, $\forall k \in \mathbb{N}^*$ (recall that for $A \subseteq \mathcal{A}$, A^k represents the set of sequences of length k using arguments from A): $\forall \alpha \in \{a_1, a_2\}^k, \alpha \rightsquigarrow_i \varphi_1$ and $\forall \alpha \in \mathcal{A}^k \mid a_e \in \alpha_{end}, \alpha \rightsquigarrow_i \varphi_2$. The latter in particular implies: $\forall \alpha \in \{a_e\}^k, \alpha \rightsquigarrow_i \varphi_2$; $\forall \alpha \in \{a_1, a_e\}^k \mid a_e \in \alpha_{end}, \alpha \rightsquigarrow_i \varphi_2$; $\forall \alpha \in \{a_2, a_e\}^k \mid a_e \in \alpha_{end}, \alpha \rightsquigarrow_i \varphi_2$. The values of $\rightsquigarrow(i, \alpha)$ for $\alpha = \emptyset$ and for $\alpha \in \mathcal{A}^k \mid a_e \notin \alpha_{end}$ are undetermined by the prediction, but whatever they are, a_e is a decisive argument for φ_2 , hence, i ’s deliberated choice is φ_2 . $\triangle$

Now that deliberated choices are defined, we need to explore what one can learn about deliberated choices using our empirically available data. This requires to define claims about deliberated choices and to explore how to validate claims.

3. Claims about deliberated choice

Given an exposure protocol $(I, \mathcal{A}, \rightsquigarrow)$, consider a claim about $\rightsquigarrow$, such as “every individuals in I choose deliberately φ ”, or, in example 1, “ a_e is a decisive argument for i in favor of φ_2 ”. Validating claims about deliberated choice is no trivial matter, for two reasons.

First, such claims must entail an infinity of constraints about $\rightsquigarrow$: in order to deduce that $p \in P_i$ for some $p \in P, i \in I$, the claim must constrain $\rightsquigarrow_i$ on every possible paths $\alpha \in \mathcal{A}^{<\mathbb{N}}$. It follows that a claim cannot be validated empirically exhaustively by observing the value of $\rightsquigarrow$ on every points that it constraints.

Second, our observations are limited in important ways: once we have shown some arguments to a given individual i , we cannot undo the effects of this action on i and show to her a different argument as if starting from scratch (unless assuming i forgets arguments after a while and waiting for sufficiently long, which would create practical and theoretical difficulties). In other words, once we have shown a path α to i , we can only continue the protocol with i by showing paths α' that extend α .

To illustrate this point with example 1, let $\alpha = (a_1, a_e, a_2)$ designate the argumentative path where the arguments are shown in the order they are described in example 1, and let $\alpha' = (a_2, a_e, a_1)$ designate the reverse order. The deliberated choice concept is defined so as to not depend on the order of the arguments shown to i , as long as the end of the path features a decisive argument: for having $p \in P_i$, it must be that directing to i the path α or the path α' both lead i to choose p . But it is impossible to observe both the value of $\rightsquigarrow(i, \alpha)$ and the value of $\rightsquigarrow(i, \alpha')$.

The first difficulty is familiar as it occurs in any endeavor to validate general claims in science: claims that cover a large (or infinite) spectrum cannot be validated by observing the system exhaustively. One must rather develop a falsification protocol, that is, a way to invalidate the claim, and then observe that the claim is not invalidated. This is the approach taken in the next section.

This approach also permits to tackle the second difficulty, by letting us study theoretically the needs and limits of falsification given our observable basis.

3.1. Claims and observable basis

Note first that given a relation $\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}}$, $\mathcal{P}(\rightsquigarrow)$ denotes the set of relations over $(I \times \mathcal{A}^{<\mathbb{N}}) \times P$ that refine $\rightsquigarrow$ (thus, each element $f \in \mathcal{P}(\rightsquigarrow)$ is right-unique but non necessarily total, meaning that it does not necessarily cover $I \times \mathcal{A}^{<\mathbb{N}}$). Given sets A and B , $A \subseteq_{\text{fin}} B$ denotes that A is a finite subset of B , thus $A \subseteq_{\text{fin}} B$ iff $A \subseteq B$ and A is finite. Given a function f on some domain $\mathcal{A}$, given $A \subseteq \mathcal{A}$, we write $f(A) = \{f(a) \mid a \in A\}$ to denote the images of each element in A . This creates no ambiguity because the functions appearing in this article apply on domains whose elements are never subsets of the domain (thus for a domain A , $a \in A \Rightarrow a \not\subseteq A$).

Given a pair $(I, \mathcal{A})$, a claim C is a set of possibilities for $\rightsquigarrow$, thus, $C \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$. Given a formula ψ , let the corresponding claim be $C_\psi = \{\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}} \mid \rightsquigarrow \text{ satisfies } \psi\}$. A claim is trivial iff it is the trivial claim, defined as $C = P^{I \times \mathcal{A}^{<\mathbb{N}}}$. A claim is finite iff its cardinal is finite (meaning that it is compatible with a finite number of possibilities). Let $C^\subseteq = \bigcup_{\rightsquigarrow \in C} \mathcal{P}(\rightsquigarrow) = \{c \subseteq (I \times \mathcal{A}^{<\mathbb{N}}) \times P \mid \exists \rightsquigarrow \in C \mid c \in \mathcal{P}(\rightsquigarrow)\}$ denote the set of non necessarily complete possibilities compatible with C (thus, each element $c \in C^\subseteq$ is right-unique but non necessarily total, meaning that it does not necessarily cover $(I \times \mathcal{A}^{<\mathbb{N}})$). Note that $C^\subseteq$ is downwards closed, meaning that $\forall c \in C^\subseteq : c' \subseteq c \Rightarrow c' \in C^\subseteq$.

Define the increasing sets of paths as the set $\mathcal{I}(\mathcal{A}^{<\mathbb{N}}) = \{H \subseteq_{\text{fin}} \mathcal{A}^{<\mathbb{N}} \mid \forall \alpha, \alpha' \in H : \# \alpha \leq \# \alpha' \Rightarrow \alpha \subseteq \alpha'\} \subseteq \mathcal{P}(\mathcal{A}^{<\mathbb{N}})$. An increasing set of paths $H \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$ can be strictly ordered by the lengths of the paths in H and can thus also be considered as a sequence of paths. Note that $\mathcal{I}(\mathcal{A}^{<\mathbb{N}})$ includes both the empty set of paths $\emptyset$ and the set of paths $\{\emptyset\}$ containing only the empty path. A non empty set of paths $H \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$ has a unique maximal element $\{\max H\} = \{\alpha \in H \mid \forall \alpha' \in H : \alpha' \subseteq \alpha\}$. A non empty set of paths $H \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$ contains all its prefixes (meaning, $\forall \alpha \in H : \{\alpha' \subseteq \alpha\} \subseteq H$) iff $H = \llbracket \max H \rrbracket$. Let $\mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I)$ denote the functions $f : I \mapsto \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$ with finite support, thus, satisfying $\#f^{-1}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}}) \setminus \emptyset) \in \mathbb{N}$. An element $Q \subseteq I \times \mathcal{A}^{<\mathbb{N}}$ is called a query iff $\forall i \in I : Q(i) = \{\alpha \in \mathcal{A}^{<\mathbb{N}} \mid (i, \alpha) \in Q\} \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$ and $\#Q \in \mathbb{N}$. Abusing notations, we also view Q as an element of $\mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I)$. An observation associates to a query an element of P per queried element, thus, the set of possible observations is $\Theta = \bigcup_{Q \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I)} P^Q$. In other words, defining a general observation O as a right unique finite subset of $(I \times \mathcal{A}^{<\mathbb{N}}) \times P$, an observation $\theta \in \Theta$ is a general observation satisfying $\forall i \in I, (i, \alpha, p), (i, \alpha', q) \in \theta : \# \alpha \leq \# \alpha' \Rightarrow \alpha \subseteq \alpha'$. Given a general observation O , let $O(P) \subseteq I \times \mathcal{A}^{<\mathbb{N}}$ denote the queried elements (the pairs (i, α)) in O and $O(\mathcal{A}^{<\mathbb{N}}, P)$ designate the observed individuals. Note that given an observation θ , $\theta(P) \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I)$. Similarly, we can view an observation θ as a function $\theta : I \mapsto \bigcup_{H \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})} P^H$ and thus $\Theta = \mathcal{F}((\bigcup_{H \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})} P^H)^I)$, where $\mathcal{F}$ designates the finite relations (considered as a set of triples). An observation θ' rightward-extends an observation θ , denoted $\theta \sqsubseteq \theta'$, iff $\theta \subseteq \theta' \wedge \forall i \in I, (i, \alpha, p) \in \theta, (i, \alpha', p') \in \theta' : \alpha \subseteq \alpha'$.

An anonymous observation $\mathbf{g} \in \bigcup_{H \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})} P^H$ associates each path from an increasing

set of paths to an element of P . We let $\{i\} \times \mathbf{g} = \{(i, \alpha, p) \mid (\alpha, p) \in \mathbf{g}\} \in \Theta$ denote the observation corresponding to applying $\mathbf{g}$ to i ; and $\mathbf{g}^{-1}(P) = \{\alpha \mid \exists p \in P \mid (\alpha, p) \in \mathbf{g}\} \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$ the used paths.

We say that an observation θ is compatible with a behavior $\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}}$ iff $\theta \subseteq \rightsquigarrow$ (seeing the behavior as a set of triples). Thus, given $\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}}$, the set $\Theta \cap \mathcal{P}(\rightsquigarrow)$ is the set of observations compatible with $\rightsquigarrow$. Similarly, an observation θ is compatible with a claim C iff $\theta \in \bigcup_{\rightsquigarrow \in C} \mathcal{P}(\rightsquigarrow)$, equivalently, iff $\theta \in C^\subseteq$. The set $\Theta \cap C^\subseteq$ is the set of observations compatible with the claim. This set is downwards-closed, as both Θ and $C^\subseteq$ are. Given a claim C and an observation θ , $\pi(\{\theta \subseteq \rightsquigarrow\} \cap C) = \{\pi(\rightsquigarrow) \mid \theta \subseteq \rightsquigarrow \in C\}$ thus denotes the DJ functions (mappings of I to $P \cup \{\emptyset\}$) compatible with the observation and the claim.

A subset of triples $T \subseteq I \times \mathcal{A}^{<\mathbb{N}} \times P$ is unobservable iff it cannot be observed together, that is, iff $T \notin \Theta$. If $\#T \notin \mathbb{N}$, T is unobservable. A finite subset of triples T is unobservable iff some subset $T' \subseteq T$ with $\#T' = 2$ is unobservable (unobservability comes from a given individual's observations, for that one, the paths are not increasing, and as the subset relation is transitive, two elements suffice to determine this fact).

It is important to realize that Θ is the set of *a priori* possible observations, meaning, those that are conceivable, possibly counter-factually. The set of observations that can result from actually observing the behavior $\rightsquigarrow$ is $\Theta \cap \mathcal{P}(\rightsquigarrow) = \{\rightsquigarrow|_Q \mid Q \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I)\}$. Such an observation $\theta \in \Theta \cap \mathcal{P}(\rightsquigarrow)$ is called a feasible observation.

Given a right unique $\tau \subseteq (I \times \mathcal{A}^{<\mathbb{N}}) \times P$, we let $\uparrow \tau = \{\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}} \mid \tau \subseteq \rightsquigarrow\}$ denote the set of “leads to” relations that complete τ (thus $\uparrow \emptyset = P^{I \times \mathcal{A}^{<\mathbb{N}}}$).

3.2. Requests

Onwards, we assume given an exposure protocol $(I, \mathcal{A}, \rightsquigarrow)$.

To be interesting for our purpose, a claim must satisfy minimal desiderata.

Most important, of course, is that the claim includes the real world. A claim C *holds* iff $\rightsquigarrow \in C$. We will study criteria to determine whether a claim holds.

Once established that a claim holds, any feasible observation can be combined with the claim to determine conclusions. We want, minimally, that at least some feasible observation permits to conclude about at least one individual's DJ. Furthermore, we want such observations to be always available even after some previous observations have occurred already, to ensure reproducibility. A claim C is *conclusive* iff $\forall \rightsquigarrow \in C, \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow), \exists \theta' \sqsubseteq \theta' \in \Theta \cap \mathcal{P}(\rightsquigarrow), i \in I \mid \pi_i(\{\theta' \subseteq \rightsquigarrow\} \cap C) = \#\{\pi_i(\rightsquigarrow) \mid \theta' \subseteq \rightsquigarrow \in C\} = 1$.

We restrict attention to claims that do not use prior knowledge about named individuals, thus, are anonymous, in the following sense. Given a permutation σ over I and any $\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}}$, let $\rightsquigarrow \circ \sigma$ denote $\{(i, \alpha, \varphi) \mid (\sigma(i), \alpha, \varphi) \in \rightsquigarrow\}$ and let $C \circ \sigma$ denote $\{\rightsquigarrow \circ \sigma \mid \rightsquigarrow \in C\}$. A claim C is *anonymous* iff $\forall \sigma : C = C \circ \sigma$. Example 27 illustrates.

3.3. Verifiability

We want to know whether a claim holds, but all we have under hand are observations. A natural starting point is to attempt to verify claims, defined as follows.

The claim C is verifiable in the world $\rightsquigarrow$ iff $\exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \forall \theta \subseteq \rightsquigarrow : \rightsquigarrow \in C$. Possible verifiability mandates that some (possibly counter-factual) observation proves the claim (makes sure that it holds), thus, C is possibly verifiable iff it is verifiable in some world; formally, iff $\exists \theta \in \Theta \mid \forall \theta \subseteq \rightsquigarrow : \rightsquigarrow \in C$. Verifiedness mandates verifiability in the actual world; formally, $\exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \forall \theta \subseteq \rightsquigarrow : \rightsquigarrow \in C$.

Verifiedness is sound (corollary 1), meaning that indeed when a claim is verified, it holds, but is too demanding for many claims. Indeed, some sufficiently simple claims, such as the claim $C = \{\rightsquigarrow \mid (a) \rightsquigarrow_i \varphi\}$ that some given path leads some given individual to some given proposition, are possibly verifiable, but such claims are not strong enough to deduce much useful conclusion concerning anyone's DJ.

Given a claim C and a world $\rightsquigarrow \in C$, the claim is *informative in* $\rightsquigarrow$ iff $\exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \pi(\{\theta \subseteq \rightsquigarrow'\} \cap C) \neq (P \cup \{\emptyset\})^I$.

It turns out that any claim that is conclusive fails verifiedness: the weaker demand of being informative in $\rightsquigarrow$ implies failure of verifiedness (corollary 4). This result may appear unsurprising, as we did not postulate yet any sort of regularity in the world, and indeed the fundamental reason for the incompatibility is that any observation can be completed in a way that contradicts any conclusion that informativeness in a world mandates (lemma 1).

We thus now turn to examining which sort of prior knowledge will suffice for ensuring the existence of a verification procedure that will allow to determine that a claim holds (when it indeed does) that is also conclusive and anonymous.

Three sorts of priors will be involved, the two first of which can be seen as applications to our context of Popperian ideas. First, we will assume that falsification tests fail falsifying the claim, under condition of sufficient observability. Second, we will assume that behaviors are in sufficient supply for being observed without limit. Third, we will assume that some behaviors that are inobservable (thus impossible to reject empirically) do not exist.

We say that a prior D proves a claim C iff $D \subseteq C$.

3.4. Verifiability under a prior

Verifiability, as defined in section 3.3, mandates that some observation proves C , in the sense that every world compatible with the observation is in C . As indicated there, this is too demanding to be applicable with no prior knowledge. In this section, we extend this idea to the case where some prior is given.

Given $C, D \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$, a claim C is verifiable at $\rightsquigarrow$ under prior D iff $\exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \cap (C \cap D)^\subseteq \mid \forall \theta \subseteq \rightsquigarrow' \in D : \rightsquigarrow' \in C$. Given $C, D \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$, a claim C is possibly verifiable under prior D iff $\exists \theta \in \Theta \cap (C \cap D)^\subseteq \mid \forall \theta \subseteq \rightsquigarrow \in D : \rightsquigarrow \in C$. Also, C is verified under prior D iff $\exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \cap (C \cap D)^\subseteq \mid \forall \theta \subseteq \rightsquigarrow \in D : \rightsquigarrow \in C$. Note that C is verified iff it is verified under the trivial prior. Finally, C satisfies established verifiability under prior D iff $\forall \rightsquigarrow \in C \cap D, \exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \forall \theta \subseteq \rightsquigarrow' \in D : \rightsquigarrow' \in C$.

Theorem 2 shows that provided the prior is true, being verified under a prior suffices for holding.

Provability by D implies established verifiability under D (just use verification $\emptyset$). For some priors, provability by D is strictly stronger than established verifiability under D (the claim C being $(a) \rightsquigarrow_i \varphi$ satisfies established verifiability under the trivial prior D but is not provable by it).

The following example illustrates that demanding that a claim be verifiable (under some prior) permits the claim to contain punctual facts, which provability under general priors to do allow.

Example 2. Consider the world $\rightsquigarrow$ constituted of two kinds of individuals, the jay's which always answer φ and the kay's which always answer φ' . Thus, $I = J \cup K$, and $\forall i \in I, \alpha \in \mathcal{A}^{<\mathbb{N}} : [i \in J \Rightarrow \alpha \rightsquigarrow_i \varphi] \wedge [i \in K \Rightarrow \alpha \rightsquigarrow_i \varphi']$. The claim $C_1 = [\exists i \mid \emptyset \rightsquigarrow_i \varphi]$ is verifiable and not conclusive. The claim $C_2 = [(\forall i : \emptyset \rightsquigarrow_i \varphi \Rightarrow \forall \alpha, \alpha \rightsquigarrow_i \varphi) \wedge (\exists i \mid \emptyset \rightsquigarrow_i \varphi)]$ is conclusive and not verifiable. $\triangle$

We will see that combining (variants of) verification procedures, which bring punctual facts, with general enough priors, lead further than procedures based only on priors or on verification could.

Established verifiability only requires *some* proving observation to exist at each $\rightsquigarrow \in C \cap D$. Mirroring how POF strengthens OF_C by requiring that *every* prefix admit a rightward falsifying extension, it is desirable that every prefix admit a rightward extension that still proves C . Given $C, D \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$ and $\rightsquigarrow \in C \cap D$, C is *permanently verifiable* (PV) under D at $\rightsquigarrow$ iff $\forall \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow), \exists \theta \sqsubseteq \theta' \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \forall \theta' \sqsubseteq \rightsquigarrow' \in D : \rightsquigarrow' \in C$. C satisfies PEV under prior D iff this holds $\forall \rightsquigarrow \in C \cap D$. Under a given prior, PEV is stronger than EV (corollary 6).

PEV and POF have a dual character as C satisfies PEV under prior D iff $\forall \rightsquigarrow \in C \cap D, \text{POF}_{D \setminus C}(\rightsquigarrow)$ (corollary 5, see also remark 10).

4. Priors

4.1. Falsifiability

Our first prior assumes that all falsification tests against C fail.

Expressing this requires properties that describe the extend to which a given (hypothetical or real) world can be observed to be not the real world. Given $\rightsquigarrow$ and C , say that C is

False in $\rightsquigarrow$ ($\text{F}_C(\rightsquigarrow)$) iff $\rightsquigarrow \notin C$;

Observably False (OF) in $\rightsquigarrow$ ($\text{OF}_C(\rightsquigarrow)$) iff C is false in this world and its falsity is observable; formally, iff $[\exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \theta \notin C^\subseteq]$, equivalently, $\Theta \cap \mathcal{P}(\rightsquigarrow) \not\subseteq C^\subseteq$;

Persistently Observably False in $\rightsquigarrow$ ($\text{POF}_C(\rightsquigarrow)$) iff C is false in this world and its falsity is observable whatever past observation have occurred already; formally, iff $\forall \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow), \exists \theta \sqsubseteq \theta' \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \theta' \notin C^\subseteq$.

We also say that C is *rejectable* iff $\text{OF}_C(\rightsquigarrow)$. We sometimes write POF_C instead of $\text{POF}_C(\rightsquigarrow)$ when the world $\rightsquigarrow$ is clear from the context, and POF when C is clear from the context (or referring to any generic claim), and similarly for OF_C and F_C .

Note that $\forall C, \rightsquigarrow : \text{POF}_C(\rightsquigarrow) \Rightarrow \text{OF}_C(\rightsquigarrow) \Rightarrow \text{F}_C(\rightsquigarrow)$ (theorem 8 proves the second implication). Also, given $C' \subseteq C$, $\forall \rightsquigarrow : \neg \text{F}_{C'}(\rightsquigarrow) \Rightarrow \neg \text{F}_C(\rightsquigarrow)$, $\neg \text{OF}_{C'}(\rightsquigarrow) \Rightarrow \neg \text{OF}_C(\rightsquigarrow)$, and $\neg \text{POF}_{C'}(\rightsquigarrow) \Rightarrow \neg \text{POF}_C(\rightsquigarrow)$. C is said to suffer visibility issues in $\rightsquigarrow$ iff $\neg \text{POF}_C(\rightsquigarrow) \wedge \text{OF}_C(\rightsquigarrow)$, which happens when some past observation (called “prefix” in this context) hides the falsity of C even though its falsity is observable when no prefix is involved. C suffers visibility issues iff $\exists \rightsquigarrow : \neg \text{POF}_C(\rightsquigarrow) \wedge \text{OF}_C(\rightsquigarrow)$. C suffers no visibility issues in $\rightsquigarrow$ iff $\text{OF}_C(\rightsquigarrow) \Rightarrow \text{POF}_C(\rightsquigarrow)$, equivalently, iff $\text{OF}_C(\rightsquigarrow) \Leftrightarrow \text{POF}_C(\rightsquigarrow)$, and suffers no visibility issues iff $\forall \rightsquigarrow : \text{OF}_C(\rightsquigarrow) \Rightarrow \text{POF}_C(\rightsquigarrow)$.

We say that C is *falsifiable* iff $C \neq P^{I \times \mathcal{A}^{<\mathbb{N}}} \wedge [\text{F}_C(\rightsquigarrow) \Rightarrow \text{OF}_C(\rightsquigarrow)]$, and *persistently falsifiable* iff $C \neq P^{I \times \mathcal{A}^{<\mathbb{N}}} \wedge [\forall \rightsquigarrow : \text{F}_C(\rightsquigarrow) \Rightarrow \text{POF}_C(\rightsquigarrow)]$.

It is important to note that (persistent) falsifiability does not depend on $\rightsquigarrow$: whether falsifiability holds can be determined analytically, meaning, by considering only the content of the claim, without considering the behavior of individuals.

We introduce the prior called C -scrutiny, which postulates that the real world is such that C is either not observably false in it, or if it is, then C suffers visibility issues in it; formally, $\neg \text{OF}_C(\rightsquigarrow) \vee (\neg \text{POF}_C(\rightsquigarrow) \wedge \text{OF}_C(\rightsquigarrow))$. This is logically equivalent to mandating $\neg \text{POF}_C(\rightsquigarrow)$, thus, excluding worlds in which C is persistently observably false.

Remark 1. Another way to describe C -scrutiny is that if the world is observably out of C , there is some prefix observation that hides this fact (formally: $\text{OF}_C(\rightsquigarrow) \Rightarrow \text{OF}_C(\rightsquigarrow) \wedge \neg \text{POF}_C(\rightsquigarrow)$).

Equivalently, if C is still observably falsifiable after any amount of observation, then no feasible observation would falsify it ($\text{POF}_C(\rightsquigarrow) \Rightarrow \neg \text{OF}_C(\rightsquigarrow)$).

Assuming, as the prior does, $\neg \text{POF}_C(\rightsquigarrow)$, leaves three possibilities: either there are visibility issues ($\text{OF}_C(\rightsquigarrow) \wedge \neg \text{POF}_C(\rightsquigarrow)$), or the claim is false, but inobservably so ($\text{F}_C(\rightsquigarrow) \wedge \neg \text{OF}_C(\rightsquigarrow)$), or the claim holds ($\neg \text{F}_C(\rightsquigarrow)$). $\triangle$

Remark 2. When considering a claim C with no visibility issues, C -scrutiny is equivalent to $\neg \text{OF}_C(\rightsquigarrow)$.

The reason for adopting the prior $\neg \text{POF}_C(\rightsquigarrow)$ instead of $\neg \text{OF}_C(\rightsquigarrow)$ is illustrated with C_1 is $(a) \rightsquigarrow_i \varphi \wedge (a') \rightsquigarrow_i \varphi$ for a given i : in some world (for example, where $(a) \rightsquigarrow_i \varphi$ even though $(a') \not\rightsquigarrow_i \varphi$), $\neg \text{POF}_{C_1}(\rightsquigarrow) \wedge \text{OF}_{C_1}(\rightsquigarrow)$, thus we cannot assume that non-observation of falsification implies not OF , because it might also be due to observing that $(a) \rightsquigarrow_i \varphi$. It also applies to $C_2 = [\forall i : (a') \rightsquigarrow_i \varphi \wedge (a) \rightsquigarrow_j \varphi]$ for a given j , which in a world satisfying $(a) \not\rightsquigarrow_j \varphi$ and $\forall i : (a') \rightsquigarrow_i \varphi$ is $\neg \text{POF}_{C_2}(\rightsquigarrow)$ (consider the prefix $\theta = \{(a') \rightsquigarrow_j \varphi\}$ and $\text{OF}_{C_2}(\rightsquigarrow)$. (Note that if C_2 fails because, for some i , $(a') \not\rightsquigarrow_i \varphi$, then θ is not a hiding prefix observation and this is what we want.) $\triangle$

Remark 9 discusses alternative formalizations of the concepts POF and C -scrutiny.

Remark 3. This prior seems unsatisfactorily not primitive enough (to be worked on...). We should obtain as a result that the reasonable prior is, if no visibility issues, then not OF , by assuming that we have observed everything we could ever dream of. $\triangle$

Theorem 18 shows that no claim is provable by $\neg \text{POF}_C$.

4.2. Repeatable observations

Our second prior assumes that whenever a behavior is exhibited by at least one individual, infinitely many individuals exhibit it.

Definition 2 (Repeatable observations). Let $E = [\forall i \in I : \#\{j \in I \mid \rightsquigarrow_j = \rightsquigarrow_i\} \notin \mathbb{N}]$, the claim that every individual's behavior is shared by infinitely many others.

Given $C \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$, combining both priors introduced so far, we obtain the prior $F_C = (\neg \text{POF}_C) \cap E$.

Intuition might suggest that under E , no visibility issues can arise. Example 13 shows that, on the contrary, a claim can be provable by $\neg \text{OF}_{C \cap E} \cap E$ yet not provable by $\neg \text{POF}_{C \cap E} \cap E$.

Theorem 22 shows that $(\neg \text{POF}_C) \cap E$ proves C iff $(\neg \text{OF}_C) \cap E$ proves C and C suffers no visibility issues in E (that is, $\forall \rightsquigarrow \in E : \text{OF}_C(\rightsquigarrow) \Rightarrow \text{POF}_C(\rightsquigarrow)$).

Theorem 24 shows that anonymous claims suffer no visibility issues in E . Thus (using theorem 22), any anonymous and falsifiable claim $C \subseteq E$ is provable by $\neg \text{POF}_{C \cap E} \cap E$ (theorem 25).

Theorem 35 shows the insufficiency of this prior: for some worlds, no anonymous claims that hold and is SAPTS and provable by $\neg \text{POF}_{C \cap E} \cap E$ exist. TODO prove this for PEV and all worlds.

5. Suitable claims

Theorem 72 and related show that anonymous and falsifiable claims that hold exist iff the situation does not satisfy $\forall \alpha : P^{\llbracket \alpha \rrbracket} \subseteq \rightsquigarrow(I)_{\llbracket \alpha \rrbracket}$.

6. Conclusion

The dependence of the deliberated judgment concept on the exposure protocol yields a first qualification about the concept of deliberated judgments when one desires to use it normatively: it is only as reasonable to call the resulting judgments “deliberate” in a normative sense as the exposure protocol is normatively relevant to deliberation. A sufficiently restricted, or even purposefully chosen, set of arguments may yield judgments that are only deliberate in the technical sense used here, but that pertain more to brain-washing than to reflective judgments. While it is indeed a long-term goal of this research program to yield a normatively relevant conception of preference (appropriate, for example, for recommendation), it is not a claim of this article that anything that corresponds to deliberated judgments as defined here qualify as being appropriate for any normative use: much depends on the chosen exposure protocol. The goal of this article is rather one of a conceptual and practical clarification. It proposes a sharp

distinction between the empirical content and the normative content of the concept of deliberated judgments: the definition proposed here transfers the full normative weight on the choice of the exposure protocol; and leaves the rest of the disagreements to empirical settling. This is methodologically akin, *mutatis mutandis*, to the axiomatic study of justice such as done in social choice, only complementing mathematical deduction with empirical claims. In social choice, the choice of axioms bears the full normative weight; once axioms are chosen, it is a matter of logic which social choice rules (if any) correspond to that choice. In our case, once an exposure protocol is chosen, it is an empirical matter which deliberated judgments result. In supplement to a possibly helpful conceptual clarification, such a sharp separation may reveal practically useful for separation of concern as it permits to study deliberated judgments empirically under various exposure protocols independently of possible disagreements about which norms are more appropriate for deliberation in which circumstances.

A. Proofs and supplementary discussion

A.1. Generalities

We say that a claim C *holds* iff $\rightsquigarrow \in C$; C *fails* otherwise.

Established verifiability mandates verifiability in all compatible worlds; formally, $\forall \rightsquigarrow \in C : \exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \forall \theta \subseteq \rightsquigarrow' : \rightsquigarrow' \in C$.

Example 3. The claim C being $[\forall i : (a) \rightsquigarrow_i \varphi] \vee (a') \rightsquigarrow_i \varphi$ is possibly verifiable but fails established verifiability. $\triangle$

A claim C is *possibly informative* iff it is informative in some world of C , namely, iff $\exists \theta \in \Theta \cap C^\subseteq \mid \pi(\{\theta \subseteq \rightsquigarrow\} \cap C) \neq (P \cup \{\emptyset\})^I$. A claim C is *informative* iff $\pi(C) \neq (P \cup \{\emptyset\})^I$. A claim C is *informative beyond counting* iff $\exists i \in I \mid \pi_i(C) = \{\pi_i(\rightsquigarrow) \mid \rightsquigarrow \in C\} \neq P \cup \{\emptyset\}$. A claim C is *a posteriori sharp* iff $\exists \theta \in \Theta, i \in I \mid \pi_i(\{\theta \subseteq \rightsquigarrow\} \cap C) = \#\{\pi_i(\rightsquigarrow) \mid \theta \subseteq \rightsquigarrow \in C\} = 1$. A claim C is *a posteriori truly sharp* iff $\exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow), i \in I \mid \pi_i(\{\theta \subseteq \rightsquigarrow\} \cap C) = \#\{\pi_i(\rightsquigarrow) \mid \theta \subseteq \rightsquigarrow \in C\} = 1$.

Table 1 lists this informativeness/sharpness family together with the related notions defined elsewhere (SAPTS, in the main text; indicative, further below in this appendix) and the renaming proposed for the two notions used in the main text (and their closest relatives), so that these read as ordinary words rather than acronyms; the more technical, appendix-only singleton-pinning family (APS, APTS, APS-P, APTS-P) keeps its acronym names.

Remark 4. Informativeness permits mere counting: the claim that exactly two persons deliberately believe φ is informative. Also, the claim $C = \{[\exists i \mid (a) \not\rightsquigarrow_i \varphi] \wedge [\forall i : (a) \not\rightsquigarrow_i \varphi \Rightarrow \varphi' \in \pi_i(\rightsquigarrow)]\}$ is informative. Informativeness may be a desirable property (rather than informativeness beyond counting) as one may, for example, want to know whether a given proposition is believed by a majority of individuals. Note that informativeness beyond counting implies informativeness. $\triangle$

Note that $\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I$ and $\mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I)$ are downwards closed, meaning that $\forall O \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I : \{O' \subseteq O\} \subseteq \mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I$ and $\forall O \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I) : \{O' \subseteq O\} \subseteq \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I)$.

Definition (summarized)	Current name	Proposed name
$\pi(C) \neq (P \cup \{\emptyset\})^I$	informative	indicative
$\exists \theta \subseteq \rightsquigarrow \mid \pi(\uparrow \theta \cap C) \neq (P \cup \{\emptyset\})^I$	informative in $\rightsquigarrow$	indicative in $\rightsquigarrow$
$\exists \theta \in C^\subseteq \mid \pi(\uparrow \theta \cap C) \neq (P \cup \{\emptyset\})^I$	possibly informative	possibly indicative
$\exists i \mid \pi_i(C) \neq P \cup \{\emptyset\}$	informative beyond counting	indicative beyond counting
$\forall i \mid \pi_i(C) \neq P \cup \{\emptyset\}$	strongly informative beyond counting	strongly indicative beyond counting
$\forall \rightsquigarrow \in C, \theta \subseteq \rightsquigarrow, \exists \theta \sqsubseteq \theta' \subseteq \rightsquigarrow, i \mid \#\pi_i(\uparrow \theta' \cap C) = 1$	SAPTS (strong APTS)	conclusive
$\exists \theta, i \mid \pi_i(\uparrow \theta \cap C) \subsetneq P$	indicative (for i) / indicativeness	informative (for i) / informativeness
$\exists \theta, i \mid \#\pi_i(\uparrow \theta \cap C) = 1$	APS (a posteriori sharp)	<i>unchanged</i>
$\exists \theta \subseteq \rightsquigarrow, i \mid \#\pi_i(\uparrow \theta \cap C) = 1$	APTS (a posteriori truly sharp)	<i>unchanged</i>
$\exists \theta, i, p \in P \mid \pi_i(\uparrow \theta \cap C) = \{p\}$	APS-P	<i>unchanged</i>
$\exists \theta \subseteq \rightsquigarrow, i, p \in P \mid \pi_i(\uparrow \theta \cap C) = \{p\}$	APTS-P	<i>unchanged</i>

Table 1: Proposed renaming of the informativeness/sharpness family.

A.2. Verifiability

Theorem 2. *Given a claim D such that $\rightsquigarrow \in D$, if a claim C is verified under prior D , then C holds.*

Proof. Consider any θ that satisfies verifiedness under D . We have $\theta \subseteq \rightsquigarrow$. By hypothesis, $\rightsquigarrow \in D$. Therefore, by verifiedness, $\rightsquigarrow \in C$. $\square$

Using the trivial prior $D = P^{I \times \mathcal{A}^{<\mathbb{N}}}$, we obtain this corollary.

Corollary 1. *If a claim is verified, it holds.*

Theorem 3. *For any $\theta \in \Theta$ and claims C, D , these properties are equivalent:*

1. $\forall \theta \subseteq \rightsquigarrow \in D : \rightsquigarrow \in C$,
2. $D \cap \{\rightsquigarrow \mid \theta \subseteq \rightsquigarrow\} \subseteq C$,
3. $\theta \notin (D \setminus C)^\subseteq$,
4. *there exists a finite disjoint partition $\{S_k\}$ of $D \setminus C$ such that $\forall k, \exists t \in \theta \mid \{t\} \notin S_k^\subseteq$ (equivalently, $t \notin \bigcup S_k$).*

Proof. Properties 1 and 2 assert that $\forall \theta \subseteq \rightsquigarrow : \rightsquigarrow \in D \Rightarrow \rightsquigarrow \in C$, equivalently, $\nexists \theta \subseteq \rightsquigarrow \mid \rightsquigarrow \in D \wedge \rightsquigarrow \notin C$, which is property 3.

To establish that property 3 implies property 4, consider any claim E and $\theta \notin E^\subseteq$. Given any $t \in \theta$, define $S_t = \{\rightsquigarrow \mid t \notin \rightsquigarrow\} \cap E$. Define $S = \bigcup_{t \in \theta} S_t$.

Consider any $\rightsquigarrow \in E$. As $\theta \not\subseteq \rightsquigarrow$, $\exists t \in \theta \mid t \notin \rightsquigarrow$, thus, $\rightsquigarrow \in S$. It follows that $S = E$.

As θ is finite, we can finitely enumerate $t \in \theta$, and define $S_1 = S_{t_1}$ and for any $2 \leq k \leq \#\theta$, $S_k = S_{t_k} \setminus \bigcup_{1 \leq k' < k} S_{t_{k'}}$. We obtain $E = S = \bigcup_{1 \leq k \leq \#\theta} S_k$, thus it is a disjoint partition of E , and as $\{t_k\} \notin S_{t_k}^\subseteq$ by construction of S_{t_k} , also $\{t_k\} \notin S_k^\subseteq$.

Finally, negating property 3, if θ is completable to some set covered by $\{S_k\}$, then θ is completable to some set S_k , which contradicts property 4. $\square$

Theorem 4. *Given any claims $C, D \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$, $[\rightsquigarrow \in D \wedge \text{OF}_{D \setminus C}(\rightsquigarrow)] \Leftrightarrow [\rightsquigarrow \in C \cap D \wedge \text{OF}_{D \setminus C}(\rightsquigarrow)]$.*

Proof. Given any $\rightsquigarrow$, $\text{OF}_{D \setminus C}(\rightsquigarrow) \Rightarrow \rightsquigarrow \notin D \setminus C$; equivalently, $\text{OF}_{D \setminus C}(\rightsquigarrow) \Rightarrow \rightsquigarrow \in C \cup \overline{D}$. $\square$

Corollary 2. *Given any claims $C, D \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$, established verifiability under prior D , that is, $\forall \rightsquigarrow \in C \cap D, \exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \forall \theta \subseteq \rightsquigarrow' \in D : \rightsquigarrow' \in C$, is equivalent to $\forall \rightsquigarrow \in C \cap D, \exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \theta \notin (D \setminus C)^\subseteq$.*

Corollary 3. *C satisfies established verifiability under prior D iff $\forall \rightsquigarrow \in C \cap D : \text{OF}_{D \setminus C}(\rightsquigarrow)$.*

Lemma 1. *If $\#\mathcal{A} \notin \mathbb{N}$, given any general observation O and $f : I \rightarrow (P \cup \{\emptyset\})$, $\exists \rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}} \mid O \subseteq \rightsquigarrow \wedge \pi(\rightsquigarrow) = f$.*

Proof. For each $i \mid f(i) \in P$, define $a_i \notin (O(P))(i)(\mathbb{N})$ as an argument not used for i ; for each $i \mid f(i) = \emptyset$, for each $p \in P$, define $k_i^p \notin (O(P))(i)$ as a path not used for i , with all k_i^p different from each other ($\forall p, p' \in P, k_i^p \neq k_i^{p'}$).

Define $f' = O \cup_i \{f(i) \in P, \alpha \mid a_i \in \alpha_{\text{end}}\} \cup_i \{f(i) = \emptyset, p \in P, a \in \mathcal{A}\} \{(i, (k_i^p, a), p)\}$. Define $\rightsquigarrow$ as any completion of f' to a total function in $P^{I \times \mathcal{A}^{<\mathbb{N}}}$. $\square$

Theorem 5. *If C is such that $\pi(C) \neq (P \cup \{\emptyset\})^I$ and is possibly verifiable under D , then*

- $\exists \theta \in \Theta, f \in (P \cup \{\emptyset\})^I \mid \forall \rightsquigarrow \in D : [\theta \subseteq \rightsquigarrow \Rightarrow \pi(\rightsquigarrow) \neq f]$;
- if $\#\mathcal{A} \notin \mathbb{N}$, D is not trivial.

Proof. By theorem 3, $D \cap \{\rightsquigarrow \mid \theta \subseteq \rightsquigarrow\} \subseteq C$, and for some f , $\rightsquigarrow \in C \Rightarrow \pi(\rightsquigarrow) \neq f$.

Also, $D \neq P^{I \times \mathcal{A}^{<\mathbb{N}}}$, as $\exists \theta \subseteq \rightsquigarrow \mid \pi(\rightsquigarrow) = f$ (by lemma 1), thus $\rightsquigarrow \notin D$. $\square$

Corollary 4. *If $\#\mathcal{A} \notin \mathbb{N}$, any informative claim is not possibly verifiable.*

Theorem 6. *If $\#\mathcal{A} \notin \mathbb{N}$, any claim verifiable in $\rightsquigarrow$ is informative in $\rightsquigarrow$ iff it is informative.*

Proof. Let $\theta_v \in \Theta \cap \mathcal{P}(\rightsquigarrow)$ witness that C is verifiable in $\rightsquigarrow$, so that every world extending θ_v lies in C ; in particular $\rightsquigarrow \in C$, hence informativeness in $\rightsquigarrow$ is defined.

If C is informative, then $\emptyset \in \Theta \cap \mathcal{P}(\rightsquigarrow)$ witnesses that it is informative in $\rightsquigarrow$, since $\{\emptyset \subseteq \rightsquigarrow\} \cap C = C$ and thus $\pi(\{\emptyset \subseteq \rightsquigarrow\} \cap C) = \pi(C) \neq (P \cup \{\emptyset\})^I$.

Conversely, assume C is not informative, that is, $\pi(C) = (P \cup \{\emptyset\})^I$; we show it is not informative in $\rightsquigarrow$. Let $\theta \in \Theta \cap \mathcal{P}(\rightsquigarrow)$. As $\theta \subseteq \rightsquigarrow$ and $\theta_v \subseteq \rightsquigarrow$, the union $\theta \cup \theta_v$ is a right unique finite subset of $(I \times \mathcal{A}^{<\mathbb{N}}) \times P$, hence a general observation. Given any $f \in (P \cup \{\emptyset\})^I$, lemma 1 yields $\rightsquigarrow' \in P^{I \times \mathcal{A}^{<\mathbb{N}}}$ with $\theta \cup \theta_v \subseteq \rightsquigarrow'$ and $\pi(\rightsquigarrow') = f$. Then $\theta_v \subseteq \rightsquigarrow'$ gives $\rightsquigarrow' \in C$, and $\theta \subseteq \rightsquigarrow'$ gives $\rightsquigarrow' \in \{\theta \subseteq \rightsquigarrow\}$, so $f \in \pi(\{\theta \subseteq \rightsquigarrow\} \cap C)$. As f is arbitrary, $\pi(\{\theta \subseteq \rightsquigarrow\} \cap C) = (P \cup \{\emptyset\})^I$; as θ is arbitrary, C is not informative in $\rightsquigarrow$. $\square$

Remark 5. The hypothesis $\#\mathcal{A} \notin \mathbb{N}$ cannot be dropped: without it, a claim may be verifiable and informative in $\rightsquigarrow$ yet not informative (the first implication, however, uses no such hypothesis). Let $\mathcal{A} = \{a\}$ and $I = \{i_0\}$, so that every nonempty path ends in $\{a\}$ and $\pi(\rightsquigarrow)(i_0) = \{p\}$ if $\rightsquigarrow$ responds p to every nonempty path and $\emptyset$ otherwise. Write $\rightsquigarrow_p$ for the world responding p everywhere, let $\theta_v = \{(i_0, (a), \varphi')\}$, and let $C = \{\theta_v \subseteq \rightsquigarrow\} \cup \{\rightsquigarrow_\varphi, \rightsquigarrow_0\}$. Then C is verifiable in $\rightsquigarrow_{\varphi'}$, witnessed by θ_v . It is also informative in $\rightsquigarrow_{\varphi'}$: as $\rightsquigarrow_\varphi, \rightsquigarrow_0 \notin \{\theta_v \subseteq \rightsquigarrow\}$, we have $\{\theta_v \subseteq \rightsquigarrow\} \cap C = \{\theta_v \subseteq \rightsquigarrow\}$, and forcing $(a) \rightsquigarrow_{i_0} \varphi'$ leaves only $\pi(\rightsquigarrow)(i_0) \in \{\{\varphi'\}, \emptyset\}$ (reaching $\{\varphi\}$ or $\{0\}$ would require every nonempty path to answer φ , resp. 0), so $\pi(\{\theta_v \subseteq \rightsquigarrow\} \cap C) = \{\{\varphi'\}, \emptyset\} \neq (P \cup \{\emptyset\})^I$. Yet C is not informative, since $\rightsquigarrow_\varphi$ and $\rightsquigarrow_0$ supply the missing profiles: $\pi(C) = \{\{\varphi'\}, \emptyset, \{\varphi\}, \{0\}\} = (P \cup \{\emptyset\})^I$. $\triangle$

This (to be proven) captures the claim “there is no decisive argument for i ”.

Theorem 7. *Given any $\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}}$: $\pi(\rightsquigarrow)(i) = \{\emptyset\} \Leftrightarrow \forall a \in \mathcal{A}, \exists p_1 \neq p_2 \in P, \alpha^1, \alpha^2 \in \mathcal{A}^{<\mathbb{N}} \mid a \in \alpha_{end}^1 \cap \alpha_{end}^2 \wedge \alpha^1 \rightsquigarrow_i p_1 \wedge \alpha^2 \rightsquigarrow_i p_2$.*

Example 4 (No decisive argument). The claim $C = \{\rightsquigarrow \mid \pi(\rightsquigarrow)(I) = \{\emptyset\}\}$ is possibly verifiable under prior D iff D is observationally completable to $\{\rightsquigarrow \mid \pi(\rightsquigarrow)(I) = \{\emptyset\}\}$. $\triangle$

Theorem 8. $\forall C, \rightsquigarrow \in C : \Theta \cap \mathcal{P}(\rightsquigarrow) \subseteq C^\subseteq$.

Proof. This holds by definition of $C^\subseteq = \bigcup_{\rightsquigarrow \in C} \mathcal{P}(\rightsquigarrow)$. $\square$

If $D \subseteq D'$, then established verifiability under prior D' implies established verifiability under prior D .

Corollary 10 shows that impossibilities propagate to stricter claims. Together with theorem 5, it shows that the prior required must be at least as strong as claiming that if some observable holds, then some configuration of DJs is excluded.

Remark 6. Suffices that the prior states something like: if almost all individuals deliberately believe φ , then everybody deliberately believe φ . This innocuous enough prior suffices because we request only informativeness, and indeed it only permits to restrict possibilities, not to determine anybody's DJ. $\triangle$

Theorem 7 and example 4 illustrate about no decisive arguments.

Remark 7. No general prior is reasonable. A prior D that states something general and reasonable such as “falsifiability implies correctness” and would be strong enough to allow determining that some substantial claim C holds would mean that later rejecting C would require rejecting the general prior. We want to keep the option of later rejecting C (because one cannot be sure), and we want this to happen without having to reject a general claim such as the correctness of the method of falsifiability, rather just a particular claim about over-generalization.

The prior must say that we can generalize this particular claim, which allow for later rejection. To prove that no general prior is reasonable, we might assume that two contradictory theories, C ((a) leads everyone to φ) and C' ((a) leads everyone to φ except if (a, a') does not), can be validated under the prior, considering some observations at some point. If under the prior, some obs proves C , then no further obs can prove C' under the same prior, so the prior must say something that is contradicted by the further obs, compatible with C' , so the prior contradicts C' . So if the prior allows to validate C , then it contradicts C' . To be checked... $\triangle$

A.3. Falsifiability

The following partitioning of claims with respect to their falsifiability status and examples might help better understand this central notion.

Any claim C satisfies exactly one of these:

falsifiable iff $\forall \rightsquigarrow \notin C : \text{OF}_C(\rightsquigarrow)$ and $\exists \rightsquigarrow \notin C$;

never falsifiable iff $\forall \rightsquigarrow \notin C : \neg \text{OF}_C(\rightsquigarrow)$, equivalently, iff $\forall \rightsquigarrow : \neg \text{OF}_C(\rightsquigarrow)$, equivalently, iff $\Theta \subseteq C^\subseteq$;

with mixed falsifiability iff $\exists \rightsquigarrow \notin C \mid \neg \text{OF}_C(\rightsquigarrow)$ and $\exists \rightsquigarrow \notin C \mid \text{OF}_C(\rightsquigarrow)$.

Indeed, first implies $\exists \rightsquigarrow \notin C \mid \text{OF}_C(\rightsquigarrow)$ which negates second, and the negation of the conjunction of the first two is $(\exists \rightsquigarrow \notin C \mid \neg \text{OF}_C(\rightsquigarrow) \vee \nexists \rightsquigarrow \notin C) \wedge (\exists \rightsquigarrow \notin C \mid \text{OF}_C(\rightsquigarrow))$, equivalent to the third.

We want to study the following dichotomy. Any claim C satisfies exactly one of these:

falsifiable or trivial iff $\forall \rightsquigarrow \notin C : \text{OF}_C(\rightsquigarrow)$, equivalently, iff $\forall \rightsquigarrow : F_C(\rightsquigarrow) \Rightarrow \text{OF}_C(\rightsquigarrow)$, still equivalently, iff $\forall \rightsquigarrow : F_C(\rightsquigarrow) \Leftrightarrow \text{OF}_C(\rightsquigarrow)$;

not falsifiable and not trivial iff $\exists \rightsquigarrow \notin C \mid \neg \text{OF}_C(\rightsquigarrow)$, that is, whenever the implication $F_C(\rightsquigarrow) \Rightarrow \text{OF}_C(\rightsquigarrow)$ fails.

Distinguishing two cases among the latter situation leads to the following partition. Any claim C satisfies exactly one of these:

falsifiable or trivial iff $\forall \rightsquigarrow \notin C : \text{OF}_C(\rightsquigarrow)$, equivalently, iff $\forall \rightsquigarrow : F_C(\rightsquigarrow) \Rightarrow \text{OF}_C(\rightsquigarrow)$, still equivalently, iff $\forall \rightsquigarrow : F_C(\rightsquigarrow) \Leftrightarrow \text{OF}_C(\rightsquigarrow)$;

never falsifiable and not trivial iff $\forall \rightsquigarrow \notin C : \neg \text{OF}_C(\rightsquigarrow)$ and $\exists \rightsquigarrow \notin C$;

with mixed falsifiability iff $\exists \rightsquigarrow \notin C \mid \neg \text{OF}_C(\rightsquigarrow)$ and $\exists \rightsquigarrow \notin C \mid \text{OF}_C(\rightsquigarrow)$.

Indeed, second implies $\exists \rightsquigarrow \notin C \mid \neg \text{OF}_C(\rightsquigarrow)$ which negates first, and the negation of the conjunction of the first two is $\exists \rightsquigarrow \notin C \mid \neg \text{OF}_C(\rightsquigarrow) \wedge (\exists \rightsquigarrow \notin C \mid \text{OF}_C(\rightsquigarrow) \vee \nexists \rightsquigarrow \notin C)$, equivalent to the third.

Here are typical examples.

Falsifiable $(a) \rightsquigarrow_i \varphi$ (satisfies established verifiability); $\forall i : (a) \rightsquigarrow_i \varphi$ (fails possible verifiability); $\forall i : a \hookrightarrow_i \varphi$ (fails possible verifiability) (example 8).

Never falsifiable $[\exists i \mid (a) \rightsquigarrow_i \varphi]$ which satisfies established verifiability; $[\forall i : [(a) \rightsquigarrow_i \varphi \vee (a') \rightsquigarrow_i \varphi]]$ (example 7) which fails possible verifiability; $[(a) \rightsquigarrow_i \varphi \vee (a') \rightsquigarrow_i \varphi]$ which satisfies established verifiability; $[\forall i : (a) \rightsquigarrow_i \varphi] \vee [\forall i : (a, a) \rightsquigarrow_i \varphi] \vee \dots$ (example 9) which fails possible verifiability.

With mixed falsifiability $[\exists i \mid (a) \rightsquigarrow_i \varphi] \wedge (a') \rightsquigarrow_i \varphi$ (proof: $\forall \rightsquigarrow \notin C : \text{OF}_C(\rightsquigarrow) \Leftrightarrow (a') \not\rightsquigarrow_i \varphi$) (satisfies established verifiability); $[\exists i \mid (a) \rightsquigarrow_i \varphi] \wedge [\forall i : (a') \rightsquigarrow_i \varphi]$ (proof: $\forall \rightsquigarrow \notin C : \text{OF}_C(\rightsquigarrow) \Leftrightarrow [\exists i \mid (a') \not\rightsquigarrow_i \varphi]$) (fails possible verifiability); “Exactly two of the three paths in $\{\emptyset, (a), (a')\}$ lead i to φ and the remaining one leads i to φ' ” (example 11) which fails possible verifiability.

Theorem 11 shows that falsifiability indeed fills the mentioned gap: falsifiability guarantees that a non trivial claim holds if, and only if, it is not rejectable.

Remark 8. Making a claim more precise (meaning going from a claim C to $C' \subsetneq C$) can make it falsifiable, or conversely, can make it unfalsifiable. Given α, i , consider the claim $C_2 = \alpha \rightsquigarrow_i \varphi$. It is falsifiable. Now consider the claim $C = C_1 \cap C_2$ with C_1 being the claim “every individual chooses φ after seeing solely either a or a' ”. It is more precise but unfalsifiable. But we can make it falsifiable again by making it even more precise (picking any single relation). (See remark 35.) $\triangle$

Theorem 9. *Given any claim C , the prior $\neg \text{OF}_C(\rightsquigarrow)$ proves C iff C is falsifiable or trivial.*

Proof. D proves C means, by definition, $\forall \rightsquigarrow : \neg \text{OF}_C(\rightsquigarrow) \Rightarrow \neg F_C(\rightsquigarrow)$. $\square$

Provability by $\neg \text{OF}_C(\rightsquigarrow)$ (equivalently, falsifiability or triviality) is strictly stronger than established verifiability under $\neg \text{OF}_C(\rightsquigarrow)$; and established verifiability is strictly stronger than established verifiability under $\neg \text{OF}_C(\rightsquigarrow)$.

Example 5. The claim $C_1 = \{\exists i \mid (a) \rightsquigarrow_i \varphi\}$, with infinitely many individuals, satisfies established verifiability (consider $\theta = \{(a) \rightsquigarrow_i \varphi\}$), hence, established verifiability under $\neg \text{OF}_C(\rightsquigarrow)$, but is not provable by $\neg \text{OF}_C(\rightsquigarrow)$. Similarly for $C_2 = \{(a) \rightsquigarrow_i \varphi \vee (b) \rightsquigarrow_i \varphi\}$ and $\theta = \{(a) \rightsquigarrow_i \varphi\}$.

Also, consider C_3 being $\forall j : (c) \rightsquigarrow_j \varphi$, C_4 being $(d) \rightsquigarrow_j \varphi \wedge (e) \rightsquigarrow_j \varphi$ and C being C_1 or C_2 together with C_3 or C_4 (assuming infinitely many individuals if using C_1 or C_3). Then, C fails established verifiability (due to C_3 or C_4) and $\neg \text{OF}_C(\rightsquigarrow)$ does not prove C (due to C_1 or C_2), but C satisfies established verifiability under prior $\neg \text{OF}_C(\rightsquigarrow)$ (with verification $(a) \rightsquigarrow_i \varphi$). $\triangle$

This suggests conjecturing that established verifiability under D' is always strictly stronger than established verifiability under D , for any priors $D \subseteq D'$. This is not the case in general; section F investigates the question and reports the current state of its characterization.

Note that if a claim is not provable by $\neg \text{OF}_C(\rightsquigarrow)$ (equivalently, some world is out of C and NOT OF), then some world is out of C and NOT POF.

It is probably useful to remember that $\neg \text{OF}_C(\rightsquigarrow)$ states that negating C , in that world, requires incompatible, or infinitely many, triples: the triples completable only out of C are all unobservable. Indeed, $\Theta \cap \mathcal{P}(\rightsquigarrow) \subseteq C^\subseteq$ is equivalent to $\forall T \in \mathcal{P}(\rightsquigarrow), T \notin C^\subseteq \Rightarrow T \notin \Theta$. When a claim is never falsifiable, thus $\forall \rightsquigarrow, \neg \text{OF}_C(\rightsquigarrow)$, we obtain $\Theta \subseteq C^\subseteq$.

A.4. Verifiability under a prior

Theorem 10. *Given $C, D \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$ and $\rightsquigarrow \in C \cap D$: C is PV under D at $\rightsquigarrow$ iff $\text{POF}_{D \setminus C}(\rightsquigarrow)$.*

Proof. By theorem 3 (items 1 and 3), for any $\theta' \in \Theta \cap \mathcal{P}(\rightsquigarrow)$: θ' proves C under D (that is, $\forall \theta' \subseteq \rightsquigarrow' \in D : \rightsquigarrow' \in C$) iff $\theta' \notin (D \setminus C)^\subseteq$. Substituting into the definition of PV at $\rightsquigarrow$ gives $\forall \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow), \exists \theta \sqsubseteq \theta' \in \Theta \cap \mathcal{P}(\rightsquigarrow) : \theta' \notin (D \setminus C)^\subseteq$, which is exactly $\text{POF}_{D \setminus C}(\rightsquigarrow)$. $\square$

Corollary 5. C satisfies PEV under prior D iff $\forall \rightsquigarrow \in C \cap D : \text{POF}_{D \setminus C}(\rightsquigarrow)$.

Corollary 6. Given $C, D \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$: PEV under D implies established verifiability under D .

Proof. $\text{POF}_S(\rightsquigarrow) \Rightarrow \text{OF}_S(\rightsquigarrow)$ for any $S, \rightsquigarrow$ (taking $\theta = \emptyset$ in the definition of POF_S already gives the witness required by OF_S), so corollary 5 together with corollary 3 gives the first implication. $\square$

Corollary 7. Given $C, D \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$: provability of C by D implies PEV under D .

Proof. If $D \subseteq C$, then $D \setminus C = \emptyset$, so $(D \setminus C)^\subseteq = \emptyset$ and $\text{POF}_\emptyset(\rightsquigarrow)$ holds trivially (any $\theta' \in \Theta \cap \mathcal{P}(\rightsquigarrow)$, in particular $\theta' = \emptyset$, already satisfies $\theta' \notin \emptyset$), giving PV under D at every $\rightsquigarrow \in C \cap D$ by corollary 5. $\square$

A.5. Observational exclusion and completion

$S \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$ is observationally excludable iff $\exists \theta \in \Theta \mid \theta \notin S^\subseteq$, equivalently, iff $\Theta \not\subseteq S^\subseteq$, equivalently, iff $\exists \rightsquigarrow \notin S \mid \text{OF}_S(\rightsquigarrow)$, equivalently, iff $\exists \rightsquigarrow \mid \text{OF}_S(\rightsquigarrow)$, equivalently, iff S is NOT never falsifiable.

D is observationally completable to C iff $\exists \theta \in (\Theta \cap D^\subseteq) \mid \theta \notin (D \setminus C)^\subseteq$. Corollary 9 shows the equivalence with possible verifiability under prior D .

D is necessarily observationally completable to C iff $\forall \rightsquigarrow \in D, \text{OF}_{D \setminus C}(\rightsquigarrow)$. Remark 15 shows the lack of equivalence with established verifiability under prior D .

Converting a claim to disjunctive normal form is useful to characterize its compatible observations, as theorem 20 indicates.

Example 6. Consider six different arguments $a_1, a'_1, a_2, a'_2, a_3, a'_3 \in \mathcal{A}$ and the following claim E : if any two arguments $a_k, a_{k'}$ fail to convince i or $a'_k, a'_{k'}$ fail to convince j , then the third argument $a_{k''}$ convinces i and $a'_{k''}$ convinces j . The claim is not observationally excludable. Indeed, we see that $E = \{\rightsquigarrow \mid [(a_1) \rightsquigarrow_i \varphi \wedge (a'_1) \rightsquigarrow_j \varphi] \vee [(a_2) \rightsquigarrow_i \varphi \wedge (a'_2) \rightsquigarrow_j \varphi] \vee [(a_3) \rightsquigarrow_i \varphi \wedge (a'_3) \rightsquigarrow_j \varphi]\}$, and $\forall 1 \leq k \leq 3$, define $S_k = \{\rightsquigarrow \mid [(a_k) \rightsquigarrow_i \varphi \wedge (a'_k) \rightsquigarrow_j \varphi]\}$, define $\mathcal{S} = \{S_1, S_2, S_3\}$, consider any $f \subseteq (I \times \mathcal{A}^{<\mathbb{N}}) \times P \mid f \notin E^\subseteq$, we have $\forall 1 \leq k \leq 3, \exists t_k \in f \mid \{t_k\} \notin S_k^\subseteq$, thus $f \notin \Theta$ (wlog, pick $t_1 = ((a_1), i, p_1 \neq \varphi)$, then for observability we need to pick $t_2 = ((a'_2), j, \cdot)$ and there is no choice left for t_3). $\triangle$

C -scrutiny is the prior $D_C = \{\rightsquigarrow \mid \exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \forall \theta \sqsubseteq \theta' \in \Theta \cap \mathcal{P}(\rightsquigarrow) : \theta' \in C^\subseteq\}$.

Remark 9. Reusing, from remark 2, the example $C_2 = [\forall i : (a') \rightsquigarrow_i \varphi \wedge (a) \rightsquigarrow_j \varphi]$ for a given j and considering again a world satisfying $(a) \not\rightsquigarrow_j \varphi$ and $\forall i : (a') \rightsquigarrow_i \varphi$, we see that formalizing $\neg \text{POF}_C$, independently of $\rightsquigarrow$, as $\exists \theta \in \Theta \mid \forall \theta \sqsubseteq \theta' \in \Theta : \theta' \in C^\subseteq$, is inadequate, as this prevents the prefix $\theta = \{(a') \rightsquigarrow_j \varphi\}$ from bringing not POF. The condition $\exists \theta \in \Theta \mid \forall \theta \sqsubseteq \theta' \in \Theta \cap \mathcal{P}(\rightsquigarrow) : \theta' \in C^\subseteq$ is also not suitable, as it is always satisfied: pick any $\theta \in \Theta$ such that $\theta \notin \mathcal{P}(\rightsquigarrow)$.

The following alternative prior, which might be considered for its apparent proximity with the chosen one, is also inadequate: $\forall \rightsquigarrow, \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow), [\exists \theta \sqsubseteq \theta' \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \theta' \notin C^\subseteq] \Rightarrow [\theta \in C^\subseteq]$. Consider again C_1 and $\rightsquigarrow$ such that $(a) \rightsquigarrow_i \varphi \wedge (a') \not\rightsquigarrow_i \varphi$. The chosen

prior ($\text{POF}_{C_1}(\rightsquigarrow) \Rightarrow \neg \text{OF}_C(\rightsquigarrow)$) holds, as intuition commands, as the antecedent fails for the reason given above. The alternative prior however fails as $\theta = \{(a') \rightsquigarrow_i \varphi\}$ is completable only out of C . $\triangle$

Remark 10. Verifiability and observable falsity have a dual character as C satisfies established verifiability under prior D iff $\forall \rightsquigarrow \in C \cap D, \text{OF}_{D \setminus C}(\rightsquigarrow)$ (corollary 3). $\triangle$

Theorem 11. *For any falsifiable or trivial claim C : $\rightsquigarrow \in C$ iff $\neg \text{OF}_C(\rightsquigarrow)$.*

Example 7 (Disjunct with incompatible paths, not rejectable). Given $a \neq a' \in \mathcal{A}$, the claim $\forall i : [(a) \rightsquigarrow_i \varphi \vee (a') \rightsquigarrow_i \varphi]$ is not rejectable. In fact, $\Theta \subseteq C^\subseteq$. Consider any $\theta \in \Theta$ and let us prove that $\theta \in \bigcup_{\rightsquigarrow \in C} \mathcal{P}(\rightsquigarrow)$. By definition of Θ , $\theta(P)$ may not contain both $((a), i)$ and $((a'), i)$, thus letting t denote one of these pairs not in $\theta(P)$ and defining $r = (t, \varphi)$, pick any $\rightsquigarrow'$ such that $\theta \subseteq \rightsquigarrow' \wedge t \in \rightsquigarrow'$. We have $\theta \in \mathcal{P}(\rightsquigarrow')$ and $\rightsquigarrow' \in C$. $\triangle$

Example 7 shows a claim that is not rejectable, irrespective of the exposure protocol. In particular, the example applies when C does not hold (that is, when $\rightsquigarrow \notin C$). We conclude that a claim may fail to hold but still be not rejectable. It follows that non-rejection cannot be a sufficient condition for validating a claim.

Remark 11. Examples 9 and 11 present situations where, similarly, the claim may fail to hold but still be not rejectable. In example 9, non rejection is due to finiteness of observations. In example 11, rejection depends on $\rightsquigarrow$. $\triangle$

Example 8. Consider any a and φ . The claim $\forall i : a \hookrightarrow_i \varphi$ is falsifiable: given any $\rightsquigarrow \notin C$, consider any $i \mid a \not\hookrightarrow_i \varphi$ and any $\alpha \mid \alpha \not\rightsquigarrow_i \varphi \wedge a \in \alpha_{\text{end}}$, and define $\theta = \{(i, \alpha, \rightsquigarrow(i, \alpha) \neq \varphi)\}$. $\triangle$

The following example presents a situation where non rejection is due to finiteness of observations.

Example 9 (Big disjunct, non rejectable). Consider $a \in \mathcal{A}$. The claim $[\forall i : (a) \rightsquigarrow_i \varphi] \vee [\forall i : (a, a) \rightsquigarrow_i \varphi] \vee \dots$, thus, $C = \{\rightsquigarrow \mid \exists n \in \mathbb{N}^* \mid \forall i : (a)^n \rightsquigarrow_i \varphi\}$, is non rejectable: $\Theta \subseteq C^\subseteq$.

Given any $\theta \in \Theta$, define $l = \max_{\alpha \mid \exists i \in I, p \in P \mid (i, \alpha, p) \in \theta} \alpha + 1$ if $\exists p \in P \mid (i, \alpha, p) \in \theta$, and $l = 1$ otherwise. [Claude: In this second condition ("if $\exists p \in P \dots$ "), i and α are free — not bound by an enclosing $\exists i$ /the $\max$'s own α binder. Is this meant to hold for the same α achieving the $\max$ above, and some/any i ? As written the condition's truth value is unclear.] Pick any $\rightsquigarrow$ such that $\theta \subseteq \rightsquigarrow \wedge (a)^l \rightsquigarrow_i \varphi$. We have $\theta \in \mathcal{P}(\rightsquigarrow)$ and $\rightsquigarrow \in C$, thus, $\theta \in C^\subseteq$. $\triangle$

Example 10 (Big disjunct unfals iff infinite). Given any $a \in \mathcal{A}$ and an infinite I , the claim $\exists i \mid a \hookrightarrow_i \varphi$ is strongly unfalsifiable: any $\theta \in \Theta$ only has finitely many i in its triples, thus, some i is not reached by θ , thus $\theta \in C^\subseteq$. If I is finite, then the claim is falsifiable: if it is wrong, observe for each i some adequate triple $(i, \alpha_i, p \neq \varphi)$. $\triangle$

Example 11 (Three functions, may be rejectable). Consider $i \in I$, $a \neq a' \in \mathcal{A}$ and the three paths $B = \{\emptyset, (a), (a')\}$. The claim "Exactly two of the three paths in B lead i to φ and the remaining one leads i to φ' " is not rejectable iff $\rightsquigarrow_i$ associates, in order, $(\emptyset, (a), (a'))$ to either $(\varphi', \varphi, \varphi)$, $(\varphi, \varphi', \varphi)$, $(\varphi, \varphi, \varphi')$, $(\varphi, \varphi, \varphi)$ or $(\varphi, \varphi', \varphi')$.

Indeed, if $\rightsquigarrow_i$ satisfies this, then either it holds and by theorem 8 it is not rejectable, or it associates this triplet to $(\varphi, \varphi, \varphi)$ or $(\varphi, \varphi', \varphi')$ in which case, as (a) and (a') cannot be together in θ , any $\theta \in \Theta$ with $\theta \subseteq \rightsquigarrow$ satisfy $\theta \in C^\subseteq$.

Conversely, if C is not rejectable, then for any θ_1 containing $\emptyset$ and (a) and θ_2 containing $\emptyset$ and (a') , with $\theta_1, \theta_2 \subseteq \rightsquigarrow$, we have $\theta_1, \theta_2 \in C^\subseteq$, which requires that $\rightsquigarrow_i$ associates $(\emptyset, (a))$ to (φ', φ) , (φ, φ') or (φ, φ) and $(\emptyset, (a'))$ to (φ', φ) , (φ, φ) or (φ, φ') . $\triangle$

Theorem 12. $D \setminus C$ is falsifiable or trivial iff $(C$ satisfies established verifiability under prior D AND $\forall \rightsquigarrow \notin D : \text{OF}_{D \setminus C}(\rightsquigarrow)$).

Proof. $D \setminus C$ is falsifiable or trivial iff $\forall \rightsquigarrow \notin (D \setminus C) : \text{OF}_{D \setminus C}(\rightsquigarrow)$, equivalently (as $\rightsquigarrow \notin D \setminus C \Leftrightarrow \rightsquigarrow \notin D \vee \rightsquigarrow \in C$), iff $[\forall \rightsquigarrow \notin D : \text{OF}_{D \setminus C}(\rightsquigarrow)] \wedge [\forall \rightsquigarrow \in C : \text{OF}_{D \setminus C}(\rightsquigarrow)]$. As $C \setminus D \subseteq \overline{D}$, the second conjunct is already implied on $C \setminus D$ by the first, so it reduces to $\forall \rightsquigarrow \in C \cap D : \text{OF}_{D \setminus C}(\rightsquigarrow)$, that is, by corollary 3, to established verifiability under prior D . $\square$

Corollary 8. $P^{I \times \mathcal{A}^{<\mathbb{N}}} \setminus C$ is falsifiable or trivial iff C satisfies established verifiability.

Theorem 13. For any $\theta \in \Theta$, the following statements are equivalent.

- $\theta \notin C^\subseteq$,
- $\forall \theta \subseteq \rightsquigarrow, \rightsquigarrow \notin C$,
- $\forall \theta \subseteq \rightsquigarrow : \text{OF}_C(\rightsquigarrow)$,
- $\theta \notin (\neg \text{OF}_C(\rightsquigarrow))^\subseteq$.

Proof. 1 and 2 are equivalent by definition. From 1 to 3, that's because for any such $\rightsquigarrow$, we have indeed $\theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \theta \notin C^\subseteq$. $\square$

Theorem 14. Given any claims $C \subseteq C'$ and any $\rightsquigarrow : \text{OF}_{C'}(\rightsquigarrow) \Rightarrow \text{OF}_C(\rightsquigarrow)$.

Proof. By definition, $\neg \text{OF}_C(\rightsquigarrow)$ means $\Theta \cap \mathcal{P}(\rightsquigarrow) \subseteq C^\subseteq$, thus the contrapositive is proven by observing that $C^\subseteq \subseteq C'^\subseteq$. $\square$

Theorem 15. For any claim C , with D being $\neg \text{OF}_C(\rightsquigarrow)$, given any $\rightsquigarrow : \text{OF}_C(\rightsquigarrow) \Leftrightarrow \text{OF}_D(\rightsquigarrow)$; and $\text{OF}_D(\rightsquigarrow) \Rightarrow \text{OF}_{D \setminus C}(\rightsquigarrow)$ (using theorem 14).

Proof. $\text{OF}_C(\rightsquigarrow) \Leftrightarrow \exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \theta \notin C^\subseteq \Leftrightarrow \exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \theta \notin (\neg \text{OF}_C(\rightsquigarrow))^\subseteq \Leftrightarrow \text{OF}_D(\rightsquigarrow) \Rightarrow \text{OF}_{D \setminus C}(\rightsquigarrow)$. $\square$

Theorem 16. Considering a claim C and prior D being $\neg \text{OF}_C(\rightsquigarrow)$, $D \setminus C$ is falsifiable or trivial iff C satisfies established verifiability under prior $\neg \text{OF}_C(\rightsquigarrow)$.

Proof. Using theorem 12, it is sufficient to prove that for any C , considering any $\rightsquigarrow \notin D$ (thus, $\text{OF}_C(\rightsquigarrow)$), $\text{OF}_{D \setminus C}(\rightsquigarrow)$ holds. This is given by theorem 15. $\square$

Given $\rightsquigarrow, \rightsquigarrow' \in P^{I \times \mathcal{A}^{<\mathbb{N}}}$, say $\rightsquigarrow \sim_{\text{fin}} \rightsquigarrow'$ iff $\rightsquigarrow$ and $\rightsquigarrow'$ differ at only finitely many pairs (i, α) , that is, iff $\#\{(i, \alpha) \in I \times \mathcal{A}^{<\mathbb{N}} \mid \rightsquigarrow(i, \alpha) \neq \rightsquigarrow'(i, \alpha)\} \in \mathbb{N}$. $\sim_{\text{fin}}$ is an equivalence relation on $P^{I \times \mathcal{A}^{<\mathbb{N}}}$: reflexivity and symmetry are immediate, and transitivity holds because, if $\rightsquigarrow \sim_{\text{fin}} \rightsquigarrow' \sim_{\text{fin}} \rightsquigarrow''$, the pairs where $\rightsquigarrow$ and $\rightsquigarrow''$ differ are among those where $\rightsquigarrow$ differs from $\rightsquigarrow'$ or $\rightsquigarrow'$ differs from $\rightsquigarrow''$, a finite union of two finite sets.

Definition 3 (Closed under finite modification). A claim C is *closed under finite modification* iff $\forall \rightsquigarrow \in C, \rightsquigarrow' \in P^{I \times \mathcal{A}^{<\mathbb{N}}} : \rightsquigarrow \sim_{\text{fin}} \rightsquigarrow' \Rightarrow \rightsquigarrow' \in C$.

Being closed under finite modification and non-empty does not entail triviality.

Example 12 (Closed under finite modification, yet non-trivial). Fix any $\rightsquigarrow^{(0)} \in P^{I \times \mathcal{A}^{<\mathbb{N}}}$ and let $C = \{\rightsquigarrow \mid \rightsquigarrow \sim_{\text{fin}} \rightsquigarrow^{(0)}\}$, the $\sim_{\text{fin}}$ -equivalence class of $\rightsquigarrow^{(0)}$. C is closed under finite modification, by transitivity of $\sim_{\text{fin}}$, and $\rightsquigarrow^{(0)} \in C$, yet $C \neq P^{I \times \mathcal{A}^{<\mathbb{N}}}$ (the world differing from $\rightsquigarrow^{(0)}$ at every pair, say, is excluded). $\triangle$

Theorem 17. *If $C \neq \emptyset$ is closed under finite modification, then C is never falsifiable.*

Proof. Fix $\rightsquigarrow^{(0)} \in C$ and let us show that $\Theta \subseteq C^\subseteq$. Given any $\theta \in \Theta$, let $\rightsquigarrow$ agree with θ on $\theta(P)$ and with $\rightsquigarrow^{(0)}$ elsewhere; then $\rightsquigarrow \sim_{\text{fin}} \rightsquigarrow^{(0)}$, as they differ only on the finite set $\theta(P)$, so $\rightsquigarrow \in C$, and $\theta \subseteq \rightsquigarrow$, giving $\theta \in C^\subseteq$. $\square$

Remark 12. The converse of theorem 17 fails: example 7 is never falsifiable ($\Theta \subseteq C^\subseteq$) yet is not closed under finite modification, as flipping a single individual's two values (making it satisfy neither disjunct) already leaves C . $\triangle$

A.6. POF

Remark 13. By corollary 5, $\rightsquigarrow \in C \cap D$ satisfies established verifiability but not PV under D iff $\text{OF}_{D \setminus C}(\rightsquigarrow) \wedge \neg \text{POF}_{D \setminus C}(\rightsquigarrow)$, that is, iff $D \setminus C$ suffers visibility issues at $\rightsquigarrow$. So, for a given claim and prior, the gap between established verifiability and PV is precisely governed by visibility issues of $D \setminus C$: no such gap arises wherever $D \setminus C$ suffers no visibility issues. $\triangle$

A claim C is persistently falsifiable or trivial iff it is provable by $\neg \text{POF}_C$.

Theorem 18. *A claim C is provable by $\neg \text{POF}_C$ iff $C = \emptyset \vee C = P^{I \times \mathcal{A}^{<\mathbb{N}}}$.*

Proof. If $C = \emptyset$, then $\forall \theta \in \Theta : \theta \notin C^\subseteq$, thus $\forall \rightsquigarrow : \text{POF}_C(\rightsquigarrow)$.

Conversely, suppose $P^{I \times \mathcal{A}^{<\mathbb{N}}} \neq C \neq \emptyset$; we show $\exists \rightsquigarrow \notin C \mid \neg \text{POF}_C(\rightsquigarrow)$.

If C is closed under finite modification (definition 3), by theorem 17, $\forall \rightsquigarrow : \neg \text{OF}_C(\rightsquigarrow)$, thus $\neg \text{POF}_C(\rightsquigarrow)$; suffices to pick any $\rightsquigarrow \notin C$.

If C is not closed under finite modification, for some $n \in \mathbb{N}$, some two worlds $\rightsquigarrow^{(0)} \in C, \rightsquigarrow^{(n)} \notin C$ differ at exactly n pairs. We can build a sequence of worlds $(\rightsquigarrow^{(k)})_{1 \leq k \leq n-1}$, by flipping one pair at a time, such that $\forall 1 \leq k \leq n$, each $\rightsquigarrow^{(k)}$ differs from $\rightsquigarrow^{(k-1)}$ at exactly one pair. Taking $j = \min\{1 \leq k \leq n \mid \rightsquigarrow^{(k)} \notin C\}$, we obtain worlds $\rightsquigarrow^{(j-1)} \in C, \rightsquigarrow^{(j)} \notin C$ that differ at exactly one pair (i, α) . To show $\neg \text{POF}_C(\rightsquigarrow^{(j)})$, pick any α' satisfying $\#\alpha < \#\alpha'$, define $\theta = \{(i, \alpha', \rightsquigarrow^{(j)}(i, \alpha'))\}$, pick any $\theta \sqsubseteq \theta' \in \Theta \cap \mathcal{P}(\rightsquigarrow^{(j)})$,

and let us show that $\theta' \in C^\subseteq$. Indeed, $(i, \alpha, \cdot) \notin \theta'$, because θ' observing i must use only extensions of α' , itself longer than α ; thus, $\theta' \subseteq \rightsquigarrow^{(j-1)}$, as $\rightsquigarrow^{(j-1)}$ agrees with $\rightsquigarrow^{(j)}$ on every other pair than (i, α) . $\square$

Theorem 19. *A claim C is PEV under $D = \neg \text{POF}_C$ iff $C = \emptyset \vee C = P^{I \times \mathcal{A}^{<\mathbb{N}}}$.*

Proof. If $C = \emptyset$, then $\forall \theta \in \Theta : \theta \notin C^\subseteq$, thus $\forall \rightsquigarrow : \text{POF}_C(\rightsquigarrow)$.

Conversely, suppose $P^{I \times \mathcal{A}^{<\mathbb{N}}} \neq C \neq \emptyset$; we show $\exists \rightsquigarrow \in C \mid \neg \text{POF}_C(\rightsquigarrow)$.

PEV demands that all worlds in C satisfy PV: $\forall \theta, \exists \theta' \subseteq \theta' \notin (D \setminus C)^\subseteq$. In other words, that $\text{POF}_{D \setminus C}(\rightsquigarrow)$.

We show $\exists \rightsquigarrow \in C \mid$ for some prefix θ , all completions θ' are completable in $D \setminus C$.

If C is closed under finite modification (definition 3), by theorem 17, $\forall \rightsquigarrow : \neg \text{OF}_C(\rightsquigarrow)$, thus $\neg \text{POF}_C(\rightsquigarrow)$, thus $D \setminus C = \overline{C}$. As $\sim_{\text{fin}}$ is an equivalence relation, C is closed under finite modification iff $\overline{C}$ is closed under finite modification. Therefore, by theorem 17 again, $\forall \rightsquigarrow : \neg \text{POF}_{\overline{C}}(\rightsquigarrow)$.

If C is not closed under finite modification, for some $n \in \mathbb{N}$, some two worlds $\rightsquigarrow^{(0)} \in C, \rightsquigarrow^{(n)} \notin C$ differ at exactly n pairs. We can build a sequence of worlds $(\rightsquigarrow^{(k)})_{1 \leq k \leq n-1}$, by flipping one pair at a time, such that $\forall 1 \leq k \leq n$, each $\rightsquigarrow^{(k)}$ differs from $\rightsquigarrow^{(k-1)}$ at exactly one pair. Taking $j = \min\{1 \leq k \leq n \mid \rightsquigarrow^{(k)} \notin C\}$, we obtain worlds $\rightsquigarrow^{(j-1)} \in C, \rightsquigarrow^{(j)} \notin C$ that differ at exactly one pair (i, α) . Pick any α' satisfying $\# \alpha < \# \alpha'$, define $\theta = \{(i, \alpha', \rightsquigarrow^{(j-1)}(i, \alpha'))\}$, pick any $\theta' \subseteq \theta' \in \Theta \cap \mathcal{P}(\rightsquigarrow^{(j-1)})$, and let us show that $\theta' \in (D \setminus C)^\subseteq$. Indeed, $(i, \alpha, \cdot) \notin \theta'$, because θ' observing i must use only extensions of α' , itself longer than α . Thus, $\theta' \subseteq \rightsquigarrow^{(j)}$, as $\rightsquigarrow^{(j)}$ agrees with $\rightsquigarrow^{(j-1)}$ on every other pair than (i, α) . Furthermore, $\neg \text{POF}_C(\rightsquigarrow^{(j)})$ because $\theta \subseteq \rightsquigarrow^{(j)}$ and $\theta' \subseteq \rightsquigarrow^{(j-1)} \in C$. $\square$

Remark 14. I believe that we're basically finding a prefix θ such that $\forall \theta' \subseteq \theta' : \theta' \subseteq \rightsquigarrow^{(j-1)} \in C \wedge \theta' \subseteq \rightsquigarrow^{(j)} \notin C$. Also, perhaps, existence of such a prefix is equivalent to the statement of the theorem. This may perhaps bring some better understanding. To be checked. $\triangle$

A.7. Observational completion

Corollary 9. *Given any claims $C, D \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$, possible verifiability under prior D , that is, $\exists \theta \in \Theta \cap D^\subseteq \mid \forall \theta' \subseteq \rightsquigarrow \in D : \rightsquigarrow \in C$, is equivalent to D being observationally completable to C , that is, $\exists \theta \in \Theta \cap D^\subseteq \mid \theta \notin (D \setminus C)^\subseteq$.*

Remark 15. Given any claims $C, D \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$, $[C$ satisfies established verifiability under prior D ($\forall \rightsquigarrow \in C \cap D, \text{OF}_{D \setminus C}(\rightsquigarrow)$) is equivalent to D is necessarily observationally completable to C ($\forall \rightsquigarrow \in D, \text{OF}_{D \setminus C}(\rightsquigarrow)$)] iff $D \setminus C = \emptyset$. Indeed, assume EV under D and consider $\rightsquigarrow \in D \cap \overline{C}$, we have $\neg \text{F}_{D \setminus C}(\rightsquigarrow)$, thus $\neg \text{OF}_{D \setminus C}(\rightsquigarrow)$. $\triangle$

Corollary 10. *Given any claims $C, D \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$, if some claim $C' \subseteq C$ is possibly verifiable under prior D , that is, $\exists \theta \in \Theta \cap D^\subseteq \mid \forall \theta' \subseteq \rightsquigarrow \in D : \rightsquigarrow \in C'$, then D is observationally completable to C , that is, $\exists \theta \in \Theta \cap D^\subseteq \mid \theta \notin (D \setminus C)^\subseteq$.*

Theorem 20. *Given any E and a cover $\mathcal{S}$ of E (a set of non-empty subsets whose union equals E) such that $\forall S \in \mathcal{S}, \exists f_S \subseteq (I \times \mathcal{A}^{<\mathbb{N}}) \times P \mid S = \uparrow f_S$ with f_S right unique, given any $f \subseteq (I \times \mathcal{A}^{<\mathbb{N}}) \times P$: $f \notin E^\subseteq$ is equivalent to $\forall S \in \mathcal{S}, \exists t \in f \mid \{t\} \notin S^\subseteq$.*

Proof. Pick any $S \in \mathcal{S}$ and $f_S \mid S = \uparrow f_S$. Given any $p \in f_S$, define $S_p = \uparrow \{p\}$, obtaining $S = \bigcap_{p \in f_S} S_p$. Thus $f \cup f_S$ is not completable to a function (if $f \cup f_S$ is completable to a function, that function is in S , thus in E , thus f compatible with E). It follows that $f \cup f_S$ is not right unique, thus, that some $t \in f$ contradicts f_S .

Conversely, if $f \in E^\subseteq$, then $f \subseteq \rightsquigarrow$ for some $\rightsquigarrow \in E$. As $\mathcal{S}$ covers E , $\rightsquigarrow \in S$ for some $S \in \mathcal{S}$. Then $\forall t \in f$: $\{t\} \subseteq \rightsquigarrow \in S$, thus $\{t\} \in S^\subseteq$. Thus, for this S , no $t \in f$ satisfies $\{t\} \notin S^\subseteq$. $\square$

Remark 16. $(a \neq b) \vee (a \wedge c)$ is equivalent to $(a \neq b) \vee (b \wedge c)$, thus there is no unique minimal DNF. $\triangle$

The following result indicates that, given a finite cover of $D \setminus C$, $\text{OF}_{D \setminus C}(\rightsquigarrow)$ requires some set of triple T that, for each S , contains a triple incompatible with S ; thus $\text{OF}_{D \setminus C}(\rightsquigarrow)$ holds iff such a T is observable in the world.

Theorem 21. *Given C, D , define $E = D \setminus C$, consider a finite cover $\mathcal{S}$ of E (cover is understood as above). The world $\rightsquigarrow$ satisfies $\text{OF}_E(\rightsquigarrow)$ iff $\exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \forall S \in \mathcal{S}, \exists t \in \theta \mid \{t\} \notin S^\subseteq$.*

Proof. By definition, $\text{OF}_E(\rightsquigarrow)$ iff $\exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \theta \notin E^\subseteq$. By theorem 20, applied to $f = \theta$, $\theta \notin E^\subseteq$ iff $\forall S \in \mathcal{S}, \exists t \in \theta \mid \{t\} \notin S^\subseteq$. $\square$

A.8. Repeatability

Theorem 22. *Given $C, D \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$: $(\neg \text{POF}_C) \cap D$ proves C iff $(\neg \text{OF}_C) \cap D$ proves C and C suffers no visibility issues in D (that is, $\forall \rightsquigarrow \in D : \text{OF}_C(\rightsquigarrow) \Rightarrow \text{POF}_C(\rightsquigarrow)$).*

Proof. The claim is equivalent to $[\forall \rightsquigarrow \notin C, \rightsquigarrow \in D : \text{POF}_C(\rightsquigarrow)] \Leftrightarrow [\forall \rightsquigarrow \notin C, \rightsquigarrow \in D : \text{OF}_C(\rightsquigarrow) \wedge \forall \rightsquigarrow \notin C, \rightsquigarrow \in D : \text{OF}_C(\rightsquigarrow) \Rightarrow \text{POF}_C(\rightsquigarrow)]$, because $\forall \rightsquigarrow \in C, \neg \text{OF}_C(\rightsquigarrow)$. $\square$

Example 13. Define $C = \alpha \rightsquigarrow_i \varphi$ for a given α and i .

We have $\neg \text{OF}_C \subseteq C$, as visible by observing that $\alpha \not\rightsquigarrow_i \varphi \Rightarrow \text{OF}_C(\rightsquigarrow)$.

Define $\rightsquigarrow$ such that $\forall \alpha', j : \alpha' \rightsquigarrow_j \varphi'$ (thus $\rightsquigarrow \in E$ and $\rightsquigarrow \notin C$) and let us show that $\neg \text{POF}_{C \cap E}(\rightsquigarrow)$. Pick any $\#\alpha < \#\alpha'$ and define $\theta = (i, \alpha', \varphi') \in \Theta \cap \mathcal{P}(\rightsquigarrow)$. [Claude: θ is an observation (a subset of $I \times \mathcal{A}^{<\mathbb{N}} \times P$), but here it is set equal to a single triple (i, α', φ') rather than to a set containing it. Should this be $\theta = \{(i, \alpha', \varphi')\}$?] Consider any $\theta \subseteq \theta' \in \Theta \cap \mathcal{P}(\rightsquigarrow)$. Define an infinitely numerous subset J among the individuals for which α was not observed in θ' such that $i \in J$, leaving another infinitely numerous subset $I \setminus J$, and define $\rightsquigarrow'$ such that $\forall j \in J : \alpha \rightsquigarrow'_j \varphi \wedge \forall \alpha'' \neq \alpha, \alpha'' \rightsquigarrow'_j \varphi'$, and $\forall j \in I \setminus J, \alpha'' : \alpha'' \rightsquigarrow'_j \varphi'$. We have $\theta' \subseteq \rightsquigarrow' \in C \cap E$, thus, $\theta' \in (C \cap E)^\subseteq$. $\triangle$

Example 13 shows that a claim can be provable by $\neg \text{OF}_C$ (thus, provable by $\neg \text{OF}_{C \cap E} \cap E$) yet not provable by $\neg \text{POF}_{C \cap E} \cap E$ (thus, also not provable by $\neg \text{POF}_C \cap E$). (Recall that $\neg \text{POF}_{C \cap E} \subseteq \neg \text{POF}_C$ and $\neg \text{OF}_{C \cap E} \subseteq \neg \text{OF}_C$.)

Say that a world admits a proof of $\neg C$ avoiding J iff $\exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \theta(\mathcal{A}^{<\mathbb{N}}, P) \subseteq I \setminus J \wedge \theta \notin C^\subseteq$. A sufficient condition for a world being POF is that for any finite subset of individuals, some proof of $\neg C$ avoids that subset.

Theorem 23. *Given any claim C and world $\rightsquigarrow$: $[\forall J \subseteq_{\text{fin}} I, \exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \theta(\mathcal{A}^{<\mathbb{N}}, P) \subseteq I \setminus J \wedge \theta \notin C^\subseteq] \Rightarrow \text{POF}_C(\rightsquigarrow)$.*

Claude proof. Fix $\tau \in \Theta \cap \mathcal{P}(\rightsquigarrow)$; let $J \subseteq I$ be its (finite) individuals, and pick, by hypothesis, $\theta \in \Theta \cap \mathcal{P}(\rightsquigarrow)$, $\theta \notin C^\subseteq$, touching only individuals outside J . As τ, θ touch disjoint individuals, $\tau' = \tau \cup \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow)$ and $\tau \sqsubseteq \tau'$ (no point is added for any individual already in τ 's domain). As $C^\subseteq$ is downwards closed and $\theta \subseteq \tau'$, $\tau' \notin C^\subseteq$. As τ was arbitrary, $\text{POF}_C(\rightsquigarrow)$ holds. $\square$

The converse of theorem 23 fails: when some given i is non avoidable for a proof of $\neg C$ but all α do prove $\neg C$, it does not bar POF.

Example 14. Fix $i^* \in I$. Let $\rightsquigarrow(i, \alpha) = \varphi$ for all $i \in I$, $\alpha \in \mathcal{A}^{<\mathbb{N}}$, and $C = \{v \in P^{I \times \mathcal{A}^{<\mathbb{N}}} \mid \forall \alpha \in \mathcal{A}^{<\mathbb{N}} : v(i^*, \alpha) = \varphi'\}$. Then $\rightsquigarrow \notin C$ and $\text{POF}_C(\rightsquigarrow)$: given any $\tau \in \Theta \cap \mathcal{P}(\rightsquigarrow)$, if i^* is not queried by τ , extend τ by one observation of i^* at $\emptyset$ to get τ' with $\tau \sqsubseteq \tau'$ and $\tau' \notin C^\subseteq$ (as $\rightsquigarrow(i^*, \emptyset) = \varphi \neq \varphi'$); if i^* is already queried, τ records φ at some path, so no $v \in C$ (requiring φ' everywhere for i^*) extends τ , and $\tau' = \tau$ works. Yet poofsoi fails for $J = \{i^*\}$: any $\theta \in \Theta \cap \mathcal{P}(\rightsquigarrow)$ querying only in $I \setminus \{i^*\}$ lies in $C^\subseteq$, since setting $v(i^*, \alpha) = \varphi'$ for all α and $v(i, \alpha) = \varphi$ for $i \neq i^*$ yields $v \in C$ with $\theta \subseteq v$. $\triangle$

A world $\rightsquigarrow$ differs for infinitely many individuals with E iff it differs for infinitely many individuals with each world in E ; formally, iff $\forall \rightsquigarrow \neq \rightsquigarrow' \in E : \#\{i \in I \mid \rightsquigarrow_i \neq \rightsquigarrow'_i\} \notin \mathbb{N}$.

Example 15 (A world differing for infinitely many individuals with E). Let $w^*(i, \alpha) = \varphi'$ for all $i \in I$, $\alpha \in \mathcal{A}^{<\mathbb{N}}$ (constant world). Then $w^* \in E$ (all individuals share the same behavior). For each $\rightsquigarrow \in E \setminus \{w^*\}$: $\rightsquigarrow \neq w^*$ means some individual j has $\rightsquigarrow(j, \cdot) \neq$ (always φ'); since $\rightsquigarrow \in E$, this behavior $\rightsquigarrow(j, \cdot)$ is shared by infinitely many individuals i , each satisfying $\rightsquigarrow(i, \cdot) \neq$ (always φ') = $w^*(i, \cdot)$. Hence $\#\{i \in I \mid \rightsquigarrow(i, \cdot) \neq w^*(i, \cdot)\} \notin \mathbb{N}$ for every $\rightsquigarrow \in E \setminus \{w^*\}$: w^* differs for infinitely many individuals with E . $\triangle$

Example 16 (A world not differing for infinitely many individuals with E). Fix $a \in \mathcal{A}$, an infinite proper subset $S \subsetneq I$ with $I \setminus S$ infinite, and $i_0 \in S$. Let $v \in E$ be the world where $v(i, (a)) = \varphi'$ for $i \in S$ and $v(i, \alpha) = \varphi$ otherwise. Let w_0 agree with v everywhere except $w_0(i_0, (a)) = \varphi$. Then $w_0 \in E$: individuals in $S \setminus \{i_0\}$ all share behavior " φ' at (a) , φ elsewhere" (infinitely many), and individuals in $\{i_0\} \cup (I \setminus S)$ all share behavior "always φ " (infinitely many). The worlds w_0 and v differ only at i_0 , so $\#\{i \in I \mid w_0(i, \cdot) \neq v(i, \cdot)\} = 1 \in \mathbb{N}$: w_0 does not differ for infinitely many individuals with $v \in E$. $\triangle$

Lemma 2. *Given any anonymous C and any world $\rightsquigarrow \in E$ such that $\text{OF}_C(\rightsquigarrow)$: $\forall J \subseteq_{\text{fin}} I, \exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \theta(\mathcal{A}^{<\mathbb{N}}, P) \subseteq I \setminus J \wedge \theta \notin C^\subseteq$.*

Proof. As $\text{OF}_C(\rightsquigarrow)$, some $\theta^{(0)} \in \Theta \cap \mathcal{P}(\rightsquigarrow)$ satisfies $\theta^{(0)} \notin C^\subseteq$. Define $J^{(0)} = \theta^{(0)}(\mathcal{A}^{<\mathbb{N}}, P)$ as the individuals queried by $\theta^{(0)}$.

Given any $J \subseteq_{\text{fin}} I$, define r as an injection from $J^{(0)}$ to $I \setminus (J \cup J^{(0)})$ such that $\rightsquigarrow_{r(i)} = \rightsquigarrow_i$ (which exists because $\rightsquigarrow \in E$), define $J' = r(J^{(0)})$ (thus r is a bijection from $J^{(0)}$ to J' disjoint from both J and $J^{(0)}$), complete r to the self-inverse permutation $\sigma = r \cup r^{-1} \cup \{(i, i) \mid i \notin J^{(0)} \cup J'\}$ over I , and define $\theta = \{(r(i), \alpha, p) \mid (i, \alpha, p) \in \theta^{(0)}\}$ as $\theta^{(0)}$ where individuals from $J^{(0)}$ are replaced by individuals determined by r . We observe that $\theta \subseteq \rightsquigarrow$ (by construction of r) and that $\theta(\mathcal{A}^{<\mathbb{N}}, P) = J' \subseteq I \setminus (J \cup J^{(0)}) \subseteq I \setminus J$. Finally, if there exists $\theta \subseteq \rightsquigarrow_J \in C$, then also $\theta^{(0)} \subseteq \rightsquigarrow_J \circ \sigma^{-1} \in C$ (by anonymity of C); as $\theta^{(0)} \notin C^\subseteq$, we obtain $\theta \notin C^\subseteq$. $\square$

Lemma 3. *Given any two claims C_1, C_2 and world $\rightsquigarrow$: if C_1 and C_2 each satisfy $\forall J \subseteq_{\text{fin}} I, \exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \theta(\mathcal{A}^{<\mathbb{N}}, P) \subseteq I \setminus J \wedge \theta \notin C^\subseteq$ (with $C = C_1$ and $C = C_2$ respectively), then so does $C_1 \cup C_2$.*

Proof. Given any $J \subseteq_{\text{fin}} I$: apply the hypothesis for C_2 to J , obtaining $\theta_2 \in \Theta \cap \mathcal{P}(\rightsquigarrow)$ with $\theta_2(\mathcal{A}^{<\mathbb{N}}, P) \subseteq I \setminus J$ and $\theta_2 \notin C_2^\subseteq$; let $J_2 = \theta_2(\mathcal{A}^{<\mathbb{N}}, P)$. Apply the hypothesis for C_1 to $J \cup J_2$, obtaining θ_1 with $\theta_1(\mathcal{A}^{<\mathbb{N}}, P) \subseteq I \setminus (J \cup J_2)$ and $\theta_1 \notin C_1^\subseteq$. Set $\theta = \theta_1 \cup \theta_2$: since θ_1 and θ_2 query disjoint individuals and are both consistent with $\rightsquigarrow$, $\theta \in \Theta \cap \mathcal{P}(\rightsquigarrow)$ and $\theta(\mathcal{A}^{<\mathbb{N}}, P) \subseteq I \setminus J$. If $\theta \subseteq v$ for some $v \in C_1$, then $\theta_1 \subseteq \theta \subseteq v$, contradicting $\theta_1 \notin C_1^\subseteq$; so $\theta \notin C_1^\subseteq$. Likewise $\theta_2 \subseteq \theta$ implies $\theta \notin C_2^\subseteq$. Hence $\theta \notin (C_1 \cup C_2)^\subseteq$. $\square$

Lemma 4. *Given any finite $F \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$ and world $\rightsquigarrow$ such that $\forall v \in F : \#\{i \in I \mid \rightsquigarrow_i \neq v_i\} \notin \mathbb{N}$: $\forall J \subseteq_{\text{fin}} I, \exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \theta(\mathcal{A}^{<\mathbb{N}}, P) \subseteq I \setminus J \wedge \theta \notin F^\subseteq$.*

Proof. Given any $J \subseteq_{\text{fin}} I$, define s as an injection from F to $I \setminus J$ such that $w_{s(w)} \neq \rightsquigarrow_i$, and $\alpha_{s(w)}$ a path such that $w(\alpha_{s(w)}, s(w)) \neq \rightsquigarrow(\alpha_{s(w)}, s(w))$. Define $\theta = \{(s(w), \alpha_{s(w)}, \rightsquigarrow(\alpha_{s(w)}, s(w))) \mid w \in F\}$.

Then $\theta \in \Theta \cap \mathcal{P}(\rightsquigarrow)$, $\theta(\mathcal{A}^{<\mathbb{N}}, P) = \{i_v \mid v \in F\} \subseteq I \setminus J$, and for each $v' \in F$: $(\alpha_{v'}, i_{v'}, \rightsquigarrow(\alpha_{v'}, i_{v'})) \in \theta$ with $\rightsquigarrow(\alpha_{v'}, i_{v'}) \neq v'(\alpha_{v'}, i_{v'})$, so $\theta \not\subseteq v'$, hence $\theta \notin F^\subseteq$. $\square$

Lemma 5. *Given any anonymous C_0 , finite $F \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$ with $\forall w \in F, \forall v \in E \setminus \{w\} : \#\{i \in I \mid v_i \neq w_i\} \notin \mathbb{N}$, $C = C_0 \cup F$, and any world $\rightsquigarrow \in E$ such that $\text{OF}_C(\rightsquigarrow)$: $\forall J \subseteq_{\text{fin}} I, \exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \theta(\mathcal{A}^{<\mathbb{N}}, P) \subseteq I \setminus J \wedge \theta \notin C^\subseteq$.*

Proof. As $\text{OF}_C(\rightsquigarrow)$, $\rightsquigarrow \notin C$, so $\rightsquigarrow \notin F$; as $\rightsquigarrow \in E$, the hypothesis on F gives $\forall v \in F : \#\{i \in I \mid \rightsquigarrow_i \neq v_i\} \notin \mathbb{N}$. Moreover, any $\theta \notin C^\subseteq$ also lies outside $C_0^\subseteq$ (as $C_0 \subseteq C$ implies $C_0^\subseteq \subseteq C^\subseteq$), so $\text{OF}_{C_0}(\rightsquigarrow)$ holds. Apply lemma 2 to C_0 , lemma 4 to F , and lemma 3: $C = C_0 \cup F$ satisfies the conclusion. $\square$

This follows from theorem 23 and lemma 5.

Theorem 24 (Finitely many, individually-spread exceptions to an anonymous claim do not register in E). *If C_0 is anonymous, $F \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$ is finite, and $\forall w \in F, \forall \rightsquigarrow \in E \setminus \{w\} : \#\{i \in I \mid \rightsquigarrow_i \neq w_i\} \notin \mathbb{N}$, then $C = C_0 \cup F$ suffers no visibility issues in E .*

Example 17 (Instance: a single named exception, escaping E). Let $C_0 = \{\rightsquigarrow \mid \forall i \in I : \rightsquigarrow(i, a) = \varphi\}$ (anonymous), fix $i^* \in I$, and let $\rightsquigarrow^{(0)}$ agree with φ everywhere except $\rightsquigarrow^{(0)}(i^*, a) = \varphi'$. By theorem 24 (with $F = \{\rightsquigarrow^{(0)}\}$), $C = C_0 \cup \{\rightsquigarrow^{(0)}\}$ suffers no visibility issues in E , yet is not anonymous (swapping i^* with any other individual moves the exception away from both C_0 and $\rightsquigarrow^{(0)}$). Notably, $\rightsquigarrow^{(0)} \notin E$: its one distinguished individual i^* has no clones. $\triangle$

Theorem 25. *If an anonymous claim $C \subseteq E$ is falsifiable or trivial, it is provable by $\neg \text{POF}_C \cap E$.*

Proof. C is falsifiable or trivial iff $\neg \text{OF}_C \subseteq C$, which implies $\neg \text{OF}_C \cap E \subseteq C$.

From theorem 24 (with $F = \emptyset$), C suffers no visibility issues in E . From theorem 22, $\neg \text{POF}_C \cap E$ proves C iff $\neg \text{OF}_C \cap E$ proves C . $\square$

The converse fails, as example 18 shows.

Theorem 26. *If $\#I \notin \mathbb{N}$, given any general observation O , $\exists O \subseteq \rightsquigarrow \in E$.*

Proof. Define for each individual observed in O a behavior that completes the observation that concern her in O . Define $\rightsquigarrow$ as the world associating infinitely many individuals to each of these behaviors. $\square$

Example 18. The claim $C = E$ is anonymous, provable by $\neg \text{POF}_C \cap E$, not trivial and not falsifiable: consider any $\rightsquigarrow \notin E$ (such as a world associating a given behavior to a single individual i and a different behavior to all others), we have $\text{F}_C(\rightsquigarrow)$ but $\neg \text{OF}_C(\rightsquigarrow)$ as any observation θ is completable in $C = E$ (theorem 26). $\triangle$

Theorem 27. *Given any $\rightsquigarrow \in E$, there exists an anonymous claim that is provable by $\neg \text{POF}_{C \cap E} \cap E$.*

Proof. Take $C = E$. E is anonymous: for any $\rightsquigarrow \in E$ and permutation σ , individual j in $\rightsquigarrow \circ \sigma$ has the same behavior as $\sigma(j)$ in $\rightsquigarrow$, so the behavior classes of $\rightsquigarrow \circ \sigma$ are those of $\rightsquigarrow$ relabeled, each still infinite, giving $\rightsquigarrow \circ \sigma \in E$. $\rightsquigarrow \in E = C$ by hypothesis. $C \cap E = E$, so $(\neg \text{POF}_{C \cap E}) \cap E = (\neg \text{POF}_E) \cap E \subseteq E = C$. $\square$

Theorem 28. *Given any $\rightsquigarrow \in E$, there exists an anonymous claim that holds, is informative and provable by $\neg \text{POF}_{C \cap E} \cap E$.*

Proof. Take $C = E$. E is anonymous (as in the previous proof), $\rightsquigarrow \in E = C$, and $(\neg \text{POF}_{C \cap E}) \cap E = (\neg \text{POF}_E) \cap E \subseteq E = C$. *Informativeness.* For any $\rightsquigarrow \in E$, each behavior class is infinite and individuals sharing a behavior share the same π -value, so every non-empty fiber of $\pi(\rightsquigarrow)$ is infinite. Hence, fixing any $i_0 \in I$ and $v \in P$, the profile assigning v to i_0 and $\emptyset$ to all other individuals has a v -fiber of size $1 \in \mathbb{N}$, so it lies outside $\pi(E)$, giving $\pi(E) \neq (P \cup \{\emptyset\})^I$. $\square$

In some worlds, the prior by $\neg \text{POF}_{C \cap E} \cap E$ is sufficient to get existence of suitable claims.

Example 19. Consider the world $\rightsquigarrow = J \times \mathcal{A}^{<\mathbb{N}} \times \{\varphi\} \cup K \times \mathcal{A}^{<\mathbb{N}} \times \{\varphi'\}$ of example 2, with infinite sets of individuals J, K . The claim $C = [(\forall i : \emptyset \rightsquigarrow_i \varphi \Rightarrow \forall \alpha, \alpha \rightsquigarrow_i \varphi) \wedge (\exists i \mid \emptyset \rightsquigarrow_i \varphi)] \cap E$ holds, is anonymous, APTS and PEF under $\neg \text{POF}_{C \cap E} \cap E$. $\triangle$

The following shows that verifiability is a mandatory part of the criterion for determining whether C holds, in that world: requiring the stronger provability instead of verifiability (under the same prior) leads to an impossibility.

Theorem 29. *Consider the world $\rightsquigarrow = J \times \mathcal{A}^{<\mathbb{N}} \times \{\varphi\} \cup K \times \mathcal{A}^{<\mathbb{N}} \times \{\varphi'\}$ of example 2, with infinite sets of individuals J, K . No anonymous claim that holds is APTS and provable by $\neg \text{POF}_{C \cap E} \cap E$.*

Proof. TODO. Either weaken the phrasing to apply example 22, or invoke a more general result coming later (and move this). $\square$

For this, see remark 24.

We are here interested in claims implying propositions of the form: any individual whose behavior satisfies some given observable condition admits a given argument being decisive. Given $T : H \rightarrow P$ for some finite $H \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$, write $T(i) = \{i\} \times T = \{(i, \alpha, p) \mid (\alpha, p) \in T\} \in \Theta$ for the observation that i conforms to T on the increasing path H .

Given $B \subseteq P^{\mathcal{A}^{<\mathbb{N}}}$ and $H \subseteq \mathcal{A}^{<\mathbb{N}}$, let $B_{\upharpoonright H} = \{f_{\upharpoonright H} \mid f \in B\}$ denote the behaviors in B restricted to H . Say a behavior g is undistinguishable from $B \subseteq P^{\mathcal{A}^{<\mathbb{N}}}$ iff $\forall H \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}}) : g_{\upharpoonright H} \in B_{\upharpoonright H}$. The claim C_B associated to $B \subseteq P^{\mathcal{A}^{<\mathbb{N}}}$ is the claim $C_B = [\rightsquigarrow(I) \subseteq B]$.

Theorem 30. *Consider $\rightsquigarrow \in E$, an anonymous claim C that holds, and a world $\rightsquigarrow'$. If $\forall b \in \rightsquigarrow'(I), b$ is undistinguishable from $\rightsquigarrow(I)$, then $\neg \text{OF}_C(\rightsquigarrow')$.*

Proof. Consider any $\theta \subseteq \rightsquigarrow'$ and let us show that $\theta \in C^{\subseteq}$.

Define the injection $s : \theta(\mathcal{A}^{<\mathbb{N}}, P) \rightarrow I$ as follows. Consider each $i \in \theta(\mathcal{A}^{<\mathbb{N}}, P)$ in turn. Let $H = \theta(i, P) \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$. As $\rightsquigarrow'_i$ is undistinguishable from $\rightsquigarrow(I)$, $\exists i' \mid \rightsquigarrow'_i \upharpoonright H = \rightsquigarrow_{i'} \upharpoonright H$. Set $s(i) = i'$. As $\rightsquigarrow \in E$, there are infinitely many suitable i' so we can pick one that has not been chosen yet, to ensure s is an injection.

The function s has domain $\theta(\mathcal{A}^{<\mathbb{N}}, P)$ and codomain $s(\theta(\mathcal{A}^{<\mathbb{N}}, P))$, both finite sets of equal cardinality. Turn s into a permutation $\sigma = s \cup s'$ over I by defining any bijection $s' : I \setminus \theta(\mathcal{A}^{<\mathbb{N}}, P) \rightarrow I \setminus s(\theta(\mathcal{A}^{<\mathbb{N}}, P))$ (TBD: argue that we can do this). [Claude: This bijection s' does exist and the "TBD" seems closeable: $\theta(\mathcal{A}^{<\mathbb{N}}, P)$ and its image under s are finite (both of size $\#\theta(\mathcal{A}^{<\mathbb{N}}, P)$), and I is infinite (any $\rightsquigarrow \in E$ has, for every i , an infinite class of individuals sharing $\rightsquigarrow_i$, so I itself is infinite); removing a finite subset from an infinite set always leaves a set of the same cardinality as the original, for either finite subset removed, so $I \setminus \theta(\mathcal{A}^{<\mathbb{N}}, P)$ and $I \setminus s(\theta(\mathcal{A}^{<\mathbb{N}}, P))$ have equal cardinality and a bijection s' exists. Worth stating explicitly instead of leaving as TBD.]

By construction, $\theta \subseteq \rightsquigarrow \circ \sigma$, which belongs to C because C holds and is anonymous. $\square$

Theorem 31. *Given $\rightsquigarrow \in E$, $a \in \mathcal{A}$, $b^{new} \in P^{\mathcal{A}^{<\mathbb{N}}}$, some finite $\emptyset \neq H \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$ and an anonymous claim $C \subseteq [\forall i \in I : b^{new}_{\upharpoonright H} \subseteq \rightsquigarrow_i \Rightarrow \exists p \in P \mid a \hookrightarrow_i p]$, if b^{new} is undistinguishable from $\rightsquigarrow(I)$ yet $b^{new} \notin \rightsquigarrow(I)$ and $\nexists p \in P \mid a \hookrightarrow_{b^{new}} p$ and C holds, it is not verifiable at $\rightsquigarrow$ under $\neg \text{OF}_C \cap E$ (thus also not verifiable at $\rightsquigarrow$ under $\neg \text{POF}_C \cap E$).*

Proof. Consider any prefix $\theta \subseteq \rightsquigarrow$. We build $\theta \subseteq \rightsquigarrow' \in (\neg \text{OF}_C \cap E) \setminus C$. Consider any $i \in \theta(\mathcal{A}^{<\mathbb{N}}, P)$. Pick any $b \in \rightsquigarrow(I) \mid \theta(i) \subseteq b$, define $\rightsquigarrow'_i = b$. Partition $I \setminus \theta(\mathcal{A}^{<\mathbb{N}}, P)$ into infinite sets, one for each behavior in $\rightsquigarrow(I)$, plus one for b^{new} , thereby creating a function $f : I \setminus \theta(\mathcal{A}^{<\mathbb{N}}, P) \rightarrow P^{\mathcal{A}^{<\mathbb{N}}}$ (TBD argue from $\#\rightsquigarrow(I) \leq \#I$ that we can do this). [Claude: This is a substantive gap, not just bookkeeping: it requires partitioning the cofinite set $I \setminus \theta(\mathcal{A}^{<\mathbb{N}}, P)$ into one infinite class per element of $\rightsquigarrow(I)$ plus one more, all pairwise disjoint. If $\rightsquigarrow(I)$ (the set of behaviors realized in $\rightsquigarrow$) is large — e.g. as large as, or larger than, I itself — such a partition need not exist (you cannot in general split a set of cardinality $\#I$ into $\#\rightsquigarrow(I)$ many infinite disjoint pieces unless $\#I \geq \#\rightsquigarrow(I) \cdot \aleph_0$, roughly). Nothing in the hypotheses bounds $\#\rightsquigarrow(I)$ relative to $\#I$, so as stated the construction is not guaranteed to go through in general.] For each $i \in I \setminus \theta(\mathcal{A}^{<\mathbb{N}}, P)$ with $f(i) \in \rightsquigarrow(I)$, define $\rightsquigarrow'_i = f(i)$, for the remaining infinite set, define $\rightsquigarrow'_i = b^{\text{new}}$.

Note that $\rightsquigarrow' \in E$. Also, $\rightsquigarrow' \notin C$: picking any $i \mid \rightsquigarrow'_i = b^{\text{new}}$, $b^{\text{new}} \upharpoonright_H \subseteq \rightsquigarrow'_i$, but $\nexists p \mid a \hookrightarrow_i p$.

Observe that each behavior in $\rightsquigarrow'(I)$ is undistinguishable from $\rightsquigarrow(I)$. By theorem 30, $\neg \text{OF}_C(\rightsquigarrow')$. $\square$

Example 20. Consider the world $\rightsquigarrow = (J \times \mathcal{A}^{<\mathbb{N}} \times \{\varphi\}) \cup (K \times \{\emptyset\} \times \{\varphi\}) \cup (K \times (\mathcal{A}^{<\mathbb{N}} \setminus \{\emptyset\}) \times \{\varphi'\})$ of example 2, with infinite sets of individuals J, K . [Claude: example 2 states the kay's "always answer φ' ", i.e. $\forall i \in K, \alpha \in \mathcal{A}^{<\mathbb{N}} : \alpha \rightsquigarrow_i \varphi'$, including at $\alpha = \emptyset$. Here the kay's instead answer φ at $\emptyset$ and φ' everywhere else, which is a different world than the one literally defined in example 2 (and the special-casing at $\emptyset$ turns out to matter: it is exactly what makes $i \in K$ fail the antecedent of C 's condition below, via the $1 \leq \# \max H$ hypothesis). Referring to it as "the world ... of example 2" may mislead a reader checking back against that example's literal statement.]

Pick some finite $\emptyset \neq H \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$ such that $1 \leq \#(\max H)$, pick $d \in \mathcal{A} \setminus \{a, (\max H)(1)\}$. Define $S = \{\emptyset \neq \alpha \mid \alpha_1 = d\} \subseteq \mathcal{A}^{<\mathbb{N}}$ as the set of paths starting with d . Let b_j be the behavior associated to $\rightsquigarrow(J)$, define b_k similarly. Define $b^{\text{new}} \in P^{\mathcal{A}^{<\mathbb{N}}}$ as $b^{\text{new}} = b_k \upharpoonright_S \cup b_j \upharpoonright_{\mathcal{A}^{<\mathbb{N}} \setminus S}$. Consider any anonymous claim $C \subseteq [\forall i \in I : b^{\text{new}} \upharpoonright_H \subseteq \rightsquigarrow_i \Rightarrow \exists p \in P \mid a \hookrightarrow_i p]$ that holds (for example, $[\forall i \in I : b^{\text{new}} \upharpoonright_H \subseteq \rightsquigarrow_i \Rightarrow \exists p \in P \mid a \hookrightarrow_i p]$). [Claude: Theorem 31 additionally requires " C holds", i.e. $\rightsquigarrow \in C$; not every anonymous C included in the bracketed set satisfies this (e.g. $C = \emptyset$ does not). It does hold for the largest such C (taking C to be exactly the bracketed set, which is itself anonymous since it quantifies over all $i \in I$ uniformly): every $j \in J$ trivially satisfies the antecedent and is decisive for φ , and every $k \in K$ fails the antecedent because $b^{\text{new}} \upharpoonright_H = b_j \upharpoonright_H$, constantly φ (since $S \cap H = \emptyset$) while $\rightsquigarrow_k$ disagrees with φ on H 's nonempty path(s) — which is exactly where the $1 \leq \# \max H$ hypothesis is needed, to guarantee H has a nonempty path on which k 's behavior (constant φ' off $\emptyset$) departs from φ . This isn't spelled out in the example; as written, "consider any anonymous claim $C \subseteq [\dots]$ " doesn't establish that C can be chosen to hold at $\rightsquigarrow$.]

Observe that $S \cap H = \emptyset$ ($\emptyset \notin S$ by definition and $\forall \emptyset \neq \alpha \in H : \alpha_1 = (\max H)(1) \neq d$) [Claude: This follows because H , being an increasing set of paths, is a chain under the prefix order, so every nonempty $\alpha \in H$ is a prefix of $\max H$ and hence shares its first argument $(\max H)(1) \neq d$; combined with $\emptyset \notin S$ by definition, no element of H lies in S . Worth a one-line justification, as it's the crux of the construction (and is also used implicitly, in the

same way, to show b^{new} is undistinguishable from $\rightsquigarrow(I)$ below.)] and as $(d, a) \in S$ and $(a, d) \notin S$, $b^{\text{new}}((d, a)) = \varphi' \neq b^{\text{new}}((a, d)) = \varphi$, thus $\nexists p \mid a \hookrightarrow_{b^{\text{new}}} p$. Also, b^{new} is undistinguishable from $\rightsquigarrow(I) = \{b_j, b_k\}$: for any $H' \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$, since H' is a chain, either all its nonempty paths start with d — in which case $b^{\text{new}}|_{H'} = b_k|_{H'}$ — or none do — in which case $b^{\text{new}}|_{H'} = b_j|_{H'}$; and $b^{\text{new}} \notin \rightsquigarrow(I)$: e.g. $b^{\text{new}}((d)) = \varphi' \neq \varphi = b_j((d))$ and $b^{\text{new}}((a)) = \varphi \neq \varphi' = b_k((a))$, using $a \neq d$. Thus, theorem 31 applies. $\triangle$

Remark 17. The $H = \emptyset$ case technically holds, but either requires to take seriously the $\exists p$ in the claim C 's deduction (i_1 has a decisive for φ and i_2 has a decisive for φ'), or if p is fixed for everybody, then the antecedent of the theorem never triggers. $\triangle$

A.9. One more prior

Given $\mathbf{g}$, $a \in \mathcal{A}$, $p \in P$, $A, B \subseteq P^{\mathcal{A}^{<\mathbb{N}}}$, define $D_{A,B}^{\mathbf{g},a,p}$ as: any behavior $b \in A \setminus B$ failing $\mathbf{g} \subseteq \rightsquigarrow_b \Rightarrow a \hookrightarrow_b p$ (thus satisfying $\mathbf{g} \subseteq \rightsquigarrow_b \wedge a \not\hookrightarrow_b p$) is distinguishable from B .

Given $\mathbf{g}$, $a \in \mathcal{A}$, $p \in P$, define $D_{P^{\mathcal{A}^{<\mathbb{N}}}, \rightsquigarrow(I)}^{\mathbf{g},a,p}$ as: any behavior $b \notin \rightsquigarrow(I)$ failing $\mathbf{g} \subseteq \rightsquigarrow_b \Rightarrow a \hookrightarrow_b p$ is distinguishable from $\rightsquigarrow(I)$.

Given $\mathbf{g}$, $a \in \mathcal{A}$, $p \in P$, define $D_C^{\mathbf{g},a,p}$ as $D_{\rightsquigarrow(I), S_C}^{\mathbf{g},a,p}$: any behavior that fails $\mathbf{g}$ leads i to (a, p) does so visibly out of B , the types of C . (We could relax the prior by adding $B \not\subseteq \rightsquigarrow(I) \vee \dots$ (or $B \subseteq \rightsquigarrow(I) \Rightarrow \dots$), because we need only verifiability when $B \not\subseteq \rightsquigarrow(I)$!)

Theorem 32. *Given $B \subseteq P^{\mathcal{A}^{<\mathbb{N}}}$ with $\rightsquigarrow(I) \subseteq B$, and $\rightsquigarrow \in D_{P^{\mathcal{A}^{<\mathbb{N}}}, B}^{\mathbf{g},a,p}$ such that for some H with $1 \leq \#(\max H)$, $\mathbf{g} : H \rightarrow P$, a, p , $\mathbf{g} \subseteq \rightsquigarrow_i \Rightarrow a \hookrightarrow_i p$, and $\exists i \mid \mathbf{g} \subseteq \rightsquigarrow_i$, there exists an anonymous claim C that holds, is SAPTS and PEV under $(\neg \text{POF}_C) \cap E \cap D_{\rightsquigarrow(I) \cap B, S_C}^{\mathbf{g},a,p}$.*

Proof. Let C_0 be the claim built in the proof of theorem 33, and set $C := C_0 \cap [\exists i \in I \mid \mathbf{g} \subseteq \rightsquigarrow_i]$, writing $S_C := \bigcup_{\rightsquigarrow' \in C} \rightsquigarrow'(I)$ for the types of this (smaller) claim, and $D := (\neg \text{POF}_C) \cap E \cap D_{\rightsquigarrow(I) \cap B, S_C}^{\mathbf{g},a,p}$ for the prior at stake (both evaluated at this C , as in theorem 33's proof).

Holds, Anonymous. $\rightsquigarrow \in C_0$, and the new conjunct holds at $\rightsquigarrow$ by hypothesis, so $\rightsquigarrow \in C$. The new conjunct has the anonymity-friendly shape $\exists i : Q(\rightsquigarrow(i))$: for any permutation σ , $\exists i : \mathbf{g} \subseteq (\rightsquigarrow \circ \sigma)(i) \Leftrightarrow \exists i : \mathbf{g} \subseteq \rightsquigarrow(\sigma(i)) \Leftrightarrow \exists j : \mathbf{g} \subseteq \rightsquigarrow(j)$, reindexing $j := \sigma(i)$; combined with C_0 's anonymity, $C \circ \sigma = C$.

SAPTS. Fix $\rightsquigarrow \in C$ and $\theta \in \Theta \cap \mathcal{P}(\rightsquigarrow)$. As $\rightsquigarrow \in C$, fix $i^* \in I$ with $\mathbf{g} \subseteq \rightsquigarrow_{i^*}$, and let $\theta' := \theta \cup (\{i^*\} \times \mathbf{g})$, so $\theta \sqsubseteq \theta'$ and $\theta' \in \Theta \cap \mathcal{P}(\rightsquigarrow)$ (as in theorem 33's *APTS-P* paragraph, now with i^*, θ in place of that paragraph's i_0, θ_0). Any $\rightsquigarrow' \in \{\theta' \subseteq \rightsquigarrow\} \cap C$ has $\mathbf{g} \subseteq \rightsquigarrow'(i^*)$, so C_0 's first conjunct at i^* gives $a \hookrightarrow_{i^*} p$ in $\rightsquigarrow'$, hence $\pi_{i^*}(\rightsquigarrow') = p$ by theorem 1. As $\rightsquigarrow$ itself lies in $\{\theta' \subseteq \rightsquigarrow\} \cap C$, this set is nonempty, so $\pi_{i^*}(\{\theta' \subseteq \rightsquigarrow\} \cap C) = \{p\}$: (θ', i^*) witnesses SAPTS at $(\rightsquigarrow, \theta)$. As both were arbitrary, C is SAPTS.

PEV under D . By corollary 5, it suffices to show $\text{POF}_{D \setminus C}(\rightsquigarrow)$ for every $\rightsquigarrow \in C \cap D$.

$D \subseteq C_0$. Steps 1, A and B of theorem 33's proof only ever use C_0 's first and second conjuncts (never a bound on C from above), so they apply verbatim to C in place of

that proof's own claim, with this C 's own S_C : any $\rightsquigarrow' \in E \cap D_{\rightsquigarrow'(I) \cap B, S_C}^{\mathfrak{g}, a, p}$ outside C_0 satisfies $\text{POF}_C(\rightsquigarrow')$, that is $\rightsquigarrow' \notin \neg \text{POF}_C$. So no $\rightsquigarrow' \in D$ lies outside C_0 : $D \subseteq C_0$.

Fix $\rightsquigarrow \in C \cap D$ and $\theta \in \Theta \cap \mathcal{P}(\rightsquigarrow)$; reuse i^*, θ' from the *SAPTS* paragraph above. Suppose $\rightsquigarrow'' \in D$ extends θ' . By $D \subseteq C_0$, $\rightsquigarrow'' \in C_0$; and $\theta' \subseteq \rightsquigarrow''$ gives $\mathfrak{g} \subseteq \rightsquigarrow''(i^*)$, so $\rightsquigarrow''$ also satisfies C 's extra conjunct (witnessed by i^*), hence $\rightsquigarrow'' \in C$. So no $\rightsquigarrow'' \in D \setminus C$ extends θ' , that is, $\theta' \notin (D \setminus C)^\subseteq$. As θ was arbitrary, $\text{POF}_{D \setminus C}(\rightsquigarrow)$ holds; as $\rightsquigarrow \in C \cap D$ was arbitrary, C satisfies PEV under D . $\square$

Theorem 33. *Given $B \subseteq P^{\mathcal{A}^{<\mathbb{N}}}$ with $\rightsquigarrow(I) \subseteq B$, and $\rightsquigarrow \in D_{P^{\mathcal{A}^{<\mathbb{N}}}, B}^{\mathfrak{g}, a, p}$ such that for some H with $1 \leq \#(\max H)$, $\mathfrak{g} : H \rightarrow P$, a, p , $\mathfrak{g} \subseteq \rightsquigarrow_i \Rightarrow a \hookrightarrow_i p$, and $\exists i \mid \mathfrak{g} \subseteq \rightsquigarrow_i$, there exists an anonymous claim C that holds, is APTS and provable by $(\neg \text{POF}_C) \cap E \cap D_{\rightsquigarrow(I) \cap B, S_C}^{\mathfrak{g}, a, p}$.*

Proof. Write $\text{Bad} := \{b \in P^{\mathcal{A}^{<\mathbb{N}}} \mid b \notin B \wedge \mathfrak{g} \subseteq b \wedge a \not\hookrightarrow_b p\}$. As $\rightsquigarrow \in D_{P^{\mathcal{A}^{<\mathbb{N}}}, B}^{\mathfrak{g}, a, p}$, fix, for each $b \in \text{Bad}$, some $H_b \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$ with $b|_{H_b} \notin B|_{H_b}$. Take

$$C := [\forall i \in I : \mathfrak{g} \subseteq \rightsquigarrow_i \Rightarrow a \hookrightarrow_i p] \cap [\forall i \in I, b \in \text{Bad} : \rightsquigarrow_i|_{H_b} \neq b|_{H_b}].$$

Holds. The first conjunct holds at $\rightsquigarrow$: this is exactly the hypothesis $\mathfrak{g} \subseteq \rightsquigarrow_i \Rightarrow a \hookrightarrow_i p$. For the second, fix $b \in \text{Bad}$: as $\rightsquigarrow(I) \subseteq B$, every $\rightsquigarrow_i \in B$, so $b|_{H_b} \notin B|_{H_b}$ gives $\forall i \in I : \rightsquigarrow_i|_{H_b} \neq b|_{H_b}$.

Anonymous. Write $Q_1(f) := [\mathfrak{g} \subseteq f \Rightarrow a \hookrightarrow_f p]$ and $Q_2(f) := [\forall b \in \text{Bad} : f|_{H_b} \neq b|_{H_b}]$, properties of a single row f not depending on i (Bad and each H_b are fixed once and for all), so that $C = [\forall i : Q_1(\rightsquigarrow(i)) \wedge Q_2(\rightsquigarrow(i))]$. Fix a permutation σ of I and $\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}}$. As $(\rightsquigarrow \circ \sigma)(i) = \rightsquigarrow(\sigma(i))$,

$$\rightsquigarrow \circ \sigma \in C \Leftrightarrow \forall i \in I : Q_1(\rightsquigarrow(\sigma(i))) \wedge Q_2(\rightsquigarrow(\sigma(i))) \Leftrightarrow \forall j \in I : Q_1(\rightsquigarrow(j)) \wedge Q_2(\rightsquigarrow(j)) \Leftrightarrow \rightsquigarrow \in C,$$

reindexing $j := \sigma(i)$, a bijection of I . As $\rightsquigarrow, \sigma$ were arbitrary, $C \circ \sigma = C$: C is anonymous.

APTS-P (hence APTS). Fix $i_0 \in I$ with $\theta_0 := \{i_0\} \times \mathfrak{g} \subseteq \rightsquigarrow$ (hypothesis). For any $\rightsquigarrow' \in \{\theta_0 \subseteq \rightsquigarrow\} \cap C$: $\theta_0 \subseteq \rightsquigarrow'$ gives $\mathfrak{g} \subseteq \rightsquigarrow'(i_0)$, so C 's first conjunct at i_0 gives $a \hookrightarrow_{i_0} p$ in $\rightsquigarrow'$ directly, that is $p \in P_{i_0}(\rightsquigarrow')$; by theorem 1, $P_{i_0}(\rightsquigarrow') = \{p\}$ exactly, that is $\pi_{i_0}(\rightsquigarrow') = p$. Since moreover $\rightsquigarrow \in \{\theta_0 \subseteq \rightsquigarrow\} \cap C$ (as $\theta_0 \subseteq \rightsquigarrow$ and $\rightsquigarrow \in C$), this set is nonempty, so $\pi_{i_0}(\{\theta_0 \subseteq \rightsquigarrow\} \cap C) = \{p\}$: as $p \in P$ and θ_0 is feasible, this is APTS-P, witnessed by (θ_0, i_0, p) ; in particular C is APTS.

Provable by $(\neg \text{POF}_C) \cap E \cap D_{\rightsquigarrow(I) \cap B, S_C}^{\mathfrak{g}, a, p}$. Write $S_C := \bigcup_{\rightsquigarrow' \in C} \rightsquigarrow'(I)$ for the types of C , evaluated at the claim C just constructed above (not circular: S_C , and the prior $D_{\rightsquigarrow(I) \cap B, S_C}^{\mathfrak{g}, a, p}$, are only ever evaluated at this C , never used to construct it).

Step 1 (S_C -types are decisive). For $h \in S_C$, some $\rightsquigarrow'' \in C$ and $i'' \in I$ satisfy $\rightsquigarrow''(i'') = h$; C 's first conjunct at i'' gives $\mathfrak{g} \subseteq h \Rightarrow a \hookrightarrow_h p$. Contrapositively, any h with $\mathfrak{g} \subseteq h \wedge a \not\hookrightarrow_h p$ satisfies $h \notin S_C$.

We show $(\neg \text{POF}_C) \cap E \cap D_{\rightsquigarrow(I) \cap B, S_C}^{\mathfrak{g}, a, p} \subseteq C$. Fix $\rightsquigarrow' \in E \cap D_{\rightsquigarrow(I) \cap B, S_C}^{\mathfrak{g}, a, p}$ with $\rightsquigarrow' \notin C$.

Case B: the second conjunct fails (the template holds, but its extra content — avoiding collision with Bad — does not). Some $i \in I$, $b \in \text{Bad}$ satisfy $\rightsquigarrow'(i)|_{H_b} = b|_{H_b}$; write $g := \rightsquigarrow'(i)$, so $g|_{H_b} = b|_{H_b}$.

Fix any $\theta \in \Theta \cap \mathcal{P}(\rightsquigarrow')$. As $\rightsquigarrow' \in E$, $J_g := \{j \in I \mid \rightsquigarrow'(j) = g\}$ is infinite; as θ is finite, $\theta(\mathcal{A}^{<\mathbb{N}}, P)$ is finite, so $J_g \setminus \theta(\mathcal{A}^{<\mathbb{N}}, P) \neq \emptyset$. Pick i' in this set and let $\theta_b^{i'} := \{(i', \alpha, g(\alpha)) \mid \alpha \in H_b\}$. As $H_b \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$, $\theta_b^{i'} \in \Theta$, and $\theta_b^{i'} \subseteq \rightsquigarrow'$ (as $\rightsquigarrow'(i') = g$).

If some $\rightsquigarrow''' \in C$ extended $\theta_b^{i'}$, then $f := \rightsquigarrow'''(i')$ would satisfy $f|_{H_b} = g|_{H_b} = b|_{H_b}$, contradicting C 's second conjunct at i', b , applied to $\rightsquigarrow'''$. So $\theta_b^{i'} \notin C^\subseteq$.

Let $\theta' := \theta \cup \theta_b^{i'}$. Since i' is untouched by θ , $\theta \sqsubseteq \theta'$, and $\theta' \in \Theta \cap \mathcal{P}(\rightsquigarrow')$. Since $\theta_b^{i'} \subseteq \theta'$ and $C^\subseteq$ is downward closed, $\theta' \in C^\subseteq$ would force $\theta_b^{i'} \in C^\subseteq$, contradicting the above; so $\theta' \notin C^\subseteq$. As θ was arbitrary (the same b, H_b serve throughout; only the clone i' depends on θ), $\text{POF}_C(\rightsquigarrow')$ holds.

Case A: the second conjunct holds but the first fails (falsifying the template itself). Some $i \in I$ has $g := \rightsquigarrow'(i)$ with $\mathfrak{g} \subseteq g$ and $a \not\rightarrow_g p$; by Step 1, $g \notin S_C$. Moreover $g \notin \text{Bad}$: otherwise the second conjunct, evaluated at this i and $b := g \in \text{Bad}$, would require $g|_{H_g} \neq g|_{H_g}$, absurd. Since g satisfies $\mathfrak{g} \subseteq g \wedge a \not\rightarrow_g p$ yet $g \notin \text{Bad} = \{b \in P^{\mathcal{A}^{<\mathbb{N}}} \setminus B \mid \mathfrak{g} \subseteq b \wedge a \not\rightarrow_b p\}$, this forces $g \in B$. So $g \in (\rightsquigarrow'(I) \cap B) \setminus S_C$, and as $\rightsquigarrow' \in D_{\rightsquigarrow(I) \cap B, S_C}^{\mathfrak{g}, a, p}$, this yields some $H \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$ with $g|_H \notin S_{C|H}$.

Fix any $\theta \in \Theta \cap \mathcal{P}(\rightsquigarrow')$. As $\rightsquigarrow' \in E$, $J_g := \{j \in I \mid \rightsquigarrow'(j) = g\}$ is infinite; as θ is finite, $\theta(\mathcal{A}^{<\mathbb{N}}, P)$ is finite, so $J_g \setminus \theta(\mathcal{A}^{<\mathbb{N}}, P) \neq \emptyset$. Pick i' in this set and let $\theta_H^{i'} := \{(i', \alpha, g(\alpha)) \mid \alpha \in H\}$. As $H \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$, $\theta_H^{i'} \in \Theta$, and $\theta_H^{i'} \subseteq \rightsquigarrow'$ (as $\rightsquigarrow'(i') = g$).

If some $\rightsquigarrow''' \in C$ extended $\theta_H^{i'}$, then $f := \rightsquigarrow'''(i')$ would satisfy $f|_H = g|_H$ while $f \in \rightsquigarrow'''(I) \subseteq S_C$, giving $g|_H \in S_{C|H}$ — contradiction. So $\theta_H^{i'} \notin C^\subseteq$.

Let $\theta' := \theta \cup \theta_H^{i'}$. Since i' is untouched by θ , $\theta \sqsubseteq \theta'$, and $\theta' \in \Theta \cap \mathcal{P}(\rightsquigarrow')$. Since $\theta_H^{i'} \subseteq \theta'$ and $C^\subseteq$ is downward closed, $\theta' \in C^\subseteq$ would force $\theta_H^{i'} \in C^\subseteq$, contradicting the above; so $\theta' \notin C^\subseteq$. As θ was arbitrary (the same g, H serve throughout; only the clone i' depends on θ), $\text{POF}_C(\rightsquigarrow')$ holds.

Either way, $\text{POF}_C(\rightsquigarrow')$ holds, so $\rightsquigarrow' \notin \neg \text{POF}_C$. As $\rightsquigarrow' \in E \cap D_{\rightsquigarrow(I) \cap B, S_C}^{\mathfrak{g}, a, p} \setminus C$ was arbitrary, $(\neg \text{POF}_C) \cap E \cap D_{\rightsquigarrow(I) \cap B, S_C}^{\mathfrak{g}, a, p} \subseteq C$. $\square$

Remark 18. Taking $B = \rightsquigarrow(I)$ trivializes the filter prior: $\rightsquigarrow(I) \subseteq S_C$ (every row of $\rightsquigarrow$ is, by *Holds*, a type of C), so $\rightsquigarrow(I) \cap B \subseteq S_C$ always, the antecedent $(\rightsquigarrow(I) \cap B) \setminus S_C$ is never met, and $D_{\rightsquigarrow(I) \cap B, S_C}^{\mathfrak{g}, a, p} = P^{I \times \mathcal{A}^{<\mathbb{N}}}$. Taking $B = P^{\mathcal{A}^{<\mathbb{N}}}$ instead makes $\text{Bad} = \emptyset$, so C_0 's second conjunct vanishes and $D_{\rightsquigarrow(I) \cap B, S_C}^{\mathfrak{g}, a, p} = D_{\rightsquigarrow(I), S_C}^{\mathfrak{g}, a, p} = D_C^{\mathfrak{g}, a, p}$, the full prior of corollary 12. This only ever used *Holds* and the definition of Bad , so it holds verbatim for theorem 32's construction too: either theorem, at $B = \rightsquigarrow(I)$, reduces to corollary 11, and at $B = P^{\mathcal{A}^{<\mathbb{N}}}$, to corollary 12. Between these extremes, the stronger the upfront prior on $\rightsquigarrow$ (the smaller B), the weaker the residual prior needed in the filter (the smaller $\rightsquigarrow(I) \cap B$ to test) — and conversely. $\triangle$

Corollary 11. *Given $\rightsquigarrow \in D^{\mathfrak{g}, a, p}$ such that for some H with $1 \leq \#(\max H)$, $\mathfrak{g} : H \rightarrow P$, a, p , $\mathfrak{g} \subseteq \rightsquigarrow_i \Rightarrow a \hookrightarrow_i p$, and $\exists i \mid \mathfrak{g} \subseteq \rightsquigarrow_i$, there exists an anonymous claim that holds, is SAPTS and PEV under $\neg \text{POF}_C \cap E$.*

Proof. Apply theorem 32 with $B := \rightsquigarrow(I)$: the hypothesis $\rightsquigarrow(I) \subseteq B$ is trivial, and $D_{P^{\mathcal{A}^{<\mathbb{N}}}, \rightsquigarrow(I)}^{\mathfrak{g}, a, p} = D^{\mathfrak{g}, a, p}$ is exactly the hypothesis here. By the remark following theorem 33, at this B the filter prior $D_{\rightsquigarrow(I) \cap B, S_C}^{\mathfrak{g}, a, p}$ trivializes to $P^{I \times \mathcal{A}^{<\mathbb{N}}}$, so PEV under $(\neg \text{POF}_C) \cap E \cap D_{\rightsquigarrow(I) \cap B, S_C}^{\mathfrak{g}, a, p}$ is exactly PEV under $\neg \text{POF}_C \cap E$. $\square$

Corollary 12. *Given, for some H with $1 \leq \#(\max H)$, $\mathbf{g} : H \rightarrow P$, a, p , $\mathbf{g} \subseteq \rightsquigarrow_i \Rightarrow a \hookrightarrow_i p$, and $\exists i \mid \mathbf{g} \subseteq \rightsquigarrow_i$, there exists an anonymous claim C that holds, is SAPTS and PEV under $(\neg \text{POF}_C) \cap E \cap D_C^{\mathbf{g},a,p}$.*

Proof. Apply theorem 32 with $B := P^{\mathcal{A}^{<\mathbb{N}}}$: $\rightsquigarrow(I) \subseteq B$ trivially, and $\rightsquigarrow \in D_{P^{\mathcal{A}^{<\mathbb{N}}}, P^{\mathcal{A}^{<\mathbb{N}}}}^{\mathbf{g},a,p}$ holds vacuously, its antecedent $P^{\mathcal{A}^{<\mathbb{N}}} \setminus B = \emptyset$ being empty — no distinguishability hypothesis on $\rightsquigarrow$ is needed at all. At this B , $\text{Bad} = \emptyset$, so C_0 's second conjunct vanishes, leaving $C = [\forall i \in I : \mathbf{g} \subseteq \rightsquigarrow_i \Rightarrow a \hookrightarrow_i p] \cap [\exists i \in I \mid \mathbf{g} \subseteq \rightsquigarrow_i]$; and $\rightsquigarrow(I) \cap B = \rightsquigarrow(I)$, so the filter prior $D_{\rightsquigarrow(I) \cap B, S_C}^{\mathbf{g},a,p}$ is the full $D_{\rightsquigarrow(I), S_C}^{\mathbf{g},a,p} = D_C^{\mathbf{g},a,p}$. (All of the distinguishing burden is now paid at each candidate world rather than upfront on $\rightsquigarrow$, in contrast to corollary 11.) $\square$

Corollary 13. *Given, for some H with $1 \leq \#(\max H)$, $\mathbf{g} : H \rightarrow P$, a, p , $\mathbf{g} \subseteq \rightsquigarrow_i \Rightarrow a \hookrightarrow_i p$, and $\exists i \mid \mathbf{g} \subseteq \rightsquigarrow_i$, there exists an anonymous claim C that holds, is SAPTS and PEV under $(\neg \text{POF}_C) \cap E \cap D$, where*

$$D := \left\{ \rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}} \mid \forall g \in \rightsquigarrow(I) \text{ with } \mathbf{g} \subseteq g \wedge a \not\hookrightarrow_g p, \right. \\ \left. \exists \alpha \in \mathcal{A}^{<\mathbb{N}} \mid \mathbf{g}^{-1}(P) \cup \{\alpha\} \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}}) \wedge a \in \alpha_{\text{end}}(\alpha) \wedge g(\alpha) \neq p \right\}.$$

Proof. Let $C := [\forall i \in I : \mathbf{g} \subseteq \rightsquigarrow_i \Rightarrow a \hookrightarrow_i p] \cap [\exists i \in I \mid \mathbf{g} \subseteq \rightsquigarrow_i]$, the claim of corollary 12, which holds, is SAPTS and PEV under $(\neg \text{POF}_C) \cap E \cap D_C^{\mathbf{g},a,p}$; it remains to show $D_C^{\mathbf{g},a,p} = D$.

As $\text{Bad} = \emptyset$ at this C (corollary 12's proof), $S_C = \{b \in P^{\mathcal{A}^{<\mathbb{N}}} \mid \mathbf{g} \subseteq b \Rightarrow a \hookrightarrow_b p\}$. A row g failing $\mathbf{g} \subseteq g \Rightarrow a \hookrightarrow_g p$ never lies in S_C , so the condition " $g \in \rightsquigarrow(I) \setminus S_C$ " in the definition of $D_C^{\mathbf{g},a,p}$ is automatic whenever $\mathbf{g} \subseteq g \wedge a \not\hookrightarrow_g p$; unfolding distinguishability, $\rightsquigarrow \in D_C^{\mathbf{g},a,p}$ says exactly: for every $g \in \rightsquigarrow(I)$ with $\mathbf{g} \subseteq g \wedge a \not\hookrightarrow_g p$, some $H_0 \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$ has no $b \in S_C$ with $b|_{H_0} = g|_{H_0}$. Fix such a g ; we show this holds iff some α as in D 's definition exists.

If such an α exists, take $H_0 := \mathbf{g}^{-1}(P) \cup \{\alpha\} \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$. If some $b \in S_C$ matched g on H_0 , then matching g on $\mathbf{g}^{-1}(P) \subseteq H_0$ would give $\mathbf{g} \subseteq b$ (as $\mathbf{g} \subseteq g$), hence $a \hookrightarrow_b p$ by $b \in S_C$, hence $b(\alpha) = p$ (as $a \in \alpha_{\text{end}}(\alpha)$); but matching g at $\alpha \in H_0$ gives $b(\alpha) = g(\alpha) \neq p$ — contradiction. So no such b exists: H_0 witnesses g distinguishable from S_C .

If no such α exists, fix any $H_0 \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$; we build $b \in S_C$ matching g on H_0 . If $\mathbf{g}^{-1}(P) \not\subseteq H_0$, pick $\gamma \in \mathbf{g}^{-1}(P) \setminus H_0$ and $q \in P \setminus \{\mathbf{g}(\gamma)\}$ ($\#P \geq 2$); the row b with $b(\gamma) = q$, $b = g$ on H_0 (consistent, as $\gamma \notin H_0$), completed arbitrarily elsewhere, fails $\mathbf{g} \subseteq b$, so satisfies the template vacuously: $b \in S_C$. Otherwise $\mathbf{g}^{-1}(P) \subseteq H_0$; since H_0 is a chain, every $\beta \in H_0$ with $a \in \alpha_{\text{end}}(\beta)$ is chain-compatible with $\mathbf{g}^{-1}(P) \subseteq H_0$, so, by hypothesis (no valid α), every such β has $g(\beta) = p$. The row b with $b = g$ on H_0 and $b(\beta) = p$ for every β with $a \in \alpha_{\text{end}}(\beta)$ outside H_0 , completed arbitrarily on the rest, then satisfies $\mathbf{g} \subseteq b$ (as $\mathbf{g}^{-1}(P) \subseteq H_0$) and $a \hookrightarrow_b p$ (using the previous sentence for the a -ending points inside H_0), so $b \in S_C$. Either way, some $b \in S_C$ matches g on H_0 ; as H_0 was arbitrary, g is not distinguishable from S_C . $\square$

Remark 19. Compare to corollary 12: writing $C' := [\forall i : Q_1(\rightsquigarrow(i))] \cap [\exists i \in I \mid \mathbf{g} \subseteq \rightsquigarrow_i]$ (corollary 12's own claim, for the same $\mathbf{g}, a, p$; Q_1 and Bad as in theorem 33's proof, instantiated at $B = \rightsquigarrow(I)$), $C = C' \cap [\forall i, b \in \text{Bad} : \rightsquigarrow_i|_{H_b} \neq b|_{H_b}] \subseteq C'$ (the extra Bad -avoidance conjunct commutes past the shared witness conjunct): the claim proved here is stronger (strictly so unless $\text{Bad} = \emptyset$), as it additionally excludes, on each realized row, collision with any $\mathbf{g}$ -matching, decisiveness-failing non-real behavior on its own witnessing window, while C' only demands decisiveness on $\mathbf{g}$ -matching rows.

$D^{\mathbf{g},a,p}$ and $D_{C'}^{\mathbf{g},a,p}$ are not comparable by inclusion: $D^{\mathbf{g},a,p}$ quantifies over behaviors $b \notin \rightsquigarrow(I)$, tested against $\rightsquigarrow(I)$ itself, whereas $D_{C'}^{\mathbf{g},a,p}$ quantifies over behaviors $h \in \rightsquigarrow(I) \setminus S_{C'}$, tested against the external set $S_{C'}$ (the types of C').

Both proofs move a witness to an untouched clone using E in the same way, but differ in when the distinguishing prior is cashed in. Here, $\neg \text{POF}_C \cap E \subseteq C$ outright: $\rightsquigarrow \in D^{\mathbf{g},a,p}$ is a hypothesis on the true world alone, spent only once, upfront, to fix B and the H_b 's, before any candidate world is examined, so it drops out of corollary 11's conclusion entirely. In corollary 12, by contrast, $\rightsquigarrow \in D_{C'}^{\mathbf{g},a,p}$ must instead be invoked afresh at each tested $\rightsquigarrow$, since its witnessing window depends on that world's own failing row $g := \rightsquigarrow(i)$, which varies with $\rightsquigarrow$; so that prior cannot similarly be dropped from the conclusion. So the asymmetry is not about E , but about when the distinguishing prior is cashed in: once, from $\rightsquigarrow$ alone, for the stronger, $\rightsquigarrow$ -dependent claim C ; or per candidate world, for the weaker, $\rightsquigarrow$ -independent claim C' .

The same two instantiations ($B := \rightsquigarrow(I)$ and $B := P^{\mathcal{A}^{<\mathbb{N}}}$) apply just as well to theorem 33 itself, giving analogues of corollaries 11 and 12 for the original, unaugmented claim C_0 : an anonymous, holding claim that is APTS and provable by $\neg \text{POF}_C \cap E$, respectively APTS and provable by $(\neg \text{POF}_C) \cap E \cap D_{C_0}^{\mathbf{g},a,p}$. $\triangle$

Remark 20. $D^{\mathbf{g},a,p}$ is not the minimal hypothesis for *some* suitable claim to exist: a differently shaped claim can hold, be anonymous and APTS-P, in a world violating both. But the obstruction they route around does not disappear there — it relocates into the "provable by" filter instead of being paid upfront on $\rightsquigarrow$ alone.

Fix $a \neq c \in \mathcal{A}$ and partition I into infinite J, K , with $\forall i \in J, \alpha \in \mathcal{A}^{<\mathbb{N}} : \alpha \rightsquigarrow_i \varphi$, and $\forall i \in K, \alpha \in \mathcal{A}^{<\mathbb{N}} : \alpha \rightsquigarrow_i \varphi$ if $c \in \alpha_{\text{end}}(\alpha)$, $\alpha \rightsquigarrow_i \varphi'$ otherwise (so $(a) \rightsquigarrow_i \varphi'$ throughout K , as $c \neq a$). Let b_j, b_k be the behaviors associated to $\rightsquigarrow(J), \rightsquigarrow(K)$, and take

$$C = [\forall i \in I : (a) \rightsquigarrow_i \varphi \Rightarrow a \hookrightarrow_i \varphi] \cap [\forall i \in I : (a) \rightsquigarrow_i \varphi' \Rightarrow c \hookrightarrow_i \varphi] = C_1 \cap C_2.$$

C holds (J satisfies C_1 and vacuously C_2 , K conversely), is anonymous, and is APTS-P (witnessed by any $i_0 \in J$ and $\theta_0 = \{(a)\} \times \{((a), \varphi)\}$, as in theorem 33).

$\rightsquigarrow \notin D^{\{((a), \varphi)\}, a, \varphi}$. Pick $d \in \mathcal{A} \setminus \{a, c\}$, $S = \{\alpha \neq \emptyset \mid \alpha_1 = d\}$, and $g := b_k|_S \cup b_j|_{\mathcal{A}^{<\mathbb{N}} \setminus S}$. Any $H \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$ is a chain, hence lies entirely inside S or entirely outside it, so g matches b_k or b_j on H : g is undistinguishable from $\{b_j, b_k\}$. As $d \neq a$, $(a) \notin S$, so $g((a)) = b_j((a)) = \varphi$: g matches $\mathbf{g} = \{((a), \varphi)\}$. As $(d, a) \in S$ and $c \notin \{d, a\}$, $g((d, a)) = b_k((d, a)) = \varphi'$, and $a \in \alpha_{\text{end}}((d, a))$: a is not decisive for φ in g . Yet $g \notin \{b_j, b_k\}$ ($g((d)) = \varphi' \neq \varphi = b_j((d))$; $g((a)) = \varphi \neq \varphi' = b_k((a))$). So g is exactly the counterexample $D^{\{((a), \varphi)\}, a, \varphi}$ forbids: the prior genuinely fails at $\rightsquigarrow$.

Yet C is not provable by $\neg \text{OF}_C \cap E$ either. Carve an infinite $I_3 \subseteq J$, leaving $J \setminus I_3$ infinite, and let $\rightsquigarrow^*$ agree with $\rightsquigarrow$ outside I_3 , while every $i \in I_3$ realizes g (answers φ' on S , φ elsewhere). Then $\rightsquigarrow^* \in E$, and $\rightsquigarrow^* \notin C$: g violates C_1 , exactly as just shown. As in remark 24, any finite $\theta \subseteq \rightsquigarrow^*$ restricted to $i \in I_3$ is a single chain, so it lies entirely inside S or entirely outside it — the two witnessing histories (a) and (d, a) are incomparable, so no observation ever sees both — and it always extends into some world of C (complete i as fully a -decisive, or route it into C_2). Individuals outside I_3 embed by relabelling. So $\neg \text{OF}_C(\rightsquigarrow^*)$ holds outright: $\rightsquigarrow^* \in E \setminus C$ with $\neg \text{OF}_C(\rightsquigarrow^*)$ shows C is not provable by $\neg \text{OF}_C \cap E$, hence not by $\neg \text{POF}_C \cap E$ either.

Where the assumption goes instead. Applying the $B := P^{\mathcal{A}^{<\mathbb{N}}}$ analogue of theorem 33 (remark 19) to C_1, C_2 separately and intersecting gives C provable by $(\neg \text{POF}_C) \cap E \cap D_1 \cap D_2$, where $D_2 = D_{C_2}^{\{(a), \varphi'\}, c, \varphi}$ is the claim-relative prior of the second conjunct, and $D_1 = D_{C_1}^{\{(a), \varphi\}, a, \varphi}$ that of the first: here, $\rightsquigarrow \in D_1$ iff

$$\forall g \in \rightsquigarrow(I) \mid (a) \rightsquigarrow_g \varphi \wedge a \not\rightsquigarrow_g \varphi : \exists (a) \subseteq \alpha \in \mathcal{A}^{<\mathbb{N}} \mid a \in \alpha_{\text{end}}(\alpha) \wedge \alpha \not\rightsquigarrow_g \varphi,$$

and $\rightsquigarrow \in D_2$ iff

$$\forall g \in \rightsquigarrow(I) \mid (a) \rightsquigarrow_g \varphi' \wedge c \not\rightsquigarrow_g \varphi : \exists (a) \subseteq \alpha \in \mathcal{A}^{<\mathbb{N}} \mid c \in \alpha_{\text{end}}(\alpha) \wedge \alpha \not\rightsquigarrow_g \varphi.$$

$\rightsquigarrow \in D_1 \cap D_2$ only vacuously, as neither b_j nor b_k violates either conjunct. But $\rightsquigarrow^* \notin D_1$: its spliced row g has $g((a)) = \varphi$ and a not decisive for φ in g (both shown above), yet g is undistinguishable from $\{b_j, b_k\} \subseteq S_{C_1}$ on every H — exactly the fact established above. So $\rightsquigarrow^*$ is excluded from $D_1 \cap D_2$ exactly as it would need to be excluded, upfront, from $D_{C_1}^{\{(a), \varphi\}, a, \varphi}$ for corollary 11's route to apply to this claim's shape.

So the two routes are not comparable in strength, and this is not a trade of one commitment for another. D^{gap} is a hypothesis *about the true world* $\rightsquigarrow$, quantified over all of $P^{\mathcal{A}^{<\mathbb{N}}}$: something that must actually hold of reality, bought in exchange for the strongest conclusion, provable by the clean $\neg \text{POF}_C \cap E$ itself, nothing left in the filter. D_1, D_2 are not a hypothesis on $\rightsquigarrow$ at all here — $\rightsquigarrow \in D_1 \cap D_2$ holds automatically, vacuously — but a restriction on the *scope* of the conclusion: worlds such as $\rightsquigarrow^*$ are simply left outside the guarantee, whatever $\rightsquigarrow$ turns out to be. That is a strictly cheaper, more reasonable kind of commitment. For this C it is, moreover, the only kind available: $\neg \text{OF}_C(\rightsquigarrow^*)$ is a structural fact about $\rightsquigarrow^*$ alone, exactly as in remark 24 — independent of $\rightsquigarrow$ — so no hypothesis on $\rightsquigarrow$, D^{gap} -shaped or otherwise, could ever repair provability by the clean $\neg \text{POF}_C \cap E$ for this branch-and-escape shape of claim. D^{gap} is not a pricier alternative route here: it is inapplicable. What survives is narrower than first stated: D^{gap} remains the minimal prior, on $\rightsquigarrow$ alone, for provability by the clean $\neg \text{POF}_C \cap E$, only for claims sharing corollary 11's single-template shape; for a claim like C here, no hypothesis on $\rightsquigarrow$ alone buys the clean conclusion, and the claim-relative route's weaker, scoped guarantee is the best available. $\triangle$

A.10. More draft results

Theorem 34. *Verifiability in $\rightsquigarrow \in C \cap D$ under D fails iff $\exists \rightsquigarrow' \in D \setminus C \mid \forall \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) : \theta \subseteq \rightsquigarrow'$ (missing some conditions...)*

Proof. By definition, verifiability fails iff $\forall \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow), \exists \theta \subseteq \rightsquigarrow' \in D \setminus C$.

Assume the condition in the theorem. Verifiability fails.

Assume the condition here above (verifiability fails). Pick $\theta = \emptyset$, pick the corresponding $\rightsquigarrow' \in D \setminus C$. Pick any other $\theta \in \Theta \cap \mathcal{P}(\rightsquigarrow)$. Let's show that it is in $\rightsquigarrow'$. Missing some not yet clearly defined hypothesis about anonymity... $\square$

These two are about the intuition that if the real world has individuals whose DJs cover $P \cup \{\emptyset\}$, provability alone cannot lead to informativeness by lack of punctual facts about a given individual.

Theorem 35. *Assume $\#\mathcal{A} \notin \mathbb{N}$. There exists $\rightsquigarrow \in E$ such that no anonymous claim that holds and is informative beyond counting is provable by $\neg \text{POF}_{C \cap E} \cap E$.*

Proof. Since $\#\mathcal{A} \notin \mathbb{N}$, the family $\mathcal{S} = \{(H, g) \mid H \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})), g : H \rightarrow P\}$ is countable. Partition I into infinite subsets $(I_{H,g})_{(H,g) \in \mathcal{S}}$ and define $\rightsquigarrow$ by $\rightsquigarrow(i, \alpha) = g(\alpha)$ for $i \in I_{H,g}$, $\alpha \in H$, and $\rightsquigarrow(i, \alpha) = \varphi$ otherwise. Since all individuals in each $I_{H,g}$ share the same behavior, $\rightsquigarrow \in E$.

Suppose for contradiction that C is anonymous, informative beyond counting, $\rightsquigarrow \in C$, and $(\neg \text{POF}_{C \cap E}) \cap E \subseteq C$.

$E \subseteq C$. Fix any $\rightsquigarrow' \in E$ and $\theta \in \Theta \cap \mathcal{P}(\rightsquigarrow')$, with queried individuals $i_1, \dots, i_m$. For each l , let $H_l = \theta(P)(i_l) \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$ and $g_l : H_l \rightarrow P$ with $g_l(\alpha) = \theta(i_l, \alpha)$. Since each I_{H_l, g_l} is infinite, pick mutually distinct $j_1 \in I_{H_1, g_1}, \dots, j_m \in I_{H_m, g_m}$, and let σ be a permutation of I with $\sigma(i_l) = j_l$ for each l . Then $(\rightsquigarrow \circ \sigma)(i_l, \alpha) = \rightsquigarrow(j_l, \alpha) = g_l(\alpha) = \theta(i_l, \alpha)$ for all $\alpha \in H_l$, so $\rightsquigarrow \circ \sigma \in C \cap E$ (by anonymity of C and E) extends θ ; hence $\theta \in (C \cap E)^\subseteq$. As θ was arbitrary, $\neg \text{OF}_{C \cap E}(\rightsquigarrow')$, hence $\neg \text{POF}_{C \cap E}(\rightsquigarrow')$, so $\rightsquigarrow' \in (\neg \text{POF}_{C \cap E}) \cap E \subseteq C$. As $\rightsquigarrow'$ was arbitrary, $E \subseteq C$.

Contradiction. Since C is informative beyond counting, $\exists i_0 \in I : \pi_{i_0}(C) \neq P \cup \{\emptyset\}$. But $C \supseteq E$, so $\pi_{i_0}(C) \supseteq \pi_{i_0}(E)$. For each $v \in P$, the world $\rightsquigarrow_v$ responding v everywhere lies in E (single infinite behavior class) with $\pi_{i_0}(\rightsquigarrow_v) = v$; and the world $\rightsquigarrow_\emptyset$ cycling through P as $\alpha \mapsto p(|\alpha| \bmod 3)$ (for some surjection $p : \{0, 1, 2\} \rightarrow P$) likewise lies in E with $\pi_{i_0}(\rightsquigarrow_\emptyset) = \emptyset$ (histories with any argument in α_{end} cover all residues, so no argument is decisive). Hence $\pi_{i_0}(E) = P \cup \{\emptyset\}$, giving $\pi_{i_0}(C) = P \cup \{\emptyset\}$. Contradiction. $\square$

Example 21 (A concrete world defeating informativeness beyond counting). Theorem 35 invokes provability only to force $E \subseteq C$; a single, concretely described world already rules out any anonymous, holding, informative-beyond-counting claim, with no provability assumption. Partition I into four infinite classes according to $i \bmod 4$, and let $\rightsquigarrow$ respond, for an individual in class r : constantly φ if $r = 0$, constantly φ' if $r = 1$, constantly 0 if $r = 2$, and, if $r = 3$, φ on paths of even length and φ' on paths of odd length. Each behavior is shared by infinitely many individuals, so $\rightsquigarrow \in E$. Class $r = 3$ is undecided: given any $a \in \mathcal{A}$, the paths (a) and (a, a) both have $a \in \alpha_{\text{end}}$ yet answer φ' (odd length) and φ (even length) respectively, so no argument is decisive and its π -value is $\emptyset$. The four classes thus realize the π -values φ , φ' , 0 , and $\emptyset$, so $\pi(\rightsquigarrow)$ ranges over all of $P \cup \{\emptyset\}$. Consequently, any anonymous C with $\rightsquigarrow \in C$ satisfies $\pi_{i_0}(C) = P \cup \{\emptyset\}$ for every i_0 : for each realized value v , permuting an individual of π -value v onto i_0 keeps

the world in C by anonymity and yields $\pi_{i_0} = v$. Hence no anonymous claim that holds is informative beyond counting. $\triangle$

Example 22 (Provability cannot conclude a decisive argument for a named value). Split I into two infinite classes by parity, odd individuals answering φ constantly and even ones φ' constantly; then $\rightsquigarrow \in E$ and every odd individual is decisive for φ . Still, no anonymous claim C that holds and satisfies $C \subseteq \{\rightsquigarrow \mid \exists i : \pi_i(\rightsquigarrow) = \varphi\}$ is provable by $\neg \text{POF}_{C \cap E} \cap E$. Let $v \in E$ answer φ' everywhere; no individual is decisive for φ in v , so $v \notin C$. But every observation of v consists of φ' -answers and embeds into $\rightsquigarrow \in C$ by permuting the queried individuals onto even ones, so $\neg \text{OF}_{C \cap E}(v)$, hence $\neg \text{POF}_{C \cap E}(v)$ and $v \in (\neg \text{POF}_{C \cap E}) \cap E \setminus C$: C is not provable. Observing an odd individual answer φ refutes “the world is v ” (OF_C of $\rightsquigarrow$ against $\{v\}$), but never makes v falsify C through its own φ' -answers. The same asymmetry defeats the local claim $C = \{\rightsquigarrow \mid \forall i, p : (a) \rightsquigarrow_i p \Rightarrow a \hookrightarrow_i p\}$, which also holds here and is anonymous: a world with an individual answering p on every path except $(a', a) \rightsquigarrow_i q$ (some $a' \neq a, q \neq p$) has $(a) \rightsquigarrow_i p$ with a non-decisive, yet as (a', a) is incomparable to (a) no observation exhibits the violation, so it escapes $\text{OF}_{C \cap E}$ and C is not provable. By contrast, the claim that every individual answers constantly holds and is provable, concluding that everyone is decided—though not for a named value (to be checked!). $\triangle$

Given $i \in I$, $c, b \in \mathcal{A}$ distinct and $p \neq p' \in P$, say a world $\rightsquigarrow (i, c, b, p, p')$ -splits iff $\forall \alpha \in \mathcal{A}^{<\mathbb{N}}$ with $\alpha_1 = c$: $\alpha \rightsquigarrow_i p$, and $\forall \alpha \in \mathcal{A}^{<\mathbb{N}}$ with $\alpha_1 \neq c$: $\alpha \rightsquigarrow_i p'$ (recalling that a query restricted to one individual is an element of $\mathcal{I}(\mathcal{A}^{<\mathbb{N}})$, hence a $\subseteq$ -chain sharing a single first argument, so any two histories with $\alpha_1 \neq \alpha'_1$ are incomparable and cannot both be queried for the same individual).

Lemma 6 (Splitting kills decisiveness). *If $\rightsquigarrow (i, c, b, p, p')$ -splits, then $\pi_i(\rightsquigarrow) = \emptyset$.*

Proof. For any $a \in \mathcal{A}$: if $a = c$, then $(b, c) \not\rightsquigarrow_i p$ while $c \in \alpha_{\text{end}}((b, c))$, so c is not decisive for p (nor, being the sole value c ever takes at length 1, for anything else); if $a \neq c$, then $(c, a) \rightsquigarrow_i p \neq p'$ while $a \in \alpha_{\text{end}}((c, a))$, so a is not decisive for p' (nor for anything else, by the same argument applied at (c, a) versus a ’s own value at length 1, which is p'). Hence no argument is decisive for i . $\square$

Theorem 36 (Anon, Prov and APS are pairwise, but not jointly, satisfiable). *Fix $\varphi \neq \varphi' \in P$, partition I into two infinite classes I_1, I_2 , and let $\rightsquigarrow$ be the world of example 22: $\forall i \in I_1, \alpha \in \mathcal{A}^{<\mathbb{N}} : \alpha \rightsquigarrow_i \varphi$, and $\forall i \in I_2, \alpha \in \mathcal{A}^{<\mathbb{N}} : \alpha \rightsquigarrow_i \varphi'$. For $C \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$, say C is:*

Anon iff C is anonymous;

Prov iff C is provable by $\neg \text{OF}_C \cap E$, that is, $(\neg \text{OF}_C) \cap E \subseteq C$;

APS iff C is a posteriori sharp, that is, $\exists \theta \in \Theta, i \in I \mid \#\pi_i(\{\theta \subseteq \rightsquigarrow\} \cap C) = 1$; call such a witness (θ, i) constant iff θ assigns every individual it queries a single, constant value from $\{\varphi, \varphi'\}$ across all the histories queried for that individual.

Among claims C with $\rightsquigarrow \in C$: any two of Anon, Prov, APS are jointly satisfiable — witnessing APS, where relevant, by a constant witness — but no C satisfies Anon, Prov, and APS via a constant witness.

Proof. Write $C_2 = \{w \in P^{I \times \mathcal{A}^{<\mathbb{N}}} \mid \forall i \in I : [\forall \alpha \in \mathcal{A}^{<\mathbb{N}} : w(i, \alpha) = \varphi] \vee [\forall \alpha \in \mathcal{A}^{<\mathbb{N}} : w(i, \alpha) = \varphi']\}$ for “every individual is constant φ or constant φ' ”, and fix $b \neq c \in \mathcal{A}$ (using $\#\mathcal{A} \geq 2$).

Anon $\wedge$ Prov, not APS. Take $C_1 = \{w \in P^{I \times \mathcal{A}^{<\mathbb{N}}} \mid \forall i \in I, \alpha \in \mathcal{A}^{<\mathbb{N}} : w(i, \alpha) \in \{\varphi, \varphi'\}\}$. It holds ($\rightsquigarrow \in C_1$) and is anonymous ($\forall i$ -quantified). It is provable by $\neg \text{OF}_C$ alone (hence by $\neg \text{OF}_C \cap E$): any $w \notin C_1$ has some (i, α) with $w(i, \alpha) \notin \{\varphi, \varphi'\}$, a single point directly refuting C_1 , so $\text{OF}_{C_1}(w)$ for every $w \notin C_1$. It is not APS: given any θ, i , if θ is not C_1 -compatible then $\{\theta \subseteq w\} \cap C_1 = \emptyset$ already has cardinality $0 \neq 1$; otherwise let c_i be θ ’s shared first argument for i (or any fixed argument, if i is untouched by θ) and $p_{c_i} \in \{\varphi, \varphi'\}$ the (θ -forced or free) value along that branch; extending θ by making i constant p_{c_i} everywhere gives $\pi_i = p_{c_i}$, while extending θ so that $\rightsquigarrow (i, c_i, b, p_{c_i}, p'_{c_i})$ -splits (writing p'_{c_i} for the other element of $\{\varphi, \varphi'\}$) gives $\pi_i = \emptyset$ by lemma 6; both completions lie in C_1 (all values used are in $\{\varphi, \varphi'\}$) and extend θ , so $\#\pi_i(\{\theta \subseteq w\} \cap C_1) \geq 2$.

Anon $\wedge$ APS, not Prov. Take C_2 . It holds and is anonymous. Fix any $i_0 \in I_1$ and $\theta = \{(i_0, (c), \varphi)\}$: any $w \in \{\theta \subseteq w\} \cap C_2$ has i_0 constant φ or constant φ' , and since $w(i_0, (c)) = \varphi \neq \varphi'$, it must be constant φ ; so $\pi_{i_0}(\{\theta \subseteq w\} \cap C_2) = \{\varphi\}$, giving APS. It is not Prov: let w^* be such that individuals in I_1 ($i, c, b, \varphi, \varphi'$)-split and individuals in I_2 ($i, c, b, \varphi', \varphi$)-split. Then $w^* \in E$ (both classes remain infinite and internally uniform) and $w^* \notin C_2$ (no individual is constant, by lemma 6 $\pi_i(w^*) = \emptyset \neq \varphi, \varphi'$ throughout — though membership in C_2 only needs constancy, which indeed fails). Yet $\neg \text{OF}_{C_2}(w^*)$: any finite observation of w^* queries each touched individual along a single chain, hence a single first argument, hence sees a constant value (φ or φ' depending on whether that chain starts at c or not), so it extends into some world of C_2 (declare every queried individual constant at its observed value, and complete the rest of I_1, I_2 arbitrarily to stay constant). So $w^* \in E \setminus C_2$ with $\neg \text{OF}_{C_2}(w^*)$: C_2 is not provable by $\neg \text{OF}_C \cap E$.

Prov $\wedge$ APS, not Anon. Fix $i_0 \in I_1$ and take $C_3 = \{w \in P^{I \times \mathcal{A}^{<\mathbb{N}}} \mid \forall \alpha \in \mathcal{A}^{<\mathbb{N}} : w(i_0, \alpha) = \varphi\}$. It holds. It is provable by $\neg \text{OF}_C$ alone: any $w \notin C_3$ has some α^* with $w(i_0, \alpha^*) \neq \varphi$, a single point refuting C_3 directly (no comparison with a second point is needed, unlike example 22’s local claim, because the target φ is named in advance). It is APS, indeed sharp unconditionally ($\theta = \emptyset$): every $w \in C_3$ has i_0 constant φ , so every argument is decisive for i_0 regarding φ , giving $\pi_{i_0}(C_3) = \{\varphi\}$. It is not anonymous: swapping i_0 with any $j_0 \in I_2$ sends C_3 to $\{w \mid \forall \alpha \in \mathcal{A}^{<\mathbb{N}} : w(j_0, \alpha) = \varphi\}$, which differs from C_3 (as $\rightsquigarrow$ itself witnesses: $\rightsquigarrow \in C_3$ but this relabelled claim fails at $\rightsquigarrow$, since $\rightsquigarrow(j_0, \alpha) = \varphi' \neq \varphi$ for every α).

Not all three. Suppose C is Anon, Prov, $\rightsquigarrow \in C$, and APS via a constant witness (θ, i, v) : since θ then assigns each queried individual a constant value from $\{\varphi, \varphi'\}$, $\{\theta \subseteq w\} \cap C_2 \neq \emptyset$. By anonymity and $\rightsquigarrow \in C$, every relabelling $\rightsquigarrow \circ \sigma \in C$; as any two infinite-cofinite bipartitions of I are related by some permutation of I , this yields $C_2 \subseteq C$. Since $\{\theta \subseteq w\} \cap C_2 \subseteq \{\theta \subseteq w\} \cap C$ and the latter has π_i -image exactly $\{v\}$, so does the former (a nonempty subset of a singleton image equals it).

If i is untouched by θ : C_2 contains both a completion making i constant φ and one making it constant φ' (everything else fixed as forced by θ , or arbitrarily), giving $\{\varphi, \varphi'\} \subseteq \pi_i$ -image, contradicting $\{v\}$.

If i is touched by θ , forced constant $p_i \in \{\varphi, \varphi'\}$: then $v = p_i$. Build $w^\dagger$: for i , and for every other individual k touched by θ , writing c_k for k 's θ -forced first argument, $p_k \in \{\varphi, \varphi'\}$ for k 's θ -forced constant value, and p'_k for the other element of $\{\varphi, \varphi'\}$, extend θ via a (k, c_k, b, p_k, p'_k) -split as in the second part above, each replicated across a dedicated fresh infinite class of clones; complete all untouched individuals to stay constant φ or φ' in infinite classes. Then $\theta \subseteq w^\dagger$, $w^\dagger \in E$, $\pi_i(w^\dagger) = \emptyset \neq p_i = v$ by lemma 6, and, exactly as in the second part above, $\neg \text{OF}_{C_2}(w^\dagger)$ hence $\neg \text{OF}_C(w^\dagger)$ (as $C_2 \subseteq C$, completability into C_2 implies completability into C). By Prov, $w^\dagger \in C$, so $w^\dagger \in \{\theta \subseteq w\} \cap C$ with $\pi_i(w^\dagger) \neq v$, contradicting APS.

Either way, contradiction. $\square$

Remark 21. It is natural to ask whether theorem 36's incompatibility extends to APS witnesses that are not constant (querying some individual a value outside $\{\varphi, \varphi'\}$, or a non-constant pattern across comparable histories). The proof above does not settle this: it is specific to $\neg \text{OF}_C \cap E$ -provability via $C_2 \subseteq C$'s relabellings of $\rightsquigarrow$, and the obstruction to generalizing is one of *richness* rather than of values as such. Building the analogous $w^\dagger$ for a non-constant witness needs, for each individual θ queries, infinitely many further individuals independently compatible with θ 's pattern there, to serve as clones; $C_2 \subseteq C$ supplies such clones for free when the pattern is constant, but for a non-constant pattern the only known instance of it is the single witness $w_0 \in \{\theta \subseteq w\} \cap C$ exhibiting v , and neither anonymity (which only relabels existing individuals, so cannot manufacture new clones of w_0 's i -row) nor lemma 2 (which upgrades OF_C to POF for worlds *already* in E , rather than establishing $\neg \text{OF}_C$ in the first place) supplies the missing repetition. Whether w_0 can always be replaced by an E -compatible witness — which would close the gap — is left open. $\triangle$

A claim can be APS via a witness whose singleton value is the sentinel $\emptyset$ rather than a genuine element of P ; the following variant excludes this.

Definition 4 (A posteriori sharp with a named value). A claim C is a *posteriori sharp with a named value* (APS-P) iff $\exists \theta \in \Theta, i \in I, p \in P \mid \pi_i(\{\theta \subseteq \rightsquigarrow\} \cap C) = \{p\}$.

APS-P implies APS (theorem 36), the singleton $\{p\}$ already witnessing cardinality 1; it is strictly stronger, ruling out the witness being uniformly non-decisive ($\pi_i(\{\theta \subseteq \rightsquigarrow\} \cap C) = \{\emptyset\}$).

Given $\theta \in \Theta$, let $\theta^\downarrow = \{(\alpha, p) \in \mathcal{A}^{<\mathbb{N}} \times P \mid \exists i \in I : (i, \alpha, p) \in \theta\}$ denote θ with individual labels erased, and, for $i \in I$, let $\theta^i = \{(i, \alpha, p) \mid (\alpha, p) \in \theta^\downarrow\}$ denote its copy at i .

Theorem 37 (Anonymous APS-P claims are uniformly sharp). *If C is anonymous and APS-P via (θ, i_0, p) , then*

$$C \subseteq \{\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}} \mid \forall i \in I : \theta^i \subseteq \rightsquigarrow \Rightarrow p \in \pi_i(\rightsquigarrow)\}.$$

Proof. Let $\rightsquigarrow \in C$ with $\forall i \in I : \theta^i \subseteq \rightsquigarrow$, and fix $j \in I$. Let σ exchange i_0 and j (the identity if $j = i_0$), and $\rightsquigarrow' = \rightsquigarrow \circ \sigma$. For every $x \in I$: $\theta(x) \subseteq \theta^\downarrow$ trivially, and $\theta^{\sigma(x)} \subseteq \rightsquigarrow$ by hypothesis, that is, $\forall(\alpha, p) \in \theta^\downarrow : \rightsquigarrow(\sigma(x), \alpha) = p$; restricting to $(\alpha, p) \in \theta(x) \subseteq \theta^\downarrow$ and using $\rightsquigarrow'(x, \alpha) = \rightsquigarrow(\sigma(x), \alpha)$ gives $\theta(x) \subseteq \rightsquigarrow'$. As this holds for every x , $\theta \subseteq \rightsquigarrow'$. By anonymity, $\rightsquigarrow' = \rightsquigarrow \circ \sigma \in C$, so $\rightsquigarrow' \in \{\theta \subseteq \rightsquigarrow'\} \cap C$, and APS-P gives $\pi_{i_0}(\rightsquigarrow') = p$. Decisiveness at i_0 in $\rightsquigarrow'$ depends only on $\rightsquigarrow'$'s i_0 -row (definition 1), which equals $\rightsquigarrow$'s $\sigma(i_0) = j$ -row, so $\pi_{i_0}(\rightsquigarrow') = \pi_j(\rightsquigarrow)$. Hence $\pi_j(\rightsquigarrow) = p$. $\square$

Remark 22. $\theta^\downarrow$ need not be right-unique, that is, may fail $\forall(\alpha, p), (\alpha, p') \in \theta^\downarrow : p = p'$ — this happens exactly when two individuals queried by θ disagree about the value at some shared history. When it fails, no row of any world can satisfy $\theta^i \subseteq \rightsquigarrow$ for any i (a row is a function), so $\forall i \in I : \theta^i \subseteq \rightsquigarrow$ is then satisfied by no $\rightsquigarrow$ at all, and theorem 37's conclusion holds vacuously. Right-uniqueness of $\theta^\downarrow$ is thus not a hypothesis of the theorem: it only governs whether the theorem's antecedent can ever be triggered non-vacuously.

Theorem 37 needs neither C to hold nor any constancy or single-individual restriction on θ : it subsumes, as a single general statement, the relabelling arguments used piecemeal in the proof of theorem 36 (*Anon* $\wedge$ *APS*, *not Prov* and *Not all three*). Whether every anonymous APS claim is in fact APS-P — whether the sentinel value $\emptyset$ can always be avoided by choosing a better witness — remains open. $\triangle$

Definition 5 (A posteriori truly sharp with a named value). A claim C is a *posteriori truly sharp with a named value* (APTS-P) iff $\exists \theta \subseteq \rightsquigarrow, i \in I, p \in P \mid \pi_i(\{\theta \subseteq \rightsquigarrow\} \cap C) = \{p\}$.

APTS-P strengthens both APS-P (definition 4, dropping the requirement that θ be feasible) and a posteriori truly sharp (APTS, dropping the named value): it requires a witness that is simultaneously actually observable ($\theta \subseteq \rightsquigarrow$) and pins a genuine element of P rather than the sentinel $\emptyset$.

Conjecture 1. Fix $\varphi \neq \varphi' \in P$, partition I into two infinite classes I_1, I_2 , and let $\rightsquigarrow$ be the world: $\forall i \in I_1, \alpha \in \mathcal{A}^{<\mathbb{N}} : \alpha \rightsquigarrow_i \varphi$, and $\forall i \in I_2, \alpha \in \mathcal{A}^{<\mathbb{N}} : [\alpha = \emptyset \Rightarrow \alpha \rightsquigarrow_i \varphi \wedge \alpha \neq \emptyset \Rightarrow \alpha \rightsquigarrow_i \varphi']$. No C of the form $[\forall i, j, \dots : \text{some obs implies that } a \text{ is a decisive argument for } i]$ that holds satisfies anonymity, provability by $\neg \text{OF}_C \cap E$, and APTS-P.

Remark 23 (A self-referential template is already excluded). Fix $a \in \mathcal{A}$, and suppose $C \subseteq D_a := \{\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}} \mid \forall i \in I : [(a) \rightsquigarrow_i \varphi \Rightarrow a \hookrightarrow_i \varphi]\}$, together with $\rightsquigarrow \in C$, Anon, Prov, and APTS-P.

Since I_2 answers φ' , not φ , at every nonempty history, D_a 's antecedent is satisfied by $\rightsquigarrow$ exactly on I_1 ; there is no analogous case to treat on I_2 's side.

Pick $a' \in \mathcal{A} \setminus \{a\}$ (using $\#\mathcal{A} \geq 2$). Carve an infinite $I_3 \subseteq I_1$, leaving $I_1 \setminus I_3$ infinite, and let w^* agree with $\rightsquigarrow$ outside I_3 , while every $i \in I_3$ answers φ' on every α with $\alpha_1 = a'$ and φ on every other α (in particular at $\emptyset$ and at (a) itself, since $a \neq a'$).

Then a is not decisive for $i \in I_3$: $a \in \alpha_{\text{end}}((a)) = \{a\}$, which answers φ , while $a \in \alpha_{\text{end}}((a', a)) = \{a', a\}$, which answers φ' (as $(a', a)_1 = a'$). Since $(a) \rightsquigarrow_i \varphi$ holds for

$i \in I_3$ but $a \not\rightarrow_i \varphi$, $w^* \notin D_a$, hence $w^* \notin C$. All three classes $(I_1 \setminus I_3, I_3, I_2)$ are infinite and internally uniform, so $w^* \in E$.

It remains to show $\neg \text{OF}_C(w^*)$, contradicting Prov together with $w^* \in E \setminus C$. Every finite $\theta_0 \subseteq w^*$ must extend into some member of C . For individuals outside I_3 this is immediate, relabelling onto fresh members of $\rightsquigarrow$'s own classes (Anon and $\rightsquigarrow \in C$). For $i \in I_3$, θ_0 restricted to i is a single chain, hence every queried history is a prefix of the deepest one queried; since any two comparable nonempty histories share their first argument, the whole chain either has $\alpha_1 = a'$ throughout its nonempty part — constant φ' , possibly with φ at $\emptyset$, embedding into I_2 's shape — or never does — constant φ throughout, embedding into I_1 's shape. Unlike a template split *past* a shared nonempty prefix, no chain can straddle both: the two histories witnessing a 's non-decisiveness, (a) and (a', a) , are incomparable, so no single observation ever sees both. Hence $\neg \text{OF}_C(w^*)$ holds outright, and $w^* \in (\neg \text{OF}_C) \cap E \setminus C$ contradicts Prov.

This settles the case of a single self-referential template: no anonymous, provable, holding claim of the form D_a is APTS-P. The next remark shows the same trick — routing the counterexample through a branch entirely foreign to the template, rather than splitting past a shared prefix — handles the fully general template as well. $\triangle$

Remark 24 (The same trick handles the fully general template). Fix $a \in \mathcal{A}$, a finite $H \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$, and $T : H \rightarrow P$, and suppose $C \subseteq D_{a,T} := \{\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}} \mid \forall i \in I : [T(i) \subseteq \rightsquigarrow \Rightarrow \exists p \in P \mid a \hookrightarrow_i p]\}$, writing $T(i) = \{(i, \alpha, p) \mid \alpha \in H, T(\alpha) = p\}$, together with $\rightsquigarrow \in C$ and Anon and Prov (APTS-P plays no role below, so the conclusion holds without it, and for the unpinned decisiveness question rather than only at φ).

Write $\alpha^{(k)} = \max H$ (or $\emptyset$ if $H = \emptyset$), and suppose T is compatible with constant φ (the case of I_2 's shape is symmetric, carving I_3 from I_2 instead and swapping the roles of φ, φ' throughout; if T is compatible with neither, no individual of $\rightsquigarrow$ ever satisfies the antecedent and the claim holds vacuously, as in the first remark below).

Pick $d \in \mathcal{A} \setminus \{a, \alpha^{(k)}(1)\}$ if $\alpha^{(k)} \neq \emptyset$, or $d \in \mathcal{A} \setminus \{a\}$ if $\alpha^{(k)} = \emptyset$ (possible since $\mathcal{A}$ is infinite). Carve an infinite $I_3 \subseteq I_1$, leaving $I_1 \setminus I_3$ infinite, and let w^* agree with $\rightsquigarrow$ outside I_3 , while every $i \in I_3$ answers φ' on every α with $\alpha_1 = d$ and φ on every other α .

Since $\alpha^{(k)} = \emptyset$ or $\alpha^{(k)}(1) \neq d$, no history in H starts with d , so w^* agrees with the constant- φ prefix throughout H : $T(i) \subseteq w^*$ holds for $i \in I_3$, exactly as it did for $\rightsquigarrow$ itself.

Then a is not decisive for $i \in I_3$, for the same reason as in the self-referential case above and independently of H, T : since $a \neq d$, (a) lies outside the d -branch, so $a \in \alpha_{\text{end}}((a)) = \{a\}$ answers φ ; while (d, a) lies in the d -branch, so $a \in \alpha_{\text{end}}((d, a)) = \{d, a\}$ answers φ' . As w^* only ever answers φ or φ' , a is decisive for no $p \in P$, so $w^* \notin D_{a,T}$, hence $w^* \notin C$. All three classes $(I_1 \setminus I_3, I_3, I_2)$ are infinite and internally uniform, so $w^* \in E$.

It remains to show $\neg \text{OF}_C(w^*)$. As in the self-referential case, every finite $\theta_0 \subseteq w^*$ restricted to $i \in I_3$ is a single chain, hence lies entirely within the d -branch (constant φ' , possibly with φ at $\emptyset$, embedding into I_2 's shape) or entirely outside it (constant φ throughout, embedding into I_1 's shape) — this dichotomy never referred to $\alpha^{(k)}$, H , or

T , so it holds regardless of their complexity. Individuals outside I_3 embed by relabelling as usual (Anon , $\rightsquigarrow \in C$). So $\neg \text{OF}_C(w^*)$ holds outright, and $w^* \in (\neg \text{OF}_C) \cap E \setminus C$ contradicts Prov .

So no anonymous, provable, holding claim of the general form $D_{a,T}$ — for any finite one-individual observation template (a, H, T) compatible with I_1 's or I_2 's shape, whatever the candidate a 's relation to H — survives, whether or not it is additionally APTS-P. (This only settles claims built from a single, uniform, one-individual template; the conjecture's own phrasing allows for claims quantifying jointly over several individuals, which is not addressed here.) The two remarks below illustrate genuine successes for weaker properties, and neither is in tension with this one: the first, using the same $\rightsquigarrow$ fixed above, pins a value (0) compatible with neither shape, hence falls in the vacuous case excluded here; the second uses a different $\rightsquigarrow$ altogether (example 22's, where I_2 answers φ' even at $\emptyset$), so its APS-only conclusion is simply not addressed by this remark. $\triangle$

Remark 25. Fix $\varphi \neq \varphi' \in P$, partition I into two infinite classes I_1, I_2 , and let $\rightsquigarrow$ be the world: $\forall i \in I_1, \alpha \in \mathcal{A}^{<\mathbb{N}} : \alpha \rightsquigarrow_i \varphi$, and $\forall i \in I_2, \alpha \in \mathcal{A}^{<\mathbb{N}} : [\alpha = \emptyset \Rightarrow \alpha \rightsquigarrow_i \varphi \wedge \alpha \neq \emptyset \Rightarrow \alpha \rightsquigarrow_i \varphi']$. The claim C stating that for all individuals, $(a) \rightsquigarrow_i 0$ implies that a is decisive holds, satisfies anonymity, provability by $\neg \text{OF}_C \cap E$, and APS-P. $\triangle$

Remark 26. Fix $\varphi \neq \varphi' \in P$, partition I into two infinite classes I_1, I_2 , and let $\rightsquigarrow$ be the world of example 22: $\forall i \in I_1, \alpha \in \mathcal{A}^{<\mathbb{N}} : \alpha \rightsquigarrow_i \varphi$, and $\forall i \in I_2, \alpha \in \mathcal{A}^{<\mathbb{N}} : \alpha \rightsquigarrow_i \varphi'$. The claim $C = \{\rightsquigarrow \mid \forall i \in I : [\emptyset \rightsquigarrow_i \varphi \Rightarrow a \hookrightarrow_{ii} \varphi]\}$ holds, is anonymous, probable by $\neg \text{OF}_C \cap E$, and APS. $\triangle$

A.11. PEV under not POF

We now turn to claims $C \subseteq E$ satisfying PEV under $F_C = (\neg \text{POF}_C) \cap E$.

Recall the definition. C satisfies PEV under prior $(\neg \text{POF}_C) \cap E$ iff $\forall \rightsquigarrow \in C, \forall \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow), \exists \theta \subseteq \theta' \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \forall \theta' \subseteq \rightsquigarrow' \in (\neg \text{POF}_C) \cap E : \rightsquigarrow' \in C$.

Theorem 38. *Given any $B \subseteq P^{\mathcal{A}^{<\mathbb{N}}}$, if $C = C_B \cap E$ satisfies PEV under F_C , B includes its undistinguishables.*

Proof. Consider any $g \in P^{\mathcal{A}^{<\mathbb{N}}}$ undistinguishable from B . Consider any $\rightsquigarrow \in C$ and any $\rightsquigarrow' \in E$ such that $\rightsquigarrow'(I) = B \cup \{g\}$. Note that $\rightsquigarrow' \in C \Leftrightarrow B \cup \{g\} \subseteq B \Leftrightarrow g \in B$.

Towards showing $\neg \text{OF}_{C \cap E}(\rightsquigarrow')$, consider any $\theta \in \Theta \cap \mathcal{P}(\rightsquigarrow')$. Given any $i \in I$, let $\mathbf{g} = \theta(i)$ and $H = \mathbf{g}^{-1}(P)$, note that $\mathbf{g} \subseteq g$ and $\mathbf{g} = g_{\upharpoonright H}$, and by definition of Θ , $H \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$, thus $\exists b_i \in B \mid \mathbf{g} = g_{\upharpoonright H} = b_{i \upharpoonright H} \subseteq b_i$. Thus, θ is completeable in $C \cap E$.

Now, to obtain $\rightsquigarrow' \in C$, pick $\theta = \emptyset$ and any $\theta' \in \Theta \cap \mathcal{P}(\rightsquigarrow)$ such that $\theta' \subseteq \rightsquigarrow' \in (\neg \text{POF}_C) \cap E \Rightarrow \rightsquigarrow' \in C$ (known to exist by hypothesis). As $\theta' \subseteq \rightsquigarrow$, $\theta'(I) \in B$, thus $\theta'(I) \in B \cup \{g\}$, hence, $\theta' \subseteq \rightsquigarrow'$. $\square$

Conjecture 2. *Partition I into two infinite classes I_1, I_2 , and define $\rightsquigarrow$ as $\rightsquigarrow(i_1) = \rightsquigarrow^\varphi$ and $\rightsquigarrow(i_2) = \rightsquigarrow^{\varphi'}$. Any claim $C \subseteq E$ about types and entailing $[(a) \rightsquigarrow_i \varphi \text{ implies that } a \text{ is a decisive argument for } i]$ that holds fails PEV under $\neg \text{OF}_C \cap E$.*

Proof to be checked. To hold, C must allow for the two types in the real world. The world where there is also a third type of individual that behaves as: for emptyset of any argument starting with a , leads to φ , otherwise, leads to φ' , is undistinguishable from the two types in C , thus must be included for satisfying PEV, but this would prevent the required entailment. $\square$

Example 23. Consider $\rightsquigarrow(I) = \{\rightsquigarrow_i^\varphi, \rightsquigarrow_i^{\varphi'}\}$. Define $\mathbf{g} = \{((a, b), \varphi)\}$. The claim $C = [(a, b) \rightsquigarrow_i \varphi \Rightarrow a \hookrightarrow_i \varphi \wedge \forall \alpha : \alpha \not\rightsquigarrow_i \varphi \Rightarrow (\alpha, a, b) \not\rightsquigarrow_i \varphi]$ holds, is anonymous, APTS-P, and provable by $(\neg \text{POF}_C) \cap E \cap D_C^{\mathbf{g}, a, p}$.

Also, any behavior $\rightsquigarrow_i$ out of $D_C^{\mathbf{g}, a, p}$ (meaning, any behavior whose presence in a world makes it fail that prior) satisfies, for some α : $(\alpha, a, b) \not\rightsquigarrow_i \varphi$ and $(a, b, \alpha) \rightsquigarrow_i \varphi$. In other words, any prior that forbids all those patterns suffice to ensure provability (together with $(\neg \text{POF}_C) \cap E$, as usual). $\triangle$

Theorem 33's proof already splits, when showing provability, into worlds falsifying the template itself (Case A: handled purely from visibility out of S_C and repeatability) and worlds satisfying the template yet failing its extra, Bad-avoidance content (Case B). There the second kind is handled too — but only because that particular extra content happens to be repairable by the very same argument (visibility out of S_C along H_b , then repeating the witness at a fresh clone). The claim in the example just above instead adds a differently-shaped extra content ($\forall \alpha : \alpha \not\rightsquigarrow_i \varphi \Rightarrow (\alpha, a, b) \not\rightsquigarrow_i \varphi$) to the template, and nothing guarantees in general that failing it is similarly repairable: the following statement isolates what is needed for a world satisfying the template but failing arbitrary extra content to still be caught.

Theorem 39. *Given $\mathbf{g}$, $a \in \mathcal{A}$, $p \in P$, $C = [\forall i : \{i\} \times \mathbf{g} \subseteq \rightsquigarrow \Rightarrow a \hookrightarrow_i p]$, $C' \subseteq C$ such that $\rightsquigarrow \in C'$, $\exists i_0 \in I \mid \{i_0\} \times \mathbf{g} \subseteq \rightsquigarrow$, C' is anonymous, and $(C \setminus C') \cap E \cap D_{C'}^{\mathbf{g}, a, p} \subseteq \text{POF}_{C'}$, then C' holds and is APTS-P and provable by $(\neg \text{POF}_{C'}) \cap E \cap D_{C'}^{\mathbf{g}, a, p}$.*

Taking $C' = C$ recovers corollary 12 (equivalently, theorem 33 at $B := P^{\mathcal{A}^{<\mathbb{N}}}$, where $\text{Bad} = \emptyset$): $C \setminus C' = \emptyset$ makes the last condition vacuous, and anonymity of C is then a consequence — shown, for the general conjunction, in theorem 33's *Anonymous* paragraph — rather than a hypothesis. For the example above, the last condition is exactly the (unproven) claim that every world agreeing with $\rightsquigarrow$'s template-decisiveness at a but violating the $\forall \alpha$ -closure is nonetheless caught, visibly, by $D_{C'}^{\mathbf{g}, a, p} \cap E$.

Proof of example 23. Write $\mathbf{g} = \{((a, b), \varphi)\}$ (so $\mathbf{g}^{-1}(P) = \{(a, b)\} \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$), $p := \varphi$, and let $C = [(a, b) \rightsquigarrow_i \varphi \Rightarrow a \hookrightarrow_i \varphi]$ be the template of corollary 12 for this $\mathbf{g}, a, p$, so that example 23's claim is $C' = C \wedge [\forall \alpha : \alpha \not\rightsquigarrow_i \varphi \Rightarrow (\alpha, a, b) \not\rightsquigarrow_i \varphi]$, manifestly $C' \subseteq C$. We check the hypotheses of theorem 39.

$\rightsquigarrow \in C'$ and a witness i_0 . Write $I_1, I_2 \subseteq I$ for the (infinite) classes behaving as $\rightsquigarrow_i^\varphi, \rightsquigarrow_i^{\varphi'}$ respectively. For $i \in I_1$: every history maps to φ , so $(a, b) \rightsquigarrow_i \varphi$ and a is trivially decisive for φ — the first conjunct holds at i — and, for any α , $\alpha \rightsquigarrow_i \varphi$, so the second conjunct's antecedent never fires at i . For $i \in I_2$: $(a, b) \not\rightsquigarrow_i \varphi$, so the first conjunct holds vacuously at i ; and for any α , $\alpha \not\rightsquigarrow_i \varphi$ while also $(\alpha, a, b) \not\rightsquigarrow_i \varphi$ (both

equal φ'), so the second conjunct holds at i too. Hence $\rightsquigarrow \in C'$, and any $i_0 \in I_1$ satisfies $\{i_0\} \times \mathbf{g} \subseteq \rightsquigarrow$.

C' anonymous. Both conjuncts of C' are of the form $\forall i : Q(\rightsquigarrow(i))$ for a property Q of a single row (the first exactly as in theorem 33's *Anonymous* paragraph; the second through $Q(f) := [\forall \alpha : f(\alpha) \neq \varphi \Rightarrow f(\alpha, a, b) \neq \varphi]$). The same relabelling computation as there, applied to each conjunct, gives $C' \circ \sigma = C'$ for every permutation σ .

The residual condition. We show the stronger $(C \setminus C') \cap E \subseteq \text{POF}_{C'}$, which contains $(C \setminus C') \cap E \cap D_{C'}^{\mathbf{g}, a, p}$. Fix $\rightsquigarrow \in (C \setminus C') \cap E$. Since $\rightsquigarrow \in C \setminus C'$, the failure of C' lies in the second conjunct: some $i \in I$ and $\alpha \in \mathcal{A}^{<\mathbb{N}}$ satisfy $\alpha \not\rightsquigarrow_i \varphi$ and $(\alpha, a, b) \rightsquigarrow_i \varphi$. Let $g := \rightsquigarrow(i)$, so $g(\alpha) \neq \varphi$ and $g(\alpha, a, b) = \varphi$, and let $H := \{\alpha, (\alpha, a, b)\} \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$ (α prefixes (α, a, b)).

Fix any $\theta \in \Theta \cap \mathcal{P}(\rightsquigarrow)$. As $\rightsquigarrow \in E$, $J_g := \{j \in I \mid \rightsquigarrow(j) = g\}$ is infinite (definition 2); as θ is finite, pick $i' \in J_g \setminus \theta(\mathcal{A}^{<\mathbb{N}}, P)$. Let $\theta_H^{i'} := \{(i', \alpha, g(\alpha)), (i', (\alpha, a, b), g(\alpha, a, b))\} \in \Theta \cap \mathcal{P}(\rightsquigarrow)$ and $\theta' := \theta \cup \theta_H^{i'}$; since i' is untouched by θ , $\theta \sqsubseteq \theta'$ and $\theta' \in \Theta \cap \mathcal{P}(\rightsquigarrow)$.

Suppose some $\rightsquigarrow''' \in C'$ extended θ' , and let $f := \rightsquigarrow'''(i')$. Then $f(\alpha) = g(\alpha) \neq \varphi$ and $f(\alpha, a, b) = g(\alpha, a, b) = \varphi$. But $\rightsquigarrow''' \in C'$ requires its second conjunct at i', α : $\alpha \not\rightsquigarrow_{i'} \varphi \Rightarrow (\alpha, a, b) \not\rightsquigarrow_{i'} \varphi$, whose contrapositive, $(\alpha, a, b) \rightsquigarrow_{i'} \varphi \Rightarrow \alpha \rightsquigarrow_{i'} \varphi$, together with $f(\alpha, a, b) = \varphi$ forces $f(\alpha) = \varphi$ — contradicting $f(\alpha) \neq \varphi$. So no such $\rightsquigarrow'''$ exists: $\theta' \notin (C')^\subseteq$.

As θ was arbitrary — the same H serves throughout, only the clone i' depending on θ — $\text{POF}_{C'}(\rightsquigarrow)$ holds. Hence $(C \setminus C') \cap E \subseteq \text{POF}_{C'}$.

All hypotheses of theorem 39 are met, so C' holds, is anonymous, is APTS-P, and is provable by $(\neg \text{POF}_{C'}) \cap E \cap D_{C'}^{\mathbf{g}, a, p}$ — exactly example 23's claim. $\square$

Proof of the second part of example 23. As above, $B = S_{C'} = \{g \in P^{\mathcal{A}^{<\mathbb{N}}} \mid Q(g)\}$ where $Q(g) := [g(a, b) = \varphi \Rightarrow a \text{ decisive for } \varphi \text{ in } g] \wedge [\forall \gamma : g(\gamma) \neq \varphi \Rightarrow g(\gamma, a, b) \neq \varphi]$: if $Q(g)$ holds, the world assigning g to every individual satisfies C' , so $g \in S_{C'}$; conversely if $g \in S_{C'}$, some C' -world has g as some individual's row, and C' 's condition at that individual gives $Q(g)$.

Let $\rightsquigarrow$ be any world and $h \in \rightsquigarrow(I) \setminus B$ a row with $\mathbf{g} \subseteq h$ (that is, $h(a, b) = \varphi$) and $a \not\rightsquigarrow_h \varphi$, undistinguishable from B (that is, $\forall H \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}}) : h|_H \in B|_H$) — exactly the situation making $\rightsquigarrow$ fail $D_{C'}^{\mathbf{g}, a, p}$ at h . Since $a \not\rightsquigarrow_h \varphi$, fix $\beta \in \mathcal{A}^{<\mathbb{N}}$ with $a \in \alpha_{\text{end}}(\beta)$ and $h(\beta) \neq \varphi$; note $\beta \neq \emptyset$, as $\alpha_{\text{end}}(\emptyset) = \emptyset$.

β is incomparable to (a, b) . If not, $H := \{\beta, (a, b)\} \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$ is a chain, so some $g \in B$ matches h on H : $g(\beta) = h(\beta) \neq \varphi$ and $g(a, b) = h(a, b) = \varphi$. But $Q(g)$'s first conjunct then forces a decisive for φ in g , in particular $g(\beta) = \varphi$ (as $a \in \alpha_{\text{end}}(\beta)$) — contradicting $g(\beta) \neq \varphi$. In particular $\beta \neq (a)$ (a prefix of (a, b)), so $\#\beta \geq 2$ (a length-1 history β with $a \in \alpha_{\text{end}}(\beta) = \beta$ would force $\beta = (a)$).

$h(a, b, \beta) = \varphi$. As $\#\beta \geq 2$, $a \in \alpha_{\text{end}}(a, b, \beta) = \alpha_{\text{end}}(\beta)$ (the last two arguments of (a, b, β) are β 's own last two, unaffected by what precedes β), so (a, b, β) is itself an a -decisiveness test point, and (a, b) prefixes it. Were $h(a, b, \beta) \neq \varphi$, the same argument as above — with $H := \{(a, b), (a, b, \beta)\}$, forcing some $g \in B$ with $g(a, b) = \varphi$ and $g(a, b, \beta) = h(a, b, \beta) \neq \varphi$, contradicting $Q(g)$'s first conjunct at the witness (a, b, β) — would fail. Hence $h(a, b, \beta) = \varphi$.

$h(\beta, a, b) \neq \varphi$. Let $H := \{\beta, (\beta, a, b)\} \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$ (β prefixes (β, a, b)). Were $h(\beta, a, b) = \varphi$, some $g \in B$ would match h on H : $g(\beta) = h(\beta) \neq \varphi$ and $g(\beta, a, b) = \varphi$. But $Q(g)$'s second conjunct at $\gamma := \beta$ gives $g(\beta) \neq \varphi \Rightarrow g(\beta, a, b) \neq \varphi$, contradicting $g(\beta, a, b) = \varphi$. Hence $h(\beta, a, b) \neq \varphi$.

Taking $\alpha := \beta$: $(\alpha, a, b) \not\rightsquigarrow_h \varphi$ (previous paragraph) and $(a, b, \alpha) \rightsquigarrow_h \varphi$ (two paragraphs above) — exactly the claimed pattern, witnessed by the very history β that exhibits a 's non-decisiveness at h . $\square$

Theorem 49 characterizes situations that admit a claim that holds and is anonymous and indicative.

NB theorem 74 has the key! And I should read section E.5.

Example 24. $C = [\exists i \mid a \hookrightarrow_i \varphi]$. Consider any world $\rightsquigarrow'$ where all i satisfy $a \not\hookrightarrow_i \varphi$ and each behavior is infinite. It is $\neg \text{OF}_C$: any observation $\theta \subseteq \rightsquigarrow'$, being finite, is completable in C . Now consider a world where a single i satisfies $a \hookrightarrow_i \varphi$. Consider $\theta = \emptyset$ and any observation $\theta' \subseteq \rightsquigarrow$. It is completed by a world $\rightsquigarrow' \in (\neg \text{POF}_C) \cap E$ but out of C . $\triangle$

Example 25. Define $\rightsquigarrow^p$ as: empty leads to φ , everything else leads to p , for all individuals. Consider C_B with $B = \{\rightsquigarrow^\varphi, \rightsquigarrow^{\varphi'}\}$. Consider the behavior g as φ if path starts with α or is empty, and φ' otherwise, and consider the world equal to g for each individual. We see that observing $(a) \rightsquigarrow_i \varphi$ and $(a) \not\rightsquigarrow_j \varphi$ rejects successfully that world. However, consider the world with three behaviors: $\rightsquigarrow^\varphi, \rightsquigarrow^{\varphi'}, g$. Using the fact that this world must specify which individuals have which behaviors, we can observe wisely individuals mapped to g to prove that they do not. But this feels like cheating. The opponent should perhaps be able to postulate that the world is made of these three types, and not specify the individuals! We could force specifying something to discriminate, like some obs implies some behavior. Though perhaps the fundamental problem is that if the world has the two postulated types of individuals, it could also have those three types of individuals! $\triangle$

We could introduce a weaker demand, a variant of the demand that $C \subseteq E$ satisfies PEV under $F_C = (\neg \text{POF}_C) \cap E$, by inverting the order of attack and defense. However, I have some doubt that it does not improve verifiability.

C satisfies IPEV iff $\forall \rightsquigarrow \in C, \rightsquigarrow' \in (\neg \text{POF}_C) \cap E \mid \rightsquigarrow' \notin C, \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow), \exists \theta \sqsubseteq \theta' \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \theta' \not\sqsubseteq \rightsquigarrow'$.

A.12. Interplay

We are here interested in the interplay between properties of holding, anonymity, informativeness and permanent falsifiability.

B. Supplementary material

B.1. Explanatory examples

Example 26 (Disjunct with incompatible paths, not falsifiable, to be deleted). Given $a \neq a' \in \mathcal{A}$, the claim appearing in example 7, namely, $\forall i : [(a) \rightsquigarrow_i \varphi \vee (a') \rightsquigarrow_i \varphi]$, is not falsifiable. Consider any $\rightsquigarrow \notin C$, thus satisfying, for some i , $(a) \not\rightsquigarrow_i \varphi \wedge (a') \not\rightsquigarrow_i \varphi$. Consider any $\theta \subseteq \rightsquigarrow$ and let us prove that $\theta \in C^\subseteq$. (This is a consequence of example 7 so this proof should not be useful.) By definition of a general observation, θ may not contain both $(i, (a), \varphi)$ and $(i, (a'), \varphi)$, thus letting t denote one of these triples not in θ , pick any $\rightsquigarrow'$ such that $\theta \subseteq \rightsquigarrow' \wedge t \in \rightsquigarrow'$. We have $\theta \in \mathcal{P}(\rightsquigarrow')$ and $\rightsquigarrow' \in C$, thus, $\theta \in C^\subseteq$. $\triangle$

Example 27. Given a and φ , the claim $a \rightsquigarrow_i \varphi$ is not anonymous iff $\#I \geq 2$. Indeed, pick any $j \neq i$ and define $\sigma(i) = j$, $\sigma(j) = i$ and $\forall k \in I \setminus \{i, j\} : \sigma(k) = k$, then $C = \{\rightsquigarrow \mid \rightsquigarrow(i, a) = \varphi\}$, which contains in particular $\rightsquigarrow$ defined as $\forall \alpha : [\forall k \in I \setminus \{j\}, \rightsquigarrow(k, \alpha) = \varphi \wedge \rightsquigarrow(j, \alpha) = \varphi']$; while $C \circ \sigma = \{\rightsquigarrow \mid \rightsquigarrow(j, a) = \varphi\}$ which does not contain $\rightsquigarrow$. $\triangle$

Example 28. Consider $I = \{i\}$ and $a_1 \neq a_2 \in \mathcal{A}$. The claim $C = \{\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}} \mid (\forall \alpha \neq \emptyset : \alpha \rightsquigarrow_i \varphi) \vee (\forall \alpha \neq \emptyset : \alpha \rightsquigarrow_i \varphi')\}$ is not falsifiable.

Consider $\rightsquigarrow$ defined as $\alpha \rightsquigarrow_i \varphi$ iff $\alpha = \emptyset \vee \alpha_1 = a_1$ and $\alpha \rightsquigarrow_i \varphi'$ otherwise. We have $\rightsquigarrow \notin C$. But any $\theta \subseteq \rightsquigarrow$ satisfies $\theta \in C^\subseteq$. $\triangle$

Example 29. Consider $I = \{i_1, i_2\}$ and $a_1 \neq a_2 \in \mathcal{A}$. Given $p_0, p, p'_0, p' \in P$, define $\rightsquigarrow^{i_1 \mapsto (p_0, p); i_2 \mapsto (p'_0, p')}$ as $\emptyset \rightsquigarrow_{i_1} p_0 \wedge \emptyset \rightsquigarrow_{i_2} p'_0 \wedge \forall \alpha \in \mathcal{A}^{<\mathbb{N}} \setminus \{\emptyset\} : \alpha \rightsquigarrow_{i_1} p \wedge \alpha \rightsquigarrow_{i_2} p'$.

The claim $C = \{\rightsquigarrow^{i_1 \mapsto (p_0, p); i_2 \mapsto (p'_0, p')} \mid p_0, p, p'_0, p' \in \{\varphi, \varphi'\}\}$ is not falsifiable. Consider $\rightsquigarrow$ defined as, $\forall i \in I$, $\alpha \rightsquigarrow_i \varphi$ iff $\alpha = \emptyset \vee \alpha_1 = a_1$ and $\alpha \rightsquigarrow_i \varphi'$ otherwise. We have $\rightsquigarrow \notin C$.

Consider now any bigger claim C' , thus with $C \subseteq C'$. I believe that C' is falsifiable iff it includes all relations that are constant on all paths starting with a given argument (thus all $\rightsquigarrow$ such that $\forall \alpha, \alpha' \in \mathcal{A}^{<\mathbb{N}} : \alpha_1 = \alpha'_1 \Rightarrow \rightsquigarrow_i(\alpha) = \rightsquigarrow_i(\alpha')$). $\triangle$

Example 30. Consider $I = \{i_{p_0, p} \mid p_0, p \in \{\varphi, \varphi'\}\}$ (thus $\#I = 4$). Given $p_0, p \in P$, define $R^{(p_0, p)} = R^{\emptyset \mapsto p_0; p} : \mathcal{A}^{<\mathbb{N}} \rightarrow P$ as $R(\emptyset) = p_0 \wedge \forall \alpha \in \mathcal{A}^{<\mathbb{N}} \setminus \{\emptyset\} : R(\alpha) = p$.

Define $\rightsquigarrow$ as $\forall p_0, p \in \{\varphi, \varphi'\} : \rightsquigarrow_{i_{p_0, p}} = R^{\emptyset \mapsto p_0; p}$.

Given $f : I \rightarrow \{\varphi, \varphi'\}^2$, define $\rightsquigarrow^f$ as $\forall i \in I : \rightsquigarrow_i^f = R^{f(i)}$. Note that any $f \in (\{\varphi, \varphi'\}^2)^I$ can be seen as a permutation σ over I and that $\rightsquigarrow^f = \rightsquigarrow \circ \sigma$.

The claim $C = \{\rightsquigarrow^f \mid f \in (\{\varphi, \varphi'\}^2)^I\}$ is falsifiable: consider any $\rightsquigarrow$ observably compatible with C ; if there is $\alpha^1, \alpha^2, \alpha^3 \in \mathcal{A}^{<\mathbb{N}} \setminus \{\emptyset\}$ and $i_1 \neq i_2 \neq i_3 \neq i_1 \in I$ with $\forall k \in [1, 3] : \alpha^k \rightsquigarrow_{i_k} \varphi$, then $\rightsquigarrow$ is not observably compatible with C , so there must be at most 2 individuals mapping some non empty path to φ (thus at least 2 mapping all non empty paths to φ') and for the same reason, at least 2 individuals mapping all non empty paths to φ ; thus (together with noting that two individuals mapping every paths to φ is not observably compatible with C and adding the missing bits), $\rightsquigarrow \in C$. $\triangle$

Example 31. Consider $I = \cup_{p_0, p \in \{\varphi, \varphi'\}} I_{p_0, p}$, with each $I_{p_0, p}$ infinite, and $a_1 \neq a_2 \in \mathcal{A}$. Given $p_0, p \in P$, define $R^{(p_0, p)} = R^{\emptyset \mapsto p_0; p} : \mathcal{A}^{<\mathbb{N}} \rightarrow P$ as $R(\emptyset) = p_0 \wedge \forall \alpha \in \mathcal{A}^{<\mathbb{N}} \setminus \{\emptyset\} : R(\alpha) = p$. Define $\rightsquigarrow$ as $\forall p_0, p \in \{\varphi, \varphi'\}, i \in I_{p_0, p} : \rightsquigarrow_i = R^{\emptyset \mapsto p_0; p}$.

The claim $C = \{\rightsquigarrow \circ \sigma \mid \sigma : I \leftrightarrow I\}$ is not falsifiable: consider $\rightsquigarrow$ defined as, $\forall i \in I$, $\alpha \rightsquigarrow_i \varphi$ iff $\alpha = \emptyset \vee \alpha_1 = a_1$ and $\alpha \rightsquigarrow_i \varphi'$ otherwise; we have $\rightsquigarrow \notin C$ but any $\theta \in \Theta, \theta \subseteq \rightsquigarrow$ satisfies $\theta \in C^\subseteq$.

Now consider $C' = \{\rightsquigarrow \mid \forall i \in I, \alpha, \alpha' \in \mathcal{A}^{<\mathbb{N}} : \alpha_1 = \alpha'_1 \Rightarrow \rightsquigarrow_i(\alpha) = \rightsquigarrow_i(\alpha')\}$. (Thus $C \subseteq C'$.) This one is falsifiable: if $\rightsquigarrow \notin C$, then $\exists i \in I, \alpha, \alpha' \mid \alpha_1 = \alpha'_1 \wedge \rightsquigarrow_i(\alpha) \neq \rightsquigarrow_i(\alpha')$, now define α'' as the common prefix to α and α' ; as $\rightsquigarrow_i(\alpha) \neq \rightsquigarrow_i(\alpha')$, we have at least one of $\rightsquigarrow_i(\alpha) \neq \rightsquigarrow_i(\alpha'')$ or $\rightsquigarrow_i(\alpha') \neq \rightsquigarrow_i(\alpha'')$, and as α'' satisfies $\alpha''_1 = \alpha_1$ and $\alpha''_1 = \alpha'_1$, observing the corresponding pair of paths for i falsifies C . $\triangle$

Example 32. Imagine that the real relation is composed of individuals who map (a_1, a_2) to φ and (a_1, a_3) to φ' and individuals converse (with both, say, mapping everything else to 0). Then the claim $C = \{\rightsquigarrow \mid \forall i \in I : \rightsquigarrow_i(a_1, a_2) \in \{\varphi, \varphi'\} \wedge \rightsquigarrow_i(a_1, a_3) \in \{\varphi, \varphi'\}\}$ is falsifiable.

Imagine that the real relation is composed of individuals who map (a_1, a_2) to φ and (a_1, a_3) to φ' and individuals converse (with both, say, mapping everything else to 0) and individuals who map (a_4, a_5) to φ and (a_6) to φ' and individuals converse, with both mapping everything else to 0. Then the claim $C = \{\rightsquigarrow \mid \forall i \in I : [\rightsquigarrow_i(a_1, a_2) \in \{\varphi, \varphi'\} \wedge \rightsquigarrow_i(a_1, a_3) \in \{\varphi, \varphi'\} \wedge \rightsquigarrow_i(H) = 0] \vee [\rightsquigarrow_i(a_4, a_5) \in \{\varphi, \varphi'\} \wedge \rightsquigarrow_i(a_6) \in \{\varphi, \varphi'\} \wedge \rightsquigarrow_i(H) = 0]\}$, where H designates all “other” paths (abusing notation), is falsifiable.

Imagine that the real relation is composed of individuals who map (a_1, a_2) to φ and (a_1, a_3) to φ' and individuals converse (with both, say, mapping everything else to 0) and individuals who map (a_1, a_4) to φ and (a_6) to φ' and individuals converse, with both mapping everything else to 0. Then the claim $C = \{\rightsquigarrow \mid \forall i \in I : [\rightsquigarrow_i(a_1, a_2) \in \{\varphi, \varphi'\} \wedge \rightsquigarrow_i(a_1, a_3) \in \{\varphi, \varphi'\} \wedge \rightsquigarrow_i(H) = 0] \vee [\rightsquigarrow_i(a_1, a_4) \in \{\varphi, \varphi'\} \wedge \rightsquigarrow_i(a_6) \in \{\varphi, \varphi'\} \wedge \rightsquigarrow_i(H) = 0]\}$, where H designates all “other” paths (abusing notation), is falsifiable: could have an individual mapping both (a_1, a_2) and (a_1, a_4) to 0 but then if she maps a_6 to 0, observe both (a_1, a_2) and a_6 and falsify C ; if she maps a_6 in $\{\varphi, \varphi'\}$, both (a_1, a_4) and a_6 and falsify C (here assuming falsely in the whole example that a_6 can be observed together with $(a_1, .)$, so we probably should replace a_6 by $\emptyset$). $\triangle$

B.2. More examples

Example 33 (Partly falsifiable is still unfalsifiable). Given $j \in I$, $\alpha \in \mathcal{A}^{<\mathbb{N}}$ and $a \neq a' \in \mathcal{A}$, the claim $C = \{\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}} \mid \alpha \rightsquigarrow_j \varphi \wedge \forall i \in I : [(a) \rightsquigarrow_i \varphi \vee (a') \rightsquigarrow_i \varphi]\}$ is unfalsifiable but not strongly unfalsifiable.

Define $C_1 = \{\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}} \mid \forall i \in I : [(a) \rightsquigarrow_i \varphi \vee (a') \rightsquigarrow_i \varphi]\}$ and $C_2 = \{\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}} \mid \alpha \rightsquigarrow_j \varphi\}$, so that $C = C_1 \cap C_2$. Observe that C_1 is the claim of example 26 and C_2 is falsifiable and coverable. Theorem 78 concludes. $\triangle$

Example 34 (Partly again). Considering again example 33, given $j \in I$, $\alpha \in \mathcal{A}^{<\mathbb{N}}$ and $a \neq a' \in \mathcal{A}$, $C = \{\rightsquigarrow \mid \alpha \rightsquigarrow_j \varphi \wedge \forall i : [(a) \rightsquigarrow_i \varphi \vee (a') \rightsquigarrow_i \varphi]\}$ is not falsifiable but corollary 19 is not applicable. Observe that $C^\subseteq = \{R \in I \times \mathcal{A}^{<\mathbb{N}} \times P \mid [(j, \alpha) \in R(P) \Rightarrow$

$(j, \alpha, \varphi) \in R] \wedge \forall i : [(i, (a)) \in R(P) \wedge (i, (a')) \in R(P)] \Rightarrow [(i, (a), \varphi) \in R \wedge (i, (a'), \varphi) \in R]\}$ and $C^\subseteq \cap \Theta = \{\theta \in \Theta \mid (j, \alpha) \in \theta(P) \Rightarrow (j, \alpha, \varphi) \in \theta\}$. The relations obs-compatible with $C^\subseteq \cap \Theta$ are those satisfying $\alpha \rightsquigarrow_j \varphi$. We conclude that C is not falsifiable as C is smaller than that set. $\triangle$

Example 35 (Come back). In example 8 we considered a given a and φ and defined the falsifiable claim $C = \{\rightsquigarrow \mid \forall i : a \hookrightarrow_i \varphi\}$. Observe that $C^\subseteq \cap \Theta = \{\theta \in \Theta \mid \forall \alpha \mid a \in \alpha_{end} : \theta(i, \alpha) = \emptyset \vee \theta(i, \alpha) = \varphi\}$. The relations obs-compatible with $C^\subseteq \cap \Theta$ are exactly those in C .

Example 10 assumed an infinite I , in which case the claim $\exists i \mid a \hookrightarrow_i \varphi$ is not falsifiable. Observe that $C^\subseteq \cap \Theta = \{\theta \in \Theta \mid \exists i \mid \forall \alpha \mid a \in \alpha_{end} : \theta(i, \alpha) = \emptyset \vee \theta(i, \alpha) = \varphi\}$. Any relation $\rightsquigarrow$ is obs-compatible with that set as $C^\subseteq \cap \Theta = \Theta$: given any $\theta \in \Theta$, there is an i such that $\forall \alpha : \theta(i, \alpha) = \emptyset$ as θ must have a finite support. $\triangle$

Example 36. Pick $i \in I$, $p \neq p' \in P$ and $\alpha, \alpha' \in \mathcal{A}^{<\mathbb{N}}$ with α and α' non prefix related. The claim $\psi = (\alpha \rightsquigarrow_i p \vee \alpha \rightsquigarrow_i p') \wedge (\alpha' \rightsquigarrow_i p \vee \alpha' \rightsquigarrow_i p') \wedge (\forall(j, \alpha'') \in (I \times h) \setminus \{(i, \alpha), (i, \alpha')\} : \alpha'' \rightsquigarrow_j p)$ is falsifiable.

Given $q, q' \in P$, let $\rightsquigarrow_i^{q, q'}$ be defined as $\alpha \rightsquigarrow_i^{q, q'} q$, $\alpha' \rightsquigarrow_i^{q, q'} q'$ and $\forall(j, \alpha'') \in (I \times h) \setminus \{(i, \alpha), (i, \alpha')\} : \alpha'' \rightsquigarrow_j^{q, q'} p$. Note that $C_\psi = \{\rightsquigarrow^{p, p}, \rightsquigarrow^{p, p'}, \rightsquigarrow^{p', p}, \rightsquigarrow^{p', p'}\}$.

Then, $C_\psi^\subseteq \cap \Theta = \{\theta \in \Theta \mid \theta(i, \alpha) \in \{\emptyset, p, p'\} \wedge \theta(i, \alpha') \in \{\emptyset, p, p'\} \wedge \forall(j, \alpha'') \in (I \times h) \setminus \{(i, \alpha), (i, \alpha')\} : \theta(j, \alpha'') \in \{\emptyset, p\}\}$.

Any obs-compatible relation has to satisfy $\forall(j, \alpha'') \in (I \times h) \setminus \{(i, \alpha), (i, \alpha')\} : \alpha'' \rightsquigarrow_j p$. It also must satisfy $\rightsquigarrow(i, \alpha) \in \{p, p'\}$, for otherwise, with $\theta = \{(i, \alpha, \rightsquigarrow(i, \alpha))\}$, we get $(\rightsquigarrow \cap \theta) \notin C_\psi^\subseteq \cap \Theta$. Similarly, it must satisfy $\rightsquigarrow(i, \alpha') \in \{p, p'\}$. $\triangle$

Example 37. Consider any i , $a_1 \neq a_2 \in \mathcal{A}$, $p, p' \in P$. The claim $C = \{\rightsquigarrow \mid (a_1) \rightsquigarrow_i p \wedge (a_2) \rightsquigarrow_i p'\}$ is falsifiable.

We have $C^\subseteq \cap \Theta = \{\theta \mid (\theta((a_1), i) = \emptyset \vee \theta((a_1), i) = p) \wedge (\theta((a_2), i) = \emptyset \vee \theta((a_2), i) = p')\}$. The obs-compatible relations are $\{\rightsquigarrow \mid (a_1) \rightsquigarrow_i p \wedge (a_2) \rightsquigarrow_i p'\}$. $\triangle$

Example 38. Assume that $\#I \geq 2$. Consider any $p, p' \in P$ and the constant functions $\rightsquigarrow^p$ to p and $\rightsquigarrow^{p'}$ to p' and the claim $C = \{\rightsquigarrow^p, \rightsquigarrow^{p'}\}$. We have $C^\subseteq \cap \Theta = \{\theta \mid \theta(\theta(P)) = \{p\} \vee \theta(\theta(P)) = \{p'\}\}$. Consider any relation $\rightsquigarrow$ that is obs-compatible with $C^\subseteq \cap \Theta$. Assume that for some α, i , $\alpha \rightsquigarrow_i p$. Then, pick any $j \neq i$, and observe that by obs-compatibility, for any α' , $\alpha' \rightsquigarrow_j p$. It follows, still by obs-compatibility, that for any α' , $\alpha' \rightsquigarrow_i p$ (otherwise pick θ including (α', i) and any (α'', j)). Conclude that $\rightsquigarrow$ is constant to p . It follows that the obs-compatible relations are $\{\rightsquigarrow^p, \rightsquigarrow^{p'}\}$, thus, C is falsifiable. $\triangle$

Example 39. Assume that $\#I = 1$. Consider any $p, p' \in P$ and the constant functions $\rightsquigarrow^p$ to p and $\rightsquigarrow^{p'}$ to p' and the claim $C = \{\rightsquigarrow^p, \rightsquigarrow^{p'}\}$. We have $C^\subseteq \cap \Theta = \{\theta \mid \theta(\theta(P)) = \{p\} \vee \theta(\theta(P)) = \{p'\}\}$. Consider any relation $\rightsquigarrow$ that is obs-compatible with $C^\subseteq \cap \Theta$. Assume that for some α, i , $\alpha \rightsquigarrow_i p$. Observe that by obs-compatibility, $\emptyset \rightsquigarrow_i p$. It follows, still by obs-compatibility, that for any α' , $\alpha' \rightsquigarrow_i p$ (otherwise pick θ including (α', i) and $(\emptyset, i)$). Conclude that $\rightsquigarrow$ is constant to p . It follows that the obs-compatible relations are $\{\rightsquigarrow^p, \rightsquigarrow^{p'}\}$, thus, C is falsifiable. $\triangle$

Example 40. Consider any $p, p' \in P$ and the claim $C = \{\rightsquigarrow \mid (\forall \alpha \neq \emptyset, i \in I : \alpha \rightsquigarrow_i p) \vee (\forall \alpha \neq \emptyset, i \in I : \alpha \rightsquigarrow_i p')\}$. That claim is falsifiable iff $\#I \geq 2$ or $\#\mathcal{A} = 1$. The proof if so is similar to example 38; otherwise, pick any $a \in \mathcal{A}$ and note that any relation satisfying $(a.*) \rightsquigarrow_i p$ and $\forall b \neq a : (b.*) \rightsquigarrow_i p'$ is obs-compatible with $C^\subseteq \cap \Theta$ but is not in C . $\triangle$

Example 41. Consider any $i, \alpha, \alpha' \in \mathcal{A}^{<\mathbb{N}}, p, p' \in P$ where α and α' are not prefix related. The claim $C = \{\rightsquigarrow \mid \alpha \rightsquigarrow_i p \vee \alpha' \rightsquigarrow_i p'\}$ is unfalsifiable.

We have $C^\subseteq \cap \Theta = \{\theta \mid (\theta(i, \alpha) = \emptyset \vee \theta(i, \alpha) = p) \vee (\theta(i, \alpha') = \emptyset \vee \theta(i, \alpha') = p')\}$. Observe that any $\theta \in \Theta$ satisfy $\theta(i, \alpha) = \emptyset \vee \theta(i, \alpha') = \emptyset$, thus, $C^\subseteq \cap \Theta = \Theta$. It follows that the obs-compatible relations are $P^{I \times \mathcal{A}^{<\mathbb{N}}}$, which is bigger than C . $\triangle$

Example 42. Consider any $i, a \neq a' \in \mathcal{A}, p, p' \in P$. The claim $C = \{\rightsquigarrow \mid (a) \rightsquigarrow_i p \vee (a') \rightsquigarrow_i p'\}$ is unfalsifiable (per example 41).

Now consider the smaller claim $C' = \{\rightsquigarrow \mid (a_1) \rightsquigarrow_i p\}$ (thus $C' \subsetneq C$). It yields a smaller set $C'^\subseteq \cap \Theta = \{\theta \mid (\theta((a_1), i) = \emptyset \vee \theta((a_1), i) = p)\}$. The obs-compatible relations are the smaller set $\{\rightsquigarrow \mid (a_1) \rightsquigarrow_i p\}$. Thus, C' is falsifiable.

Note that any claim C'' bigger than C is unfalsifiable as well, unless trivial: from $C \subseteq C''$, we deduce $\Theta = C^\subseteq \cap \Theta \subseteq C''^\subseteq \cap \Theta$, thus $C''^\subseteq \cap \Theta = \Theta$ and the obs-compatible relations are $P^{I \times \mathcal{A}^{<\mathbb{N}}}$, which equals C'' only if C'' is trivial.

Now consider the alternative smaller claim of example 38, $C^{(2)} = \{\rightsquigarrow^p, \rightsquigarrow^{p'}\}$ (thus $C^{(2)} \subsetneq C$). It yields a smaller set $C^{(2)\subseteq} \cap \Theta$. $\triangle$

Example 43. Consider any i and $p, p' \in P$. The claim $C = \{\rightsquigarrow \mid (\forall \alpha : \#\alpha \in 2\mathbb{N} \Rightarrow \alpha \rightsquigarrow_i p) \vee (\forall \alpha : (\#\alpha \in 2\mathbb{N} + 1 \vee \alpha = \emptyset) \Rightarrow \alpha \rightsquigarrow_i p')\}$, namely, all paths of even lengths lead i to p OR all paths of odd lengths or empty lead i to p' , is falsifiable.

Note that $C^\subseteq \cap \Theta = \{\theta \in \Theta \mid (\forall \alpha : \#\alpha \in 2\mathbb{N} \Rightarrow \theta(\alpha) \in \{\emptyset, p\}) \vee (\forall \alpha : (\#\alpha \in 2\mathbb{N} + 1 \vee \alpha = \emptyset) \Rightarrow \theta(\alpha) \in \{\emptyset, p'\})\}$. Consider any obs-compatible $\rightsquigarrow$. Assume that $\exists \alpha \mid \#\alpha \in 2\mathbb{N} \wedge \alpha \not\rightsquigarrow_i p$. Pick $\theta = \{(i, \emptyset, \rightsquigarrow_i(\emptyset)), (i, \alpha, \rightsquigarrow_i(\alpha))\}$. We have $(\rightsquigarrow \cap \theta) \in C^\subseteq \cap \Theta$, and $\rightsquigarrow \cap \theta = \theta$, and from $\theta \in C^\subseteq \cap \Theta$ we obtain $\rightsquigarrow_i(\emptyset) = p'$. Similarly, using $\theta = \{(i, \emptyset, \rightsquigarrow_i(\emptyset)), (j, \alpha', \rightsquigarrow_j(\alpha'))\}$, we obtain that $\rightsquigarrow$ conforms to the relation in C . $\triangle$

Remark 27. Intuition (to be explored) is that when $C^\subseteq \cap \Theta$ has the covering property, then the obs-compatible relations are easily deduced from the set (something like, just consider compatibility point per point, one singleton observation at a time). Covering is what is used above, something like $\exists i, \alpha, \alpha'$ such that $\forall \alpha'' : [\forall j \neq i : \{(i, \alpha, \cdot), (j, \alpha', \cdot)\} \in C^\subseteq \cap \Theta] \wedge [\exists j \neq i \mid \{(i, \alpha', \cdot), (j, \alpha'', \cdot)\} \in C^\subseteq \cap \Theta]$. $\triangle$

B.3. Behavior function as a forest

Define the *suffix* relation $R = \{(\alpha, \alpha') \in \mathcal{A}^{<\mathbb{N}} \times \mathcal{A}^{<\mathbb{N}} \mid \alpha' = (\alpha, \cdot)\}$ as the (infinite) tree whose set of nodes is $\mathcal{A}^{<\mathbb{N}}$ and which links argumentative paths to all of their direct suffixes. Note that a path in such a tree (an increasing sequence of elements in $\mathcal{A}^{<\mathbb{N}}$) corresponds to an argumentative path, that is, to an element of $\mathcal{A}^{<\mathbb{N}}$.

Any function $\rightsquigarrow_i : \mathcal{A}^{<\mathbb{N}} \mapsto P$ can be considered as having this tree as a domain. Any relation $\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}}$ can then be viewed as a forest associating each $i \in I$ to a tree labeled by the function $\rightsquigarrow_i$ (see fig. 1).

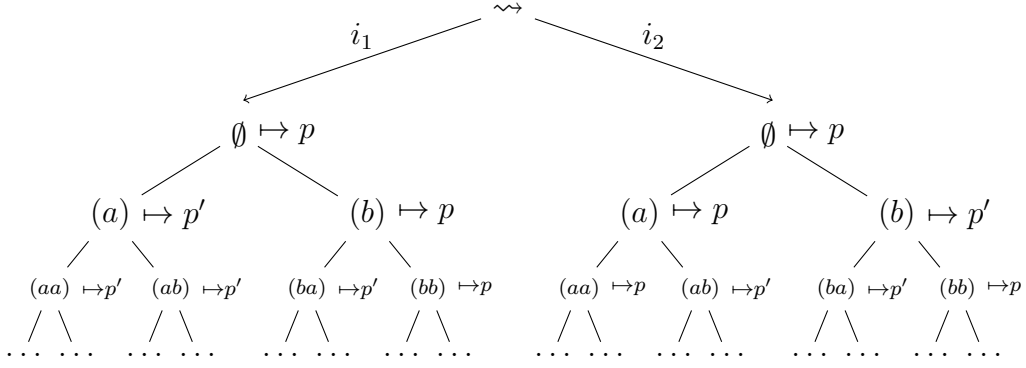


Figure 1: A forest

B.4. Compatibility of claim properties

We are here interested in the interplay between properties of holding, anonymity, (strong) informativeness beyond counting, and falsifiability. Theorems 40 to 42 show their compatibility when one of holding or anonymity is not required (except that informativeness beyond counting cannot be made strong in the edge case where $\forall i \in I : P_i = \emptyset$).

We thus turn to requiring both anonymous and holding. They are always satisfiable when only these two are required (theorem 43). Theorem 45 (and extensions, theorems 46 and 47) shows that informativeness beyond counting (and variants) is too demanding if one wants to investigate situations where individuals have differing deliberate preferences.

It is possible under reasonable conditions of regularity to satisfy anonymous and holding together with indicativeness. This happens for example when everybody has a deliberate judgment and a finite set of decisive arguments suffices to lead everybody to their deliberate judgment (corollary 14).

Theorem 40. *For all exposure protocols, there exists a claim that holds and is strongly informative beyond counting and falsifiable iff $\exists i \mid P_i \neq \emptyset$.*

Proof. Define $C = \{\rightsquigarrow\}$. □

Theorem 41. *For all exposure protocols, there exists a claim that holds and is informative beyond counting and falsifiable.*

Proof. Define $C = \{\rightsquigarrow\}$. □

Theorem 42. *For all exposure protocols, there exists a claim that is anonymous, strongly informative beyond counting and falsifiable.*

Proof. Define the claim as $\forall i \in I, \alpha \in \mathcal{A}^{<\mathbb{N}} : \alpha \rightsquigarrow_i \varphi$. □

Theorem 43. *For all exposure protocols, there exists a claim that holds and is anonymous.*

Proof. Define the claim as $\forall i \in I : \rightsquigarrow_i \in \{\rightsquigarrow_j \mid j \in I\}$. $\square$

Theorem 44. *Given any $Q \subseteq P \cup \{\emptyset\}$, the following two conditions on $\rightsquigarrow$ are equivalent: there exists a claim that holds and is anonymous and satisfies $\exists j \in I \mid \pi(C)(j) \subsetneq Q$; and $\{P_i, i \in I\} \subsetneq Q$.*

Proof. Consider any claim C that holds and is anonymous, pick any $j \in I$, and let us show that $\{P_i, i \in I\} \subseteq \pi(C)(j)$. Pick any $p \in \{P_i, i \in I\}$. Pick any $i \in I$ such that $\pi(\rightsquigarrow)(i) = p$. Define σ as the permutation exchanging i and j . As C holds, it contains $\rightsquigarrow$, and by anonymity, it contains $\rightsquigarrow \circ \sigma$. Observing that $\pi(\rightsquigarrow \circ \sigma)(j) = \pi(\rightsquigarrow)(i) = p$, we obtain $p \in \pi(C)(j)$.

It follows that for any claim C that holds and is anonymous, if $\exists j \in I \mid \pi(C)(j) \subsetneq Q$ then $\{P_i, i \in I\} \subseteq \pi(C)(j) \subsetneq Q$ thus $\{P_i, i \in I\} \subsetneq Q$.

Turning to the other direction, if $\{P_i, i \in I\} \subsetneq Q$, then pick any $p \in Q \setminus \{P_i, i \in I\}$; the claim $\forall i \in I : P_i \neq p$ satisfies the desired properties. $\square$

Theorem 45. *There exists a claim that holds and is anonymous and informative beyond counting iff $\{P_i, i \in I\} \neq P \cup \{\emptyset\}$.*

Proof. From theorem 44 with $Q = P \cup \{\emptyset\}$, we have that there exists a claim that holds and is anonymous and satisfies $\exists j \in I \mid \pi(C)(j) \subsetneq P \cup \{\emptyset\}$ iff $\{P_i, i \in I\} \subsetneq P \cup \{\emptyset\}$. $\square$

Theorem 46. *There exists a claim that holds and is anonymous and strongly informative beyond counting iff $\{P_i, i \in I\} \subsetneq P$.*

Proof. From theorem 44 with $Q = P$, we have that there exists a claim that holds and is anonymous and satisfies $\exists j \in I \mid \pi(C)(j) \subsetneq P$ iff $\{P_i, i \in I\} \subsetneq P$. $\square$

Theorem 47. *Given $Q = \{\varphi, \varphi'\}$, the following two conditions on $\rightsquigarrow$ are equivalent: there exists a claim that holds and is anonymous and satisfies $\exists j \in I \mid \pi(C)(j) \subsetneq Q$; and $\{P_i, i \in I\} \subsetneq Q$.*

Proof. From theorem 44 with $Q = \{\varphi, \varphi'\}$, we have that there exists a claim that holds and is anonymous and satisfies $\exists j \in I \mid \pi(C)(j) \subsetneq \{\varphi, \varphi'\}$ iff $\{P_i, i \in I\} \subsetneq \{\varphi, \varphi'\}$. $\square$

Define the claim $C^{\text{HA}} = \{\rightsquigarrow \circ \sigma \mid \sigma : I \Leftrightarrow I\}$. Under anonymity, requesting that a claim holds is the same as requesting that it includes C^{HA} .

Theorem 48. *A claim holds and is anonymous iff it includes C^{HA} and is anonymous. In particular, C^{HA} holds and is anonymous.*

Remark 28. Define $C^{\text{A}} = \{\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}} \mid \rightsquigarrow(I) = \rightsquigarrow(I)\}$. Observe that C^{A} is anonymous, that $C^{\text{HA}} \subseteq C^{\text{A}}$, and that $C^{\text{HA}} \neq C^{\text{A}}$ iff $\exists i, j \in I : \rightsquigarrow_i \neq \rightsquigarrow_j$ (to be checked). $\triangle$

Theorem 49. *The following two conditions on $\rightsquigarrow$ are equivalent:*

T49-1 for some $H \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}}))$ and $i_1 \in I \mid P_{i_1} \in P$, we have $\forall i_2 \mid P_{i_2} \neq P_{i_1}$ that $\exists \alpha \in H \mid \rightsquigarrow_{i_1}(\alpha) \neq \rightsquigarrow_{i_2}(\alpha)$;

T49-2 there exists a claim that holds and is anonymous and indicative.

Proof. For $T49-1 \Rightarrow T49-2$, define $p = P_{i_1}$. Consider any claim C such that $C \subseteq \{\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}} \mid \rightsquigarrow_{i_1} \in \rightsquigarrow(I) \subseteq \rightsquigarrow(I)\}$ and C is anonymous (for example, C^{HA} from theorem 48 is suitable). The claim is indicative for $i = i_1$ and $\theta = \{(i_1, \alpha, \rightsquigarrow_{i_1}(\alpha)) \mid \alpha \in H\}$ for the following reason. The attitudes in C compatible with θ are $\{\rightsquigarrow \mid \theta \subseteq \rightsquigarrow\} \cap C \subseteq \{\rightsquigarrow \mid \rightsquigarrow_{i_1} \in \{\rightsquigarrow_j \mid P_j = p\}\}$, and $\pi(\{\rightsquigarrow \mid \rightsquigarrow_{i_1} \in \{\rightsquigarrow_j \mid P_j = p\}\})(i_1) = p$.

For item $T49-2 \Rightarrow$ item $T49-1$, pick an indicative claim C and consider $i \in I$ and $\theta \in \Theta \cap (\{i\} \times \mathcal{A}^{<\mathbb{N}} \times P) \cap C^{\subseteq}$ such that $\pi(\{\rightsquigarrow \mid \theta \subseteq \rightsquigarrow\} \cap C)(i) \subsetneq P$. Define $H = \theta(i, P)$. $\square$

Corollary 14. *Assume that for some $p \in P$, we have $\exists i \in I \mid P_i = p$ and $\forall i_2 \mid P_{i_2} \neq p$ we have that $\exists A \in \mathcal{F}(\mathcal{A}) \mid \forall i \in I \mid P_i \neq p, \exists a \in A \mid \hookrightarrow_i(a) \in P \setminus \{p\}$, then there exists a claim that holds and is anonymous and indicative.*

Proof. We know that i_1 has some decisive argument a_1 in favor of p . Assume (wlog) that $A = \{2, \dots, \#A + 1\}$. Define $\alpha_1 = \emptyset$ and for $i \in \llbracket 2, \#A + 1 \rrbracket$, $\alpha^{(i)} = (\alpha_{i-1}, a_1, i)$. Define $H = \{\alpha^{(i)} \mid i \in \llbracket 2, \#A + 1 \rrbracket\}$. Apply theorem 49. $\square$

Example 44. Consider I_p such that $\forall i \in I_p$: i constantly says p . Consider $J = I \setminus I_p$ whose DJ is $p' \neq p$, and such that each $j \in J$ has a single decisive argument a_j for p' , and when a_j is not in α , $\alpha \rightsquigarrow_j p$. Then, the condition is not satisfied. $\triangle$

Theorem 50. *The following conditions on $\rightsquigarrow$ are equivalent:*

T50-1 there exists no claim that holds and is anonymous and indicative;

T50-2 all paths are such that for all i_1 having a DJ, there is an i_2 with another deliberated judgement that behaves identically on the whole path (formally: $\forall \alpha \in \mathcal{A}^{<\mathbb{N}}, i_1 \in I \mid \pi(i_1) \neq \emptyset : \exists i_2 \in I \mid \pi(i_1) \neq \pi(i_2) \wedge \forall \alpha' \subseteq \alpha : \rightsquigarrow_{i_1}(\alpha') = \rightsquigarrow_{i_2}(\alpha')\}$);

T50-3 all paths are such that for all i_1 , either at least one individual without a DJ, or infinitely many individuals i_2 with another deliberated judgement than i_1 , behave identically to i_1 on the whole path (formally: $\forall \alpha \in \mathcal{A}^{<\mathbb{N}}, i_1 \in I : [\exists i_2 \in I \mid \pi(i_2) = \emptyset \wedge \forall \alpha' \subseteq \alpha : \rightsquigarrow_{i_1}(\alpha') = \rightsquigarrow_{i_2}(\alpha')] \vee [\#\{i_2 \in I \mid \pi(i_1) \neq \pi(i_2) \wedge \forall \alpha' \subseteq \alpha : \rightsquigarrow_{i_1}(\alpha') = \rightsquigarrow_{i_2}(\alpha')\} = \infty]$)

Proof. Let us show that NOT T50-2 implies T49-1. Consider: some individual i and some path α are such that $\pi(i) \neq \emptyset$ and $\forall i_2 \in I \mid \pi(i) \neq \pi(i_2) : \exists \alpha' \subseteq \alpha \mid \rightsquigarrow_i(\alpha') \neq \rightsquigarrow_{i_2}(\alpha')$. Define $H = \{\alpha' \subseteq \alpha\}$ and $i_1 = i$.

Conversely, T49-1 implies NOT T50-2. Consider: for some $H \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}}))$ and $i \in I \mid P_i \in P$, we have $\forall i_2 \mid P_{i_2} \neq P_i$ that $\exists \alpha' \in H \mid \rightsquigarrow_i(\alpha') \neq \rightsquigarrow_{i_2}(\alpha')$. Define α as the maximal path (in cardinal) in H .

Let us now prove that T50-2 implies T50-3. Consider any $\alpha \in \mathcal{A}^{<\mathbb{N}}, i_1 \in I$. Define $J = \{i_2 \in I \mid \forall \alpha' \subseteq \alpha : \rightsquigarrow_{i_1}(\alpha') = \rightsquigarrow_{i_2}(\alpha')\}$. Assume that $\forall i_2 \in J : \pi(i_2) \neq \emptyset$ (otherwise, the first disjunct of T50-3 holds), and let us show that $\#\{i_2 \in J \mid \pi(i_1) \neq \pi(i_2)\} = \infty$. From $\forall i_2 \in J : \pi(i_2) \neq \emptyset$, we obtain $\pi(i_1) \neq \emptyset$, thus, $\exists a_{i_1} \in \mathcal{A} \mid a_{i_1} \hookrightarrow_{i_1} \pi(i_1)$.

Apply T50-2 to obtain some $i_2 \in J$ such that $\pi(i_1) \neq \pi(i_2)$. Also, by our assumption, $\pi(i_2) \neq \emptyset$, thus, $\exists a_{i_2} \in \mathcal{A} \mid a_{i_2} \hookrightarrow_{i_2} \pi(i_2)$. Consider the path $\alpha' = (\alpha, a_{i_1}, a_{i_2})$. We have $\alpha' \rightsquigarrow_{i_1} \pi(i_1) \neq \alpha' \rightsquigarrow_{i_2} \pi(i_2)$, thus $\rightsquigarrow_{i_1}(\alpha') \neq \rightsquigarrow_{i_2}(\alpha')$. Apply T50-2 on α', i_1 . We obtain some $i_3 \in I$ such that $\pi(i_1) \neq \pi(i_3)$ and $\forall \alpha'' \subseteq \alpha' : \rightsquigarrow_{i_1}(\alpha'') = \rightsquigarrow_{i_3}(\alpha'')$. In particular, $\rightsquigarrow_{i_1}(\alpha') = \rightsquigarrow_{i_3}(\alpha')$, thus, $i_3 \neq i_2$. Also, $i_3 \in J$. Iterating this process, we obtain an infinite sequence of distinct individuals in J . In other words, we have shown that if some $i_2 \in J, \alpha^{(2)} \in \mathcal{A}^{<\mathbb{N}}$ with $\pi(i_2) \neq \emptyset$ and i_1 and i_2 agree everywhere on $\alpha^{(2)}$, then there exists $i_3 \in J, \alpha^{(3)} \in \mathcal{A}^{<\mathbb{N}}$ with $\pi(i_3) \neq \emptyset$ and $\alpha^{(2)} \subsetneq \alpha^{(3)}$ and i_1 and i_2 disagree somewhere on $\alpha^{(3)}$ and i_1 and i_2 agree everywhere on $\alpha^{(3)}$. (Should check this summary.) Could also use the following probably simpler proof. Consider any finite subset of j individuals, build a H' with all the test paths containing decisive arguments, namely $\alpha^{(2)} = (\alpha, a_1, a_2)$, $\alpha^{(3)} = (\alpha^{(2)}, a_1, a_3)$, $\dots$, and show that there's another j in fact. $\square$

Example 45 (no HAI, with DJ). The following situation illustrates theorem 50 and involve only individuals with a DJ.

Given any infinite set $\mathcal{A}$, define $I = I^{\text{DJ}} = \{g \in P^{\mathcal{A}^{<\mathbb{N}}} \mid \pi(g) \neq \emptyset\}$ and $\forall i \in I : \rightsquigarrow_i = i$. Considering any $\alpha \in \mathcal{F}(\mathcal{A}^{<\mathbb{N}})$ and $\rightsquigarrow_i \in I$, pick another relation that coincides on α but with another DJ than $\rightsquigarrow_i$. This is possible because α is finite, thus there is an argument that is not involved in any of its elements, which can serve as decisive one. Per theorem 50, there is no claim that holds and is anonymous and indicative. $\triangle$

Example 46 (no HAI). Define $I = \{i_1, i_2, i_3\}$ and $\rightsquigarrow$ as follows: $.* \rightsquigarrow_{i_1} \varphi, \emptyset \rightsquigarrow_{i_2} \varphi, a.* \rightsquigarrow_{i_2} \varphi, a'.* \rightsquigarrow_{i_2} \varphi', \emptyset \rightsquigarrow_{i_3} \varphi, a.* \rightsquigarrow_{i_3} \varphi', a'.* \rightsquigarrow_{i_3} \varphi$. Then $\pi(i_1) = \varphi$ and $\pi(i_2) = \pi(i_3) = \emptyset$. This satisfies T50-2: considering any $\alpha \in \mathcal{A}^{<\mathbb{N}}$, if $(1, a) \in \alpha$, then i_2 collides with i_1 on α , and otherwise, i_3 does. $\triangle$

Remark 29 (no HAI condition required). The condition $\pi(i_1) \neq \emptyset$ in T50-2 is required: the following example satisfies T50-2 but fails the extended version of that condition without that condition. In other words, defining $Q(i_1, \alpha)$ as $\exists i_2 \in I \mid \pi(i_1) \neq \pi(i_2) \wedge \forall \alpha' \subseteq \alpha : \rightsquigarrow_{i_1}(\alpha') = \rightsquigarrow_{i_2}(\alpha')$, the example satisfies $\forall \alpha \in \mathcal{A}^{<\mathbb{N}}, i_1 \in I \mid \pi(i_1) \neq \emptyset : Q(i_1, \alpha)$, but does not satisfy $[\forall \alpha \in \mathcal{A}^{<\mathbb{N}}, i_1 \in I : Q(i_1, \alpha)]$, thus, satisfies $\exists \alpha, i_0 \mid \pi(i_0) = \emptyset \wedge \neg Q(i_0, \alpha)$.

Define $I = \{i_1, \dots, i_4\}$, $\rightsquigarrow_{i_1}, \rightsquigarrow_{i_2}, \rightsquigarrow_{i_3}$ as in example 46 and $\rightsquigarrow_{i_4}$ as: $.* \rightsquigarrow_{i_4} \varphi', \emptyset \rightsquigarrow_{i_4} \varphi', a.* \rightsquigarrow_{i_4} \varphi, a'.* \rightsquigarrow_{i_4} \varphi'$. This satisfies T50-2 for the reason given in example 46, and fails $Q(i_4, \emptyset)$. $\triangle$

TODO Check example 29. Check any example which seemed non admitting a falsifiable claim.

B.5. Ideas for no suitable

C is *suitable* iff it holds and is anonymous, informative and falsifiable.

Example 47. Assume that $\exists a, \varphi \mid \forall i : a \hookrightarrow_i \varphi$. Then $\forall i : a \hookrightarrow_i \varphi$ is suitable. Note that it is falsifiable but not verifiable (that is, no O implies C). $\triangle$

Our goal now is to find an exposure protocol $\rightsquigarrow$ with no suitable claim.

Could we have that if it holds then it is not informative? i_1 thinks p and i_2 thinks $\neg p$.

Given $\alpha^{(k)} \in \mathcal{A}^{<\mathbb{N}}$ and $a_l \in \mathcal{A}$, let $(\alpha^{(k)}, \dots, a_l, ?)$ denote the set $\{\alpha^{(l)} \mid \alpha^{(k)} \subsetneq \alpha^{(l)} \wedge a_l \in \alpha_{end}^{(l)}\}$. Given $\alpha \in \mathcal{A}^{<\mathbb{N}}$, $\rightsquigarrow_i$ and $a_k, a_l \in \mathcal{A}$, let $(\alpha, a_k, ?, \rightsquigarrow_i \varphi, \dots, a_l, ?)$ denote the set $\{\alpha^{(l)} \mid \exists \alpha^{(k)} \mid \alpha \subsetneq \alpha^{(k)} \subsetneq \alpha^{(l)} \wedge a_k \in \alpha_{end}^{(k)} \wedge \alpha^{(k)} \not\rightsquigarrow_i \varphi \wedge a_l \in \alpha_{end}^{(l)}\}$. Given $a_k, a_l \in \mathcal{A}$, let $(\dots, a_k, ?, \rightsquigarrow_i \varphi, \dots, a_l, ?)$ denote the set $\bigcup_{\alpha \in \mathcal{A}^{<\mathbb{N}}} (\alpha, a_k, ?, \rightsquigarrow_i \varphi, \dots, a_l, ?) = \{\alpha^{(l)} \mid \exists \alpha, \alpha^{(k)} \mid \alpha \subsetneq \alpha^{(k)} \subsetneq \alpha^{(l)} \wedge a_k \in \alpha_{end}^{(k)} \wedge \alpha^{(k)} \not\rightsquigarrow_i \varphi \wedge a_l \in \alpha_{end}^{(l)}\} = \{\alpha^{(l)} \mid \exists \alpha^{(k)} \mid \alpha^{(k)} \subsetneq \alpha^{(l)} \wedge a_k \in \alpha_{end}^{(k)} \wedge \alpha^{(k)} \not\rightsquigarrow_i \varphi \wedge a_l \in \alpha_{end}^{(l)}\}$. Given $\rightsquigarrow_i$, $p \in P$, $s \in \mathcal{A}$ and $S \subseteq \mathcal{A}$, note that $s \in \rightsquigarrow_i^{-1}(p)$ stand for $s \rightsquigarrow_i p$ and $S \subseteq \rightsquigarrow_i^{-1}(p)$ stand for $\forall s \in S : s \rightsquigarrow_i p$.

Theorem 51. *Given any $p \in P, a_k, a_l \in \mathcal{A}$, consider the following claim C :*

$$\begin{aligned} \forall i \in I, \forall \{a_1, a_2\} \mid \{a_1, a_2\} = \{a_k, a_l\}, \forall \alpha \in (\dots, a_1, ?) \cap \rightsquigarrow_i^{-1}(p) : \\ (\alpha, \dots, a_2, ?) \subseteq \rightsquigarrow_i^{-1}(p). \end{aligned} \quad (1)$$

Assume that $\rightsquigarrow \in C$. Then $\exists i \in I \mid [a_k \not\rightsquigarrow_i p \wedge a_l \not\rightsquigarrow_i p]$ is equivalent to:

$$\begin{aligned} \exists i \in I \mid \forall \{a_1, a_2\} \mid \{a_1, a_2\} = \{a_k, a_l\} : \\ \exists \alpha \in (\dots, a_1, ?) \cap \rightsquigarrow_i^{-1}(p) \mid (\alpha, \dots, a_2, ?) \subseteq \rightsquigarrow_i^{-1}(p). \end{aligned} \quad (2)$$

Proof. $a_k \not\rightsquigarrow_i p$ iff $\exists \alpha \mid a_k \in \alpha_{end} \wedge \alpha \not\rightsquigarrow_i p$. □

Remark 30. Idea. That consequence is unfalsifiable so we cannot include anything like this in C (only something stronger, but of course a stronger statement could be false). Therefore, with no addition, we are doomed that exists some situations where no claim is suitable. △

Theorem 52. *Pick any $p \in P$ and $a_k \neq a_l \in \mathcal{A}$ and consider the claim (1) and not (2), thus, (1) and $\forall i, a_k \rightsquigarrow_i p \vee a_l \rightsquigarrow_i p$. It is not falsifiable.*

Proof. We need to exhibit some $\rightsquigarrow \notin C$ such that $\forall \theta \subseteq \rightsquigarrow : \theta \in C^\subseteq$. Define $\rightsquigarrow$ as, $\forall i \in I, (a_k) \not\rightsquigarrow_i p, (a_l) \not\rightsquigarrow_i p, \forall \alpha \in \mathcal{A}^{<\mathbb{N}} \setminus \{(a_k), (a_l)\} : \alpha \rightsquigarrow_i p$. It follows that $a_k \not\rightsquigarrow_i p \wedge a_l \not\rightsquigarrow_i p$. Thus, $\rightsquigarrow \notin C$. Now pick any $\theta \subseteq \rightsquigarrow$. Define $\rightsquigarrow'$ as, $\forall i \in I : \forall \alpha \in \mathcal{A}^{<\mathbb{N}} \setminus \{(a_k), (a_l)\}, \alpha \rightsquigarrow'_i p$ and $\forall a \in \{a_k, a_l\}, (a) \rightsquigarrow'_i (\rightsquigarrow(a, i))$ if $(a, i, \rightsquigarrow(a, i)) \in O$, $(a) \rightsquigarrow'_i p$ otherwise. The set O can't have both $(a_k) \not\rightsquigarrow_i p$ and $(a_l) \not\rightsquigarrow_i p$, thus, $\rightsquigarrow' \in C$. We conclude that $\rightsquigarrow' \in S_\theta \cap C$. □

Theorem 53 (Informal). *Assume that the NOT consequence holds and the claim holds. Prove that there is no suitable claim. If the claim is just the NOT cons and claim, then it is not fals. If the claim is weaker, it does not permit to prove that a_k or a_l is decisive. (BUT does it possibly prove that some arg is decisive? But it may be false.) If the claim is stronger, it may be false.*

Theorem 54 (Formal). *Pick any $p \in P$ and $a_k \neq a_l \in \mathcal{A}$ and assume that $\rightsquigarrow$ fails eq. (2), that is, satisfies:*

$$\forall i \in I, \exists \{a_1, a_2\} = \{a_k, a_l\} \mid \forall \alpha \in (\dots, a_1, ?) \cap \rightsquigarrow_i^{-1}(p) : (\alpha, \dots, a_2, ?) \cap \rightsquigarrow_i^{-1}(p) \neq \emptyset. \quad (3)$$

Assume further that $\rightsquigarrow$ satisfies (1), thus: $\forall i \in I, \forall \{a_1, a_2\} \mid \{a_1, a_2\} = \{a_k, a_l\}, \forall \alpha \in (\dots, a_1, ?) \cap \rightsquigarrow_i^{-1}(p) : (\alpha, \dots, a_2, ?) \subseteq \rightsquigarrow_i^{-1}(p)$. Then, there is no suitable claim.

Proof. Note that $\rightsquigarrow$ satisfies, $\forall i \in I, \exists \{a_1, a_2\} \mid \{a_1, a_2\} = \{a_k, a_l\} \wedge \forall \alpha \in (\dots, a_1, ?) : \alpha \rightsquigarrow_i p$; equivalently, $a_k \hookrightarrow_i p \vee a_l \hookrightarrow_i p$.

Assume the claim is just (1) $\wedge$ (3). Then if is not fals.

If the claim is sufficiently stronger (to be falsifiable) than this (thus with $C' \subsetneq C$, thus with $\rightsquigarrow \notin C'$ as defined above), then it may be false. Indeed, to be falsifiable, we need $\rightsquigarrow' \notin C'$. Now we need C' to hold so it must contain $\rightsquigarrow$ but not contain $\rightsquigarrow'$. It does not contain $\rightsquigarrow'$ iff it implies $\neg(\wedge_{\alpha \in \mathcal{A}^{<\mathbb{N}} \setminus \{(a_l)\}} \alpha \rightsquigarrow'_i p \wedge (a_l) \rightsquigarrow'_i p')$.

Intuitively, need that when we do not observe (a_k) for a given i , but only (a_l) for that i , the claim says something contrary about that (a_l, i) . $\square$

Remark 31. Idea: then use contrapositive. Not cons implies some dec arg. Prove equivalence to establish that we need to assume not cons to make such claims informative (in such a way). We could also attempt to prove that $\exists a \mid a \hookrightarrow_i \varphi$ but assuming the claim C is everything we know, there is no hope in proving something about some argument that we ignore.

As an example, with the shortest possible paths, we get $\alpha^{(k)} = (a_k)$ and $\alpha^{(l)} = (a_l)$, thus:

$$\begin{cases} (a_k) \not\rightsquigarrow_i \varphi \wedge \{\alpha' \mid (a_k) \subseteq \alpha' \wedge a_l \in \alpha'_{end}\} \rightsquigarrow_i \varphi; \\ (a_l) \not\rightsquigarrow_i \varphi \wedge \{\alpha' \mid (a_l) \subseteq \alpha' \wedge a_k \in \alpha'_{end}\} \rightsquigarrow_i \varphi. \end{cases}$$

Idea: first consider that a_l equals $a_j \cup a_k$ so that we can reasonably use an axiom. Then in the more general case, assume φ is not in DJ at all, which also implies that $a = a_k \cup a_l$ fails; and we fall back to the same. $\triangle$

Remark 32. The theorem is realistic in the following sense. Some need a_k only; others need $a_j \cup a_k$ (ignorant of a_j at start). Consider C saying that if a_k not convincing, then $a_j \cup a_k$ is convincing afterwards and if $a_j \cup a_k$ not convincing, then a_k convincing afterwards. Define $a_l = a_j \cup a_k$.

In other words. Assume that knowing both a_j and a_k is required to get to φ . Assume that J knows a_j and when told it, gets bored, thus not convincing. For j , only a_k is decisive. But K ignores a_j . For k , only $a_j \cup a_k$ is decisive. $\triangle$

Example 48 (To be completed). Assume that $\exists a_j \neq a_k, J, K \subsetneq I \mid J \cup K = I \wedge \#J \notin \mathbb{N} \wedge \#K \notin \mathbb{N} \wedge \forall j \in J : a_j \hookrightarrow_j \varphi \wedge \forall k \in K : a_k \hookrightarrow_k \varphi \wedge \forall a \in \mathcal{A} \setminus \{a_j, a_k\}, i \in I : a \not\rightsquigarrow_i \varphi$. Then no suitable C exists?

Let's find some suitable C that shows that $\varphi \in P_i$. $\triangle$

Intuition: if a_j and afterwards a_k is convincing, then either a_j is unuseful in which case a_k is decisive; or a_j is useful in which case $a_j \cup a_k$ is decisive. However, could also be that a_j is essential as first argument; thus a_j is useful but $a_j \cup a_k$ is not decisive as a' then $a_j \cup a_k$ is not convincing. Note that the latter part can be excluded by a falsifiable claim: $\nexists a' \mid a', a_j \cup a_k \not\rightsquigarrow_i \varphi$.

The following antecedent excludes two of these three hypothesis, leaving only the one that a_j is useful. $((a_j), \dots, a_k)$ convincing ; a_k not decisive (or (a_k) not convincing?); $\nexists a \mid [(a, \dots, a_k) \text{ not convincing} \wedge (a, \dots, a_j \cup a_k) \text{ not convincing}]$. The conclusion is: $a_j \cup a_k$ decisive.

Is this an axiom or is this provable? And: is the part with $\nexists a$ a falsifiable claim?

B.6. Alternative definition of syntactic falsifiability

A claim C is attempt-falsifiable iff $\exists \theta \mid \theta \Rightarrow \neg C$, by which I mean $\theta \notin C^\subseteq$.

Example 49. $(a) \rightsquigarrow_i \varphi \vee (a, a) \rightsquigarrow_i \varphi \vee \dots$ is not falsifiable. Given any O , define $l = \max_{\alpha \mid \exists p \in P \mid (i, \alpha, p) \in O} \# \alpha + 1$. Pick any $\rightsquigarrow$ such that $\theta \subseteq \rightsquigarrow \wedge (a)^l \rightsquigarrow_i \varphi$. We have $\rightsquigarrow \in S_\theta \cap C \neq \emptyset$. $\triangle$

Example 50. $(a) \rightsquigarrow_i \varphi \vee (a') \rightsquigarrow_i \varphi$ is not falsifiable. Given any O , it may not contain both $(i, (a), \varphi)$ and $(i, (a'), \varphi)$, thus letting t denote one of these triples not in O , pick any $\rightsquigarrow$ such that $\theta \subseteq \rightsquigarrow \wedge t \in \rightsquigarrow$. We have $\rightsquigarrow \in S_\theta \cap C \neq \emptyset$. $\triangle$

Example 51 (Fals. but shouldn't). $\alpha \rightsquigarrow_j \varphi \wedge [(a) \rightsquigarrow_i \varphi \vee (a') \rightsquigarrow_i \varphi]$ is falsifiable. Consider $\theta = \{(j, \alpha, \varphi')\}$. $\triangle$

Example 52. $\forall i : a \hookrightarrow_i \varphi$ is falsifiable. Consider any $i \in I$ and $\theta = \{(i, (a), \varphi')\}$. $\triangle$

Example 51 shows that this definition of falsifiability is wrong.

C. Argumentative theories of deliberated choice

An argumentative theory of deliberated choice γ is defined as a tuple (p_γ, d_γ) , where $p_\gamma \in P$ is the proposition supported by the theory and d_γ is a defense relation.

The defense relation $d_\gamma \subseteq \mathcal{A}^{<\mathbb{N}} \times \mathcal{A}$ defines the argumentation strategy of the theory. Informally speaking, it defines the arguments that the theory claims will convince the individuals in I to choose p_γ . Assuming that some path α challenges the theory's intended conclusion in favor of p_γ (as judged by the individual), $d_\gamma(\alpha)$ is the argument that the theory plays as a counter-argument to the path α .

Formally speaking, a defense relation $d_\gamma \subseteq \mathcal{A}^{<\mathbb{N}} \times \mathcal{A}$ is a right-unique relation satisfying bounded depth (to be defined shortly). Right-unique means that given any $\alpha \in \mathcal{A}^{<\mathbb{N}}$, $d_\gamma(\alpha) = \{a \in \mathcal{A} \mid (\alpha, a) \in d_\gamma\}$ is empty or is a singleton. As is usual for right-unique relations, we write $d_\gamma(\alpha) = a$ instead of $d_\gamma(\alpha) = \{a\}$ when $d_\gamma(\alpha)$ is a singleton (and $d_\gamma(\alpha) = \emptyset$ otherwise). Given $H \subseteq \mathcal{A}^{<\mathbb{N}}$, define $d_\gamma(H) = \{d_\gamma(\alpha) \mid \alpha \in H\} \subseteq \mathcal{A}$. Given $a_0, a_1 \in \mathcal{A}$, say that a_1 defends a_0 iff $\exists \alpha \in \mathcal{A}^{<\mathbb{N}} \mid a_0 \in R(\alpha) \wedge d_\gamma(\alpha) = a_1$, equivalently, iff $a_0 \in R(d_\gamma^{-1}(a_1))$, equivalently, iff $a_1 \in d_\gamma(\{\alpha \mid a_0 \in R(\alpha)\})$.

Say that γ has depth at least $k \in \mathbb{N}^*$ iff $\exists(\alpha_i, 1 \leq i \leq k) \mid \forall 1 \leq i \leq k, d_\gamma(\{\alpha_j \mid 1 \leq j \leq i-1\}) \subsetneq d_\gamma(\{\alpha_j \mid 1 \leq j \leq i\}) \wedge \forall 1 \leq i \leq k-1, \alpha_i \cup d_\gamma(\alpha_i) \subseteq \alpha_{i+1}$. The bounded depth condition on the defense relation imposes that for some $k \in \mathbb{N}^*$, γ does not have depth at least k . The depth of a theory is defined as the smallest value $k \in \mathbb{N}$ such that it does not have depth at least $k+1$. For example, γ has depth 0 iff $d_\gamma = \emptyset$; and γ has depth 1 iff $[\exists \alpha_1 \mid d_\gamma(\alpha_1) \neq \emptyset] \wedge [\forall \alpha_1, \alpha_2 \mid d_\gamma(\alpha_1) \neq \emptyset \wedge \alpha_1 \cup d_\gamma(\alpha_1) \subseteq \alpha_2 : d_\gamma(\alpha_2) \subseteq d_\gamma(\alpha_1)]$.

[OC: Intuition. Invoke d_γ at α only when α fails. Permit to invoke a previous defense again as maybe the counter was too long to perform when summarized. If a_2 fails, we must invoke $d_\gamma(\dots, a_2, x)$; failures of $d_\gamma(\dots, a_2, \dots, x)$ should not count. We should invoke d_γ only when α fails with some defense produced by $d_\gamma()$ belongs to α_{end} .]

An argumentative theory claims that its argumentation strategy, embodied by a_γ and d_γ , will convince the individuals in I . More precisely, given $i \in I$, it claims that a_γ either is a decisive argument for p_γ , or, if a_γ is not, that the theory will defend the argument by replying to every counter-argument to a_γ with an argument that either is a decisive argument for p_γ or, in turn, will be defended by the theory, and so on as long as the theory has new counter-arguments to produce; the theory claiming that some argument along that chain is decisive.

[OC: Condition: $\alpha \not\rightsquigarrow_i p_\gamma \wedge \exists \alpha' \subseteq \alpha \mid d_\gamma(\alpha') \in \alpha_{end} \wedge d_\gamma(\alpha)$ is not decisive ($\exists \beta \subseteq \alpha \mid d_\gamma(\alpha) \in \beta_{end} \wedge \beta \rightsquigarrow_i p_\gamma$).]

To define this formally, note that either a_γ is a decisive argument for p_γ , or some paths challenge this claim, namely those $\alpha \in \mathcal{A}^{<\mathbb{N}}$ such that $[a_\gamma \in \alpha_{end} \wedge \alpha \not\rightsquigarrow_i p_\gamma]$. Generalizing this observation, given $i \in I$, define $d_{\gamma,i}(a) = \{d_\gamma(a, \alpha) \mid a \in \alpha_{end} \wedge \alpha \not\rightsquigarrow_i p_\gamma\} \subseteq d_\gamma(a)$ as the subset of defenders of a required for convincing i of choosing p_γ . Define $A_{\gamma,i} \subseteq \mathcal{A}$ as the transitive closure of a_γ under $d_{\gamma,i}$: $A_{\gamma,i} = \cup_{k \in \mathbb{N}} (d_{\gamma,i})^k(a_\gamma)$. The set $A_{\gamma,i}$ is the set of arguments required to convince i of choosing p_γ .

An argumentative theory claims that whenever some of its arguments fail to convince i of choosing p_γ , it can argue something else. It is said to be truthful iff this claim is correct. More precisely, its claims are about paths of the sort $\alpha = (a_\gamma, \dots, a_2, \dots, a_3, \dots)$ where $a_2 \in d_{\gamma,i}(a_\gamma)$, $a_3 \in d_{\gamma,i}(a_2)$, $\dots$. The following definition makes this precise by mandating that whenever increasing subsequences of α , involving arguments a_j proposed by the theory, are not convincing, then the theory is able to produce yet another argument in defense of its claim.¹

Definition 6 (Truthfulness). An argumentative theory $(p_\gamma, a_\gamma, d_\gamma)$ for the exposure protocol $(I, \mathcal{A}, \rightsquigarrow)$ is *truthful* iff $\forall i \in I, \alpha \in \mathcal{A}^{<\mathbb{N}}, n \in \mathbb{N}, k \in \llbracket 1, \#\alpha \rrbracket^n, a \in \mathcal{A}^n$:

$$\left. \begin{array}{l} a_1 = a_\gamma \\ \forall 1 \leq j < n : a_{j+1} = d_\gamma(a_j, \alpha_{\llbracket 1, k_j \rrbracket}) \in \alpha_{\llbracket 1, k_{j+1} \rrbracket, end} \\ \forall 1 \leq j \leq n : \alpha_{\llbracket 1, k_j \rrbracket} \not\rightsquigarrow_i p_\gamma \end{array} \right\} \Rightarrow d_\gamma(a_n, \alpha) \neq \emptyset.$$

¹Intuition. If a_1 , used by the theory, is not decisive, then the theory has an answer, namely, a_2 , which works at least temporarily. If a_1 used by th is not dec, and a_2 used by th is not dec *in the same sequence*, then a_3 needs to defend *in the same sequence*. $(\dots, a_1)$ NO and $(\dots, a_1, \dots, a_2)$ NO then a_3 defends it. α of the form (α^1, α^2) where a_1 in end of α^1 NO and a_2 in end of α^2 NO.

Given the paths α, α' , the sequence (α, α') denotes a path constituted of the arguments in α followed by the arguments of α' .

Axiom 1 (Answerability-1). $\forall \dots$:

$$\left. \begin{array}{l} (a_4, a_2, a) \not\rightarrow_i p_\gamma \\ (a_4, a_2, a, a_3) \hookrightarrow_i p_\gamma \\ (a_5, a_2, a) \not\rightarrow_i p_\gamma \end{array} \right\} \Rightarrow (a_5, a_2, a, a_3) \hookrightarrow_i p_\gamma.$$

Requires excluding:

$$\left. \begin{array}{l} (a_4, a_2, a) \not\rightarrow_i p_\gamma \\ (a_4, a_2, a, a_3, a') \rightsquigarrow_i p_\gamma \\ (a_5, a_2, a) \not\rightarrow_i p_\gamma \\ (a_5, a_2, a, a_3, a') \not\rightarrow_i p_\gamma. \end{array} \right\}$$

To exclude:

$$\left. \begin{array}{l} (a_4, a_2, a) \not\rightarrow_i p_\gamma \\ (a_4, a_2, a, a_3) \hookrightarrow_i p_\gamma \\ (a_5, a_2, a) \rightsquigarrow_i p_\gamma \\ (a_5, a_2, a_3, a') \not\rightarrow_i p_\gamma \end{array} \right\}$$

Using:

$$\left. \begin{array}{l} (a_4, a_2, a) \not\rightarrow_i p_\gamma \\ (a_4, a_2, a, a_3) \hookrightarrow_i p_\gamma \\ (a_5, a_2, a) \rightsquigarrow_i p_\gamma \end{array} \right\} \Rightarrow (a_5, a_2, a_3) \hookrightarrow_i p_\gamma.$$

Axiom 2 (Answerability). $\forall i \in I, \alpha \in \mathcal{A}^{<\mathbb{N}}, a \in A_{\gamma, i}$:

$$\left. \begin{array}{l} \alpha \not\rightarrow_i p_\gamma \\ (\alpha, a) \rightsquigarrow_i p_\gamma \\ \forall a' \in \mathcal{A} : (\alpha, a, a') \rightsquigarrow_i p_\gamma \end{array} \right\} \Rightarrow a \hookrightarrow_i p_\gamma.$$

2 3

² a_1, a_2, a_3 NOT and a_1, a_2, a_3, a_4 YES and a'_1, a_2, a_3, a_4 NO imply that a_1, a_2, a_3, a_4, a'_1 NO. Generalizing: α NO and α, a_4 YES and α', a_4 NO imply that $\exists a' \in \alpha' \mid \alpha, a_4, a'$ NO. Let's try with this too powerful axiom, but then we should weaken it, as it could be that a_1 AND a_4 TOGETHER are decisive, whence the axiom fails. We should have then some argument $a_1 + a_4$ to replace them, and...? In summary, we want to avoid that a is not decisive but we can't observe it when starting with some α even though α has also no decisive arguments: if α has no decisive argument, then it must be possible to counter it.

In fact, we might need (a weakening of) the following. If a_1 is "decisive" when starting with some α (meaning (α, a_1, a') YES for all arguments a'), then a_1 is decisive. Hopefully: if (α, a') YES for all arguments a' , then some $a \in \alpha$ is decisive.

³There is a problem. Consider the following may hold for some i . a_1, a_5, a_2 not convincing, a_1, a_4 not convincing, while a_1, a_5 and a_1, a_4, a_2 and other variants convincing. Then for all i one of a_1 or a_2 will appear to be decisive although for those people neither is.

Axiom 3 (Answerability - using newly built arguments (newest so far)). $\forall i \in I, \alpha^j \in (\mathcal{A}^{<\mathbb{N}})^n$:

$$[\forall j : \alpha^j \hookrightarrow_i p_\gamma] \Rightarrow [\exists a \in \mathcal{A} \mid \forall j : (\alpha^j, a) \hookrightarrow_i p_\gamma].$$

Axiom 4 (Reinstatement). $\forall i \in I, \alpha \in \mathcal{A}^{<\mathbb{N}}, a_1, a_2 \in A_{\gamma,i}, a \in \mathcal{A}$:

$$\left. \begin{array}{l} (\alpha, a_1, a) \not\rightsquigarrow_i p_\gamma \\ (\alpha, a_1, a, a_2) \rightsquigarrow_i p_\gamma \\ (\alpha, a_1, a, a_2, a) \rightsquigarrow_i p_\gamma \end{array} \right\} \Rightarrow \left\{ \begin{array}{l} (\alpha, a_2) \rightsquigarrow_i p_\gamma \\ (\alpha, a_2, a) \rightsquigarrow_i p_\gamma \\ \forall a' : (\alpha, a_2, a') \not\rightsquigarrow_i p_\gamma \Rightarrow (\alpha, a_1, a') \not\rightsquigarrow_i p_\gamma. \end{array} \right.$$

Example 53 (Reinstatement is required). Consider a_1 not decisive because of a and then $(a_1, a, a_2) \rightsquigarrow_i p$. But still a_2 not decisive: $a_2 \not\rightsquigarrow_i p$. $\triangle$

A path can be shortened to its first $k \in \llbracket 0, \#\alpha \rrbracket$ elements, which is denoted by $\alpha_{\llbracket 1, k \rrbracket}$, and “shortening” a path beyond its length yields the same path again, thus given $k \in \mathbb{Z}$, $\alpha_{\llbracket 1, k \rrbracket} = \alpha_{\llbracket 1, \min\{\#\alpha, k\} \rrbracket}$. Define a path as being decisive for p , $\alpha \hookrightarrow_i p$, iff $\alpha \rightsquigarrow_i p \wedge \forall a \in \mathcal{A} : (\alpha, a) \rightsquigarrow_i p$, or equivalently, iff $\forall \alpha' \mid \#\alpha \leq \#\alpha' \leq \#\alpha + 1 \wedge \alpha'_{\llbracket 1, \#\alpha \rrbracket} = \alpha : \alpha' \rightsquigarrow_i p$. Thus, $\alpha \not\hookrightarrow_i p \Leftrightarrow \exists a \in \mathcal{A} \mid (\alpha, a) \not\rightsquigarrow_i p$ (or...?).

Axiom 5 (Self-sufficiency of positive arguments). $\forall a_1, a_2 \in A_{\gamma,i}, a \in \mathcal{A}, \alpha \in \mathcal{A}^{<\mathbb{N}}$:

$$\left\{ \begin{array}{l} (\alpha, a_1) \not\hookrightarrow_i p_\gamma \\ (\alpha, a_2) \not\hookrightarrow_i p_\gamma \end{array} \right\} \Rightarrow (\alpha, a_1, a_2) \not\hookrightarrow_i p_\gamma.$$

Theorem 55 (Validity). *If an argumentative theory $(p_\gamma, a_\gamma, d_\gamma)$ is truthful and satisfy axioms 2 and 4 and $\mathcal{A}$ is finite, then the theory is correct.*

Proof. Consider any $i \in I$. We have to show that $p_\gamma \in \Phi_i$. Assume depth 3. Assume that a_γ and a_2 are not decisive. Let's show that a_3 is decisive. Assume that $(a_4, a_\gamma, a_2, a) \not\rightsquigarrow_i p_\gamma$. By TF, $(a_4, a_\gamma, a_2, a, a_3) \hookrightarrow_i p_\gamma$. By Reinst, $(a_4, a_\gamma, a_2, a_3) \hookrightarrow_i p_\gamma$. By Self-s, $(a_4, a_\gamma, a_3) \hookrightarrow_i p_\gamma$. By Self-s, $(a_4, a_3) \hookrightarrow_i p_\gamma$.

Now consider $(a_5, a_2) \rightsquigarrow_i p_\gamma$. (Or any such path not ending with a_3 .) We want $(a_5, a_2, a_3) \hookrightarrow_i p_\gamma$. By Answ, $\exists a \mid (a_5, a_2, a) \not\rightsquigarrow_i p_\gamma$. $\square$

Sketch of proof. Consider any i . We have to show that $p_\gamma \in \Phi_i$.⁴

Problem might be solved by observing that (a_1, a_5, a_2, a_1) convincing means that a_1 is decisive after all; and not convincing means that we have observed not truthfulness.

We want that if the theory is invalid (thus some i is not convinced), then either axiom fails or the theory is experimentally falsifiable. In the following example, the theory seems invalid but the axiom holds and the theory can't seem to be defeated experimentally.

Imagine i thinks as follows. a_1 counter a_5 ; a_2 counters a_4 ; but a_4 and a_5 together make a_1 and a_2 invalid, so that (a_4, a_5) counter both a_1 and a_2 (and also a_1 and a_2 together). Also assume that i has short memory so that a_1, a_4, a_5 does not convince (a_4 and a_5 win) while a_4, a_5, a_1 do (a_1 wins as a_4 is too far away). BUT a_5, a_4, a_1 should then not convince.

The solution is: after a_1, a_5, a_2 , a_1 appears decisive, thus also a_1, a_5, a_2, a_1, a_4 ok. Then some subset a of $a_1 \cup a_5 \cup a_2$ should satisfy a, a_4 OK and replace a_1 . And a is decisive.

⁴Assume $(a_4, a_\gamma, a) \not\rightsquigarrow_i p_\gamma$ (thus, a attacks a_γ). By tf, $(a_4, a_\gamma, a, a_2, \dots) \rightsquigarrow_i p_\gamma$. We get $(a_3, a_\gamma, a, a_2) \rightsquigarrow_i p_\gamma$ by answ. By reinst: (a_2) YES.

Assume that a_γ is not decisive (otherwise, done). Thus, for some $\alpha : (\alpha, a_\gamma, a) \not\rightsquigarrow_i p_\gamma$ or (α, a, a_γ) . Consider that $\alpha : (\alpha, a_\gamma, a) \not\rightsquigarrow_i p_\gamma$. Consider α' and wonder whether $(\alpha', a_\gamma, a) \rightsquigarrow_i p_\gamma$. If so, By Answ, this means that $\forall \alpha, (\alpha, a_\gamma, a) \not\rightsquigarrow_i p_\gamma$. By truthfulness, $a_2 = d_\gamma(a_1, (\alpha, a_\gamma, a)) \neq \emptyset \wedge (\alpha, a_\gamma, a, a_2) \rightsquigarrow_i p_\gamma$. Assume now that $(\alpha, a_\gamma, a, a_2, \dots) \rightsquigarrow_i p_\gamma$. Then by Reinstatement, $(\alpha, a_2) \rightsquigarrow_i p_\gamma$. $\square$

This definition of truthfulness is adopted because of our goal of distinguishing empirical claims. Truthfulness, so defined, can be falsified empirically: if it does not hold, it is possible to prove this empirically by exhibiting a counter-example in the form of a tuple $i, k, (a_j)_{j \in \llbracket 1, k \rrbracket}, \alpha$ such that ... (to be completed).

Note that a static theory (p, a) can be considered as an argumentative theory $(p, a, \emptyset)$, and (p, a) is truthful in the sense of section D.2 iff $(p, a, \emptyset)$ is truthful in the sense of definition 6.

Associated with an argumentative theory also comes the following axiom, that mandates a more general version of the following requirement: if some argument a_2 defends some argument a_1 , and if a_2 leads i to opt for p_γ whenever it immediately follows a_1 , then omitting a_1 would still have lead i to opt for p_γ .

Axiom 6 (Omitting defended arguments). *If the theory is truthful, then $\forall i \in I, k \in \mathbb{N}^*, l \in \llbracket 2, k \rrbracket, (a_j)_{j \in \llbracket 1, k \rrbracket} \in \mathcal{A}^k, \alpha, \beta \in \mathcal{A}^{<\mathbb{N}} \mid [a_k \in \alpha_{end} \wedge \forall j \in \llbracket 2, k \rrbracket : a_j \in d_{\gamma, i}(a_{j-1})] : (\beta, a_{l-1}, a_l, \alpha) \rightsquigarrow_i p_\gamma \Rightarrow (\beta, a_l, \alpha) \rightsquigarrow_i p_\gamma$.*

This is called an axiom because it cannot be falsified using observables. It has to be accepted on the basis of the rest of the observations and on the contents of the defending arguments. As example 54 suggests, theories can be built so as to make this axiom plausible by embedding within defending arguments the content of the defended argument.

An argumentative theory is said to be correct iff $\forall i \in I : p_\gamma \in \Phi_i$. As previously, it is said to be valid iff its truthfulness leads to its correctness, except that the above axiom needs now be included as a price for the added flexibility of the theory.

Theorem 56 (Validity). *If an argumentative theory is truthful, then axiom 6 implies that the theory is correct.*

Sketch of proof. Assume, to start with, that the theory is of depth 2, more precisely, a_γ is not decisive because $\alpha \not\rightsquigarrow_i p_\gamma, a_\gamma \in \alpha_{end}$ and $d_{\gamma, i}(a_\gamma) = a_2$ with no further hist against it.

Consider any $i \in I$. Assume a_γ is not decisive. Consider any $\alpha \mid a_2 \in \alpha_{end}$. We want to show that $\alpha \rightsquigarrow_i p_\gamma$.

IF α contains a_γ before a_2 , done by TF. Otherwise, define $hist'$ which has a_1 just before a_2 . Apply TF on $hist'$ then apply axiom.

Consider now a theory of depth 3. Consider any $i \in I$. Assume a_γ not decisive, a_2 not decisive. Consider any $\alpha \mid a_3 \in \alpha_{end}$. We want to show that $\alpha \rightsquigarrow_i p_\gamma$.

If it includes a_1 then a_2 , ok thanks to truthfulness. Otherwise, build $hist3$ that has a_1 then a_2 just before a_3 . That's ok thanks to TF. By axiom, $hist2$ with only a_2 just before a_3 is thus ok. By axiom, $hist$ is thus ok. $\square$

D. Theories of deliberated choice

This section assumes that an exposure protocol $(I, \mathcal{A}, \rightsquigarrow)$ is defined, and defines theories of deliberated choice associated with that exposure protocol. We will start with very simple theories (section D.2) in order to build an intuition about the conditions required on theories of deliberated choice, then justify the need for more complex theories (section D.3) and finally define more general theories of deliberated choice (section C).

D.1. Future

Remark 33. We could weaken indicativeness by allowing for disjuncts: observing either that $\alpha \rightsquigarrow_i \varphi$ or that $\alpha \rightsquigarrow_i \psi$ permits to deduce something about P_i . (Or more complex formulas under the form of a set of possible functions in P^H ?) (Actually I am not sure that this really weakens the condition.) $\triangle$

D.2. Theories using a static decisive argument

Example 1 suggests a simple form for theories of deliberated choice: suffice to determine an argument $a \in \mathcal{A}$ and a preference $p \in P$ such that the theory claims that $\forall i \in I : a \hookrightarrow_i p$ and that $\forall i \in I : p \in P_i$.

Definition 7 (Static theory). A static theory is a pair (p, a) for some $p \in P$ and some $a \in \mathcal{A}$.

A static theory (p, a) is said to be truthful iff $\forall i \in I : a \hookrightarrow_i p$, and to be correct iff $\forall i \in I : p \in P_i$. We can then deduce that such theories are valid, meaning that their truthfulness implies their correctness. Let us make this (very simple) result stand out as a theorem, because a parallel will be usefully made with more complicated theories.

Theorem 57 (Static theories are valid). *Any truthful static theory is correct.*

Proof. This follows immediately from definition 1. $\square$

Importantly, truthfulness can indeed be checked using our observational basis, that is, if $\exists i \in I \mid a \not\hookrightarrow_i p$, then it is possible to exhibit this fact experimentally.

Such theories have the merit of simplicity, but there is a need to design more adaptive theories: in many interesting cases, no single argument will be considered decisive by all individuals in I . We then need theories that argue with individuals by reacting to what they consider good arguments. The following section illustrates this need.

D.3. Some decision problems with no single decisive argument

One ambition of the theories to be proposed here is to permit exhibiting the existence of a deliberated consensus among a large set of persons about issues that involve numerous arguments and counter-arguments.

In many decision problems, individuals will not be attracted by the same set of arguments and counter-arguments, even though they may finally settle to the same preference. Imagine that some individual i_1 considers an argument a_1 for φ' relevant and some individual i_2 considers an argument a_2 for φ' relevant, but i_1 does not consider a_2 relevant and i_2 does not consider a_1 relevant (for example, because they already have some counter-arguments in mind to those arguments). Imagine further that there is some counter-argument a'_1 that can be given to i_1 to rebut a_1 , and a different counter-argument a'_2 that can be given to i_2 to rebut a_2 , so that they both have φ in their deliberated judgment when considering all these arguments. In such situation, static theories as defined in section D.2 are ill suited: these theories need to present the same argument to all individuals and would thus need to define a bigger argument a containing both a'_1 and a'_2 . Such an argument could be too long to be convincing to i_1 or i_2 , as it needlessly spends efforts countering an argument that the individual considers irrelevant to start with. This will become an important problem if there are not only two but many a priori possible arguments a_k for φ' , a small subset of which is considered relevant by any given individual, each with their own counter-argument a'_k for φ . This section is devoted to describing two examples to illustrate that point. Bear in mind that these examples may look somehow naïve on the aspect of the hypothesis they make about what convinces individuals. Real-world complex decision situations will require complex theories that do not fit the purpose of simple illustrative examples. Their goal is to argue that non-static theories are required, but not to fully describe such theories.

One of the settings discussed by Thaler and Sunstein [2009] permits such an illustration. Let I be a set of workers interested in establishing a savings plan. The authors claim that workers tend to not save enough because of psychological biases. It is therefore tempting to conjecture that the deliberated choice of those workers is to save more, in the sense defined here, than what their natural choice (the one revealed in absence of arguments) leads to. A difficulty related to this decision problem is that the complete set of all arguments that will be considered relevant by at least one individual for such a complex issue may well be very large. It may also be that, for each individual who considers relevant some arguments A for saving not much (i.e., spending more in the short term), it is possible to give counter-arguments A' that will convince the individual to opt for saving more. But no single argument could be decisive for each $i \in I$: it could be that only counter-arguments built specifically according to what a given individual considers relevant could convince her, given the complexity of the issue. This illustrates that the deliberated choice of each individuals in I may be to save more even though no static theory (definition 7) may achieve showing it, as it could be that no such theory is truthful.

The following example illustrates this same point in a constrained setting which permits a complete definition of $\mathcal{A}$ and $\rightsquigarrow$.

Example 54 (A decision problem about two lotteries). In a famous article about preference reversals, Lichtenstein and Slovic [2006] discuss an example involving two lotteries. The lottery L_P lets the subject win four dollars with 99% chance and lose ten cents with 1% chance; in summary, $L_P = (.99, +\$4; .01, -\$0.1)$. The lottery L_D is, using the same

notations: $(.33, +\$16; .67, -\$2)$. The letter P stands for probability: one salient aspect of L_P is the very high chance of winning. The letter D stands for dollar: one salient aspect of L_D is the high gain it gives (compared to P) in dollar amount if the subject is lucky.

This example is interesting because when asked to bid for lotteries, the theory discussed by the authors predicts, and their experiments confirm, that subjects tend to bid more for L_D than for L_P , because the bidding question tend to make individuals focus on dollar amounts; whereas when asked to choose one lottery, the lotteries being presented in pairs, subjects tend to choose L_P among the pair $\{L_P, L_D\}$, giving greater attention to probabilities.

Define $P = \{L_P, L_D, 0\}$.

Define a_e as a text arguing that the choice between L_P and L_D should be determined by their expected revenue, namely, \$3.959 for L_P and \$4 for L_D .⁵ Define a_u as a text arguing that the utility of money should matter, not the revenue, and arguing that a loss hurts more than a gain.⁶ Define a counter-argument a_l to a_u , indicating that for small monetary amounts, it is perfectly reasonable to consider utility as linear, especially given that in this experiment, multiple lotteries will be played.⁷ Define a so as to include the contents of a_e , a_u and a_l .⁸ Define $\mathcal{A} = \{a_e, a_u, a_l, a\}$.

Define I as a set of individuals facing a choice between L_P and L_D in a laboratory experiment. Define $\rightsquigarrow_i$ as exposing i to α and observing the resulting choice by asking i to indicate which lottery she prefers playing, or declare being indifferent.

Assume that some individuals $i \in I$ do not see the point of argument a_u and some do but are convinced by the counter-argument a_l . Accordingly, define $\rightsquigarrow_e$, representing the first kind of individuals, as follows. $\forall \alpha \in \mathcal{A}^{<\mathbb{N}} : \alpha \rightsquigarrow_e L_D \Leftrightarrow [a_e \in R(\alpha)]$, and $\alpha \rightsquigarrow_e L_P$ otherwise. Note that in particular, if α contains a but not a_e , $\alpha \rightsquigarrow_e L_P$, as we assume such individuals are confused by the longer argument a referring to abstract notions and therefore opt for L_P , falling back to the common psychological bias. The second kind of individuals is represented by $\rightsquigarrow_u$ defined as $\forall \alpha \in \mathcal{A}^{<\mathbb{N}} : \alpha \rightsquigarrow_u L_D \Leftrightarrow [a_u \notin R(\alpha) \vee a \in R(\alpha) \vee \{a_u, a_l\} \subseteq R(\alpha)]$, and $\alpha \rightsquigarrow_u L_P$ otherwise. We assume here that such individuals think about a_u as a correct argument in favor of L_P until shown

⁵Here is a possible text for a_e . “Imagine you play L_P 100 times. On average, you would win 99 times and lose once, which means gaining $99 \times \$4 = \396 and losing $1 \times \$0.10 = \0.10 , hence a total net gain of \$395.90. Thus, your average net gain for the L_P bet would be \$3.9590. [Similar text for L_D .] Therefore, we suggest you choose L_D for a higher expected revenue.”

⁶Here is a possible text for a_u . “ L_D is better than L_P in terms of expected gain, but it is in general absurd to reason in terms of expected gain. For example, consider a choice between winning 10^6 dollars with probability 0.8 and winning 2×10^6 dollars with probability 0.4. These two lotteries have the same expected gain, but the first one is clearly preferable. Therefore, we recommend to rather pick L_P as it presents a much better chance of earning money”.

⁷Here is a possible text for a_l . “The lottery L_D has a higher expected revenue, and thus, a higher utility than L_P , because for small monetary amounts, it is perfectly reasonable to consider utility as linear, especially given that in this experiment, multiple lotteries will be played.”

⁸Here is a possible text for a . [Text of a_e] “Note that one could say that the utility of money should matter, but for small monetary amounts, it is perfectly reasonable to consider utility as linear, especially given that in this experiment, multiple lotteries will be played.”

the rebuttal a_l (or a , which contains a similar rebuttal). Assuming finally $\rightsquigarrow$ as equal to $\rightsquigarrow_e$ for individuals of the first kind and $\rightsquigarrow_u$ otherwise concludes the definition of the exposure protocol $(I, \mathcal{A}, \rightsquigarrow)$.

It follows from this definition that $\forall i \in I : L_D \in P_i$, and that no static theory of that exposure protocol is truthful: only a_e is a decisive argument for L_D for the first kind of individuals whereas for the second kind, only a is. $\triangle$

This example illustrates the need for theories that are able to react to arguments depending on what a given individual considers relevant. This is the object of the next section.

D.4. Form of theories (draft)

Given an exposure protocol $(I, \mathcal{A}, \rightsquigarrow)$, we want to phrase some claims under the form of a logical sentence C such that: if C holds, then it implies something about the dj of some i ; if C fails, it is possible to falsify it.

We can observe set O that forms a function $f(i) = \{\alpha \rightsquigarrow_i \varphi, \alpha' \rightsquigarrow_i \varphi', \dots\}$, with $\alpha \subsetneq \alpha' \subsetneq \dots$ and with finite O . Our second condition is that $\neg C \Rightarrow \exists \theta \mid \theta \Rightarrow \neg C$. For example, if C is $\alpha \rightsquigarrow_i \varphi \vee \beta \rightsquigarrow[j] \varphi$, O is $\alpha \rightsquigarrow_i \varphi' \wedge \beta \rightsquigarrow \varphi'$, then O implies $\neg C$. [Claude: $\rightsquigarrow$ takes no optional argument (unlike $\rightsquigarrow_i$), so `\allleadsto[j]` renders as $\rightsquigarrow$ followed by the literal text "[j]", and the later bare `\allleadsto` similarly drops the implicit individual. Given the surrounding pattern ($\alpha \rightsquigarrow_i \varphi$, infix, individual-indexed), should both be $\rightsquigarrow_j / \rightsquigarrow_i$ instead?] As another example, C being $\exists \alpha \mid \alpha \rightsquigarrow_i \varphi$ is unfalsifiable, as its negation is $\forall \alpha : \alpha \not\rightsquigarrow_i \varphi$, which requires infinite O .

Consider $\rightsquigarrow$ such that $\exists a, \varphi \mid \forall i : a \rightsquigarrow_i \varphi$. Then some suitable C exists. It is $\forall i : a \hookrightarrow_i \varphi$. If false, it is falsifiable. Even though it is not provable by any O .

Now consider $\rightsquigarrow$ such that $\exists J, K \subsetneq I \mid a \hookrightarrow_j \varphi, a' \hookrightarrow_k \varphi$. Then no suitable C exists?

If C has the form, whenever a fails, a' is then decisive, it is falsifiable, but does not lead to a decisive argument. Want to show: $j \hookrightarrow_i \varphi$. Need to show: ar or ar' is decisive. Assume we know who is a j . Then we can! Claim: $a \rightsquigarrow_i \varphi$. Now assume that our claim has to be anonymous. Also, assume the formulas involve one i at a time (not needed, probably, but not much restrictive so okay for now). So it is of the form $(i, \alpha, \varphi) \vee (i, \alpha, \varphi') \wedge (i, \alpha', \varphi) \dots$ implies something decisive. So it must be of the form: all α ending with a do $\rightsquigarrow_i \varphi$. Or must imply this. Let's make it very strong, as much as possible, thus equal to $\rightsquigarrow_i \cap \rightsquigarrow_j$. That's not strong enough!

Escaping? Assume that we know that 50% are J and 50% are K.

E. Technical notes about falsification

Definition 8. A claim is *unfalsifiable* iff $C = P^{I \times \mathcal{A}^{<\mathbb{N}}} \vee \exists \rightsquigarrow \notin C \mid \theta \cap \mathcal{P}(\rightsquigarrow) \subseteq C^\subseteq$.

A claim C is *falsifiable* iff $\forall \rightsquigarrow \notin C : \exists \theta \subseteq \rightsquigarrow \mid \theta \notin C^\subseteq$, equivalently iff $\exists \rightsquigarrow \notin C \mid \forall \theta \in \Theta, \theta \subseteq \rightsquigarrow : \theta \in C^\subseteq$. (To be corrected for the trivial claims.)

E.1. Comparison with rejection

The following theorem (whose condition is that C may fail to hold and still be not rejectable) is a consequence of the contrapositive of theorem 11.

Corollary 15. *A claim C that satisfies $\exists \rightsquigarrow \notin C$ such that C is not rejectable given $\rightsquigarrow$ is unfalsifiable and not trivial.*

The following theorem (an immediate consequence of definition 8) is the converse of corollary 15. It shows that falsifiability or triviality is exactly the property that we need to fill the gap.

Theorem 58. *For any unfalsifiable and not trivial claim C : $\exists \rightsquigarrow \notin C$ such that C is not rejectable given $\rightsquigarrow$.*

Remark 34. It is interesting to note that strong unfalsifiability entails necessary non rejection whereas strict unfalsifiability entails possible non rejection. Perhaps we should exploit this by showing that a given relation not in C entails non rejection using that relation, or something like that? $\triangle$

E.2. Kinds

Definition 9. A claim is *strongly unfalsifiable* iff $\Theta \subseteq C^\subseteq$.

Examples 7 and 9 illustrate, as well as example 10 when I is infinite.

Definition 10. A claim is *strictly unfalsifiable* iff it is unfalsifiable and not strongly unfalsifiable.

Remark 35. Making a strongly unfalsifiable claim more precise ($C = (C_1 \cap C_2) \subseteq C_1$ in theorem 78) can make it strictly unfalsifiable. Also, a strictly unfalsifiable claim can be made falsifiable by picking a single relation from the claim.

But, it does not hold that more precise claims are always more falsifiable: making a falsifiable claim more precise ($C = (C_1 \cap C_2) \subseteq C_2$ in example 33) can make it strictly unfalsifiable. (In fact, example 33 shows that the relation is not one-way: going from a strongly unfalsifiable (C_1), or from a falsifiable (C_2), claim, to a more precise one ($C = C_1 \cap C_2$) yields a strictly unfalsifiable claim.) $\triangle$

E.3. Coverability

Definition 11. A claim C is *coverable* iff $\forall \theta \in \Theta \cap C^\subseteq : \exists \theta' \in \Theta \cap C^\subseteq \mid \theta \subseteq \theta' \wedge \{\rightsquigarrow \in PI^{\times \mathcal{A}^{<\mathbb{N}}} \mid \theta' \in \mathcal{P}(\rightsquigarrow)\} \subseteq C$.

Theorem 59. *For any coverable claim C and strongly unfalsifiable claim C' : $\Theta \cap C^\subseteq \subseteq (C' \cap C)^\subseteq$.*

Proof. Consider any $\theta \in \Theta \cap C^\subseteq$. Let us show that $\theta \in \bigcup_{\rightsquigarrow \in (C' \cap C)} \mathcal{P}(\rightsquigarrow)$.

Consider any $\theta' \in \Theta \cap C^\subseteq \mid \theta \subseteq \theta' \wedge \{\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}} \mid \theta' \in \mathcal{P}(\rightsquigarrow)\} \subseteq C$. As $\Theta \subseteq C'^\subseteq$, we have $\theta' \in \bigcup_{\rightsquigarrow \in C'} \mathcal{P}(\rightsquigarrow)$. Pick any such $\rightsquigarrow$ (thus with $\theta' \in \mathcal{P}(\rightsquigarrow)$ and $\rightsquigarrow \in C'$). Thus, $\theta \in \mathcal{P}(\rightsquigarrow)$. Also, as $\{\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}} \mid \theta' \in \mathcal{P}(\rightsquigarrow)\} \subseteq C$, we have $\rightsquigarrow \in C$. $\square$

A claim C is not coverable iff $\exists \theta \in \Theta \cap C^\subseteq \mid \forall \theta' \in \Theta \cap C^\subseteq, \theta \subseteq \theta' : \exists \rightsquigarrow \notin C \mid \theta' \in \mathcal{P}(\rightsquigarrow)$.

A claim C is not coverable because of the empty set iff $\forall \theta \in \Theta \cap C^\subseteq : \exists \rightsquigarrow \notin C \mid \theta \in \mathcal{P}(\rightsquigarrow)$.

Remark 36. We do not have that when $C_1 \cap C_2 \neq \emptyset$, then $\theta \in C_1^\subseteq \wedge \theta \in C_2^\subseteq \Rightarrow \theta \in (C_1 \cap C_2)^\subseteq$. For a counter-example, consider $\alpha, \alpha_1, \alpha_2 \in \mathcal{A}^{<\mathbb{N}}$, $i, i_1, i_2 \in I$ and $p \neq p', p_1, p_2 \in P$ and $C_1 = \{\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}} \mid (\alpha_1 \rightsquigarrow_{i_1} p_1 \Rightarrow \alpha \rightsquigarrow_i p) \wedge (\alpha_2 \rightsquigarrow_{i_2} p_2 \Rightarrow \alpha \rightsquigarrow_i p')\}$, $C_2 = \{\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}} \mid (\alpha \rightsquigarrow_i p \Rightarrow \alpha_2 \rightsquigarrow_{i_2} p_2)\}$ and $\theta = \{(i_1, \alpha_1, p_1)\}$. $\triangle$

Remark 37. The above remark extends to the case where C_1 is strongly unfalsifiable. For a counter-example, consider $C_1 = \{\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}} \mid (a) \rightsquigarrow_i \varphi \vee (a' \rightsquigarrow_i \varphi)\}$ and C_2 containing a condition requiring $(a) \not\rightsquigarrow_i \varphi \wedge (a' \not\rightsquigarrow_i \varphi)$. Add implications if desired to make C_1 and C_2 partially compatible, for example, $C_1 = \{\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}} \mid (\alpha \rightsquigarrow_j p) \Rightarrow [(a) \rightsquigarrow_i \varphi \vee (a' \rightsquigarrow_i \varphi)]\}$, so that C_1 and C_2 require their condition only when some trigger holds. Note that the C_2 condition may be arbitrarily complex so I see no hope of characterizing these situations. $\triangle$

Remark 38. Apparently (from example 33) we can have $C_1 \cap C_2 \neq \emptyset$ and $\overline{C_1} \cap C_2 \neq \emptyset$ but still $C_1^\subseteq \cap C_2^\subseteq \subseteq (C_1 \cap C_2)^\subseteq$? Though the above remark suggests that non inclusion will yield some similar incompatibility. $\triangle$

E.4. Closure under undistinguishability

The following procedure permits to determine whether a claim is falsifiable. Given a set of observations $S \subseteq \Theta$, define the relations that are observation-compatible with S as $E(S) = \{\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}} \mid \forall \theta \in \Theta : \rightsquigarrow \cap \theta \in S\}$. Given a claim C , define the set of its observation-compatible relations as $E(C) = E(C^\subseteq \cap \Theta)$ (reusing the letter E by abuse of notation). Note that this set extends C , in the sense that $C \subseteq E(C)$. The claim C is falsifiable iff it is closed under extension to observation-compatible relations, that is, iff $E(C) \subseteq C$.

Theorem 60. *A claim C is falsifiable or trivial iff $E(C) \subseteq C$, or equivalently, iff $\forall \rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}} :$*

$$\{\rightsquigarrow|_O \mid O \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I)\} \subseteq C^\subseteq \Rightarrow \rightsquigarrow \in C.$$

Proof. Note that $E(C) = E(C^\subseteq \cap \Theta) = \{\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}} \mid \forall O \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I) : \rightsquigarrow|_O \in C^\subseteq\}$.

The condition $E(C) \subseteq C$ is equivalent to $\forall \rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}} : (\rightsquigarrow \in E(C) \Rightarrow \rightsquigarrow \in C)$. Note that $\rightsquigarrow \in E(C) \Leftrightarrow \{\rightsquigarrow|_O \mid O \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I)\} \subseteq C^\subseteq$.

By definition, C is not falsifiable iff $\exists \rightsquigarrow \notin C \mid \Theta \cap \mathcal{P}(\rightsquigarrow) \subseteq C^\subseteq$; equivalently, iff $\exists \rightsquigarrow \notin C \mid \{\theta \subseteq \rightsquigarrow\} \subseteq C^\subseteq$. And, $\forall \rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}} : \{\rightsquigarrow|_O \mid O \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I)\} = \{\theta \subseteq \rightsquigarrow\}$. $\square$

Let the *simple a priori observations* be defined as $\cup_{J \subseteq \mathcal{F}I} (\mathcal{A}^{<\mathbb{N}})^J$, equivalently, as the right-unique finite relations over $I \times \mathcal{A}^{<\mathbb{N}}$. Given a simple a priori observation $O : J \mapsto \mathcal{A}^{<\mathbb{N}}$, we can define the a priori observation $O_O = \{(i, \alpha) \in J \times \mathcal{A}^{<\mathbb{N}} \mid \forall i \in J : \alpha \subseteq O(i)\}$ as the completion of O to all sub-paths in O .

Let $S_\theta = \{\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}} \mid \theta \subseteq \rightsquigarrow\}$ denote the possibilities for $\rightsquigarrow$ compatible with the observations. Given O and $\rightsquigarrow$, it follows that $S_{\rightsquigarrow \upharpoonright O_O}$ denotes the “leads to” relations that complete O_O using $\rightsquigarrow$. Define that a relation $\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}}$ is *obs-compatible* with a set S iff $\forall \theta \in \Theta : (\rightsquigarrow \cap \theta) \in S$.

Remark 39. Define the closure under combinations of C as the relations that can be formed by picking parts of relations in C , formally, $(\cup_{S \in C \subseteq \Theta} S) \cap P^{I \times \mathcal{A}^{<\mathbb{N}}}$. We have $E(C) \subseteq (\cup_{S \in C \subseteq \Theta} S) \cap P^{I \times \mathcal{A}^{<\mathbb{N}}}$.

Being closed under combinations is sufficient to be falsifiable, but not necessary, as seen with example 38 (in other words, for some claims, $(\cup_{S \in C \subseteq \Theta} S) \cap P^{I \times \mathcal{A}^{<\mathbb{N}}} \subseteq E(C)$ does not hold). $\triangle$

E.5. Another closure

Given a set of attitudes $S \subseteq P^{\mathcal{A}^{<\mathbb{N}}}$, the claim associated to it designates the claim $S^I = \{\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}} \mid \forall i \in I : \rightsquigarrow_i \in S\}$. Say that a claim C is a claim about types iff $\exists S \subseteq P^{\mathcal{A}^{<\mathbb{N}}} \mid C = S^I$. Informally, such a claim claims that the set of attitudes in I is contained in S (though might be a strict subset of S), or equivalently, that no attitude out of S exists among I .

The following definition assumes a set $B \subseteq P^{\mathcal{A}^{<\mathbb{N}}}$ of behaviors. We take $B = P^{\mathcal{A}^{<\mathbb{N}}}$ if nothing is specified about B .

Definition 12. Say that an attitude $g \in B$ is observably included in a set of attitudes $S \subseteq B$ iff $\forall H \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})) : \exists f \in S \mid g_{\upharpoonright H} = f_{\upharpoonright H}$.

Define the extension of a set of attitudes $S \subseteq B$ to its observably included attitudes as the set of attitudes $R(S) = \{g \in B \mid \forall H \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})) : \exists f \in S \mid g_{\upharpoonright H} = f_{\upharpoonright H}\}$. Note that $S \subseteq R(S)$. Also note that $S_1 \subseteq S_2 \Rightarrow R(S_1) \subseteq R(S_2)$.

Theorem 61. $\forall S \subseteq B : R(R(S)) \subseteq R(S)$.

Proof. Consider any $g \in R(R(S))$. Consider any $H \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}}))$. There exists $f \in R(S)$ such that $g_{\upharpoonright H} = f_{\upharpoonright H}$. There exists $f' \in S$ such that $f_{\upharpoonright H} = f'_{\upharpoonright H}$. Thus, $g_{\upharpoonright H} = f'_{\upharpoonright H}$. Thus, $g \in R(S)$. $\square$

Define the closure under observable inclusion of a claim about types S as $R^\infty(S)$, equivalently, as the smallest set S' such that $S \subseteq S' = R(S')$, equivalently, as indicated by theorem 61, as $R(S)$.

Let $\llbracket \alpha \rrbracket = (\alpha_i)_{i \in \llbracket 0, \# \alpha \rrbracket} \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}}))$ denote the increasing sequence of paths from $\emptyset$ to α .

Theorem 62. The S -labeling function associated to an attitude $f \in P^{\mathcal{A}^{<\mathbb{N}}}$ is the function $l_f^S : \mathcal{A}^{<\mathbb{N}} \rightarrow \mathcal{P}(P^{\mathcal{A}^{<\mathbb{N}}})$ defined as $l_f^S(\alpha) = \{g \in S \mid g(\alpha) = f(\alpha)\}$.

We have: $R(S) = \{f \mid \forall \alpha \in \mathcal{A}^{<\mathbb{N}} : \cap_{\alpha' \subseteq \alpha} l_f^S(\alpha') \neq \emptyset\}$.

Proof. Consider f with $\forall \alpha \in \mathcal{A}^{<\mathbb{N}} : \cap_{\alpha' \subseteq \alpha} l_f^S(\alpha') \neq \emptyset$. Consider any $H \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}}))$. We know that $\cap_{\alpha' \subseteq \alpha} l_f^S(\alpha') \neq \emptyset$, thus in particular, defining $\alpha = \max H$ and observing that $H \subseteq \{\alpha' \mid \alpha' \subseteq \alpha\}$, we have that $\cap_{\alpha' \in H} l_f^S(\alpha') \neq \emptyset$. Pick any $g \in \cap_{\alpha' \in H} l_f^S(\alpha')$. Thus, $g \in S$. Also, $\forall \alpha \in H : g \in l_f^S(\alpha)$, thus, $\forall \alpha \in H : g(\alpha) = f(\alpha)$.

First observe that given any $S \subseteq P^{\mathcal{A}^{<\mathbb{N}}}$, $f \in P^{\mathcal{A}^{<\mathbb{N}}}$ and $\alpha \in \mathcal{A}^{<\mathbb{N}}$, as $\alpha' \subseteq \alpha$ iff $\alpha' \in \llbracket \alpha \rrbracket$, we have that $\cap_{\alpha' \subseteq \alpha} l_f^S(\alpha') = \{g \in S \mid \forall \alpha' \in \llbracket \alpha \rrbracket : g(\alpha') = f(\alpha')\}$.

It follows that if $\forall \alpha \in \mathcal{A}^{<\mathbb{N}} : \cap_{\alpha' \subseteq \alpha} l_f^S(\alpha') \neq \emptyset$, then $\forall H \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})) : \emptyset \neq \{g \in S \mid \forall \alpha' \in \llbracket \max H \rrbracket : g(\alpha') = f(\alpha')\} \subseteq \{g \in S \mid \forall \alpha' \in H : g(\alpha') = f(\alpha')\}$.

Conversely, if $\forall H \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})) : \{g \in S \mid \forall \alpha' \in H : g(\alpha') = f(\alpha')\} \neq \emptyset$, then $\forall \alpha \in \mathcal{A}^{<\mathbb{N}}$, defining $H = \llbracket \alpha \rrbracket$, we have that $\cap_{\alpha' \subseteq \alpha} l_f^S(\alpha') \neq \emptyset$. $\square$

Lemma 7. *A set of attitudes $S \subseteq P^{\mathcal{A}^{<\mathbb{N}}}$ satisfies $\forall g \in P^{\mathcal{A}^{<\mathbb{N}}} \setminus S : \exists H \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})) \mid \forall f \in S : g_{\upharpoonright H} \neq f_{\upharpoonright H}$, iff its associated claim is falsifiable or trivial.*

Proof. Pick any $\rightsquigarrow \notin C$. By definition of C , $\exists i \in I \mid \rightsquigarrow_i \notin S$. As C holds, $\rightsquigarrow_i \notin \{\rightsquigarrow_j \mid j \in I\}$ (lemma 8). Pick any H such that $\forall \rightsquigarrow_j \in S : \exists \alpha \in H \mid \rightsquigarrow_i(\alpha) \neq \rightsquigarrow_j(\alpha)$. Define $\theta = \{(i, \alpha, \rightsquigarrow_i(\alpha)) \mid \alpha \in H\} \in \Theta \cap \mathcal{P}(\rightsquigarrow)$. Then, $\theta \notin C^\subseteq$, because $\forall \rightsquigarrow_j \in S : \theta \not\subseteq \rightsquigarrow_j$.

Conversely, assume that for some $\rightsquigarrow_i \in P^{\mathcal{A}^{<\mathbb{N}}} \setminus S$, no $H \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}}))$ satisfies the indicated condition. Define $\rightsquigarrow$ as including $\rightsquigarrow_i$ and for all other individuals, relations from S . Then, $\rightsquigarrow \notin C$, but any observation will be in C , thus the claim is not falsifiable and not trivial. $\square$

Lemma 8. *Consider a set of attitudes $S \subseteq P^{\mathcal{A}^{<\mathbb{N}}}$. Its associated claim holds iff $\{\rightsquigarrow_j \mid j \in I\} \subseteq S$.*

Lemma 9. *Given any set of attitudes $S \subseteq P^{\mathcal{A}^{<\mathbb{N}}}$, its associated claim is anonymous.*

The following result follows from lemmas 7 to 9.

Theorem 63. *Assume that $\forall \rightsquigarrow_i \in P^{\mathcal{A}^{<\mathbb{N}}} \setminus \{\rightsquigarrow_k \mid k \in I\} : \exists H \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})) \mid \forall j \in I : \exists \alpha \in H \mid \rightsquigarrow_i(\alpha) \neq \rightsquigarrow_j(\alpha)$. Then, the claim C associated to the set of attitudes $S = \{\rightsquigarrow_j \mid j \in I\}$ holds, is anonymous, and is falsifiable or trivial.*

Proof. For another proof, note that $C^\subseteq \cap \Theta = \{\theta \in \Theta \mid \forall i \in I : \exists j \in I \mid \theta \cap (\{i\} \times \mathcal{A}^{<\mathbb{N}} \times P) \subseteq \rightsquigarrow_j\}$. Pick any $\rightsquigarrow \in E(C^\subseteq \cap \Theta)$ and let us show that $\rightsquigarrow \in C$. Note that $\rightsquigarrow \notin C \Leftrightarrow \exists i \in I \mid \rightsquigarrow_i \in P^{\mathcal{A}^{<\mathbb{N}}} \setminus \{\rightsquigarrow_k \mid k \in I\}$, in which case the consequent in the theorem holds. We need to show that $\nexists i \in I, H \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})) \mid \forall j \in I : \exists \alpha \in H \mid \rightsquigarrow_i(\alpha) \neq \rightsquigarrow_j(\alpha)$. In other words, given any $i \in I$, we need to show that $\forall H \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})) : \exists j \in I \mid \forall \alpha \in H : \rightsquigarrow_i(\alpha) = \rightsquigarrow_j(\alpha)$. $\square$

The following result follows from lemma 7.

Theorem 64. *If a set of attitudes $S \subseteq P^{\mathcal{A}^{<\mathbb{N}}}$ is not closed under observable inclusion, then it is not falsifiable and not trivial.*

Theorem 65. *Given any situation, if the claim associated to the set of attitudes $R(\rightsquigarrow(I))$ is trivial, then there exists no claim associated to a set of attitudes that holds and is falsifiable.*

Proof. Consider a claim related to a set of attitudes S that holds and that is falsifiable or trivial. For it to hold, we need $\rightsquigarrow(I) \subseteq S$. From lemma 7, for it to be falsifiable or trivial, we need that $\forall g \in P^{\mathcal{A}^{<\mathbb{N}}} \setminus S : \exists H \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})) \mid \forall f \in S : g|_H \neq f|_H$. In other words, all g must be observably excludable from S . In other words, S must be closed under observable inclusion. Thus, we need $R(S) \subseteq S$. Also, because $S_1 \subseteq S_2 \Rightarrow R(S_1) \subseteq R(S_2)$, we need $R(\rightsquigarrow(I)) \subseteq R(S) \subseteq S$. It follows that S is trivial. $\square$

Theorem 66. *Given any situation, the claim associated to the set of attitudes $R(\rightsquigarrow(I))$ holds, is anonymous, and is falsifiable or trivial.*

Example 55. Define $f^{(p,p')} \in P^{\mathcal{A}^{<\mathbb{N}}}$ as follows: $f^{(p,p')}(\emptyset) = p$ and $\forall \alpha \in \mathcal{A}^{<\mathbb{N}} : \alpha \neq \emptyset \Rightarrow f^{(p,p')}(\alpha) = p'$.

Consider a situation such that $\rightsquigarrow(I) = \{f^{(p,p')} \mid p, p' \in \{\varphi, \varphi'\}\}$, thus, containing only people constant to φ and people constant to φ' , except that they map $\emptyset$ to anything in $\{\varphi, \varphi'\}$. Define $F^p = \{f^{(p,p')} \mid p' \in \{\varphi, \varphi'\}\}$.

Theorem 62 indicates that $R(\rightsquigarrow(I)) = \{f \mid l_f^S(\emptyset) \in \{F^\varphi, F^{\varphi'}\} \wedge \forall a \in \mathcal{A} : [\forall \alpha \mid \alpha_1 = a : l_f^S(\alpha) \in l_f^S(\emptyset)] \wedge [\forall \alpha, \alpha' \mid \alpha_1 = \alpha'_1 = a : l_f^S(\alpha) = l_f^S(\alpha')]\}$.

Thus, $R(\rightsquigarrow(I))$ contains, for each mapping $m : \mathcal{A} \rightarrow \{\varphi, \varphi'\}$, the two functions $f_m^\varphi, f_m^{\varphi'}$ defined as $f_m^\varphi(\emptyset) = \varphi$ and $\forall a \in \mathcal{A} : f_m^\varphi((a, \dots)) = m(a)$, and similarly for $f_m^{\varphi'}$.

The claim associated to $R(\rightsquigarrow(I))$ is not trivial: $\forall f \mid [\exists \alpha, \alpha' \subsetneq \alpha \mid f(\alpha') \neq f(\alpha)] : f \notin R(\rightsquigarrow(I))$. $\triangle$

Example 56. Consider $\mathcal{A} = \{a_1, a_2\}$ and $\rightsquigarrow(I) = \{f \mid f(a_1) \neq \varphi \vee f(a_2) \neq \varphi\}$, thus, $f \notin \rightsquigarrow(I)$ iff $f(a_1) = f(a_2) = \varphi$.

Consider any $f \notin \rightsquigarrow(I)$. We have $f \in R(\rightsquigarrow(I))$. Indeed, define $g_1 \in P^{\mathcal{A}^{<\mathbb{N}}}$ as $g_1 = f$ except that $g_1(a_2) = \varphi'$, and $g_2 \in P^{\mathcal{A}^{<\mathbb{N}}}$ as $g_2 = f$ except that $g_2(a_1) = \varphi'$. Then, $g_1, g_2 \in \rightsquigarrow(I)$. Now consider any $H \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}}))$. If $(a_2) \notin H$, then $f|_H = g_1|_H$; otherwise, $f|_H = g_2|_H$.

It follows that the claim associated to $R(\rightsquigarrow(I))$ is trivial. $\triangle$

Remark 40. A few more, maybe not useful, definitions.

Given a set of attitudes S and $\alpha \in \mathcal{A}^{<\mathbb{N}}$, define the equivalence relation $=_\alpha$ on S as $f =_\alpha g$ iff $f(\alpha) = g(\alpha)$. This also defines the quotient set $S / =_\alpha$. (For the readers used to that terminology: if defining $\beta_\alpha \in P^S$ as $\beta_\alpha(f) = f(\alpha)$, then $=_\alpha$ is the equivalence kernel of β_α and $S / =_\alpha$ is the coimage of β_α .) Similarly, given S and $H \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}}))$, define the equivalence relation $=_H$ on S as $f =_H g$ iff $f|_H = g|_H$. This also defines the quotient set $S / =_H$.

A function $l : \mathcal{A}^{<\mathbb{N}} \rightarrow \mathcal{P}(P^{\mathcal{A}^{<\mathbb{N}}})$ is decreasing iff $\forall \alpha \subseteq \alpha' : l(\alpha') \subseteq l(\alpha)$. Let $S \subseteq P^{\mathcal{A}^{<\mathbb{N}}}$ be a set of attitudes. Say that a function $l : \mathcal{A}^{<\mathbb{N}} \rightarrow \mathcal{P}(P^{\mathcal{A}^{<\mathbb{N}}})$ is an S -point-labeling function iff $\forall \alpha \in \mathcal{A}^{<\mathbb{N}} : l(\alpha) \in S / =_\alpha$. Say that a function $l : \mathcal{A}^{<\mathbb{N}} \rightarrow \mathcal{P}(P^{\mathcal{A}^{<\mathbb{N}}})$ is an S -sequence-labeling function iff $\forall \alpha \in \mathcal{A}^{<\mathbb{N}} : l(\alpha) \in S / =_{[\alpha]}$. $\triangle$

Theorem 67. Given $\alpha \in \mathcal{A}^{<\mathbb{N}}$, define $S_{\upharpoonright[\alpha]} = \{f_{\upharpoonright[\alpha]} \mid f \in S\}$. Given a non empty set of attitudes $S \subseteq P^{\mathcal{A}^{<\mathbb{N}}}$, the following statements are equivalent.

- The claim associated to $R(S)$ is not trivial
- $R(S) \neq P^{\mathcal{A}^{<\mathbb{N}}}$
- $\exists \alpha \in \mathcal{A}^{<\mathbb{N}}, f \in P^{[\alpha]} \mid \forall g \in S : g_{\upharpoonright[\text{hist}]} \neq f$
- $\exists \alpha, f \in P^{[\alpha]} \mid f \notin S_{\upharpoonright[\alpha]}$
- $\exists \alpha \mid P^{[\alpha]} \not\subseteq S_{\upharpoonright[\alpha]}$.

Also, the following statements are equivalent.

- The claim associated to $R(S)$ is trivial
- $R(S) = P^{\mathcal{A}^{<\mathbb{N}}}$
- $\forall \alpha, f, \exists g \in S \mid g_{\upharpoonright[\alpha]} = f$
- $\forall \alpha : P^{[\alpha]} \subseteq S_{\upharpoonright[\alpha]}$.

Furthermore, $\forall \alpha : P^{[\alpha]} \subseteq S_{\upharpoonright[\alpha]} \Rightarrow \#S \notin \mathbb{N}$.

Proof. This follows from $R(S) \neq P^{\mathcal{A}^{<\mathbb{N}}} \Leftrightarrow \exists f \in P^{\mathcal{A}^{<\mathbb{N}}}, \alpha \in \mathcal{A}^{<\mathbb{N}} \mid \cap_{\alpha' \subseteq \alpha} l_f^S(\alpha') = \emptyset$. $\square$

To recap, we now know that with $S = \rightsquigarrow(I)$, when $R(S) \neq P^{\mathcal{A}^{<\mathbb{N}}}$, there is a claim that holds and is anonymous and falsifiable. This includes all situations with finite behaviors ($\#S \in \mathbb{N}$), thus includes all situations with finite I . Consider now situations such that $R(\rightsquigarrow(I)) = P^{\mathcal{A}^{<\mathbb{N}}}$.

Define $I = P^{\mathbb{Z}}$ and $\rightsquigarrow_i = i \circ \#$, so that $S = \rightsquigarrow(I) = \{l \circ \# \in P^{\mathcal{A}^{<\mathbb{N}}} \mid l \in P^{\mathbb{Z}}\}$. We have $R(S) = P^{\mathcal{A}^{<\mathbb{N}}}$. The function $f(a) \neq f(a')$, for example, is not in S .

Theorem 68. Given a claim C , define $S_C \subseteq P^{\mathcal{A}^{<\mathbb{N}}}$ as $S_C = \bigcup_{\rightsquigarrow \in C} (\rightsquigarrow(I))$. A claim C is about types iff $C = (S_C)^I$.

Proof. If the consequent holds, then C is about types, by definition. If a claim is about types S , then consider any $f \in S$, we have $C = S^I$ thus for $\rightsquigarrow$ defined as $\rightsquigarrow(I) = \{f\}$ we have $\rightsquigarrow \in C$, thus $f \in S_C$; and consider any $f \in S_C$, consider any $\rightsquigarrow \in C \mid f \in \rightsquigarrow(I)$, we have $\rightsquigarrow \in S^I$, thus $f \in S$. $\square$

Theorem 69. Under any $B \subseteq P^{\mathcal{A}^{<\mathbb{N}}}$ (thus recall $R(S) \subseteq B$), for any anonymous claim $C \subseteq B^I$ that is falsifiable or trivial, $\forall \rightsquigarrow \in C, F \subseteq \rightsquigarrow(I) \mid [\forall f \in F : \#(\rightsquigarrow^{-1}(f)) \notin \mathbb{N}] : R(F)^I \subseteq C$.

$\exists \rightsquigarrow \in C, \emptyset \neq F \subseteq \rightsquigarrow(I) \mid \forall f \in F : \#(\rightsquigarrow^{-1}(f)) \notin \mathbb{N}$. Then, $R(F)^I \subseteq C$.

Proof. Consider any anonymous claim C , any $\rightsquigarrow \in C, F \subseteq \rightsquigarrow(I) \mid [\forall f \in F : \#(\rightsquigarrow^{-1}(f)) \notin \mathbb{N}]$ and any $\rightsquigarrow' \in R(F)^I \setminus C$. To show that C is not falsifiable, consider any observable $\theta \subseteq \rightsquigarrow'$ and let us show that it is completable in C . Let $\theta(P^{\mathcal{A}^{<\mathbb{N}}})$ denote the (finite) set of individuals that θ interrogates.

From the antecedent we see that I is infinite. We know that $\forall i \in \theta(P^{\mathcal{A}^{<\mathbb{N}}}) : \exists g \in R(F) \mid \theta(i) \subseteq g$, thus, by definition of R , $\exists f \in F \mid \theta(i) \subseteq f$. Define $\gamma : \theta(P^{\mathcal{A}^{<\mathbb{N}}}) \rightarrow F$ that picks one such f for each i . Thus, $(\gamma^{-1}(f), f \in F)$ is a partition of $\theta(P^{\mathcal{A}^{<\mathbb{N}}})$ such that $\forall i \in \gamma^{-1}(f) : \theta(i) \subseteq f$.

Consider sets of same sizes I'_f among the non observed individuals (meaning those in $I \setminus \theta(P^{\mathcal{A}^{<\mathbb{N}}})$) and with corresponding types in $\rightsquigarrow$ (so that $\forall f \in F, i \in I'_f : \rightsquigarrow_i = f$). Define a bijection σ that for each $f \in F$ bijects I_f to I'_f and that sends the elements of $I \setminus (\cup_f I_f \cup I'_f)$ to themselves. The relation $\rightsquigarrow \circ \sigma$ is in C and is compatible with the observable. $\square$

Theorem 70. *The two following statements are equivalent.*

- $\forall \alpha : P^{\llbracket \alpha \rrbracket} \subseteq \rightsquigarrow(I)_{\llbracket \alpha \rrbracket}$
- *There exists no anonymous claim that holds and is falsifiable.*

Proof. If the first one is false, then the claim $R(\rightsquigarrow(I))$ is not trivial, thus is falsifiable.

If the first one holds, let us show that starting from C^{HA} and integrating all relations that are not observably distinguishable, we obtain the trivial claim.

Consider any observable whatsoever, $O \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I)$, and let us show that for some bijection σ over I , $O \subseteq (\rightsquigarrow \circ \sigma)$.

For each of the (finitely many) individuals that O contains, we can find a different behavior f_i that is in $\rightsquigarrow(I)_{\llbracket \alpha \rrbracket}$, thus, with some $j_i \in I \mid \rightsquigarrow(j_i) = f_i$, and that is compatible with what O sees of i . We define σ so that it maps each of those individuals i to the individual j_i with the corresponding behavior. The resulting relation $\rightsquigarrow \circ \sigma$ is compatible with the observable.

It follows that if the claim is not trivial, it is not falsifiable: pick any relation missing in the claim; it cannot be rejected by any observable. $\square$

Theorem 71. *The following statements are equivalent.*

- $\exists \alpha \in \mathcal{A}^{<\mathbb{N}}, f \in P^{\llbracket \alpha \rrbracket} \mid f \notin \rightsquigarrow(I)_{\llbracket \alpha \rrbracket}$.
- $\exists \emptyset \neq H \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}}), f \in P^H \mid f \notin \rightsquigarrow(I)_H$.
- *There exists an anonymous claim that holds and is falsifiable.*

Proof. The first one is the complementary condition to theorem 70. It implies the second one. The second one implies the first one as f can be extended to $\llbracket \alpha \rrbracket = \llbracket \max H \rrbracket$. $\square$

Theorem 72 (Direct proof). *If $\forall \alpha : P^{\llbracket \alpha \rrbracket} \subseteq \rightsquigarrow(I)_{\llbracket \alpha \rrbracket}$, then any anonymous claim that holds is trivial or strongly unfalsifiable.*

Proof. To hold and be anonymous, we need C includes C^{HA} (theorem 48). Pick any such non trivial C and let us show that $\Theta \subseteq C^{\subseteq}$. Pick any $\theta \in \Theta$.

We define a bijection σ over I as follows. For each of the (finitely many) individuals that θ contains, we can find a different behavior f_i that is in $\rightsquigarrow(I)_{\llbracket \alpha \rrbracket}$, thus, with some $j_i \in I \mid \rightsquigarrow(j_i) = f_i$, and that is compatible with what θ sees of i . We define σ so that it maps each of those individuals i to the individual j_i with the corresponding behavior.

The resulting relation $\rightsquigarrow \circ \sigma$ is compatible with the observable. $\square$

E.6. Indicativeness

Let us determine which situations admit anonymous and falsifiable claims that hold and are indicative.

From theorem 49 we know that we require at least for some $H \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}}))$ and $i_1 \in I \mid P_{i_1} \in P$, we have $\forall i_2 \mid P_{i_2} \neq P_{i_1}$ that $\exists \alpha \in H \mid \rightsquigarrow_{i_1}(\alpha) \neq \rightsquigarrow_{i_2}(\alpha)$.

Theorem 73. Assume that for some $H \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}}))$ and $i_1 \in I \mid p = P_{i_1} \in P$, we have $\forall i_2 \mid P_{i_2} \neq P_{i_1}$ that $\exists \alpha \in H \mid \rightsquigarrow_{i_1}(\alpha) \neq \rightsquigarrow_{i_2}(\alpha)$.

Then $\forall H' \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}}), P^{H'} \subseteq \rightsquigarrow(I)_{\upharpoonright H'}$ iff $\forall H \subseteq H', f \in P^{H'} : \exists i \in I \mid f \subseteq \rightsquigarrow_i \wedge [f]_{\upharpoonright H} = \rightsquigarrow_{i_1 \upharpoonright H} \Rightarrow \exists a \in \mathcal{A} \mid a \hookrightarrow_i p$.

Proof. Not exactly correct, should be corrected and proven but does not seem crucial fow now. Anyway, it seems possible (see below). $\square$

Note that this is possible iff $\mathcal{A}$ is infinite. If so, suffices to take systematically $a \notin H'$. If not, then some H' exhausts all arguments and with f constant to $p' \neq p$, there is no a satisfying $a \hookrightarrow_i p$.

Theorem 74. If $a_1 \hookrightarrow_{f_1} \varphi$ and $a_2 \hookrightarrow_{f_2} \varphi'$ with $a_1 \neq a_2$ and $f_1(\emptyset) = f_2(\emptyset)$ and $\forall f \in \{f_1, f_2\} : \#\rightsquigarrow^{-1}(f) \notin \mathbb{N}$, then there is no anonymous, falsifiable and indicative about i_1 claim that holds (here, considering indicativeness as mandating to know that $P_{i_1} = \varphi$).

Proof. Any claim that holds must include $\rightsquigarrow$. Define $F = \{f_1, f_2\}$.

From indicativeness about i_1 , $\exists \theta \in \Theta \cap (\{i_1\} \times \mathcal{A}^{<\mathbb{N}} \times P) \subseteq \rightsquigarrow \mid \emptyset \neq \pi(\{\rightsquigarrow \mid \theta \subseteq \rightsquigarrow\} \cap C)(i_1) = \varphi$. Define $H \in \mathcal{I}(\mathcal{A}^{<\mathbb{N}})$ as the query corresponding to θ , thus $H = \theta(i_1, P)$.

Define $g \in P^{\mathcal{A}^{<\mathbb{N}}}$ as $g = \{(\alpha, f_1(\alpha)) \mid \exists \emptyset \neq \alpha' \in \llbracket \alpha \rrbracket \cap H\} \cup \{(\alpha, f_2(\alpha)) \mid \llbracket \alpha \rrbracket \cap H \subseteq \{\emptyset\}\}$. Then, $g \in R(F)$. To show this, we need to show that $\forall H' \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})) : \exists f \in F \mid g_{\upharpoonright H'} = f_{\upharpoonright H'}$. If $\llbracket \max H' \rrbracket \cap H \subseteq \{\emptyset\}$, then f_2 satisfies the condition (consider any $\alpha \in H'$, we have $\llbracket \alpha \rrbracket \cap H \subseteq \{\emptyset\}$); otherwise (thus $\emptyset \neq \alpha' \in \llbracket \max H' \rrbracket \cap H$), f_1 does (consider any $\alpha' \in H'$).

Now let us show that $P_g \neq \varphi$. That is because at least one of $\alpha = (a_2, a_1)$ or $\alpha = (a_1, a_2)$ satisfies $\llbracket \alpha \rrbracket \cap H \subseteq \{\emptyset\}$ and in both cases, $\alpha \rightsquigarrow_{f_2} \varphi'$, thus, $g(\alpha) = \varphi'$.

Define $\rightsquigarrow'$ as $\forall i \in I : \rightsquigarrow'_i = g$. By theorem 69, any claim that is anonymous and falsifiable must include $\rightsquigarrow'$.

Also, g is compatible with θ . It follows that $\pi(\{\rightsquigarrow \mid \theta \subseteq \rightsquigarrow\} \cap C)(i_1) \neq \varphi$. $\square$

I need some restriction on the set of a priori possible behaviors. Define $B \subseteq P^{\mathcal{A}^{<\mathbb{N}}}$ as the set of a priori possible behaviors. For example, define B_1 as the set of behaviors such that $\forall a_1, a_2, \alpha : (\alpha, a_1, a_2) = (\alpha, a_2, a_1)$ (where equality, abusing notation, means that they lead to the same result). Let $\overline{B} = P^{\mathcal{A}^{<\mathbb{N}}} \setminus B$ denote the set of behaviors that fail the condition.

Then the a priori possible situations are only B^I . A claim is falsifiable iff the situations in B^I not in C are rejectable. The other properties (holds, indicativeness, anonymity) do not change. We used to consider also situations $\overline{S} = \{\rightsquigarrow \mid \exists i \in I \mid \rightsquigarrow_i \in \overline{B}\}$ where some i is in $\overline{B}$.

Using B_1 alone does not solve the problem, as we can still show that $P_g \neq \varphi$ by using $\alpha = (a_3, a_2, a_1)$, assuming that H does not start with a_3 .

I might need something like the following. a is apparently innocuous iff $\forall \alpha, \rightsquigarrow : \rightsquigarrow(\alpha, a) = \rightsquigarrow(\alpha)$. And a is innocuous iff $\forall \alpha, \alpha' : \alpha, a, \alpha' = \alpha, \alpha'$. The required hypothesis might be B_2 : if a is apparently innocuous, then it is innocuous.

Theorem 75. *Under some B , if $a_1 \hookrightarrow_{f_1} \varphi$ and $a_2 \hookrightarrow_{f_2} \varphi'$ and $\rightsquigarrow_{f_1}(\emptyset) = \rightsquigarrow_{f_2}(\emptyset)$ and infinitely many individuals have behavior f_1 and infinitely many have behavior f_2 , then there is an anonymous, falsifiable and indicative about i_1 claim that holds (here, considering indicativeness as mandating to know that $P_{i_1} = \varphi$).*

Proof. Pick $B = \{\rightsquigarrow_{i_1}, \rightsquigarrow_{i_2}\}$ and $C = C^{\text{HA}}$. □

Conjecture 3. *Assume $\mathcal{A} = \{a_1, a_2\}$. Write them a and b . Let us find some B where theorem 74 fails.*

Proof. To block the proof of theorem 74, we need to ensure that $P_g = \varphi$. Say $H = \{\emptyset, a, ab, aba\}$. Define a non intersecting path (meaning not intersecting H) as any path such as ba . We want B such that we may not include f_2 on some non intersecting path with a failing. For example, ba may not be f_2 . For this, we need B of the form: $f(a) = f(ab) = f(aba) = \varphi$ implies $f(ba.*) = \varphi$, where $ba.*$ represents some completion of ba . □

E.7. Extensions

Theorem 76. *If C is strongly unfalsifiable, then any wider claim $C \subseteq C'$ is strongly unfalsifiable.*

Proof. $\Theta \subseteq C^\subseteq \subseteq C'^\subseteq$. □

Theorem 77. *If $C = C_1 \cap C_2$ with C_1 strongly unfalsifiable and C_2 coverable, then C is unfalsifiable.*

Proof. From C_1 being strongly unfalsifiable and C_2 being coverable, we obtain $\Theta \cap C_2^\subseteq \subseteq (C_1 \cap C_2)^\subseteq$ (theorem 59).

Pick any $\rightsquigarrow \in C_2$ with $\rightsquigarrow \notin C_1$ (thus $\rightsquigarrow \notin C$) and any $\theta \in \Theta \cap \mathcal{P}(\rightsquigarrow)$, and let us show that $\theta \in C^\subseteq$.

From $\theta \subseteq \rightsquigarrow$ we have $\theta \in C_2^\subseteq$. Thus, $\theta \in (C_1 \cap C_2)^\subseteq = C^\subseteq$. □

Theorem 78. *If $C = C_1 \cap C_2$ with C_1 strongly unfalsifiable and C_2 falsifiable and coverable, then C is unfalsifiable but not strongly unfalsifiable.*

Proof. The claim C_2 is falsifiable, thus, is not strongly unfalsifiable. Thus, the more precise claim C is not strongly unfalsifiable (theorem 76). Theorem 77 concludes. $\square$

E.8. Draft material about falsification

Remark 41. Any relation $\rightsquigarrow$ is equivalent to $\cup\{\rightsquigarrow_{|O} \mid O \in \cup_{J \subseteq \mathcal{F}I} (\mathcal{A}^{<\mathbb{N}})^J\}$. $\triangle$

Remark 42. Any relation $\rightsquigarrow$ is equivalent to $\cup\{\rightsquigarrow_{|O} \mid O \in (\mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}}))^I\}$. Conversely, any set S of observations such that $\{\theta(P) \mid \theta \in S\} = \mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I$ and $\cup S$ is right-unique corresponds to a unique relation $\cup S = \rightsquigarrow$. Thus, there is a bijection between the set $\mathcal{S}$ of such sets S and the set of leads to relations.

This is the intuitive reason that, as we will see, a claim C is falsifiable iff $\forall S \in \mathcal{S} : S \subseteq C^\subseteq \Rightarrow \cup S \in C$. $\triangle$

Definition 13. A claim C is necessarily verifiable iff $\forall \rightsquigarrow \in C : \exists \theta \in \Theta \cap \mathcal{P}(\rightsquigarrow) \mid \{\rightsquigarrow' \in P^{I \times \mathcal{A}^{<\mathbb{N}}} \mid \theta \in \mathcal{P}(\rightsquigarrow')\} \subseteq C$.

Example 57 (Bigger disjunct, strongly unfalsifiable). Pick any $i \in I$ and $a \in \mathcal{A}$. The claim $[(a) \rightsquigarrow_i \varphi] \vee [(a, a) \rightsquigarrow_i \varphi] \vee \dots$, thus, $C = \{\rightsquigarrow \mid \exists n \in \mathbb{N}^* \mid (a)^n \rightsquigarrow_i \varphi\}$, is strongly unfalsifiable.

Let $C' = \{\rightsquigarrow \mid \exists n \in \mathbb{N}^* \mid \forall j : (a)^n \rightsquigarrow_j \varphi\}$ denote the claim of example 9. We have $C' \subseteq C$. It follows that $C'^\subseteq \subseteq C^\subseteq$. We know from example 9 that $\Theta \subseteq C'^\subseteq$, thus, $\Theta \subseteq C^\subseteq$. $\triangle$

Remark 43. I think that theorem 59 is not true when replacing the assumption that C is easy with assuming that C is necessarily verifiable.

Assume that C is $\alpha_1 \rightsquigarrow_i p_1 \vee (\alpha_2 \rightsquigarrow_i p_2 \wedge \alpha_3 \rightsquigarrow_i p_3)$. Say that $\theta = (\alpha_2 \rightsquigarrow_i p_2), (\alpha_4 \rightsquigarrow_i p_4)$. Now we could have that together with α_3 , this is not observable. And together with α_3 it could be incompatible with C' : we only have guaranteed compatibility with observables. $\triangle$

Can I make a strictly unfalsifiable claim more precise and obtain a strongly unfalsifiable claim?

Remark 44. The claim “For each glass, it will not break on a monday or not break on a tuesday, whatever happens to it.” is not falsifiable. But one can make it stronger and falsify it, by giving a general theory of how glasses break, proving that it holds, and that it is not compatible with the claim. I should think about this. $\triangle$

Conjecture 4. *C is falsifiable iff exists two compatible paths in two different such forests, for a given i , that are not in a single forest, for that i , of C . This seems incorrect; need to be able to build a union relation! Prove that it is false with claim “ i goes to ϕ OR i goes to not ϕ except for empty set going to ϕ ”? Actually I need to be able to merge two relations that agree on everything but some part of the tree, in such a way that all paths crossing the boundary must be in C , but not the merge. TO be done: caract with singleton I .*

Remark 45. Say that $(r1, o1)$ is determined in C (to p) iff all relations in C including $r1$ agree on $\rightsquigarrow(o)$ (satisfy $\rightsquigarrow(o) = p$). Let $C(r, o)$ equal $\{\rightsquigarrow(o) \mid \rightsquigarrow \in C \wedge r \in \rightsquigarrow\}$.

A claim satisfying, $\forall \rightsquigarrow \in C: \forall r1 = (o1, p1) \in \rightsquigarrow, \exists o2$ with $o1$ and $o2$ obs-compatible and $(r1, o2)$ determined in C such that $\forall o3 : (o2, o3)$ is obs-compatible and $(r2 = (o2, C(r1, o2)), o3)$ is determined in C is falsifiable.

NOPE. Need to say that C is a union of disjoint subsets such that each subset C' include all relations that are specified on a subset of observations (and left free on everything else) which satisfy something like the above condition: all points related to a point that is related to all others. $\triangle$

Theorem 79. *Given $R \subseteq I \times \mathcal{A}^{<\mathbb{N}}$, say that C is determined on R iff $\exists f \in P^R \mid C = \{f \cup g \mid g \in P^{(I \times \mathcal{A}^{<\mathbb{N}}) \setminus R}\}$.*

If $\exists R \subseteq I \times \mathcal{A}^{<\mathbb{N}}$ such that C is determined on R and satisfying $\exists o \in R \mid \forall o' \in R : (\{o, o'\} \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I))$, then C is falsifiable.

Proof. Note that as C is closed-under-combinations, this follows from remark 39, but let's design a more direct proof in hope of weakening the condition.

Consider any $\rightsquigarrow$ obs-compatible with $C^\subseteq \cap \Theta$ and let us prove that $\rightsquigarrow \in C$.

Pick some $o \in R$ as in the hypothesis. Consider any $o' \in R$. As $\{o, o'\} \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I)$ and $\rightsquigarrow$ being obs-compatible, we have $\rightsquigarrow|_{\{o, o'\}} \in C^\subseteq$. Thus, $\rightsquigarrow(o') = f(o')$.

We obtain that $\forall o' \in R : \rightsquigarrow(o') = f(o')$, therefore, $\rightsquigarrow \in C$. $\square$

Conjecture 5 (draft). *Given $R \subseteq I \times \mathcal{A}^{<\mathbb{N}}$, say that C is determined on R iff $\exists f \in P^R \mid C = \{f \cup g \mid g \in P^{(I \times \mathcal{A}^{<\mathbb{N}}) \setminus R}\}$. Say that R is coverable iff $\exists o_1, \dots, O_1, \dots \mid \forall o \in O_1 : \{o_1, o\} \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I) \wedge \forall o \in O_2 : \{o_2, o\} \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I) \wedge \dots \wedge \{o_i, o_{i+1}\} \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I) \wedge \bigcup O_i = R \wedge o_i \in R$.*

If $\exists R \subseteq I \times \mathcal{A}^{<\mathbb{N}}$ coverable and such that C is determined on R , then C is falsifiable.

Proof. Consider any $\rightsquigarrow$ obs-compatible with $C^\subseteq \cap \Theta$ and let us prove that $\rightsquigarrow \in C$.

Pick some $o_1 \in R$ as in the hypothesis. Consider any $o \in O_1 \cup \{o_2\}$. As $\{o_1, o\} \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I)$ and $\rightsquigarrow$ being obs-compatible, we have $\rightsquigarrow|_{\{o_1, o\}} \in C^\subseteq$. As also $o_1, o \in R$, $\rightsquigarrow(o) = f(o)$. In particular, $\rightsquigarrow(o_2) = f(o_2)$. Iterate with o_2 and $o \in O_2$, and so on. We obtain that $\forall o \in R : \rightsquigarrow(o) = f(o)$, therefore, $\rightsquigarrow \in C$. $\square$

Example 58. Pick $i \in I$ and define $o_1 = ((a), i)$, $o_2 = (\emptyset, i)$ and $o_3 = ((a'), i)$. Define $R = \{o_1, o_2, o_3\}$. Define $f_1, f_2, f_3 \in P^R$ as follows. $f_1(o_1) = \varphi$, $f_1(o_2) = \varphi'$, $f_1(o_3) = \varphi$. $f_2(o_1) = \varphi$, $f_2(o_2) = \varphi$, $f_2(o_3) = \varphi'$. $f_3(o_1) = \varphi'$, $f_3(o_2) = \varphi$, $f_3(o_3) = \varphi$. Define $C = \{f \cup g \mid f \in \{f_1, f_2, f_3\} \wedge g \in P^{(I \times \mathcal{A}^{<\mathbb{N}}) \setminus R}\}$. This is the claim of example 11.

Note that R is covered by $O_1 = \{o_1\}$, $O_2 = \{o_2\}$, $O_3 = \{o_3\}$. Note that C is determined on R . Yet, C is not falsifiable.

Note that

$$C^\subseteq \cap \Theta = \left\{ \theta \in \Theta \mid \begin{array}{l} (\theta(o_1) \in \{\emptyset, \varphi\} \wedge \theta(o_2) \in \{\emptyset, \varphi'\} \wedge \theta(o_3) \in \{\emptyset, \varphi\}) \vee \\ (\theta(o_1) \in \{\emptyset, \varphi\} \wedge \theta(o_2) \in \{\emptyset, \varphi\} \wedge \theta(o_3) \in \{\emptyset, \varphi'\}) \vee \\ (\theta(o_1) \in \{\emptyset, \varphi'\} \wedge \theta(o_2) \in \{\emptyset, \varphi\} \wedge \theta(o_3) \in \{\emptyset, \varphi\}) \end{array} \right\}.$$

Also, $\theta(o_1) \neq \emptyset \Rightarrow \theta(o_3) = \emptyset$ (and equivalently, $\theta(o_3) \neq \emptyset \Rightarrow \theta(o_1) = \emptyset$), whence:

$$C^\subseteq \cap \Theta = \left\{ \theta \in \Theta \left| \begin{array}{l} (\theta(o_1) = \theta(o_3) = \emptyset \wedge \theta(o_2) \in \{\emptyset, \varphi'\}) \vee \\ (\theta(o_1) = \varphi \wedge \theta(o_2) \in \{\emptyset, \varphi'\}) \vee \\ (\theta(o_3) = \varphi \wedge \theta(o_2) \in \{\emptyset, \varphi'\}) \vee \\ (\theta(o_1) = \theta(o_3) = \emptyset \wedge \theta(o_2) \in \{\emptyset, \varphi\}) \vee \\ (\theta(o_1) = \varphi \wedge \theta(o_2) \in \{\emptyset, \varphi\}) \vee \\ (\theta(o_3) = \varphi' \wedge \theta(o_2) \in \{\emptyset, \varphi\}) \vee \\ (\theta(o_1) = \theta(o_3) = \emptyset \wedge \theta(o_2) \in \{\emptyset, \varphi\}) \vee \\ (\theta(o_1) = \varphi' \wedge \theta(o_2) \in \{\emptyset, \varphi\}) \vee \\ (\theta(o_3) = \varphi \wedge \theta(o_2) \in \{\emptyset, \varphi\}) \end{array} \right. \right\}.$$

Factorizing:

$$C^\subseteq \cap \Theta = \left\{ \theta \in \Theta \left| \begin{array}{l} (\theta(o_1) = \theta(o_3) = \emptyset \wedge \theta(o_2) \in \{\emptyset, \varphi, \varphi'\}) \vee \\ (\theta(o_1) = \varphi \wedge \theta(o_2) \in \{\emptyset, \varphi, \varphi'\}) \vee \\ (\theta(o_3) = \varphi \wedge \theta(o_2) \in \{\emptyset, \varphi, \varphi'\}) \vee \\ (\theta(o_3) = \varphi' \wedge \theta(o_2) \in \{\emptyset, \varphi\}) \vee \\ (\theta(o_1) = \varphi' \wedge \theta(o_2) \in \{\emptyset, \varphi\}) \end{array} \right. \right\}.$$

Note that the complement is as follows.

$$\overline{C^\subseteq \cap \Theta} = \left\{ \theta \in \Theta \left| \begin{array}{l} \theta(o_2) = 0 \vee \\ \theta(o_2) = \varphi' \wedge (\theta(o_1) \in \{\varphi', 0\} \vee \theta(o_3) \in \{\varphi', 0\}) \vee \\ \theta(o_1) \in \{\varphi', 0\} \wedge \theta(o_3) \neq \varphi' \vee \\ \theta(o_1) = \emptyset \vee \\ \theta(o_3) = \emptyset \vee \\ \theta(o_3) \in \{\varphi', 0\} \wedge \theta(o_1) \neq \varphi' \end{array} \right. \right\}.$$

This seems incorrect: if $\theta = \emptyset$, then it belongs, even though it satisfies $\theta(o_1) = \emptyset$.

The relations that are obs-compatible with $C^\subseteq \cap \Theta$ are the ones that, for (o_1, o_2, o_3) , include two φ and one φ' , the one that equals all φ , and the one that assigns o_1 and o_3 to φ' and o_2 to φ .

It follows that the relations that are obs-compatible with $C^\subseteq \cap \Theta$ and that are in $C^\subseteq \cap \Theta$ are the ones that, for (o_1, o_2, o_3) , include two φ and one φ' ; and the relations that are obs-compatible with $C^\subseteq \cap \Theta$ and that are not in $C^\subseteq \cap \Theta$ are the one that equals all φ and the one that assign o_1 and o_3 to $\emptyset$ and o_2 to φ .

Pick any $g \in P^{(I \times \mathcal{A}^{<\mathbb{N}}) \setminus R}$ and define $\rightsquigarrow = \{(i, \emptyset, \varphi), (i, (a), \varphi), (i, (a'), \varphi)\} \cup g$. This relation is obs-compatible with $C^\subseteq \cap \Theta$, but $\rightsquigarrow \notin C$. $\triangle$

Conjecture 6. *Given $R \subseteq I \times \mathcal{A}^{<\mathbb{N}}$, say that C is determined on R iff $\exists f \in P^R \mid C = \{f \cup g \mid g \in P^{(I \times \mathcal{A}^{<\mathbb{N}}) \setminus R}\}$. Say that R is coverable iff $\exists k \in \mathbb{N}, (O_l)_{l \in \llbracket 1, k \rrbracket}, (o_l)_{l \in \llbracket 1, k \rrbracket} \mid \cup_{l \in \llbracket 1, k \rrbracket}$*

$O_l = R \wedge \forall l \in \llbracket 1, k \rrbracket : o_l \in O_l \wedge \forall o \in O_l : \{o_l, o\} \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I) \wedge \forall l \in \llbracket 1, k-1 \rrbracket : \{o_l, o_{l+1}\} \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I)$.

If $\exists R \subseteq I \times \mathcal{A}^{<\mathbb{N}}$ coverable and such that C is determined on R , then C is falsifiable.

Proof. Consider any $\rightsquigarrow$ obs-compatible with $C^\subseteq \cap \Theta$ and let us prove that $\rightsquigarrow \in C$.

Pick some $k \in \mathbb{N}$, $(O_l)_{l \in \llbracket 1, k \rrbracket}, (o_l)_{l \in \llbracket 1, k \rrbracket}$ as in the hypothesis. Consider any $l \in \llbracket 1, k-1 \rrbracket$ and any $o \in O_l \cup \{o_{l+1}\}$. As $\{o_l, o\} \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I)$ and $\rightsquigarrow$ being obs-compatible, we have $\rightsquigarrow|_{\{o_l, o\}} \in C^\subseteq$. As also $o_l, o \in R$ (per $\cup_{l' \in \llbracket 1, k \rrbracket} O_{l'} = R$), $\rightsquigarrow(o) = f(o)$. In particular, $\rightsquigarrow(o_{l+1}) = f(o_{l+1})$. Iterate with o_2 and $o \in O_2$, and so on. We obtain that $\forall o \in R : \rightsquigarrow(o) = f(o)$, therefore, $\rightsquigarrow \in C$. $\square$

Conjecture 7. Given $R \subseteq I \times \mathcal{A}^{<\mathbb{N}}$, define $\sim_R \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}} \times P^{I \times \mathcal{A}^{<\mathbb{N}}}$ as $f \sim_R f'$ iff $f|_R = f'|_R$. This is an equivalence relation. Given $f \in P^{I \times \mathcal{A}^{<\mathbb{N}}}$, define the equivalence class of f for $\sim_R$ as $\sim_R(f) = \{f' \in P^{I \times \mathcal{A}^{<\mathbb{N}}} \mid f \sim_R f'\} = \{f' \in P^{(I \times \mathcal{A}^{<\mathbb{N}}) \setminus R} \mid f|_R = f'|_R\}$.

Given $R \subseteq I \times \mathcal{A}^{<\mathbb{N}}$, define the a priori observable subsets of R as $Z_R = \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I) \cap \mathcal{P}(R)$.

Consider $R \subseteq I \times \mathcal{A}^{<\mathbb{N}}$, $F \subseteq P^R$ and $C = \{f \cup g \mid f \in F \wedge g \in P^{(I \times \mathcal{A}^{<\mathbb{N}}) \setminus R}\}$. The claim C is unfalsifiable iff $\exists \rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}} \mid \bigcap_{O \in Z_R} \rightsquigarrow_O(f) \cap C = \emptyset$.

Say that R is coverable iff $\exists k \in \mathbb{N}, (O_l)_{l \in \llbracket 1, k \rrbracket}, (o_l)_{l \in \llbracket 1, k \rrbracket} \mid \cup_{l \in \llbracket 1, k \rrbracket} O_l = R \wedge \forall l \in \llbracket 1, k \rrbracket : o_l \in O_l \wedge \forall o \in O_l : \{o_l, o\} \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I) \wedge \forall l \in \llbracket 1, k-1 \rrbracket : \{o_l, o_{l+1}\} \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I)$.

If $\exists R \subseteq I \times \mathcal{A}^{<\mathbb{N}}$ coverable and such that C is determined on R , then C is falsifiable.

Consider: extend this so as to cover eg C containing f_1 and f_2 constant functions as in example; but not covering when the domain is not suitable! Use proof similar to example 43, getting started with observing that any obs-compatible $\rightsquigarrow$ failing every element but one of C at least at one point must then satisfy the last element of C at some point (?).

Conjecture 8. If $\forall \rightsquigarrow \in C : \exists R \mid \rightsquigarrow = f \cup g$ with $f \in P^R, g \in P^{(I \times \mathcal{A}^{<\mathbb{N}}) \setminus R}$ and R coverable, then C is falsifiable. Should be specified better, but also, not sure that this holds: consider C being a leads i to ϕ OR b leads i to ϕ (given i). Perhaps something like $\exists F \mid C = \cup_{f \in F} \{f \cup g\}$ where $f(P)$ being coverable.

Proof. Problem is: how to get started? $\square$

Example 59. Consider any $i, a_1 \neq a_2 \in \mathcal{A}, p \neq p' \in P$ and any $\rightsquigarrow^1, \rightsquigarrow^2$ such that $(a_1) \rightsquigarrow_i^1 p$ and $(a_2) \rightsquigarrow_i^2 p'$. Consider the claim $C = \{\rightsquigarrow^1, \rightsquigarrow^2\}$. When is it falsifiable considering these two relations? $\triangle$

Consider: investigate claims that are non falsifiable but with $C^\subseteq \cap \Theta \neq \Theta$. Then exhibit a claim such that a smaller one is not the same status and a bigger one is not the same status.

Any claim containing two relations that differ only on incompatible observations is not falsifiable.

Theorem 80. Any C such that $\exists \rightsquigarrow^1, \rightsquigarrow^2 \in C \mid$ some combination of them $\rightsquigarrow \notin C$ (meaning that $\rightsquigarrow \subseteq \rightsquigarrow^1 \cup \rightsquigarrow^2$) and the set over which they differ, $\{(i, \alpha) \mid \rightsquigarrow^1(i, \alpha) \neq \rightsquigarrow^2(i, \alpha)\}$, does not contain two elements that are simultaneously observable (meaning that no θ contains more than one of these elements), is not falsifiable.

Proof. Consider any $\theta \subseteq \rightsquigarrow$. If θ is fully compatible with $\rightsquigarrow^1$, then $\rightsquigarrow^1 \in S_\theta \cap C \neq \emptyset$. Otherwise, θ contains some element $(i, \alpha, \rightsquigarrow^2(i, \alpha))$ with $\rightsquigarrow^2(i, \alpha) \neq \rightsquigarrow^1(i, \alpha)$. Then, $\rightsquigarrow^2 \in S_\theta \cap C \neq \emptyset$, for otherwise, for at least one other element (i', α') , θ contains $(i', \alpha', \rightsquigarrow^1(i', \alpha'))$ with $\rightsquigarrow^1(i', \alpha') \neq \rightsquigarrow^2(i', \alpha')$, which violates the hypothesis. $\square$

Example 36 illustrates the need for the condition about the combination in the theorem.

Conjecture 9. *If a claim C is not falsifiable, then $\exists \rightsquigarrow^1, \rightsquigarrow^2 \in C \mid$ some combination of them $\rightsquigarrow \notin C$ (meaning that $\rightsquigarrow \subseteq \rightsquigarrow^1 \cup \rightsquigarrow^2$) and the set over which they differ, $\{(i, \alpha) \mid \rightsquigarrow^1(i, \alpha) \neq \rightsquigarrow^2(i, \alpha)\}$, does not contain two elements that are simultaneously observable (meaning that no θ contains more than one of these elements).*

Conjecture 10. *If $\forall \rightsquigarrow^1, \rightsquigarrow^2 \in C \mid$ some combination of them $\rightsquigarrow \notin C$ (meaning that $\rightsquigarrow \subseteq \rightsquigarrow^1 \cup \rightsquigarrow^2$), the set over which they differ, $\{(i, \alpha) \mid \rightsquigarrow^1(i, \alpha) \neq \rightsquigarrow^2(i, \alpha)\}$, contains at least two elements that are simultaneously observable (meaning that some θ contains more than one of these elements), then C is falsifiable.*

Proof. We need a stronger hypothesis. Consider *one* and *two* differ on $(i, (a))$, $(i, (b))$ and $(i, (a, a))$. And some *three* not in C has $(i, (a))$, $(i, (a, a))$ as in *one* and $(i, (b))$ as in *two*. Then C is not falsifiable. $\square$

Conjecture 11. *If $\forall \rightsquigarrow^1, \rightsquigarrow^2 \in C \mid$ some combination of them $\rightsquigarrow \notin C$ (meaning that $\rightsquigarrow \subseteq \rightsquigarrow^1 \cup \rightsquigarrow^2$), the set over which they differ, $\{(i, \alpha) \mid \rightsquigarrow^1(i, \alpha) \neq \rightsquigarrow^2(i, \alpha)\}$, we can find some $(i, \alpha) \in \rightsquigarrow^1, (j, \alpha') \in \rightsquigarrow^2$ obs-compatible, then C is falsifiable.*

Proof. This fails. Could have *one*, *two*, *three*, each sending everything to p , $\neg p$ and 0, respectively, and any combination of two of these in C , but no comb of three in C . Then *four* a combination of the three $(i, a, p, i, b, \neg p, i, c, 0)$, but C not falsifiable. Actually, even if we can observe two among the three (using $i, a, p, i, aa, \neg p, i, b, 0$), we can't falsify *four*. $\square$

Remark 46. Odd length to p and even length to $\neg p$. Then we can know that $P_i = \emptyset$. Or, if i finds $a_1 > a_2 > a_3 > a_1$ (more convincing relation), then $P_i = \emptyset$ and we can know it. $\triangle$

E.9. Probably redundant

NB the following results do not consider trivial claims.

Theorem 81. *A claim C is falsifiable iff its complement $\overline{C} = P^{I \times \mathcal{A}^{<\mathbb{N}}} \setminus C$ satisfies: $\forall \rightsquigarrow \in \overline{C}, \exists \theta \subseteq \rightsquigarrow \mid S_\theta \subseteq \overline{C}$, or equivalently: $\forall \rightsquigarrow \in \overline{C}, \exists O \in (\mathcal{A}^{<\mathbb{N}})^I \mid S_{\rightsquigarrow|O_O} \subseteq \overline{C}$.*

Proof. The first phrasing follows from the definition of falsifiability. Also, the second phrasing implies the first one: pick $\theta = \rightsquigarrow|O_O$.

Let us prove that the first phrasing implies the second one. Given θ , define $O = \{(i, \arg \max_{\alpha \in (\theta(i) \cup \emptyset)} \# \alpha)\}$ as associating to each i the longest corresponding α from θ , or the empty sequence if $\theta(i)$ is empty. It follows that $\theta \subseteq \rightsquigarrow|O_O$. Thus, $S_{\rightsquigarrow|O_O} \subseteq S_\theta$. As $S_\theta \subseteq \overline{C}$, we obtain $S_{\rightsquigarrow|O_O} \subseteq \overline{C}$. $\square$

Corollary 16. *A claim C is falsifiable iff $\forall f \in C^{\mathcal{F}}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I) \mid \forall O, O' \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I) : f(O)_{\upharpoonright O \cap O'} = f(O')_{\upharpoonright O \cap O'}$:*

$$\cup_{O \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I)} f(O)_{\upharpoonright O} \in C.$$

Proof. Consider a falsifiable claim C , consider such an f , and let us prove that $\cup_{O \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I)} f(O)_{\upharpoonright O} \in C$. Define $\rightsquigarrow \subseteq (I \times \mathcal{A}^{<\mathbb{N}}) \times P$ as $\cup_{O \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I)} f(O)_{\upharpoonright O}$. Its domain equals $I \times \mathcal{A}^{<\mathbb{N}}$, and it is right-unique thanks to the condition on f . Thus, theorem 60 applies and $\rightsquigarrow \in C$.

Conversely, assume that the statement about all f holds and let us prove that C is falsifiable. Consider any $\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}}$ such that $\{\rightsquigarrow_{\upharpoonright O} \mid O \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I)\} \subseteq C^{\subseteq}$. Given any O , we have $\rightsquigarrow_{\upharpoonright O} \in C^{\subseteq}$ and therefore there is some $e_O \in C$ such that $\rightsquigarrow_{\upharpoonright O} = e_{O \upharpoonright O}$. Define f as $f(O) = e_O$. As this f satisfies the condition in the hypothesis, we obtain that $\cup_{O \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I)} f(O)_{\upharpoonright O} \in C$, hence, $\cup_{O \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I)} \rightsquigarrow_{\upharpoonright O} \in C$. Therefore, $\rightsquigarrow \in C$. Apply theorem 60 to conclude that C is falsifiable. $\square$

Corollary 17. *A claim C is falsifiable iff $\forall f \in (C^{\subseteq})^{\mathcal{F}}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I) \mid \forall O, O' \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I) : f(O)_{\upharpoonright O \cap O'} = f(O')_{\upharpoonright O \cap O'}$:*

$$\cup_{O \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I)} f(O)_{\upharpoonright O} \in C.$$

Corollary 18. *A claim C is falsifiable iff $\forall f \in \cup_{O \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I)} \{\{O\} \times (P^O \cap C^{\subseteq})\} \mid \forall O, O' \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I) : f(O)_{\upharpoonright O \cap O'} = f(O')_{\upharpoonright O \cap O'}$:*

$$\cup_{O \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I)} f(O) \in C.$$

Corollary 19. *Any non-trivial claim C satisfying $\Theta \subseteq C^{\subseteq}$ is not falsifiable.*

Proof. Pick any $\rightsquigarrow \notin C$. Per hypothesis, $\forall O \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I) : \rightsquigarrow_{\upharpoonright O} \in C^{\subseteq}$. Define f as $\forall O \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I) : f(O) = \rightsquigarrow_{\upharpoonright O}$. Observe that $\cup_{O \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I)} f(O) = \rightsquigarrow$ and hence $\cup_{O \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I)} f(O) \notin C$. Apply corollary 18. $\square$

Corollary 20. *A claim C is falsifiable iff $\{\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}} \mid \forall \theta \in \Theta : (\rightsquigarrow \cap \theta) \in (C^{\subseteq} \cap \theta)\} = C$.*

A relation $\rightsquigarrow$ is obs-compatible with C iff $\forall O \in \mathcal{F}(\mathcal{I}(\mathcal{A}^{<\mathbb{N}})^I) : \rightsquigarrow_{\upharpoonright O} \in C^{\subseteq}$, equivalently, iff $\forall \theta \in \Theta : (\rightsquigarrow \cap \theta) \in (C^{\subseteq} \cap \theta)$.

Remark 47. The question of falsifiability is whether the set of relations $\rightsquigarrow$ “compatible” with $C^{\subseteq} \cap \Theta$ equals C . Compatibility needs to consider the observables, not a subset of it. Define $\Theta_1 \subsetneq \Theta$ as the observations that observe only one pair (i, α) . Define $R_1 = \{\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}} \mid \forall \theta \in \Theta_1 : (\rightsquigarrow \cap \theta) \in (C^{\subseteq} \cap \theta)\}$. Note that $R \subseteq R_1$. We know that C is falsifiable iff $R \subseteq C$. Thus, C is not falsifiable implies $C \subsetneq R \subseteq R_1$ thus $C \subsetneq R_1$. But we do not have that $C \subsetneq R_1$ implies C not falsifiable: could have $C \subsetneq R_1$ but still $R = C$. $\triangle$

F. Strictness of established verifiability across nested priors

F.1. The question

Given priors $D \subseteq D'$ (that is, $D, D' \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$), established verifiability under D' always implies established verifiability under D : restricting, in the witness required for a world $\rightsquigarrow \in C \cap D'$, the universal quantifier over D' to the subset D , already gives a witness relative to D . This suggested the following conjecture.

Conjecture 12. *Given priors $D \subseteq D'$, established verifiability under D' is strictly stronger than established verifiability under D . In other words, some claim satisfies established verifiability under D but not established verifiability under D' .*

This appendix reports the current state of conjecture 12: it is false in general, but an exact (if not externally simple) reduction is available; several increasingly general sufficient conditions are known for both possible outcomes — strictness, and its failure, which we call *collapse*; one natural family of priors is completely characterized; and the reasons further generalizations fail are identified precisely, without the general question being settled.

F.2. The conjecture fails, and an exact reduction

Example 60. Pick any $\rightsquigarrow^{(1)} \in P^{I \times \mathcal{A}^{<\mathbb{N}}}$ and, using $\#P \geq 2$, let $\rightsquigarrow^{(0)} \in P^{I \times \mathcal{A}^{<\mathbb{N}}}$ disagree with $\rightsquigarrow^{(1)}$ on every pair. Set $D = \{\rightsquigarrow^{(1)}\} \subsetneq D' = \{\rightsquigarrow^{(1)}, \rightsquigarrow^{(0)}\}$. Established verifiability under D holds for every claim C (witness $\emptyset$ if $\rightsquigarrow^{(1)} \in C$, vacuously otherwise); established verifiability under D' also holds for every claim C : given $\rightsquigarrow \in C \cap D'$, the singleton observation $\theta = \{(i, \alpha, \rightsquigarrow(i, \alpha))\}$, for any (i, α) , already excludes the other element of D' , since $\rightsquigarrow^{(0)}, \rightsquigarrow^{(1)}$ disagree everywhere. So established verifiability under D and under D' coincide for *every* claim: conjecture 12 is false as stated. $\triangle$

Established verifiability under a prior D , for a claim C , is equivalent (corollary 3) to $\forall \rightsquigarrow \in C \cap D : \text{OF}_{D \setminus C}(\rightsquigarrow)$. This gives an exact, if self-referential, reduction of conjecture 12.

Proposition 1 (Exact reduction). *Given priors $D \subseteq D'$, established verifiability under D' is strictly stronger than established verifiability under D iff there is $T \subseteq D'$ such that $\forall \rightsquigarrow \in T \cap D : \text{OF}_{D \setminus T}(\rightsquigarrow)$ and $\exists \rightsquigarrow \in T : \neg \text{OF}_{D \setminus T}(\rightsquigarrow)$.*

Proof. ($\Leftarrow$) Take $C = T$; as $T \subseteq D'$, $C \cap D = T \cap D$ and $C \cap D' = T$, so by corollary 3 the two displayed conditions are exactly established verifiability under D for C , and its failure under D' . ($\Rightarrow$) Given a claim C satisfying established verifiability under D but not under D' , let $T = C \cap D'$; as $D \subseteq D'$, $C \cap D = T \cap D$, and $C \cap D' = T$, so corollary 3 turns the two hypotheses on C into the two displayed conditions on T . $\square$

Proposition 1 settles the question in principle, but the search over $T \subseteq D'$ it requires is not, by itself, informative: the rest of this appendix develops external, checkable conditions covering many — but provably not all — cases.

F.3. Sufficient conditions for strictness

Given $D \subseteq D'$, let $R = D' \setminus D$. Restricting proposition 1 to $T \subseteq D$ (so $T \cap D = T$) gives a first, directly checkable, family of witnesses.

Corollary 21 (Finite subset isolation criterion). *Given priors $D \subseteq D'$ and $T \subseteq D$, if $\forall \rightsquigarrow \in T : \text{OF}_{D \setminus T}(\rightsquigarrow)$ but $\exists \rightsquigarrow \in T : \neg \text{OF}_{D' \setminus T}(\rightsquigarrow)$, then $C = T$ satisfies established verifiability under D but not under D' .*

Proof. Immediate from proposition 1, since $T \subseteq D$ gives $T \cap D = T$. $\square$

Corollary 22 (Triple-obstruction form of the isolation criterion). *Given priors $D \subseteq D'$, $C \subseteq D$, and finite covers $\mathcal{S}$ of $D \setminus C$, $\mathcal{S}'$ of $D' \setminus C$ (as in theorem 21), if*

- *every $\rightsquigarrow \in C$ observes some finite $U \subseteq \rightsquigarrow$, $U \in \Theta$, with $\forall S \in \mathcal{S}, \exists u \in U : \{u\} \notin S^\subseteq$, and*
- *some $\rightsquigarrow \in C$ observes no finite $U \subseteq \rightsquigarrow$, $U \in \Theta$, with $\forall S' \in \mathcal{S}', \exists u \in U : \{u\} \notin S'^\subseteq$,*

then C satisfies established verifiability under D but not under D' .

Proof. By theorem 21, the first bullet is $\forall \rightsquigarrow \in C : \text{OF}_{D \setminus C}(\rightsquigarrow)$ and the second is $\exists \rightsquigarrow \in C : \neg \text{OF}_{D' \setminus C}(\rightsquigarrow)$; conclude by corollary 21 (with $T = C$). $\square$

Remark 48. The finite covers required by corollary 22 need not exist: even $\neg \text{OF}_C$ itself, for a claim C , is not always finitely coverable. For instance, fix infinitely many individuals $i_1, i_2, \dots \in I$, and let $C = \{\rightsquigarrow \mid \forall n \geq 1 : \rightsquigarrow(i_n, \emptyset) \in \{\varphi, \varphi'\}\}$. By theorem 8, $C \subseteq \neg \text{OF}_C$; conversely, if $\rightsquigarrow \notin C$, some n_0 has $\rightsquigarrow(i_{n_0}, \emptyset) = 0$, and $\theta = \{(i_{n_0}, \emptyset, 0)\} \subseteq \rightsquigarrow$ satisfies $\theta \notin C^\subseteq$ (no completion of θ can avoid 0 at i_{n_0}), so $\text{OF}_C(\rightsquigarrow)$. Hence $\neg \text{OF}_C = C$. Yet C has no countable $\uparrow f_S$ -cover: $\uparrow f_S \subseteq C$ forces f_S to fix *every* $(i_n, \emptyset)$ to a value in $\{\varphi, \varphi'\}$ (leaving any i_{n_0} unconstrained lets a completion set 0 there), so distinct valid f_S correspond to distinct elements of $\{\varphi, \varphi'\}^\mathbb{N}$, giving continuum-many pairwise disjoint pieces, none of which a countable subfamily can cover. $\triangle$

For $\rightsquigarrow \in S \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$, say $\rightsquigarrow$ is *isolated in S* iff $\text{OF}_{S \setminus \{\rightsquigarrow\}}(\rightsquigarrow)$, that is, iff some single observation excludes every other element of S at once (so, by theorem 14, $S \subseteq S'$ and $\rightsquigarrow$ isolated in S' implies $\rightsquigarrow$ isolated in S). The case $\#T = 1$ of corollary 21 already gives strictness whenever some $\rightsquigarrow \in D$ is isolated in D but not in D' — no shared coordinate for all of D' is needed, only a single point losing its isolation.

The next example needs the full strength of corollary 21: no single point of D is isolated in D at all, yet a pair of points is.

Example 61. Fix one individual i , three pairwise distinct arguments a, b, c (so $(a), (b), (c)$ are pairwise incomparable), and, using $\#P \geq 2$, two values $p \neq q \in P$. Let $\rightsquigarrow^{(1)}, \dots, \rightsquigarrow^{(5)} \in P^{I \times \mathcal{A}^{<\mathbb{N}}}$ take value p everywhere except: $\rightsquigarrow^{(2)}(i, (a)) = q$; $\rightsquigarrow^{(3)}(i, (a)) = q$; $\rightsquigarrow^{(3)}(i, (b)) = q$; $\rightsquigarrow^{(4)}(i, (b)) = q$; $\rightsquigarrow^{(5)}(i, (c)) = q$. Set $D = \{\rightsquigarrow^{(1)}, \rightsquigarrow^{(2)}, \rightsquigarrow^{(3)}, \rightsquigarrow^{(4)}\}$ (a “square”: $\rightsquigarrow^{(1)}, \rightsquigarrow^{(3)}$ and $\rightsquigarrow^{(2)}, \rightsquigarrow^{(4)}$ are its two diagonals, differing at both (a) and (b) ; the remaining pairs differ at (a) or (b) only), $R = \{\rightsquigarrow^{(5)}\}$, $D' = D \cup R$. No

element of D is isolated in D : excluding, from $\rightsquigarrow^{(1)}$ say, its two neighbours $\rightsquigarrow^{(2)}$ (differing only at (a)) and $\rightsquigarrow^{(4)}$ (differing only at (b)) at once would need both $(i, (a), p)$ and $(i, (b), p)$ in the same observation, impossible as $(a), (b)$ are incomparable; symmetrically for $\rightsquigarrow^{(2)}, \rightsquigarrow^{(3)}, \rightsquigarrow^{(4)}$. Yet, for $T = \{\rightsquigarrow^{(1)}, \rightsquigarrow^{(2)}\}$: $D \setminus T = \{\rightsquigarrow^{(3)}, \rightsquigarrow^{(4)}\}$, both of which have value q at $(i, (b))$, unlike $\rightsquigarrow^{(1)}, \rightsquigarrow^{(2)}$; so $\theta = \{(i, (b), p)\}$ witnesses $\text{OF}_{D \setminus T}(\rightsquigarrow^{(1)})$ and $\text{OF}_{D \setminus T}(\rightsquigarrow^{(2)})$: established verifiability under D holds for $C = T$. But $D' \setminus T = \{\rightsquigarrow^{(3)}, \rightsquigarrow^{(4)}, \rightsquigarrow^{(5)}\}$, and excluding all three from a completion of some $\theta \subseteq \rightsquigarrow^{(1)}$ would need both $(i, (b), p)$ (against $\rightsquigarrow^{(3)}, \rightsquigarrow^{(4)}$) and $(i, (c), p)$ (against $\rightsquigarrow^{(5)}$, the only coordinate where $\rightsquigarrow^{(1)}, \rightsquigarrow^{(5)}$ differ) — by theorem 20, using the cover of $D' \setminus T$ by the two single-coordinate constraints $(i, (b), q)$ and $(i, (c), q)$ — impossible as $(b), (c)$ are incomparable; so established verifiability under D' fails for $C = T$, by corollary 21. $\triangle$

The obstruction throughout example 61 is that Θ only allows, for each individual, a single chain of increasingly long queried paths, so two exclusion requirements can conflict when they need incomparable histories of the same individual. The next lemma and theorem isolate and generalize this mechanism.

Lemma 10 (Differing-set necessity). *If $\rightsquigarrow, \rightsquigarrow' \in P^{I \times \mathcal{A}^{<\mathbb{N}}}$ differ exactly on a set $Z \subseteq I \times \mathcal{A}^{<\mathbb{N}}$ (that is, $\rightsquigarrow(i, \alpha) \neq \rightsquigarrow'(i, \alpha)$ for $(i, \alpha) \in Z$, and they agree elsewhere), every $\theta \in \Theta \cap \mathcal{P}(\rightsquigarrow)$ with $\theta \not\subseteq \rightsquigarrow'$ contains some pair $(i, \alpha, \rightsquigarrow(i, \alpha))$ with $(i, \alpha) \in Z$.*

Proof. This is theorem 20 for the singleton cover $\{\{\rightsquigarrow'\}\}$ of $\{\rightsquigarrow'\}$: if θ contained no such pair, every pair of θ would sit at some $(i, \alpha) \notin Z$, where $\rightsquigarrow, \rightsquigarrow'$ agree, giving $\theta \subseteq \rightsquigarrow'$. $\square$

Theorem 82 (Coordinate-conflict criterion for strictness). *Given priors $D \subseteq D'$, $R = D' \setminus D$, and $T \subseteq D$ satisfying established verifiability under D . Suppose some $\rightsquigarrow^{(0)} \in T$ has a neighbour $v_1 \in D \setminus T$ differing from $\rightsquigarrow^{(0)}$ at exactly one pair (i_0, x) , and some $v_2 \in R$ differing from $\rightsquigarrow^{(0)}$ exactly on a set $Z_2 \subseteq \{i_0\} \times \mathcal{A}^{<\mathbb{N}}$ with every $z \in Z_2$ incomparable to x . Then established verifiability under D' fails for $C = T$.*

Proof. Suppose $\theta \in \Theta \cap \mathcal{P}(\rightsquigarrow^{(0)})$ witnessed $\text{OF}_{D \setminus T}(\rightsquigarrow^{(0)})$, so $\theta \not\subseteq v_1$ and $\theta \not\subseteq v_2$. By lemma 10 (with $Z = \{(i_0, x)\}$), θ contains $(i_0, x, \rightsquigarrow^{(0)}(i_0, x))$; applied again (with $Z = Z_2$), θ contains some pair $(i_0, z, \rightsquigarrow^{(0)}(i_0, z))$ with $z \in Z_2$. As $\theta \in \Theta$, its pairs for i_0 form an increasing chain, so x, z would have to be comparable, contradicting the hypothesis. So no such θ exists, and established verifiability under D' fails for $C = T$ while holding under D . $\square$

Example 61 is the case $\#Z_2 = 1$ ($\rightsquigarrow^{(0)} = \rightsquigarrow^{(1)}$, $v_1 = \rightsquigarrow^{(4)}$, $v_2 = \rightsquigarrow^{(5)}$).

F.4. Sufficient conditions for collapse

The simplest collapse mechanism uses one coordinate shared by, and injective on, all of D' .

Theorem 83 (Collapse via one coordinate injective on all of D'). *Given priors $D \subseteq D'$, if $\exists(i_0, \alpha_0) \in I \times \mathcal{A}^{<\mathbb{N}} \mid [\rightsquigarrow \mapsto \rightsquigarrow(i_0, \alpha_0)]$ is injective on D' (in particular, this requires $\#D' \leq \#P$), then established verifiability under D and under D' coincide, and in fact both hold, for every claim.*

Proof. Write $\iota(\rightsquigarrow) = \rightsquigarrow(i_0, \alpha_0)$, injective on D' . For $\rightsquigarrow \in D'$, the singleton $\theta_{\rightsquigarrow} = \{(i_0, \alpha_0, \iota(\rightsquigarrow))\}$ trivially satisfies the increasing-chain condition, so $\theta_{\rightsquigarrow} \in \Theta \cap \mathcal{P}(\rightsquigarrow)$. For any $\rightsquigarrow' \in D'$, $\theta_{\rightsquigarrow} \subseteq \rightsquigarrow'$ iff $\iota(\rightsquigarrow') = \iota(\rightsquigarrow)$, iff $\rightsquigarrow' = \rightsquigarrow$, by injectivity. Hence, for any $D'' \subseteq D'$ (injectivity of ι on D'' following from injectivity on D') and any claim C : given $\rightsquigarrow \in C \cap D''$, $\theta_{\rightsquigarrow}$ witnesses established verifiability under D'' . Taking $D'' = D$ and $D'' = D'$ in turn, established verifiability under D and under D' both hold, for every claim. $\square$

This is exactly what happens in example 60 ($\rightsquigarrow^{(1)}, \rightsquigarrow^{(0)}$ disagree everywhere, so any coordinate is injective on D'), but it is restrictive as soon as D is not small (it forces $\#D' \leq \#P$). The next result weakens it: the shared coordinate need only separate D from R , not every element of D' from every other, provided it tracks every other coordinate at which D actually varies.

Theorem 84 (Disjoint-range decomposition). *Given priors $D \subseteq D'$ and $R = D' \setminus D$, suppose there are $i_0 \in I$ and $\alpha_0 \in \mathcal{A}^{<\mathbb{N}}$ such that D, R have disjoint ranges at (i_0, α_0) (that is, $\{\rightsquigarrow(i_0, \alpha_0) \mid \rightsquigarrow \in D\} \cap \{\rightsquigarrow(i_0, \alpha_0) \mid \rightsquigarrow \in R\} = \emptyset$), and that this also holds at every other $\alpha \neq \alpha_0$ at which D is not constant (some two elements of D differ there). Then, for every claim C : established verifiability under D' holds for C iff established verifiability under D holds for C and established verifiability under R holds for C .*

Proof. ($\Rightarrow$) If θ traps some $\rightsquigarrow \in C \cap D'$ within D' , restricting the universal quantifier to $D \subseteq D'$ (if $\rightsquigarrow \in D$) or to $R \subseteq D'$ (if $\rightsquigarrow \in R$) shows θ traps it within D , respectively within R .

($\Leftarrow$) Take $\rightsquigarrow \in C \cap D$ (the case $\rightsquigarrow \in C \cap R$ is symmetric); let θ witness established verifiability under D at $\rightsquigarrow$. Let $\theta_0 = \theta \setminus \{(i_0, \alpha, v) \in \theta \mid \alpha \neq \alpha_0 \text{ and } D \text{ is constant at } \alpha\}$: every dropped pair is satisfied by every element of D (as D is constant there), so $\theta_0 \subseteq \rightsquigarrow' \Leftrightarrow \theta \subseteq \rightsquigarrow'$ for every $\rightsquigarrow' \in D$; hence $\theta_0 \in \Theta \cap \mathcal{P}(\rightsquigarrow)$ still traps $\rightsquigarrow$ within D , and its remaining pairs for i_0 are only at α_0 or at histories where D is not constant. If θ_0 contains some pair (i_0, α, v) : as $\rightsquigarrow \in D$, v lies in D 's range at α , disjoint from R 's range there by hypothesis, so θ_0 already excludes all of R . Otherwise, $\theta'_0 = \theta_0 \cup \{(i_0, \alpha_0, \rightsquigarrow(i_0, \alpha_0))\}$ is still in $\Theta \cap \mathcal{P}(\rightsquigarrow)$, still traps within D , and excludes all of R as above. Either way, some $\theta'' \in \Theta \cap \mathcal{P}(\rightsquigarrow)$ traps $\rightsquigarrow$ within D and excludes all of R , hence traps it within D' . $\square$

Corollary 23. *Under the hypothesis of theorem 84, if moreover every $\rightsquigarrow \in R$ is isolated in R , established verifiability under D and under D' coincide for every claim.*

Proof. If every point of R is isolated in R , established verifiability under R holds for every claim; by theorem 84, established verifiability under D' for C then reduces to established verifiability under D for C , for every claim C . $\square$

Example 62. Reusing the square D from example 61, replace the singleton R by one point $\rightsquigarrow^{(5)}$ chosen so that $\rightsquigarrow^{(5)}(i, (a)) = \rightsquigarrow^{(5)}(i, (b)) = s$ for a third value $s \notin \{p, q\}$ (using $\#P \geq 3$), agreeing with the constant p elsewhere. D, R have disjoint ranges at $(i, (a))$ ($\{p, q\}$ versus $\{s\}$) and at $(i, (b))$, and D is constant elsewhere, so theorem 84 applies; as R is a singleton, trivially isolated in itself, corollary 23 gives established verifiability

under D and under $D \cup \{\rightsquigarrow^{(5)}\}$ coinciding for *every* claim — even though D is the very same non-isolated square that, against the singleton $R = \{\rightsquigarrow^{(5)}\}$ of example 61, gave strictness. So D 's internal complexity is not what matters: what matters is whether D, R overlap in the values they use, and, recursively, whether R itself is simple. $\triangle$

This locates exactly where the difficulty in conjecture 12 lies: once D, R are told apart (via theorem 83 or the more general theorem 84), collapse reduces exactly to whether R , taken as its own ambient set, is itself collapse-trivial — which can fail recursively, exactly as it can for D . Whether this recursion can be unwound into an external condition covering every R is open.

F.5. A fully resolved case, and why it resists generalization

For a full hypercube D against a same-individual singleton R , the tools above combine into a complete dichotomy.

Theorem 85 (Complete dichotomy: a hypercube against a same-individual singleton). *Fix an individual i , a finite set $H \subseteq \mathcal{A}^{<\mathbb{N}}$ of pairwise incomparable one-step histories with $\#H \geq 2$, values $p \neq q \in P$, and a baseline $\rightsquigarrow^{(\text{base})} \in P^{I \times \mathcal{A}^{<\mathbb{N}}}$. Let $D = \{\rightsquigarrow \in P^{I \times \mathcal{A}^{<\mathbb{N}}} \mid \rightsquigarrow = \rightsquigarrow^{(\text{base})} \text{ off } \{i\} \times H, \text{ and } \rightsquigarrow(i, \alpha) \in \{p, q\} \forall \alpha \in H\}$ (so $\#D = 2^{\#H}$), and $R = \{r\}$ for some r agreeing with $\rightsquigarrow^{(\text{base})}$ off $\{i\} \times H$, $r \notin D$.*

1. *If $r(i, \alpha) \notin \{p, q\}$ for every $\alpha \in H$, established verifiability under D and under $D \cup R$ coincide for every claim.*
2. *Otherwise (some $\alpha \in H$ has $r(i, \alpha) \in \{p, q\}$), established verifiability under $D \cup R$ is strictly stronger than under D .*

Proof. (1) D, R have disjoint ranges at (i, α) for every $\alpha \in H$, and D is constant off $\{i\} \times H$; theorem 84 applies (with $i_0 = i$ and any $\alpha_0 \in H$ as the pivot); as R is a singleton, corollary 23 gives the conclusion.

(2) Let $z^* \in H$ with $r(i, z^*) \in \{p, q\}$. Let $\rightsquigarrow^{(0)} \in D$ agree with r at every $\alpha \in H$ with $r(i, \alpha) \in \{p, q\}$ (in particular at z^*), taking either value in $\{p, q\}$ at the remaining histories of H . Let $v_1 \in D$ differ from $\rightsquigarrow^{(0)}$ exactly at (i, z^*) , and let $T = \{\rightsquigarrow \in D \mid \rightsquigarrow(i, z^*) = \rightsquigarrow^{(0)}(i, z^*)\}$ (so $\rightsquigarrow^{(0)} \in T$, $v_1 \in D \setminus T$). The single pair $\{(i, z^*, \rightsquigarrow^{(0)}(i, z^*))\}$ excludes all of $D \setminus T$ from every $\rightsquigarrow \in T$: established verifiability under D holds for $C = T$. By construction, $\rightsquigarrow^{(0)}$ and r differ exactly on $Z_2 = \{i\} \times \{\alpha \in H \mid r(i, \alpha) \notin \{p, q\}\}$ — nonempty, as $r \notin D$ — which does not contain z^* ; as H is pairwise incomparable, every $z \in Z_2$ is incomparable to z^* . Theorem 82 (with $x = z^*$, $v_2 = r$) then gives that established verifiability under $D \cup R$ fails for $C = T$, while it holds under D . $\square$

Theorem 85 fully settles conjecture 12 for this entire family: a full hypercube against any same-individual singleton, with no residual gap (clause 2 is the plain negation of clause 1's hypothesis). Example 62 is an instance of clause 1; example 61 is an instance of clause 2, with Z_2 a singleton.

What remains open is what happens once R is not a singleton (the recursive question above) and once D is not a full hypercube. The following example shows the second

direction is not a routine extension: the dichotomy is special to H being *pairwise incomparable*, not to D, R merely sharing or not sharing values.

Example 63. Let $D = \{\rightsquigarrow^{(1)}, \rightsquigarrow^{(4)}\}$, agreeing on the baseline p except $\rightsquigarrow^{(1)}(i, (b)) = p$, $\rightsquigarrow^{(4)}(i, (b)) = q$, and let r agree with $\rightsquigarrow^{(1)}$ at $(i, (b))$ (so $r(i, (b)) = p$, matching D 's range there) but differ at the longer, *comparable* history $(i, (b, c))$: $r(i, (b, c)) = q \neq p = \rightsquigarrow^{(1)}(i, (b, c))$. Established verifiability under D for $T = \{\rightsquigarrow^{(1)}\}$ is witnessed by $\theta = \{(i, (b), p), (i, (b, c), p)\}$ — a valid observation, as $(b) \subseteq (b, c)$ form an increasing chain — and this same θ excludes both $\rightsquigarrow^{(4)}$ (differing at (b)) and r (differing at (b, c)) at once: established verifiability under $D \cup \{r\}$ holds for T too, despite r sharing D 's value at $(i, (b))$. $\triangle$

So sharing a value with D at an active coordinate does not, by itself, force strictness: what made theorem 85 work is specifically that every "extra" coordinate r might need is incomparable to every coordinate D might need. Once R 's point can reach past D 's coordinates along an extension of the same chain, no amount of value-sharing forces strictness. Extending theorem 85 to arbitrary finite D therefore requires tracking the prefix structure of whichever coordinates a witness would need, not merely value overlap — a different, more delicate bookkeeping problem than any solved above, left open together with the case of a non-singleton R .

G. Satisfiability of PEV under $\neg \text{POF}_C$

Theorem 86 (Cufm claims fail PEV under $\neg \text{POF}_C$). *If $\emptyset \neq C \neq P^{I \times \mathcal{A} < \mathbb{N}}$ is closed under finite modification, then C does not satisfy PEV under $D = \neg \text{POF}_C$.*

Proof. PEV demands that all worlds in C satisfy PV: $\forall \theta, \exists \theta' \sqsubseteq \theta' \notin (D \cap \overline{C})^\subseteq$. In other words, that $\text{POF}_{D \cap \overline{C}}(\rightsquigarrow)$. By theorem 17, $\forall \rightsquigarrow : \neg \text{POF}_C(\rightsquigarrow)$, thus $D \cap \overline{C} = \overline{C}$.

As $\sim_{\text{fin}}$ is an equivalence relation, C is closed under finite modification iff $\overline{C}$ is closed under finite modification. Therefore, by theorem 17 again, $\forall \rightsquigarrow : \neg \text{POF}_{\overline{C}}(\rightsquigarrow)$. $\square$

Conjecture 13 (PEV outside the trivial cases). *If C is not closed under finite modification, then C does not satisfy PEV under $D = \neg \text{POF}_C$.*

Both $\emptyset$ and $P^{I \times \mathcal{A} < \mathbb{N}}$ are, trivially, closed under finite modification (definition 3 holds of them vacuously), so conjecture 13's hypothesis already excludes them: by theorem 18, C is provable by $\neg \text{POF}_C$ iff $C \in \{\emptyset, P^{I \times \mathcal{A} < \mathbb{N}}\}$, and provability by D already implies PEV under D (corollary 7); so whenever C is not closed under finite modification, C is in particular nontrivial, hence not provable by $\neg \text{POF}_C$ either. Conjecture 13 asks whether PEV can nonetheless survive where provability cannot.

A single hypothesis, " C not closed under finite modification," already rules out the two known dead ends at once: it excludes triviality directly (both trivial claims are cufm, as just noted), and it excludes the case where C is nonempty and cufm, which forces $D = P^{I \times \mathcal{A} < \mathbb{N}}$ by theorem 17 (since C is then never even falsifiable, hence never POF_C). It does not, however, rule out *every* route to $D = P^{I \times \mathcal{A} < \mathbb{N}}$: example 7 is never falsifiable without being closed under finite modification (remark 12 — flipping a single

individual's two values already leaves C), so such claims remain within conjecture 13's scope. PEV is expected to fail there for the same underlying reason as the cufm case: no finite prefix can ever force membership in C .

Proving the conjecture directly is awkward, since “ C fails PEV” unpacks to a $\exists \rightsquigarrow, \exists \theta, \forall \theta'$ statement: ruling out *every* rightward extension θ' , for some world and some prefix, is hard to attack head-on. We instead argue by contradiction: fix C not closed under finite modification (so $C \notin \{\emptyset, P^{I \times \mathcal{A}^{<\mathbb{N}}}\}$, hence $D \setminus C \neq \emptyset$ by theorem 18), assume toward contradiction that C satisfies PEV under D , and aim for an outright impossibility. This hands us an actual escaping witness to reason about — one guaranteed to exist by the assumption — rather than requiring an upfront construction ruling out all of them. Lemma 11 extracts the first consequence forced on this witness; settling conjecture 13 means either pushing that consequence to an outright impossibility, completing the reductio, or finding a case where it can be consistently satisfied after all — producing a genuine example of a non-cufm claim satisfying PEV, and refuting the conjecture instead.

Lemma 11 (Escaping PEV witnesses must diverge). *Suppose C is not closed under finite modification and satisfies PEV under $D = \neg \text{POF}_C$; let $\rightsquigarrow^{(0)} \in D \setminus C$ be arbitrary ($D \setminus C \neq \emptyset$, since C not cufm gives $C \notin \{\emptyset, P^{I \times \mathcal{A}^{<\mathbb{N}}}\}$, hence $D \setminus C \neq \emptyset$ by theorem 18). As $\rightsquigarrow^{(0)} \in D = \neg \text{POF}_C$, some $\theta_1 \in \Theta \cap \mathcal{P}(\rightsquigarrow^{(0)})$ witnesses $\neg \text{POF}_C(\rightsquigarrow^{(0)})$ (a hiding prefix, echoing the visibility-issues terminology above): every rightward $\theta_1 \subseteq \theta' \subseteq \rightsquigarrow^{(0)}$ satisfies $\theta' \in C^\subseteq$. In particular, the trivial case $\theta' = \theta_1$ gives $\theta_1 \in C^\subseteq$: pick any completion $\theta_1 \subseteq \rightsquigarrow$, $\rightsquigarrow \in C$. Now $\rightsquigarrow \in C \cap D$, so this is where the PEV hypothesis is actually used: testing it at $\rightsquigarrow$ for the prefix θ_1 gives a rightward extension $\theta_1 \subseteq \theta'_1 \subseteq \rightsquigarrow$, with $\theta'_1 \notin (D \setminus C)^\subseteq$. This θ'_1 satisfies $\theta'_1 \not\subseteq \rightsquigarrow^{(0)}$.*

Proof. If $\theta'_1 \subseteq \rightsquigarrow^{(0)}$, then $\rightsquigarrow^{(0)}$ is a completion of θ'_1 lying in $D \setminus C$, contradicting $\theta'_1 \notin (D \setminus C)^\subseteq$. $\square$

The most direct route to conjecture 13 would go as follows. $\rightsquigarrow^{(0)}$ is one specific witness of how bad $D \setminus C$ is: whatever finite amount of $\rightsquigarrow^{(0)}$'s own values, beyond θ_1 , an observer is shown, this always remains compatible with C — that is exactly what makes θ_1 a *hiding* prefix for $\rightsquigarrow^{(0)}$. So a natural hope would be that PEV's own escaping witness, tested at some $\rightsquigarrow \in C$ extending θ_1 , could simply be built by tracking $\rightsquigarrow^{(0)}$'s pattern itself: reveal more and more of what $\rightsquigarrow^{(0)}$ would say beyond θ_1 , and hope this eventually excludes $D \setminus C$.

Lemma 11 rules this hope out directly. If the escaping witness θ'_1 merely reproduced $\rightsquigarrow^{(0)}$'s own values beyond θ_1 (that is, $\theta'_1 \subseteq \rightsquigarrow^{(0)}$), then $\rightsquigarrow^{(0)}$ itself would be a completion of θ'_1 ; as $\rightsquigarrow^{(0)} \in D \setminus C$, this would mean θ'_1 fails to exclude $D \setminus C$ — contradicting the very thing θ'_1 was supposed to achieve. So the escaping witness cannot simply reproduce what $\rightsquigarrow^{(0)}$ hides behind θ_1 : at some point beyond θ_1 , it must show a value genuinely different from what $\rightsquigarrow^{(0)}$ would show there, actively diverging from it rather than passively following it.

Two routes to settling conjecture 13 remain open: a compactness argument, iterating the construction of lemma 11 along an increasing sequence of prefixes agreeing with $\rightsquigarrow^{(0)}$

further and further, to show the required divergence cannot be sustained indefinitely; and a direct construction, a claim C , not closed under finite modification, for which $D \setminus C$ is spread across many, individually escapable, worlds rather than concentrated in one $\rightsquigarrow^{(0)}$, so that every world in C can use a private, ever-available escape route. A König-type argument for the first route does not directly apply: the tree of rightward extensions is not finitely branching whenever I is infinite, since introducing a fresh individual is itself an unbounded branching choice; a compactness argument here would need a different tool, such as an ultrafilter construction over I . Even that alternative is not straightforward, and for a sharper reason than branching: $P^{I \times \mathcal{A}^{<\mathbb{N}}} = P^{I \times \mathcal{A}^{<\mathbb{N}}}$ is compact (Tychonoff, as P is finite), but standard compactness arguments — net convergence, or the finite intersection property for closed families — require the family of test extensions to be *directed*, meaning any two admit a common refinement. For a fixed $\rightsquigarrow^{(0)}$, two valid rightward extensions of the same prefix, both in $\Theta \cap \mathcal{P}(\rightsquigarrow^{(0)})$, can commit to different, incomparable branches for the same individual (extending, say, (a) to (a, a) in one and to (a, a') in the other); their union is not chain-respecting, hence not itself in Θ . So the family of Θ -valid rightward extensions of a prefix is not directed, even restricted to a single fixed $\rightsquigarrow^{(0)}$, and restricting $\mathcal{A}$ to be finite does not repair this: the non-directedness comes from any two incomparable branches existing at all, not from unboundedly many of them. This means $\neg \text{POF}_S$ is not literally a topological closure condition to begin with, since it tests strictly fewer, non-directed extensions than compactness arguments assume — precisely because Θ encodes that incomparable branches cannot be probed simultaneously, the realistic-protocol constraint the whole framework is built on. A genuine compactness proof of conjecture 13, if one exists, would need a substitute for both König’s lemma and standard filter or net arguments.

Example 64 (A non-degenerate $D \setminus C$). Every example checked so far was closed under finite modification, hence excluded by conjecture 13’s hypothesis outright (trivially so for $\emptyset$ and $P^{I \times \mathcal{A}^{<\mathbb{N}}}$ themselves, giving $D = C$; nontrivially so otherwise, giving $D = P^{I \times \mathcal{A}^{<\mathbb{N}}}$ by theorem 17); this one is not. Fix infinitely many individuals $i_1, i_2, \dots \in I$ and let $C = \{\rightsquigarrow \mid \forall n : \rightsquigarrow(i_n, \emptyset) = \varphi\}$: not closed under finite modification (flipping a single $\rightsquigarrow(i_n, \emptyset)$ already leaves C), yet still falsifiable or trivial ($\neg \text{OF}_C = C$). For $\rightsquigarrow \notin C$ failing at only finitely many individuals, an adversarial prefix can touch each failing individual at some other argument first, blocking recovery of its $\emptyset$ -value at each, and $\rightsquigarrow$ offers no other route: $\text{POF}_C(\rightsquigarrow)$ fails. For $\rightsquigarrow \notin C$ failing at infinitely many individuals, no finite prefix can touch them all, so some failing individual always remains fresh and reachable: $\text{POF}_C(\rightsquigarrow)$ holds. Hence $D = \neg \text{POF}_C = C \cup \{\rightsquigarrow \mid \rightsquigarrow \text{ fails } C \text{ at finitely many, at least one, individuals}\}$, genuinely strictly between C and $P^{I \times \mathcal{A}^{<\mathbb{N}}}$.

Yet PEV (already EV) collapses on this D : $D \setminus C$ is exactly “fails at finitely many individuals,” and any finite θ' only constrains finitely many of them, so a completion matching θ' exactly and failing at one further, untouched individual always exists — $\theta' \in (D \setminus C)^\subseteq$ regardless of θ' . So $\text{OF}_{D \setminus C}(\rightsquigarrow)$ already fails at every $\rightsquigarrow \in C$: PEV fails here too, confirming conjecture 13 through a genuinely non-degenerate $D \setminus C$. $\triangle$

Remark 49 (A duality behind the recurring collapse). A pattern underlies every example

checked: POF_C is hard to satisfy off C precisely when $\overline{C}$ offers no witness that stays redundantly available no matter how a prefix is spent — and that same difficulty is what makes $D \setminus C = (\neg \text{POF}_C) \setminus C$ rich, since it collects exactly what escaped robust exclusion. A set that is rich in that sense is, symmetrically, hard for any finite θ' to exclude. So the same mechanism — finite information cannot pin down infinitely many independent possibilities — cuts both ways: it is what keeps $D \setminus C$ non-empty whenever C is nontrivial (theorem 18), and, on the same worlds, what stops that same $D \setminus C$ from ever being excludable by a finite prefix (breaking PEV). Making this duality precise, rather than case by case, would settle conjecture 13. $\triangle$

Remark 50 (The singleton case). A tractable test: could $D \setminus C$ ever be a single point $\{\rightsquigarrow^{(0)}\}$? $\text{POF}_{\{\rightsquigarrow^{(0)}\}}(\rightsquigarrow)$, for $\rightsquigarrow \in C$, only requires every prefix to be rightward-extendable to some disagreement with $\rightsquigarrow^{(0)}$. If $\rightsquigarrow$ differs from $\rightsquigarrow^{(0)}$ at only finitely many points, an adversarial prefix can permanently block each of them (touching, for each differing individual, some other history first — always available once $\mathcal{A}$ has at least two elements, even for a difference located at the root $\emptyset$). So PV can only survive at an $\rightsquigarrow \in C$ that differs from $\rightsquigarrow^{(0)}$ at infinitely many independent individuals, mirroring the “fresh reserve” move used in the proof of lemma 11 — and PEV under D requires this of *every* $\rightsquigarrow \in C$, not just one.

This looks structurally difficult to arrange: $D \setminus C = \{\rightsquigarrow^{(0)}\}$ means C excludes only one point, which typically leaves plenty of “near misses” inside C — worlds differing from $\rightsquigarrow^{(0)}$ at only finitely many coordinates, such as a single flipped coordinate — exactly the worlds at which PV was just shown to fail. So a singleton $D \setminus C$ seems to force the very structural conflict described by remark 49: the infinite redundancy PEV needs everywhere is in tension with $D \setminus C$ being minimal. This is not yet a proof that $D \setminus C$ cannot be a singleton for a C not closed under finite modification, only a further, structurally distinct, piece of evidence for it. $\triangle$

Lemma 12 (Near misses force the escape witness through the flip). *Suppose $D = \neg \text{POF}_C$ and $D \setminus C = \{\rightsquigarrow^{(0)}\}$ exactly, with hiding prefix $\theta_1 \subseteq \rightsquigarrow^{(0)}$. Given a fresh individual $i^* \in I$ (untouched by θ_1), a history h^* , and a value $v \neq \rightsquigarrow^{(0)}(i^*, h^*)$, let $\rightsquigarrow^*$ agree with $\rightsquigarrow^{(0)}$ everywhere except $\rightsquigarrow^*(i^*, h^*) = v$. If $\rightsquigarrow^* \notin C$ (hence $\rightsquigarrow^* \neq \rightsquigarrow^{(0)}$ forces $\text{POF}_C(\rightsquigarrow^*)$, as $D \setminus C$ has no second element), then any rightward $\theta_1 \subseteq \psi$, $\psi \subseteq \rightsquigarrow^*$, witnessing $\text{POF}_C(\rightsquigarrow^*)$ (that is, $\psi \notin C^\subseteq$) must touch (i^*, h^*) .*

Proof. As $(i^*, h^*) \notin \theta_1(P)$, $\theta_1 \subseteq \rightsquigarrow^*$. If ψ did not touch (i^*, h^*) , then $\psi \subseteq \rightsquigarrow^{(0)}$ (it would agree with $\rightsquigarrow^*$ everywhere it does touch, and $\rightsquigarrow^*$ agrees with $\rightsquigarrow^{(0)}$ off (i^*, h^*)), and ψ is a rightward extension of θ_1 within $\rightsquigarrow^{(0)}$; the hiding-prefix property then gives $\psi \in C^\subseteq$, contradicting $\psi \notin C^\subseteq$. $\square$

Lemma 12 pins down where the escape witness for each near miss must go, but not yet an outright contradiction: reverting ψ 's value at (i^*, h^*) back to $\rightsquigarrow^{(0)}$'s gives a different finite specification, one already known to lie in $C^\subseteq$ via $\rightsquigarrow^{(0)}$ itself, and C remains free to be sensitive to that single coordinate — so this alone does not rule out the singleton case. Closing this gap, or exhibiting a genuine example of a claim, not closed under finite

modification, that satisfies PEV instead, remains open; three independent attempts at the latter (remark 49’s pattern, an OR-cell conjunction across individuals, and an infinitely-often tail claim) have all instead made C itself unfalsifiable, collapsing D to $P^{I \times \mathcal{A}^{<\mathbb{N}}}$ before the intended tension could be set up — the same kind of collapse discussed above, whether or not C happened to be cufm this time (theorem 17 and remark 12) — a third instance of the same failure mode, and further indirect evidence for conjecture 13.

Lemma 13 (Excluding a finite set needs individually-unblockable disagreement). *Given a finite $S = \{\rightsquigarrow^{(0)}, \dots, \rightsquigarrow^{(k-1)}\} \subseteq P^{I \times \mathcal{A}^{<\mathbb{N}}}$ and $\rightsquigarrow \notin S$: if $\text{POF}_S(\rightsquigarrow)$ holds, then for each j , $\rightsquigarrow$ differs from $\rightsquigarrow^{(j)}$ at infinitely many distinct individuals.*

Proof. Fix j , and suppose instead $\rightsquigarrow$ differs from $\rightsquigarrow^{(j)}$ at only finitely many individuals. An adversarial prefix can, for each such individual, touch it on a branch other than the one carrying its disagreement with $\rightsquigarrow^{(j)}$ there, permanently blocking recovery of that disagreement (available as soon as $\#\mathcal{A} \geq 2$, even for a disagreement located at the root $\emptyset$). After this prefix, $\rightsquigarrow^{(j)}$ remains a valid completion of every further rightward extension, so no such extension can exclude $\rightsquigarrow^{(j)}$, hence none can exclude S . $\square$

Lemma 13 generalizes the single-point mechanism behind remark 50 to any finite target: whenever $D \setminus C$ is finite, PEV can only hold by requiring every world in C to differ from *every* point of $D \setminus C$ at infinitely many individuals at once — a demand no construction tried so far can reconcile with $D \setminus C$ actually being finite. Two further, structurally distinct attempts to isolate a single hiding point — the complement of a single point, $C = P^{I \times \mathcal{A}^{<\mathbb{N}}} \setminus \{\rightsquigarrow^{(0)}\}$, and the claim “differs from $\rightsquigarrow^{(0)}$ at exactly one coordinate” — both collapse instead of isolating $\rightsquigarrow^{(0)}$. The latter yields a clean bonus characterization: exclusion of a single point by a cover of “exactly n differences”-style claims holds exactly when the disagreement spans infinitely many individuals, turning lemma 13’s necessary condition into an iff in that case — but $D \setminus C$ still comes out rich rather than a singleton, and PEV collapses on it just as in example 64. Five independent constructions now confirm conjecture 13 without exception: not one claim, not closed under finite modification, has been found to satisfy PEV under $\neg \text{POF}_C$; a general proof or a genuine example satisfying PEV both remain open.

References

- D. E. Bell, H. Raiffa, and A. Tversky, editors. *Descriptive, normative, and prescriptive interactions in decision making*, pages 9–30. Cambridge University Press, 1988. doi:10.1017/CBO9780511598951.003.
- O. Cailloux and Y. Meinard. A formal framework for deliberated judgment. *Theory and Decision*, Mar. 2020. ISSN 1573-7187. doi:10.1007/s11238-019-09722-7.
- J. S. Fishkin. *When the People Speak: Deliberative Democracy and Public Consultation*. Oxford University Press, 2011. ISBN 978-0-19960-443-2. doi:10.1093/acprof:osobl/9780199604432.001.0001.

- N. Goodman. *Fact, Fiction, and Forecast*. Harvard University Press, 1983. URL <http://www.hup.harvard.edu/catalog.php?isbn=9780674290716>.
- D. Kahneman. *Thinking, fast and slow*. Penguin Press, 2012. ISBN 978-0-141-03357-0.
- S. Lichtenstein and P. Slovic. Reversals of preferences between bids and choices gambling decisions. In S. Lichtenstein and P. Slovic, editors, *The Construction of Preference*, pages 52–68. Cambridge University Press, 1 edition, 2006. ISBN 0521834287.
- Y. Meinard and O. Cailloux. On justifying the norms underlying decision support. *European Journal of Operational Research*, 2020. doi:10.1016/j.ejor.2020.02.022. In press.
- P. Railton. *Facts, values, and norms: Essays toward a morality of consequence*. Cambridge University Press, 2003.
- J. Rawls. *A Theory of Justice*. Harvard University Press, 2nd edition, 1999. URL <http://www.hup.harvard.edu/catalog.php?isbn=9780674000780>.
- P. A. Samuelson. *Foundations of Economic Analysis*. Harvard University Press, enlarged edition, 1983. ISBN 978-0-674-31303-3. URL <https://www.hup.harvard.edu/catalog.php?isbn=9780674313033>.
- R. H. Thaler and C. R. Sunstein. *Nudge: Improving decisions about health, wealth, and happiness*. Penguin, 2009. ISBN 978-0-141-04001-1. URL <https://www.penguin.co.uk/books/56784/nudge>.